PETW Whole Project Technical Report

A Multi-Agent Human-in-the-loop Scientific Research Framework from Thunderstorm-Wind Event Observation to Design Load

English v2 Final Revision Candidate

PETW Coordinator (independent review and integration)

Scientific status: `not_scientifically_approved`

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Abstract

This study establishes the PETW (Parametric Engineering Thunderstorm Wind) scientific framework, which organizes the engineering problem of thunderstorm winds along seven scientific steps: event observation, event classification, meteorological simulation, wind field characterization, extreme value analysis, aerodynamic and structural response, and design loads. PETW adopts physical mechanisms and event classification as its organizing principles, with Multi-Agent Human-in-the-loop as its implementation method layer. This study describes a layered architecture consisting of one Coordinator, nine scientific agents, and fifteen support/collaboration agents, as well as the Researcher's continuous participation in research question formulation, method selection, remote computation authorization, result interpretation, and scientific conclusion formation.

The report provides a continuous technical narrative covering the completed work, current evidence status, and unresolved gaps of nine scientific lanes: Events, CM1, WRF, Windfield, EVA, WindTunnel, CFD, Structural, and DesignLoad, and presents multi-case combined workflow evaluations from events to design loads. The full manuscript retains the principal scientific limitations of each lane: CM1 results are model-internal Level-1 screening evidence, WRF reconstruction status is `PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION`, EVA has no formal event samples or return levels, WindTunnel is in `DESIGN_ONLY_NOT_ACTIVE` status, CFD results are bounded exploratory results, and DesignLoad is candidate interpretation.

The status of the full manuscript is `not_scientifically_approved`. Correct framework, target word count, number of figures, or successful compilation do not constitute scientific approval.

Contents

I PETW Scientific Framework **9**

1 Introduction and related research **9**
1.1 Problem, concepts, and methods . . . 9
1.1.1 Background of the thunderstorm wind engineering problem . . . 9
1.1.2 Evidence accumulation across observation, simulation, and engineering applications . . . 10
1.1.3 Scientific reading approach for this study report . . . 11

2 Event classification and scientific status description **12**
2.1 Problem, concepts, and methods . . . 12
2.1.1 Classification-first principle . . . 12
2.1.2 Six levels of classification . . . 12
2.1.3 Researcher claim gate for classification outputs . . . 13

3 Scientific process from Step 0 to Step 5 **14**
3.1 Problem, concepts, and methods . . . 14
3.1.1 Step 0: Events, event classification and identification . . . 14
3.1.2 Step 5: DesignLoad, design loads and code interface . . . 15
3.1.3 Minimum reporting unit for process status . . . 15

4 Uncertainty and scope of scientific interpretation **16**
4.1 Problem, concepts, and methods . . . 16
4.1.1 Nine categories of uncertainty across PETW stages . . . 16
4.1.2 Cross-step propagation of uncertainty . . . 18
4.1.3 Research significance of open states . . . 18

II Agentic Pipeline and Human-in-the-loop Research **19**

5 Definition and composition of agents **20**
5.1 Evidence, results, and interpretation . . . 20
5.1.1 Definition of agents in PETW . . . 20
5.1.2 Ten components of an agent . . . 21
5.1.3 Position of agents in the evidence chain . . . 21

6 Comparison with traditional research approaches **22**
6.1 Evidence, results, and interpretation . . . 22
6.1.1 Actual form of traditional working methods . . . 22
6.1.2 Applicable advantages and practical limitations . . . 23
6.2 Limitations, interfaces, and Researcher judgment . . . 24
6.2.1 Differences in handling failure states and open questions . . . 24

7 Natural language-driven research work **24**
7.1 Evidence, results, and interpretation . . . 24
7.1.1 Ten elements of a scientific task . . . 24
7.1.2 Five intervention periods of the Researcher . . . 25
7.2 Limitations, interfaces, and Researcher judgment . . . 26
7.2.1 Applicable conditions and limitations of the natural language-driven approach 26

8 Multi-model collaboration **26**
8.1 Evidence, results, and interpretation . . . 26
8.1.1 Basis for selecting main models . . . 26
8.1.2 Role division among models . . . 27
8.1.3 Role isolation of generation, review, and editing . . . 27

9 Human-in-the-loop research process **28**
9.1 Problem, concepts, and methods . . . 28
9.1.1 Full scientific responsibility of the Researcher . . . 28
9.1.2 Criteria for layered intervention . . . 29
9.1.3 Governance value of open states . . . 29

III PETW Agent Architecture and Shared Research Resources **30**

10 Layered architecture of PETW **30**
10.1 Problem, concepts, and methods . . . 30
10.1.1 Layering principles . . . 30
10.1.2 Support and collaboration agents . . . 30
10.1.3 Scientific role boundaries of the project records layer . . . 32

11 Knowledge base of each agent **32**
11.1 Problem, concepts, and methods . . . 32
11.1.1 Prerequisite status of knowledge preparation . . . 32
11.1.2 Five categories of knowledge resources and their provenance chains . . . 33
11.1.3 Distinction between knowledge coverage and scientific sufficiency . . . 34

12 Usage and coordination of agents **34**
12.1 Problem, concepts, and methods . . . 34
12.1.1 Workflow for using a single agent . . . 34
12.1.2 Information flow among scientific agents . . . 35
12.1.3 Handoff semantics, fail-closed principle, and cross-lane consistency . . . 36

IV Detailed Agent Descriptions and Results **37**

13 PETW Coordinator **37**
13.1 Evidence, results, and interpretation . . . 37

13.1.1 Responsibilities and scientific positioning . . . 37
13.1.2 Continuous interaction with the Researcher . . . 37
13.1.3 Current issues and future work . . . 38

14 Events Agent **38**
14.1 Scientific questions, knowledge, and Researcher decisions . . . 38
14.1.1 Scientific purpose . . . 38
14.1.2 Knowledge base . . . 39
14.1.3 Researcher participation and decisions . . . 40
14.2 Methods, results, and figures . . . 40
14.2.1 Workflow and interactions . . . 40
14.2.2 Completed work and main results . . . 42
14.2.3 Figures . . . 44
14.3 Limitations, diagnostics, and interfaces . . . 46
14.3.1 Current scientific limitations and future research . . . 46
14.3.2 Evidence association, detectability, and downstream qualification . . . 46

15 CM1 Agent **52**
15.1 Scientific questions, knowledge, and Researcher decisions . . . 52
15.1.1 Scientific purpose . . . 52
15.1.2 Knowledge base . . . 52
15.1.3 Researcher participation and decisions . . . 53
15.2 Methods, results, and figures . . . 53
15.2.1 Workflow and interactions . . . 53
15.2.2 Completed work and main results . . . 54
15.2.3 Figures . . . 58
15.3 Limitations, diagnostics, and interfaces . . . 58
15.3.1 Current scientific limitations and future research . . . 58
15.3.2 Diagnostic tiers, comparison design, and interface closure . . . 58

16 WRF Agent **70**
16.1 Scientific Questions, Knowledge, and Researcher Decisions . . . 70
16.1.1 Scientific Purpose . . . 70
16.1.2 Knowledge Base . . . 70
16.1.3 Researcher Participation and Decisions . . . 70
16.2 Methods, Results, and Figures . . . 71
16.2.1 Workflow and Interactions . . . 71
16.2.2 Schumacher 2023 Reconstruction Configuration . . . 71
16.2.3 Reconstruction Results and Literature Comparison . . . 74
16.3 Limitations, Diagnostics, and Interfaces . . . 77
16.3.1 Current Scientific Limitations and Future Research . . . 77
16.3.2 Hierarchy of Reconstruction Diagnostics and Error Attribution . . . 77

17 Windfield Agent **84**
17.1 Scientific Questions, Knowledge, and Researcher Decisions . . . 84
17.1.1 Scientific Purpose . . . 84
17.1.2 Knowledge Base . . . 84
17.1.3 Researcher Participation and Decisions . . . 84
17.2 Methods, Results, and Figures . . . 85
17.2.1 Workflow and Interactions . . . 85
17.2.2 Multi-scale Wind Field Components and Methods . . . 85
17.2.3 Current Scientific Limitations and Future Research . . . 90
17.3 Limitations, Diagnostics, and Interfaces . . . 90
17.3.1 Scale Transfer, Data Contract, and Comparability Gates . . . 90
17.3.2 Packet Observation Operators and Reproduction Boundaries . . . 92

18 EVA Agent **95**
18.1 Scientific Questions, Knowledge, and Researcher Decisions . . . 95
18.1.1 Scientific Purpose . . . 95
18.1.2 Knowledge Base . . . 96
18.1.3 Researcher Participation and Decisions . . . 96
18.2 Methods, Results, and Equations . . . 96
18.2.1 Main Methods . . . 96
18.2.2 Wind Speed Quantity Definition and Standardization . . . 96
18.2.3 Mechanism-aware Event Population and Observational Process . . . 97
18.2.4 Candidate Extreme-value Routes . . . 98
18.2.5 Conditional Synthesis for Mixed Wind Climates . . . 99
18.2.6 Six Categories of Uncertainty . . . 99
18.2.7 Equations . . . 99
18.2.8 Completed Work . . . 102
18.3 Evidence Audit, Interfaces, and Limitations . . . 104
18.3.1 Current Scientific Limitations and Future Research . . . 104
18.3.2 Admission Schema and Statistical Units . . . 104
18.3.3 Endpoint Conversion and Model Route Diagnostics . . . 105
18.3.4 Uncertainty Ledger and Stopping Rules . . . 106

19 WindTunnel Agent **109**
19.1 Scientific Questions, Knowledge, and Researcher Decisions . . . 109
19.1.1 Scientific Purpose . . . 109
19.1.2 Knowledge Base . . . 110
19.1.3 Researcher Participation and Decisions . . . 111
19.2 Experiment Design, Measurement, and Literature Evidence . . . 112
19.2.1 Model and Tap Design . . . 112
19.2.2 Target Velocity Time Histories and YM Conditions . . . 112
19.2.3 Instruments, Synchronization, and PIV . . . 115

19.2.4 Separation-reattachment Mechanism and Time Cells . . . 115
19.2.5 Literature Evidence . . . 116
19.3 Stage Gates, Data Products, and Limitations . . . 117
19.3.1 G0 through G9 Stage Gates . . . 117
19.3.2 Completed Work . . . 120
19.3.3 Verification Path and Planned Data Products . . . 120
19.3.4 Current Scientific Limitations and Future Research . . . 121

20 CFD Agent **127**
20.1 Scientific Questions, Knowledge, and Researcher Decisions . . . 127
20.1.1 Scientific Purpose . . . 127
20.1.2 Knowledge Base . . . 128
20.1.3 Researcher Involvement and Decisions . . . 129
20.2 Methods, Results, and Formulas . . . 129
20.2.1 Workflow and Interactions . . . 129
20.2.2 Completed Work and Results . . . 130
20.2.3 Key Formulas . . . 132
20.3 Evidence Audit, Interfaces, and Limitations . . . 135
20.3.1 Refinement of Input Contracts, Numerical Diagnostics, and Evidence Transfer 135
20.3.2 Interpretation Framework for Case-Level Diagnostics . . . 141
20.3.3 Current Scientific Limitations and Future Research . . . 144

21 Structural Agent **144**
21.1 Scientific Questions, Knowledge, and Researcher Decisions . . . 145
21.1.1 Scientific Purpose . . . 145
21.1.2 Knowledge Base . . . 145
21.1.3 Researcher Involvement and Decisions . . . 145
21.2 Methods, Results, and Formulas . . . 146
21.2.1 Workflow and Interactions . . . 146
21.2.2 Completed Work and Results . . . 147
21.2.3 Key Formulas . . . 148
21.3 Evidence Audit, Interfaces, and Limitations . . . 151
21.3.1 Load Identity, Response Targets, and Verification Levels . . . 151
21.3.2 Per-Item Interpretation of Model-Internal Benchmark . . . 155
21.3.3 Current Scientific Limitations and Future Research . . . 158

22 DesignLoad Agent **158**
22.1 Scientific Questions, Knowledge, and Researcher Decisions . . . 159
22.1.1 Scientific Purpose . . . 159
22.1.2 Knowledge Base . . . 159
22.1.3 Researcher Involvement and Decisions . . . 159
22.2 Methods, Results, and Formulas . . . 160
22.2.1 Workflow and Interactions . . . 160

22.2.2 Completed Work and Results . . . 161
22.2.3 Key Formulas . . . 164
22.3 Evidence Audit, Interfaces, and Limitations . . . 165
22.3.1 Evidence Ownership, Closure Levels, and Audit of Candidate Expressions . . . 165
22.3.2 Conditioned Interpretation of Branch-Level Results . . . 171
22.3.3 Current Scientific Limitations and Future Research . . . 173

23 Knowledge Agents 174
23.1 Evidence, Results, and Interpretation . . . 175
23.1.1 Common Purpose and Inter-Layer Division of Labor . . . 175
23.1.2 The Cycle of Knowledge Entering Research . . . 176
23.1.3 Relationship with the Researcher . . . 177

24 Writing and Communication Agents 179
24.1 Limitations, Interfaces, and Researcher Judgment . . . 179
24.1.1 Manuscript and Figures . . . 179
24.1.2 Daily Briefing and External Collaboration Channels . . . 179
24.1.3 Researcher Involvement . . . 180

25 Computation, Remote and Project Support Agents 181
25.1 Evidence, Results, and Interpretation . . . 181
25.1.1 Common Role and Functional Classification . . . 181
25.1.2 Support Relationships with Scientific Agents . . . 182
25.1.3 Researcher Authorization and Decisions . . . 182

V Integrated PETW Workflows and Evaluation 185

26 Complete workflow from events to design loads 186
26.1 Evidence, results and interpretation . . . 186
26.1.1 Pipeline interface structure and transfer conditions . . . 186
26.1.2 Current case combinations and end-to-end coverage . . . 188
26.2 Limitations, interfaces and Researcher judgment . . . 189
26.2.1 Distinction between reproducibility and the scientific framework . . . 189

27 Complete workflow of knowledge, computation, writing and external review 191
27.1 Evidence, results and interpretation . . . 192
27.1.1 Knowledge workflow . . . 192
27.1.2 Computation workflow . . . 193
27.1.3 Closed-loop handling of failures, conflicts and review comments . . . 193

28 Practical value of the agentic approach for PETW 195
28.1 Problems, concepts and methods . . . 195
28.1.1 Practical role in the continuity of scientific steps . . . 195

28.1.2 Researcher focus . . . 195
28.1.3 Currently supportable value conclusions . . . 196

VI Concluding Remarks and Outlook 197

29 Current understanding 197
29.1 Limitations, interfaces and Researcher judgment . . . 197
29.1.1 Scientific framework level . . . 197
29.1.2 Evidence status of each work lane . . . 198
29.1.3 Research methodology level . . . 200

30 Near-term research directions 201
30.1 Evidence, results and interpretation . . . 201
30.1.1 Expansion and verification of event types . . . 201
30.1.2 Conditional definition of Windfield scale conversion . . . 201
30.2 Limitations, interfaces and Researcher judgment . . . 202
30.2.1 Near-term handling of uncertainties and conflicts . . . 202

31 Improvements to the Human-in-the-loop approach 203
31.1 Limitations, interfaces and Researcher judgment . . . 205
31.1.1 Decision gate grading and consolidation . . . 205
31.1.2 Boundary between automated checks and scientific review . . . 205
31.1.3 Verification of improvement effectiveness . . . 206

32 Extensions beyond PETW 207
32.1 Evidence, results and interpretation . . . 207
32.1.1 Structural conditions for method applicability . . . 207
32.1.2 Potentially applicable research fields . . . 208
32.1.3 Limitations of PETW as a case study . . . 208

33 Concluding remarks 209

Part I
PETW Scientific Framework

1 Introduction and related research

1.1 Problem, concepts, and methods

1.1.1 Background of the thunderstorm wind engineering problem

Extreme wind loads are among the controlling factors in the safe design of buildings and infrastructure. In mid-latitude regions worldwide, near-surface strong winds produced by thunderstorms are an important component of the extreme wind climate; their proportion in the annual maximum gust sample varies with region and height, but extensive records exist in areas such as the U.S. central plains, eastern Australia, the South African plateau, and the middle and lower reaches of the Yangtze River in China. Compared with synoptic wind systems that persist for hours to days, thunderstorm outflows exhibit four distinct characteristics: short duration, with strong winds typically lasting from minutes to tens of minutes; strong non-stationarity, with wind speed progressing from the background level to a peak and then decaying within tens of seconds to minutes; pronounced spatial locality, with the horizontal scale of the strong-wind core typically ranging from hundreds of meters to several kilometers; and diverse parent-storm organization, encompassing isolated convective cells, multicell systems, squall lines, and bow echoes that produce different near-surface wind mechanisms. Near-surface wind mechanisms can vary among locations within the same event: the divergent outflow of a downburst forms a radially expanding wind field near the touchdown point, the leading-edge structure of a gust front is characterized by frontal lifting and the horizontal advance of cold air, momentum transfer caused by gravity-wave breaking can produce brief strong winds far from the precipitation core, and the superposition of these mechanisms with the background environmental wind further alters the spatiotemporal pattern of near-surface wind direction and velocity profiles.

The traditional unified framework developed for stationary synoptic winds, including a single averaging time (typically 10 minutes or 1 hour), a single vertical profile (logarithmic or power law), and a single extreme value population (annual maxima or threshold exceedances), cannot fully represent these differences. The stationarity assumption requires the statistical properties of wind speed to remain constant within the selected time window, whereas the rate of change in the mean thunderstorm-outflow wind speed before and after the peak can be far greater than the fluctuation variance within that window. The single-profile assumption presumes that near-surface wind speed increases monotonically with height, whereas the vertical profile of a downburst increases below the low-level jet height and decreases above it, with the peak height typically below 100 meters and as low as 30 to 50 meters in strong events. The single-population assumption presumes that all annual maxima arise from the same physical process, whereas the tail behavior of different mechanisms may differ significantly: the extreme value distribution of synoptic winds typically approaches the Gumbel type (shape parameter tending to zero), while the possibility that the conditional tail of thunderstorm outflow has a different shape parameter remains open. These distinctions show that thunderstorm wind engineering requires a research framework centered on physical mechanisms and event types rather than on statistical tools.

1.1.2 Evidence accumulation across observation, simulation, and engineering applications

A body of interrelated, but individually limited, evidence has accumulated across observation, simulation, and engineering applications.

For observations, surface station networks (such as ASOS and the Oklahoma Mesonet in the United States), towers, Doppler radar networks (NEXRAD), satellite observations (GLM lightning mapping), and reanalysis products record thunderstorm events from different spatiotemporal scales. However, measurement heights, averaging times, instrument response, and spatial support all differ: the standard measurement height for ASOS stations is 10 meters, the reporting interval is 5 minutes, and the gust window has historically transitioned from an approximately 5-second equivalent response to a 3-second moving average; the Oklahoma Mesonet likewise operates at 10-meter height with 5-minute reporting intervals, where WMAX is defined as the maximum of 3-second observations within that 5-minute period, but this maximum is not equivalent to a recovered continuous 3-second time series. NEXRAD provides radial velocity information at beam scan heights rather than at the surface; beam height increases with range, and at distances of 50 to 100 kilometers the beam base can be above 1 to 3 kilometers, where low-level divergence structures may have already decayed significantly. MRMS (Multi-Radar Multi-Sensor) provides higher temporal resolution composite products, but its gridding process introduces additional spatial smoothing. Lightning mapping (GLM) characterizes parent convective activity and does not directly provide near-surface wind information. Because the amount of data recorded for the same event varies among measurement chains, an event's engineering wind speed cannot simply be obtained directly from any single source.

For simulations, high-resolution idealized models (such as CM1) can reproduce the dynamical structure of downbursts, including cold pool formation, downdraft touchdown, divergent outflow expansion, and ring vortex evolution, under controlled initial conditions, but the resulting evidence is constrained by two-dimensional or quasi-three-dimensional idealized setups, initial perturbation methods, and microphysics parameterization schemes; the output consists of model-internal quantities rather than directly usable engineering wind fields. Mesoscale models (such as WRF) can reconstruct the genesis and development of convective systems under real initial and boundary conditions, but resolution, physics schemes, lateral boundary update frequency, and model version collectively determine the reconstruction fidelity, and the accuracy of near-surface wind fields is typically lower than that of upper-level quantities. The complementary use of these two simulation types, CM1 for mechanism prototypes and WRF for real event reconstruction, is the basic organizational approach of the PETW scientific chain; the two do not replace each other but also cannot be simply spliced together.

For engineering applications, wind tunnel experiments can study surface pressures, aerodynamic forces, and flow field structure of building models under controlled inflow conditions, but traditional stationary wind tunnel experiments cannot directly simulate the non-stationarity of thunderstorm outflow; accelerating-flow facilities provide an experimental pathway for this purpose, but the correspondence between their target time histories, Reynolds number ranges, and boundary layer development and those of natural thunderstorm outflows must be specified item by item. Computational fluid dynamics (CFD) can study building-scale transient flow and pressure distributions, but inlet condition construction, turbulence treatment (RANS, URANS, or LES), grid resolution, and time step collectively determine the reliability of results, and differences in how

different turbulence models handle separation zones and near-wall regions can lead to significantly different local pressure coefficients. Structural dynamic analysis converts loads into displacements, accelerations, internal forces, and base reactions, but under non-stationary loading the definitions of equivalent static expressions, peak factors, and gust response factors require re-examination.

The organizational principle of PETW is to arrange this observational, simulation, and engineering evidence as a continuous scientific chain in which the output of each stage can serve as a bounded input to the next. Rather than a simple linear pipeline, this chain is a mutually constraining evidence network: the evidence ceiling of any downstream conclusion is determined by the weakest upstream stage.

1.1.3 Scientific reading approach for this study report

This report brings together literature synthesis, project status, model-internal results, engineering artifacts, experimental plans, and candidate scientific interpretations. Each object should be interpreted according to its source and status rather than inferring evidence level from section position. A design figure located in a results-type chapter may still be only generated input; a registry count located in a methods-type chapter may still be a data-supported project result; a process overview primarily expresses dependency relationships. Figure titles, body text, and ledgers use the same status terms to reduce misreading.

Numerical values in the report are not ranked by visual prominence. Run counts, file counts, grid counts, and QA pass counts describe the scale and completeness of engineering objects; velocity, pressure, force, response, and statistical parameters are the physical or statistical quantities; candidate category counts are further constrained by screening rules. Different numbers cannot substitute for one another. An abundance of engineering artifacts does not reduce the need for scientific verification; computability of physical quantities does not reduce the need for source identity and quantity contracts.

The chapters use forward-referencing interfaces but do not claim causality by narrative order. The chapters on Events, CM1, WRF, Windfield, EVA, WindTunnel, CFD, Structural, and DesignLoad describe their respective inputs, products, and stopping points; subsequent workflow chapters then summarize dependencies. Listing one chapter's output as a candidate input for another chapter means only that the variables and responsibility boundaries have been defined, not that handoff has been executed or approved by the Researcher.

The primary purpose of this report is therefore to provide a scientific working document for review. It presents existing evidence, engineering preparation, source conflicts, incomplete verifications, and the next decision together. The report does not elevate any scientific claim through length, number of figures, or successful compilation; these metrics only check whether expression and artifacts meet task requirements. Scientific adoption still depends on subsequent data, comparisons, and PI explicit decision.

Evidence and formula boundaries of Introduction and related research Chapter 1 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Introduction and related research." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 1 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

2 Event classification and scientific status description

2.1 Problem, concepts, and methods

2.1.1 Classification-first principle

PETW begins by classifying events and mechanisms rather than applying a unified statistical treatment. The classification-first principle reflects a fundamental requirement: the correct method selection at every subsequent step depends on mechanism identity. Different mechanisms have different vertical profile shapes; the low-level jet profile of a downburst and the logarithmic profile of synoptic wind lead to different floor force distributions in structural load calculations. Different mechanisms have different turbulence structures; the turbulence intensity profile at the leading edge of a gust front differs from that at the core of a divergent outflow, affecting peak factor estimation. Different mechanisms have different spatial footprints; the difference in spatial coherence and equivalent load width between the radially symmetric footprint of a divergent outflow and the linear leading edge of a gust front is significant. Different mechanisms require different simulation configurations; the idealized setup of CM1 is suited for mechanism prototype studies, while real event reconstruction with WRF requires completely different initial conditions and domain configurations. Different mechanisms require separate statistical populations; directly mixing extreme value samples from different mechanisms may produce apparent heavy-tail characteristics, misleading return level estimation. Different mechanisms require different conditional relationships in engineering conversion; the definitions of gust factors, duration corrections, and dynamic amplification coefficients may require different expressions under different mechanism conditions.

Classification is intended not merely to assign a single label to an event, but to determine which observational evidence, simulation methods, statistical populations, and engineering conversion methods should be used. Classification is therefore a foundational step, the starting point of the entire scientific chain, rather than a post-processing option. When classification has not been completed or classification uncertainty has not been quantified, subsequent steps can proceed, but their results must retain conditional dependence on the classification assumption.

2.1.2 Six levels of classification

The classification framework has six levels, each addressing a different question; a label at any one level cannot automatically determine the others.

The first level identifies the parent storm family. The parent storm family determines the climatological and mesoscale context of the event. Parent organizational modes, including isolated convective cells, multicell systems, squall lines, bow echoes, and supercells, correspond to different environmental conditions (CAPE, shear, moisture), different life cycle time scales, and different near-surface wind production mechanisms. Determination of the parent family typically relies on radar volume scan observations, satellite imagery, and mesoscale analysis; its uncertainty arises from observational temporal resolution, radar coverage, and analyst judgment.

The second level identifies the surface event class. This level describes the strong-wind event category recorded at the surface, such as thunderstorm outflow, gust front, mixed event, or unresolved event. Determination of the surface event class depends on the spatiotemporal density of station network observations: in a network with 30-kilometer station spacing, the wind speed, wind direction, temperature, and pressure joint time series at stations can identify some event categories, but microburst-scale events may fall entirely between stations and cannot be identified from surface records.

The third level identifies the local wind mechanism. The station-scale local wind mechanism describes the source of the wind field experienced at a specific location at a specific time. Within the same downburst, a station near the divergence center experiences divergent outflow, a station farther from the center may experience the gust front leading edge, and a station even farther away may experience only a brief enhancement of the environmental wind. Therefore, the local wind mechanism is not the same as the event-level mechanism label: an event classified as a downburst can produce different local wind mechanisms at different stations.

The fourth level identifies engineering wind-field features. The wind-field features used in engineering analysis include the shape and low-level jet height of the vertical profile, the radius and decay rate of the horizontal footprint, the gust ramp rate (the time and rate at which wind speed increases from background value to peak), near-surface turbulence intensity and gust factor, integral scale and spectral characteristics, and the movement speed and direction of the outflow. These features are direct inputs to engineering analysis; their definitions must be specified under particular time averaging, spatial support, and height endpoint conditions.

The fifth level assesses evidence reliability. This level stratifies the reliability of the observational and simulation evidence supporting the above judgments. Evidence reliability depends on the quality of the data source (instrument calibration, QC status, spatial coverage), the robustness of the analysis method (sensitivity to parameter choices, cross-validation), and the degree of independent verification (multi-source consistency). Evidence reliability is not the same as classification probability; a classification judgment with high evidence reliability may still have uncertainty due to the complexity of the physical process.

The sixth level specifies downstream scientific use. It defines which downstream analyses the event is permitted to enter under current evidence. If the mechanism attribution of an event is uncertain, it may be permitted to enter certain analyses that do not depend on mechanism assumptions (such as statistical description of surface wind speed) but not permitted to enter engineering analyses requiring mechanism-specific profile models. The permitted range of downstream use is determined by the combined state of the above five levels.

2.1.3 Researcher claim gate for classification outputs

Classification statements are separated into at least four tiers: recorded facts, event associations, candidate mechanisms, and permitted uses. Recorded facts describe quantities observed by a given platform under specified conditions; event associations describe multi-platform records that are compatible in time and space with the same convective process; candidate mechanisms describe which physical interpretation is better supported by existing evidence; permitted uses specify which analyses that interpretation may enter. Each tier requires separate review; the establishment of a preceding tier does not automatically approve a subsequent tier.

The claim gate first checks whether the stated subject matches the identity of the evidence. Station records cannot be written as event-wide maxima, radar morphology cannot be written as surface mechanism ground truth, classifier labels cannot be written as physical occurrence probabilities, and model compatibility cannot be written as observational confirmation. The gate then checks whether the quantity is accompanied by its qualifying conditions, including time, location, height, platform, algorithm, and event version. Finally, the gate checks whether restrictions on extrapolation have been transmitted with the output.

Figures and summaries are subject to the same gate. Colored categories on a map may create an impression of certainty; therefore, legends should distinguish confirmed, candidate, and

unresolved, and should annotate observational coverage and data-void regions. Time series plots should show raw quantities and processing windows, so that smoothed curves are not mistaken for direct observations. Flowcharts can represent only the judgment sequence and do not indicate that every event has completed all evidence steps.

When new data become available, the gate allows a status to be upgraded, maintained, or downgraded. An upgrade requires new evidence that genuinely increases the discriminability between candidates; additional data of the same type that remain at an unobservable scale only increase background and do not upgrade the mechanism. Maintenance indicates that the data improve the description but do not change the strongest proposition. Downgrade occurs upon temporal alignment errors, source corrections, sensor issues, or reopening of alternative mechanisms. All status changes must preserve the old version and rationale.

Cross-chapter use must rely on the usage field, not merely the label. The simulation and statistical processes of the third chapter read different fields; the specific Events chapter reads observational and adjudication bases; Windfield reads extractable features; EVA reads members and independence; DesignLoad does not directly consume classification labels but can only receive conditional statistics and response evidence formed through intermediate steps. This interface prevents labels from passing through scale and load conversions that have not yet been closed.

Final scientific responsibility rests with the Researcher, who must make traceable decisions about propositions and uses rather than filling in missing observations on behalf of algorithms. When evidence is insufficient, the correct output is a clearly bounded unresolved state and the next discriminating requirement. The classification and status description contract established in this chapter does not claim to have formed a complete event catalog, nor does it change the existing evidence status of any specific event or downstream result.

Evidence and formula boundaries of Event classification and scientific status description Chapter 2 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Event classification and scientific status description." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 2 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

3 Scientific process from Step 0 to Step 5

The PETW scientific chain comprises six steps: Step 0 event classification and identification, Step 1 meteorological simulation (two complementary lanes of CM1 and WRF), Step 2 wind field characterization, Step 3 extreme value analysis, Step 4 wind tunnel, CFD, and structural response, and Step 5 design loads and code interface. This chapter defines the scientific questions, inputs, outputs, and interconnections of each step, and describes the feedback between steps as an integral part of the scientific chain design. The six steps and the three branches of Step 4 each address different scientific questions and cannot substitute for one another.

3.1 Problem, concepts, and methods

3.1.1 Step 0: Events, event classification and identification

Step 0 establishes event identity, temporal and spatial extent, parent-storm context, surface evidence, and a mechanism hypothesis from multi-source observations, thereby providing traceable event data for downstream use. This step initiates the entire scientific chain: the quality of event

data determines the interpretation ceiling of all subsequent steps.

The inputs include atmospheric radar data (NEXRAD Level II volume scan data and MRMS composite products), satellite observations (GLM lightning mapping), surface station networks (ASOS 5-minute reports and Mesonet continuous records), reanalysis products (providing environmental soundings and thermodynamic indices), and historical storm reports (Storm Events Database). The intended output is a classified event catalog, with each event carrying mechanism labels and evidence status.

A complete event record includes event identification (event_id and episode_id), raw observational quantities (wind speed, wind direction, temperature, pressure, precipitation), peak window and averaging algorithm (WMAX defined as the maximum of 3-second observations within a 5-minute period), measurement height (standard 10 meters), exposure conditions (open terrain or sheltered), instrument type (cup or ultrasonic), QC status (pass, flagged, or excluded), mechanism label (current state of the six-level classification), classification confidence (confirmed, probable, possible, or unresolved), independence rules (parent separation criteria for subsequent de-clustering), and permitted downstream use (which analyses the event may enter under current evidence).

Step 0 produces a complete record of this information rather than a single label. Its quality determines the interpretation ceiling of all subsequent steps; therefore, the Events Agent prioritizes record completeness over label certainty, preferring to mark an event as unresolved rather than assign a definitive label when evidence is insufficient.

3.1.2 Step 5: DesignLoad, design loads and code interface

Step 5 links event mechanisms, extreme value statistics, aerodynamic forces, and structural response to design wind speed and design load expressions. The design-load stage receives mechanism-conditioned wind speed statistical objects, event rates, aerodynamic evidence (wind tunnel or CFD), and structural response. Outputs are candidate gust response factors or non-stationary response corrections, equivalent static loads, and draft code interface language.

This study organizes the discussion using the following general form:

$$V_{\text{design}} = F\left[V_k(T),\, \lambda_k,\, \eta_k,\, \theta\right], \tag{1}$$

where $V_k(T)$ is the conditional wind speed statistic for mechanism k (such as the T-year return level), λ_k is the event rate for mechanism k, η_k is the response correction factor for mechanism k (including gust response factor, non-stationary effects, and dynamic amplification), and θ is the parameter set (including exposure conditions, height, direction, structural properties, and other applicable conditions). This equation is neither a verified PETW design formula nor a code recommendation. It is an organizing framework that allows every factor in the design expression to be traced to a specific stage and evidence source in the scientific chain.

The principal theoretical reference is the generalized wind-loading chain of Kareem, Hu, Guo, and Kwon (2019), which extends the Davenport stationary wind-loading chain to non-stationary winds and expresses the relationships among wind, force, and response using time-frequency evolutionary power spectra.

3.1.3 Minimum reporting unit for process status

For each step, the minimum reporting unit includes the question, input objects, processing version, output objects, quality checks, evidence status, limitations, permitted uses, and next decision.

The question defines what the output should answer; input and version establish reproducibility; quality checks indicate whether execution is complete; evidence status and limitations control interpretation; permitted uses specify what downstream can do; the next decision identifies the recovery or advancement path. An output filename alone without these fields does not constitute a scientific handoff.

The minimum reporting unit is generated at the case level and can be aggregated to the batch level. Batch counts must preserve denominators, exclusion items, and category definitions, rather than reporting only success ratios; case anomalies must be traceable to original configurations and logs and cannot be masked by batch averaging. Every point in a statistical or summary figure should be traceable to case identity and the observation operator.

For plans that have not yet been executed, the reporting unit records the target, generated inputs, expected criteria, resource requirements, and stopping conditions, but does not include assumed results. WindTunnel designs, CFD long jobs, unapproved structural or EVA routes are all recorded under this status. The completeness of a plan can be subject to engineering review; physical results can only be produced after actual execution and QA.

For existing model-internal output, the reporting unit further distinguishes among raw results, derived diagnostics, and candidate interpretations. Raw results preserve original variables and spatiotemporal identity; derived diagnostics record transformations and observation operators; candidate interpretations list supporting, contrary, and missing evidence. With the three layers separated, even if an interpretation is withdrawn, raw and diagnostic objects can still be reused under appropriate conditions.

The unified reporting unit allows Steps 0 through 5 to use the same receiving logic without erasing disciplinary differences. Events emphasizes events and evidence identity; meteorological simulation emphasizes configuration and field quantities; Windfield emphasizes scales and features; EVA emphasizes samples and statistical objects; Step 4 emphasizes inflow, pressure, and response; DesignLoad emphasizes conversion and approval boundaries. Common fields are used for traceability; specialized fields are used for scientific judgment.

Evidence and formula boundaries of Scientific process from Step 0 to Step 5 Chapter 3 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Scientific process from Step 0 to Step 5." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 3 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

4 Uncertainty and scope of scientific interpretation

4.1 Problem, concepts, and methods

4.1.1 Nine categories of uncertainty across PETW stages

Uncertainty across PETW is divided into nine categories that act at different stages of the chain, cannot be assumed mutually independent by default, and cannot be compressed into a single overall interval.

Measurement uncertainty reflects differences in instrument response, sampling rate, measurement height, and exposure conditions. Different instruments can record different values for the same physical wind speed; this is not measurement error in the usual sense (i.e., deviation from

the true value), but rather a systematic difference caused by different measurement definitions. For example, cup anemometers have a low-pass filtering effect on rapidly changing wind speeds due to rotational inertia, with an equivalent response time of approximately 5 seconds; ultrasonic anemometers at 10 to 20 hertz sampling rates can resolve sub-second wind speed fluctuations. In the same event, the peak values recorded by the two types of instruments can differ significantly; this difference is not random error but a systematic characteristic of the measurement chain.

Classification uncertainty concerns event identity, mechanism attribution, and the handling of unresolved events. Classification labels carry their own uncertainty, and this uncertainty propagates through population construction to extreme value analysis: if an event that actually belongs to a gust front is incorrectly assigned to the downburst category, it may alter the conditional distribution tail estimate for that category. Classification uncertainty and detection-process uncertainty need to be distinguished conceptually: the former concerns the mechanism attribution of already detected events, while the latter concerns whether potential events were detected at all.

Model-form uncertainty reflects the choice of functional form for distribution families (GEV vs. GPD vs. XIMIS), parameterization schemes (turbulence models, microphysics schemes), and structural models (linear vs. nonlinear). Different functional forms may yield different fitting results for the same dataset, especially when extrapolating to extreme quantiles.

Numerical uncertainty reflects discretization (grid resolution, time step), integration methods (explicit vs. implicit), solver convergence, and numerical diffusion. In CFD, grid convergence studies are the standard method for assessing numerical uncertainty, but under non-stationary conditions the evaluation criteria for grid convergence must simultaneously consider temporal and spatial resolution.

Sampling uncertainty reflects sample size, record length, and extrapolation distance. In extreme value analysis, when extrapolating from limited record lengths to long return periods (such as 50 years or 500 years), sampling uncertainty is the dominant source of uncertainty. The confidence interval of the return level widens with increasing extrapolation distance; when the confidence interval width exceeds the physical meaning of the estimate, extrapolation should stop.

Parameter uncertainty reflects estimation errors of model parameters. In wind field parameterization, the estimation of parameters such as peak height and decay rate depends on limited calibration data. In extreme value analysis, the estimation of the GEV shape parameter ξ has a decisive effect on tail behavior, but under small samples the standard error of ξ can be large.

Scale-transfer uncertainty reflects the conversion procedure between meteorological-grid and engineering scales. Scale conversion involves not only interpolation between spatial resolutions, but also the representation of sub-grid physical processes, the applicability of boundary-layer parameterization, and the treatment of nonlinear effects. There is currently no universally accepted general scale-transfer procedure.

Experimental uncertainty reflects wind tunnel facilities, calibration, and repeatability conditions. In accelerating-flow experiments, the repeatability of inflow time histories, boundary layer development state, influence of model natural frequencies, and Reynolds number effects all need to be considered in experimental design.

Scenario uncertainty reflects the choice of historical and future time periods, downscaling methods, and environmental proxy quantities. The impact of climate change on thunderstorm frequency and intensity is estimated through global models and regional downscaling methods, but the parameterization of convective processes makes these estimates subject to considerable uncer-

tainty. Changes in convective environment proxies (such as CAPE) cannot be directly translated into future 3-second gust changes.

4.1.2 Cross-step propagation of uncertainty

Uncertainty propagates through the scientific chain. Consider the general expression

$$\mathcal{U}_{j+1} = \mathcal{T}_j\left(\mathcal{U}_j, \mathcal{U}_{\text{method},j}, \mathcal{U}_{\text{data},j}\right) \tag{2}$$

In this organizing framework, the existing uncertainty $\mathcal{U}_j$ at step j, the method uncertainty $\mathcal{U}_{\text{method},j}$ of this step, and the data uncertainty $\mathcal{U}_{\text{data},j}$ of this step enter the next step through the transformation $\mathcal{T}_j$. This equation organizes the sources of uncertainty and does not imply that their distributions are known or that they are mutually independent. When $\mathcal{T}_j$ is a nonlinear transformation, upstream uncertainty cannot be propagated through simple variance addition; when different uncertainty sources are dependent, intervals calculated under an independence assumption may underestimate or overestimate the actual uncertainty.

Four key disciplines govern this propagation in practice. First, classification uncertainty must enter EVA membership explicitly; if the uncertainty of event classification is ignored when entering extreme value analysis, the return level estimate will reflect only sampling and model form uncertainty while omitting potentially more important membership uncertainty. Second, simulation uncertainty must enter the applicability conditions of Windfield features and CFD inlet; the accuracy of simulation output cannot be assumed to be maintained during scale conversion, and the decay or increase of accuracy must be explicitly stated at each step. Third, experimental and numerical uncertainty must enter the input specification of structural response; the uncertainty of load time histories propagates through structural dynamic equations to the response, and the contribution of load uncertainty at different frequency bands to the response differs. Fourth, statistical uncertainty must enter the applicability boundary of design expressions; the confidence interval of the return level determines the reliability level of the design wind speed and cannot be forgotten in the transformation from return level to design wind speed.

Discarding upstream uncertainty at any stage would incorrectly raise the evidence ceiling of downstream conclusions. This is the core reason PETW maintains uncertainty transparency.

4.1.3 Research significance of open states

An open state does not mean that no results exist. It can indicate that sources have been recovered but scientific identity still awaits confirmation, methods have been implemented but target data are insufficient, models have produced internal quantities but lack observational comparison, experimental plans have been frozen but not yet executed, or statistical routes have been defined but samples do not qualify. Precisely identifying the stopping position protects existing work and also prevents downstream from treating preparation status as verified results.

Open states should specify recovery conditions. Source problems require original file, configuration, or record confirmation; quantity problems require common definitions and conversion verification; mechanism problems require observations or three-dimensional diagnostics capable of distinguishing candidates; verification problems require matching inputs and common endpoints; design use requires the complete scientific chain and PI approval. If recovery conditions cannot be satisfied by existing data, the status remains open rather than forcing a conclusion.

Open states also preserve alternative routes. If a mechanism cannot be uniquely classified, the unresolved population and sensitivity branches can be retained; if real event inputs are insufficient,

Table 1: Roles of the nine uncertainty categories across scientific chain stages

Uncertainty category	Active stages	Cross-step propagability
Measurement	Events, Windfield	Propagates through wind speed quantity definitions
Classification	Events, EVA	Propagates through membership
Model form	CM1/WRF, EVA, Structural	Propagates through method selection
Numerical	CM1/WRF, CFD, Structural	Propagates through output accuracy
Sampling	EVA	Propagates through sample size
Parameter	Windfield, EVA, Structural	Propagates through parameter estimation
Scale transfer	Windfield, CFD	Propagates through scale conversion
Experimental	WindTunnel	Propagates through comparison benchmarks
Scenario	EVA, DesignLoad	Propagates through time period selection

an independent benchmark testing procedure can be used, but the research question shift must be explicitly stated; if a particular model is not applicable, a different scale or method can be selected without erasing the failure record. Alternative routes require new input and usage contracts and do not inherit the claims of the original route.

For readers, open states delineate the most important scientific boundaries of this report. Figures, tables, and chapters can fully present existing evidence and methods, but their value lies in identifying which relationships have data, which have only model-internal support, and which are still generated input or planned work. Such a report can guide subsequent research but cannot be directly translated into design winds, design loads, or code recommendations.

The final product of uncertainty governance is therefore not a single overall interval but a set of traceable objects, conditioned propositions, stopping points, and recovery paths. This structure allows precise project updates as evidence grows, and also allows partial interpretations to be withdrawn when new evidence produces conflicts. Under this framework, the full manuscript remains scientifically reviewable and engineeringly auditable, and explicitly maintains not_scientifically_approved.

Evidence and formula boundaries of Uncertainty and scope of scientific interpretation Chapter 4 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Uncertainty and scope of scientific interpretation." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 4 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

Part II

Agentic Pipeline and Human-in-the-loop Research

5 Definition and composition of agents

5.1 Evidence, results, and interpretation

5.1.1 Definition of agents in PETW

Within PETW, an Agent is an AI-assisted research unit with clearly defined professional responsibilities, inputs and outputs, tool permissions, workspace, persistent knowledge, and scientific boundaries. This definition distinguishes three concepts.

An Agent is distinct from a language model. A language model supplies reasoning and generation capabilities and is one component of the computational infrastructure on which an Agent operates, but the same language model can support multiple Agents with different responsibilities, and the same Agent may use different language models at different times. Switching the language model does not change the Agent's scientific responsibilities: the Events Agent, regardless of which language model it uses, retains its responsibility for event classification and identification, and its product claim ceiling remains unchanged. Confusing the brand or version of a language model with the scientific role of an Agent is a common misconception to be avoided in agentic research.

An Agent is also distinct from a chat interface. Chat provides one mode of interaction between the Researcher and an Agent, but the scope of an Agent's work extends far beyond conversation: file reading and processing, data analysis, code execution, computation coordination, figure generation, and result recording are all components of an Agent's work. Understanding an Agent as a chat interface underestimates its actual role in the research process and also misrepresents its evidence boundaries; an Agent produces not only text replies but also files, data, figures, and executable computation configurations.

An Agent is also distinct from a one-time ad hoc subagent call. When handling complex tasks, an Agent may create temporary workers (subagents) in isolated contexts to accomplish limited auxiliary tasks, such as executing code in an independent sandbox or conducting literature retrieval in a separate context. These temporary workers do not retain persistent knowledge after task completion and do not assume ongoing scientific responsibilities. The distinction between an Agent and a subagent lies in persistence and responsibility: an Agent is a research unit that continuously bears responsibility for a particular scientific lane, while a subagent is a one-time auxiliary tool.

This definition separates organizational concepts (scientific responsibilities) from model capabilities (reasoning and generation), and avoids three common misidentifications: confusing model brand with scientific role (using a more powerful model does not automatically elevate the conclusion level), confusing a conversation interface with scientific product (conversation records do not equal tested scientific results), and confusing a temporary call with a persistent commitment (a subagent's output does not serve as an independent source of conclusions until verified by the owner agent).

5.1.2 Ten components of an agent

Ten components define an Agent and collectively establish checkable boundaries, allowing the provenance and limitations of each product to be independently verified.

Identity specifies who the Agent is and what scientific lane it bears. For example, the Events Agent bears Step 0 event classification and identification; the CM1 Agent bears Step 1 mechanism-oriented simulation. Identity is persistent: as long as the structure of the scientific chain does not change, the Agent's identity does not change.

Responsibility specifies what the Agent does and does not do. Positive responsibility definitions ensure the Agent provides complete work within its domain; negative responsibility definitions (what it does not do) prevent the Agent from overstepping, for example, the Events Agent does not estimate return levels, and the CM1 Agent does not produce design loads.

Input specifies the data, evidence, and task descriptions accepted by the Agent. The definition of input includes source (from which upstream Agent or data source), format (file type, field definitions), and quality requirements (QC status, evidence level).

Output specifies the products generated by the Agent and their formats. Output includes not only result data but also source paths, units, limitation statements, and items requiring Researcher decision.

Tools specify the programs and interfaces available to the Agent: code execution environments, file system access, computation resources, web search, figure generation, and database queries, among others. Tool permissions are layered: certain operations (such as local file reading and writing) are automatically executed within authorized scope, while certain operations (such as remote computation submission and external writing) require the Researcher's explicit decision.

The knowledge base specifies which literature, Wikis, and historical results the Agent has access to. The content of the knowledge base is detailed in Chapter 11.

The workspace defines the directories the Agent reads from and writes to. Workspace isolation prevents different Agents' files from interfering with each other, while controlled cross-Agent data transfer is achieved through handoff mechanisms.

Relationships with other agents specify upstream, downstream, and support relationships: what inputs the upstream Agent provides, what outputs the downstream Agent requires, and which support agents provide assistance.

Allowed actions specify which operations the Agent can perform under what conditions. PETW adopts layered automation levels: A0 for read-only and analysis, A1 for local file writing, A2 for supervised computation execution, A3 for remote or facility operations within explicitly approved scope, and A4 for closed-loop autonomy (currently deferred, not enabled). Each Agent's allowed action level is determined by the Researcher.

The claim ceiling defines the maximum scope of claims of the Agent's products. The claim ceiling prevents any output from exceeding its evidence boundary; for example, the CM1 Agent's products cannot claim to be validated downbursts or engineering inflow, and the EVA Agent cannot provide return levels without formal samples. The claim ceiling can be raised by the Researcher as evidence accumulates, but upgrades can only be decided by the Researcher.

5.1.3 Position of agents in the evidence chain

An Agent's position in the evidence chain depends jointly on the evidence it consumes and the evidence it can produce. When the Events Agent processes observations and event data, it can form event packages with reliability ratings but cannot confirm a unique mechanism solely on

the basis of organizational completeness; when CM1 and WRF process model configurations and outputs, they can form named-case diagnostics but cannot write model-internal consistency as observational verification; when Windfield, EVA, and engineering lanes process derived quantities, they must inherit the conditions and open states of upstream objects. No support Agent's file processing capability, however strong, can alter this evidence inheritance relationship.

Every derivation in the evidence chain must preserve its source edges. Source edges point not only to files but also specify which variables, time periods, spatial extents, and method versions from those files were used. A single figure may simultaneously depend on data, screening scripts, and plotting configurations; a single synthesis may simultaneously depend on several sources and their mutual inconsistencies. If a derived product preserves only final numerical values or conclusions, the Agent can continue to operate on the carrier but cannot determine which content needs to be withdrawn when upstream conditions change.

Different forms of Agent verification cannot be collapsed into an undifferentiated "verified." Platform verification confirms files and execution; method verification confirms that implementation is consistent with established procedures; source verification confirms that propositions actually appear in named materials; comparison verification checks the consistency and deviation between models, observations, or benchmarks; scientific adoption judges whether these pieces of evidence are sufficient to support the current use. The first several types of checks can be performed and recorded by Agents; the last type is still decided by the Researcher. Clarifying verification types prevents a genuine technical PASS from being propagated as a higher-level scientific status.

When counterevidence emerges or a source is downgraded, the Agent should mark affected objects along derivation relationships rather than only adding a limitation note in a new report. Affected analyses can still be preserved as historical artifacts, but the current status should indicate that their inputs have been superseded or their claim ceilings have narrowed. Thus, Agents bear the responsibility of maintaining the continuity of evidence relationships and candidate products, while the Researcher bears the responsibility of interpreting conflicts, accepting downgrades, and deciding subsequent uses. Both types of responsibility working together constitute the persistent research units used by PETW.

Evidence and formula boundaries of Definition and composition of agents Chapter 5 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Definition and composition of agents." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 5 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

6 Comparison with traditional research approaches

6.1 Evidence, results, and interpretation

6.1.1 Actual form of traditional working methods

In traditional research, researchers typically switch manually among multiple disconnected tools. Literature management takes place in Zotero, EndNote, or Mendeley; searching takes place in databases and search engines; the connection between the two depends on the researcher's manual annotation. Code development takes place in version control systems (Git), but the connections between code and the data it analyzes and the literature assumptions it depends on typically

exist only in the researcher's memory or scattered notes. Computation is executed on clusters through submission scripts, run status is monitored through query commands, and results are obtained through manual download; the correspondence between submission configurations and results must be maintained by the researcher. Data processing takes place in local or remote environments, with intermediate results saved in directory structures where naming conventions are the sole organizational method. Writing takes place in document tools (Word, LaTeX), figures are manually copied from result directories, and citations are manually inserted from literature management tools. Collaboration proceeds through email, instant messaging, and meetings, and the connection between discussion content and research decisions depends on participants' memory.

The cost of switching extends beyond the time consumed. The core problem is the disruption of context and provenance: how a literature source supports a computation configuration, how a configuration produces a result, and how a result enters a conclusion form a causal chain that the researcher must maintain through personal memory and notes. The larger the project and the more parallel directions, the more disruptions and the harder the checking. When a project involves multiple disciplinary directions (as in PETW with meteorology, wind engineering, statistics, structural engineering, and design), how to aggregate the evidence accumulated in each direction into a unified conclusion framework is a systematic challenge facing traditional working methods.

6.1.2 Applicable advantages and practical limitations

The agentic approach offers applicable advantages in four areas. Natural language operation reduces the interaction cost between the researcher and tools; the researcher can describe scientific objectives in natural language rather than manually writing scripts or filling in forms. Cross-tool connection allows literature, computation, data, and writing to be executed along the same task chain; a single task can simultaneously accomplish literature retrieval, data processing, result generation, and report writing. Multi-model comparison provides complementary roles and independent checks; different models reviewing the same result from different angles can help discover problems that a single model might miss. Repetitive checking allows format, citation, and limitation statements to be programmatically verified; whether every citation in the report exists in the BibTeX file and whether every numerical value has units and a source can be checked automatically.

The practical limitations are equally clear. Model misreading may broaden (ignoring limitations) or narrow (over-restricting) the applicability conditions of a source, producing incorrect synthesis judgments. Source errors may be masked by fluent prose; the text produced by an Agent may be grammatically and logically flawless, but the underlying scientific judgment may be based on incorrect source understanding. Automation bias may cause the researcher to over-trust the coherent output of the Agent; when the system consistently produces correct results, the researcher may reduce checking intensity, failing to detect precisely when the system errs. Context loss may cause later execution to deviate from the initial settings in long tasks; when processing large amounts of information, the Agent's effective context window is limited, and later processing may not fully account for earlier constraints.

These limitations motivate three PETW disciplines: sources take priority over summaries (an Agent's synthesis cannot substitute for primary source verification), evidence levels cannot be automatically crossed (model output does not equal verified results), and the Researcher decides all scientific statements (an Agent's text output does not carry scientific validity until accepted by the Researcher).

6.2 Limitations, interfaces, and Researcher judgment

6.2.1 Differences in handling failure states and open questions

In traditional work, explorations that do not yield the expected results often remain in personal directories or memory, and whether they are revisited depends on the researcher. The agentic approach can save failure checks, stopping reasons, and unresolved questions in a searchable state, enabling subsequent workers to avoid repeating the same attempts. What is valuable here is the scientifically relevant stopping rationale and its impact on existing results, not writing every runtime error into the reader's body text.

Status recording can nevertheless create pressure for false closure. Task systems tend to categorize objects as complete or incomplete, but scientific questions often need to remain open due to insufficient sources, non-unique methods, or contrary evidence. If an Agent selects an interpretation to satisfy a completion condition, the structured workflow may actually reinforce over-determination. PETW allows actions to be complete while scientific questions remain open, and allows technical checks to pass while downstream qualification remains closed. The hesitation that the researcher intuitively preserves in traditional approaches must become an explicit status in the agentic approach.

An open status is useful only when it specifies evidence gaps and review trigger conditions. A generic "more research is needed" cannot guide subsequent action; specifying what type of observation is missing, what variable definition, what comparison, or what authorization is needed enables future tasks to judge whether restart conditions are met. This structure does not shorten the time required to obtain new evidence, but it prevents the system from mistaking execution idleness for problem resolution, and allows the Researcher to focus attention on the objects that truly limit scientific progress.

Evidence and formula boundaries of Comparison with traditional research approaches
Chapter 6 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Comparison with traditional research approaches." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 6 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

7 Natural language-driven research work

7.1 Evidence, results, and interpretation

7.1.1 Ten elements of a scientific task

To become an executable scientific task, a natural-language request must contain ten elements. The scientific question states what the task should answer, not "analyze the BESSIE event" but "does the near-surface strong wind of the BESSIE event support a divergent outflow mechanism." Existing evidence states what the task is based on: existing observational data, completed simulations, and established literature foundations. New evidence to be produced states what the task should produce: new data analyses, new simulation runs, new comparisons, or new synthesis judgments. Established methods state what procedures to follow: whether to use an established benchmark protocol, analysis script, or comparison framework. Required checks state how to verify: reasonableness checks of results, comparisons with known benchmarks, and consistency verification of units and coordinates. Completion conditions define when the task can stop: when

all target quantities have been produced, all checks have passed, and all limitations have been stated. Stopping conditions define when the task must stop: when computational resources are exhausted, results are clearly unreasonable, or task scope needs modification. Permitted interpretation defines the product's claim ceiling: what results may claim at most and what they may not claim. Researcher subsequent decisions state which matters require human judgment: method disagreements, result interpretation, and downstream use qualification. Output location ensures products are traceable: in what directory and in what format results are saved.

A task description that omits any element may allow the executor to improvise, producing unverifiable results. In practice, the Researcher does not need to explicitly list all ten elements in every request; commonly used task types have established defaults, and the Coordinator can supplement missing elements, but the Researcher needs to confirm whether the supplemented defaults are appropriate.

7.1.2 Five intervention periods of the Researcher

The Researcher's intervention is distributed across five periods rather than concentrated at the end of the process.

Before a task begins, the Researcher determines the scientific question (what to answer), scope (within what boundaries to answer), and method (what tools and procedures to use). The judgment at this stage determines the direction of all subsequent work and is the most critical intervention period.

When key choices arise, the Researcher determines the event (which event to study), cases (what configurations to use), models (which language model and which Agents to use), and resources (what computational resources to allocate). These choices are locked before execution begins; modifications during execution require the Researcher to re-decide.

When anomalies or disagreements arise, the Researcher determines their meaning and how they should be handled. For CM1 simulation results that do not match expectations, the Researcher must distinguish a simulation-configuration problem from a potential new finding about the physical mechanism. For systematically low WRF-reconstructed wind speeds, the Researcher must distinguish a model-version difference from an initial-condition problem. These judgments require evidence from multiple sources.

Before results are used across Agents, the Researcher decides whether upstream results are sufficient to enter the next step. This decision covers whether CM1 model-internal diagnostics may enter Windfield feature extraction, whether WRF reconstruction output may constitute a candidate input for a CFD inlet, and whether EVA statistical-object definitions may be used in DesignLoad design expressions. These cross-step qualification judgments are critical to the integrity of the scientific chain.

When results are formulated, the Researcher distinguishes evidence levels and makes final statements. The required distinctions are between model output as candidate diagnostic evidence and as a verified scientific result, between statistical inference as an exploratory estimate and as a baseline estimate that can enter design, and between engineering conversion as a conditional possibility and as confirmed applicability. These distinctions determine how results are expressed in the report and used in subsequent research.

Intervention does not obstruct the process; it identifies the natural location for scientific judgment. People handle stages that require judgment, the system handles stages that can be programmatized, and risk and responsibility determine the allocation between them.

7.2 Limitations, interfaces, and Researcher judgment

7.2.1 Applicable conditions and limitations of the natural language-driven approach

Natural language is well suited to expressing cross-tool objectives, scientific context, exception conditions, and expected interpretations, especially for tasks where methods are already established and execution steps are numerous. It can reduce the cost for the researcher of switching between the syntax of various tools, and can simultaneously convey purpose and limitations to the executor. Its advantage depends on stable terminology, bindable objects, controlled tools, and clear acceptance criteria. If the concepts used in the request have no common definition within the project, the more fluent the language, the more likely the executor is to fill gaps with default meanings.

Natural language does not replace formal data schemas, computation configurations, experimental safety procedures, and design responsibility documents. It can generate or select these objects, but final execution should consume frozen versions. For high-cost, high-risk, or irreversible actions, the Researcher's natural language intent must still be converted into explicitly scoped authorization; historical authorization and similar wording do not automatically carry over. For scientific conclusions, natural language generation only forms candidate statements; source verification and interpretation ownership do not change because of expression quality.

Context capacity further constrains long tasks. When a task undergoes compression or cross-session recovery, early negative constraints, minority counterexamples, and version relationships are the most easily lost. The solution is not to extend conversations indefinitely but to write the conditions that determine result identity into external contracts and artifacts. On system recovery, these control objects are read first, followed by necessary sources and results; conversation summaries aid navigation only and do not serve as the sole basis for current facts.

The natural-language-driven research approach is therefore a controlled front end, not an autonomous scientific method. It connects the Researcher's scientific intent to checkable materials, procedures, and execution, but does not determine whether sources are correct, methods are applicable, or results are sufficient to support downstream use. Only when contracts, artifacts, checks, error recovery, and Human ownership are simultaneously present does language convenience avoid sacrificing evidence boundaries.

Evidence and formula boundaries of Natural language-driven research work Chapter 7 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Natural language-driven research work." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 7 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

8 Multi-model collaboration

8.1 Evidence, results, and interpretation

8.1.1 Basis for selecting main models

PETW does not rely on a single model for every task. The main model is selected according to task type, based on eight dimensions: tool use capability (Agents need to read and write files, execute code, and call APIs), context length (ability to handle long documents and large datasets), scientific reasoning (understanding physical concepts, evaluating evidence, recognizing uncertainty), programming capability (writing and debugging analysis code), Chinese and English

writing quality (accuracy and readability of technical narrative), speed (latency of task completion), cost (economics of API calls), and project-internal performance (historical performance records on actual PETW tasks).

Models are evaluated using actual PETW tasks rather than model-brand assumptions or general benchmark rankings. Outstanding performance on a single task does not directly result in a permanent role; it is first recorded with a provisional label and confirmed after multiple consistent performances. A model's advantage on one type of task does not automatically extend to other task types: a model that performs excellently in long-document synthesis may not be as effective at code debugging as other models, and vice versa.

8.1.2 Role division among models

The GPT family functions as the primary coordinator, programmer, tool user, file handler, structured synthesizer, and full-text editor. As the earliest model family established in PETW, the GPT series has accumulated the most task history and configuration experience in the project.

Claude Opus supports large-context writing, structural reorganization, and external review. The Claude series has specific advantages in handling long documents and maintaining global consistency, making it suitable for chapter-level rewriting and cross-chapter consistency checking.

Fable provides an independent, science-first review lane, used for critical checking. The use case for Fable is when existing conclusions need to be challenged and alternative interpretations offered; its role is as a challenger rather than a confirmer.

Kimi K3 (whose formal model name is `kimi-k3`, provided by Kimi/Moonshot's `kimi-coding` provider) is used for alternative writing, programming, Chinese technical expression, and second opinions. K3's performance in Chinese technical writing has been verified through PETW tasks, making it suitable as an alternative or supplement to the primary writing lane.

GLM supports alternative writing, programming, and Chinese technical expression, providing another independent Chinese technical writing lane.

Gemini supports web search, topic expansion, candidate material discovery, and supplementary analysis, playing a role in information frontier expansion.

The DeepSeek series supports technical reasoning, hypothesis checking, and independent writing, with project tasks providing verification in logically rigorous technical analysis and hypothesis evaluation.

Role division is based on complementarity: one model is responsible for generation and another for checking; the checker does not share the same defaults and biases as the generator. Complementarity is not voting; agreement among multiple models does not constitute scientific establishment; differences discovered by different models are used to locate positions requiring review.

8.1.3 Role isolation of generation, review, and editing

Role isolation, rather than the number of models, is central to the multi-model process. The generation stage forms a complete structure within frozen sources; the review stage does not inherit the generator's conclusions and focuses on finding source mismatches, evidence upgrades, undefined variables, and omitted limitations; the editing stage modifies based on explicit action items without adding unreviewed scientific content. Each stage saves its inputs and outputs, enabling conflicts to be traced to specific sources rather than resolved by voting.

When models disagree, the first step is to determine whether the disagreement belongs to

Table 2: PETW multi-model roles and usage conditions

Model	Primary role	Usage conditions
GPT family	Coordination, programming, tool use, structured synthesis, full-text editing	Editing and checking after structure stabilization
Claude Opus	Large-context writing, structural reorganization, external review	When rewriting and deep synthesis are needed
Fable	Independent science-first review	When an independent critical lane is needed
Kimi K3	Alternative writing, programming, Chinese expression, second opinions	When alternative lane comparison is needed
GLM	Alternative writing, programming, Chinese expression	When alternative lane comparison is needed
Gemini	Web search, topic expansion, candidate materials	During information frontier expansion
DeepSeek	Technical reasoning, hypothesis checking, independent writing	When independent writing or adversarial checking is needed

language choice, method choice, or scientific interpretation. Language differences can be unified by editing; method differences require comparing assumptions and diagnostics; scientific interpretation differences must go back to primary sources, data, or model outputs. The Researcher closes a disagreement only when sufficient evidence exists; otherwise, the conflict is preserved as a limitation. This process makes multi-model collaboration a tool for discovering problems rather than a tool for manufacturing apparent consensus.

Evidence and formula boundaries of Multi-model collaboration Chapter 8 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Multi-model collaboration." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 8 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

9 Human-in-the-loop research process

9.1 Problem, concepts, and methods

9.1.1 Full scientific responsibility of the Researcher

The Researcher (Liang Hu, PI bears full responsibility for research questions, data, methods, computation, experiments, result interpretation, and formal statements. This responsibility extends beyond nominal authorship to substantive participation throughout the entire research process.

Research questions establish the scientific direction of the project by defining what PETW does and does not study, the questions it addresses, and the boundaries of those questions. Data determine the quality boundary of evidence by specifying the observational data, simulation outputs, and literature sources used. Methods determine the comparability of results through the selected model configurations, statistical methods, and comparison frameworks. Computation and experiments determine the physical basis of products through the jobs submitted, experiments

conducted, and resources authorized for execution. Result interpretation determines what the evidence does and does not represent and the conditions under which it holds. Formal statements determine the scientific content available for external use, the content included or excluded from the report, and its stated evidence level.

Agents provide execution and support without assuming these responsibilities. The non-transferability of responsibility is the core of the Human-in-the-loop principle: any Agent output must be interpreted by the Researcher in combination with sources, methods, limitations, and scientific questions, and cannot automatically become an official scientific conclusion of PETW.

9.1.2 Criteria for layered intervention

The Researcher intervenes in layers rather than through step-by-step approval. Layering is based on risk and responsibility.

Internal synthesis based on clearly identified sources does not require sentence-by-sentence approval. For example, organizing already verified literature into a review draft according to an established structure is an operation whose inputs (verified literature) and methods (established structure) are both determined; the Agent's work is primarily organizational and expressive, and errors are easily discovered by subsequent checking.

Intervention is required in six situations. First, a scientific conflict that cannot be resolved with existing evidence; for example, two papers give different conclusions about the parameterization of the same mechanism, and it cannot be determined which is more applicable to current conditions. Second, extrapolation beyond the applicable range of sources; for example, using a profile model calibrated on limited events to predict wind fields under different conditions. Third, formation of a major conclusion; for example, synthesizing a judgment about the event mechanism from multi-source evidence. Fourth, making a verification claim; for example, judging whether simulation results pass an established verification standard. Fifth, involvement of design, code, remote, or facility actions; for example, deciding whether to submit HPC jobs or whether to initiate wind tunnel experiments. Sixth, preparation for external release; for example, deciding which content in the report can be shared with the External Reviewer.

Layering does not imply uncontrolled delegation. All Agent products remain in candidate status, and the Researcher's acceptance is their only upgrade pathway. The purpose of layering is to let the Researcher focus attention on positions most in need of judgment rather than consuming cognitive resources on routine operations.

9.1.3 Governance value of open states

Maintaining open states allows the project to acknowledge the ceiling of current sources. Open items should specify what evidence is lacking, what uses are affected, what existing products remain usable, and under what conditions re-review occurs. It differs from a general to-do item because its stopping reason is insufficient scientific evidence rather than unscheduled execution.

An open status also protects the coexistence of multiple interpretations. When observations support propagating convective features but are insufficient to uniquely determine the local mechanism, the preceding level of statement can be accepted while the subsequent level remains open. Layered acceptance is more consistent with the evidence structure of cross-scale research than wholesale acceptance or wholesale rejection, and also allows Agents to continue work that does not depend on the disputed interpretation.

Evidence and formula boundaries of Human-in-the-loop research process Chapter 9 retains only content directly related to the scientific questions, methods, evidence, limitations, and

interfaces of “Human-in-the-loop research process.” Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 9 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

Part III
PETW Agent Architecture and Shared Research Resources

10 Layered architecture of PETW

10.1 Problem, concepts, and methods

10.1.1 Layering principles

The PETW research system organizes the relationship between scientific decision-making and technical execution through a layered structure. Layering reflects the different degrees of human judgment required by different types of scientific work: the formulation of scientific questions, method selection, result interpretation, and final scientific statements are non-delegable researcher responsibilities; task decomposition, execution coordination, and status tracking are intermediate-layer responsibilities that can be delegated under clear rules; and specialized Agents can autonomously complete technical execution within specific domains, including literature retrieval, data processing, simulation configuration, and figure preparation, over a wider scope, provided that their scientific boundaries are clearly defined.

This arrangement is not a simple hierarchy, but a functional stratification based on decision authority and information flow characteristics. Scientific directives are transmitted downward from the top layer, while scientific results and decision requests are reported upward from the execution layer. The vertical information flow has a clear directionality: items requiring scientific judgment are formed only at the Human and academic layer, transmitted through the Coordinator down to the execution layer; any Agent at any level that encounters a problem exceeding its scientific boundaries must escalate the problem upward rather than making its own judgment. Horizontal information flow, that is, data transfer between different Agents at the same level, is relayed and coordinated through the Coordinator to prevent unsupervised direct dependency relationships between Agents. This relay mechanism ensures completeness checking during cross-Agent data transfers, including whether evidence status labels and usage restrictions are transmitted together with the data.

The PETW architecture comprises five functional layers: Human and academic layer, Coordinator layer, Scientific agent layer, Support and collaboration agent layer, and Project records layer. Table 3 lists the membership, Agent count, authority scope, and information interfaces of each layer.

10.1.2 Support and collaboration agents

Fifteen support and collaboration agents form four functional groups, providing non-scientific support services to all scientific agents.

Table 3: Layered composition of the PETW architecture

Layer	Members	Authority scope	Primary responsibilities
Human and academic	3 persons	Final scientific decision review	Question formulation, method selection, claim adjudication, cross-step use decisions
Coordinator	1 agent	Task routing and status governance	Task decomposition, Agent selection, execution coordination, result aggregation, decision request escalation
Scientific agents	9 agents	Technical execution within respective lanes	Analysis, simulation, and result production for Steps 0 through 5
Support/collaboration	15 agents	Service-oriented execution	Literature management, knowledge organization, writing and figures, computation coordination, remote support, external collaboration
Project records	–	Storage and traceability	Subproject workspaces, reports, handoffs, project status files

The knowledge group comprises three Agents. The Zotero Literature Agent [path] provides the sole authoritative access point to Zotero and is responsible for literature identity verification, PDF management, and bibliographic metadata; it is the authoritative layer for source verification. The PETW LitWiki Agent [path] maintains the PETW literature Wiki in a transactional and role-isolated manner, managing paper cards, claim and citation alignment, evidence status labels, source hashes, and a governed registry apply process; it is the claim-level evidence layer. The Web and WeChat Knowledge Agent handles knowledge resources obtained from web pages, WeChat official accounts, and other external channels; it is the supplementary layer for information frontiers. The division of labor among the three Agents reflects the principle that identifying literature identity and understanding literature content are two different tasks.

The writing and communication group comprises five Agents. The Manuscript Agent handles report organization, manuscript formatting, and language expression. The Figures Agent handles figure formatting, source annotation, and visualization quality. The Daily Briefing Agent generates periodic progress summaries. The Collaboration Test Bot receives explicit PETW requests from the External Reviewer through the Lark Buildings group and routes them through a fixed path to the Coordinator for processing. The Claude Analysis Bot performs file analysis, report comparison, and reviewer-style commentary within the same group. The roles of these two collaboration Bots differ: the former is a request entry point and the latter is an analysis tool; the output of both belongs to external scientific input, which is converted into specific research work only after the Researcher decides.

The computation and remote group comprises four Agents. The MATLAB Agent provides script support for numerical computation, data processing, and plotting. The ComputeOps Agent handles scheduling and monitoring of local, remote, and HPC computation tasks; in the persistent call corpus it received 128 inbound requests, of which 63 from CM1 to ComputeOps and 60 from CFD to ComputeOps constitute the heaviest single edges. The RemoteOps Agent provides file

synchronization and status retrieval support for remote hosts. The CAD Agent provides geometry model preparation and pressure tap layout drawings, having produced R1 pressure tap layout packages for three models (M11/M31/M51).

The project and model support group comprises three Agents. The ModelOps Agent maintains model service availability and health status. The IncidentOps Agent handles runtime incidents and records recovery processes. The Vault Manager Agent manages secure storage and access control of project files.

Of these fifteen Agents, five are PETW-exclusive profiles, eight are shared services across ResearchOS (with PETW-specific or adjacent evidence), and one is a PETW-scoped entry point on a shared collaboration channel. Shared services must be described as shared services and must not be presented as PETW-exclusive Bots. Support agents do not produce scientific conclusions; their products serve the work of scientific agents.

10.1.3 Scientific role boundaries of the project records layer

The project records layer stores tasks, configurations, results, handoffs, and statuses, enabling conclusions to be traced back to sources. The records layer can prove when a particular file was produced by what task, using which version, and through which engineering checks. It cannot prove that a model is physically correct, that an experiment is representative of natural conditions, or that a statistical interpretation holds. Traceability is a prerequisite for review, not a review conclusion.

Records must also distinguish current from historical status. The current status points to the effective version; historical status preserves superseded paths. Judging the current status simply by the most recent modification time may misread rapidly changing directories; therefore, important packages should specify the controlling report or inheritance relationship. This rule applies to scientific results, knowledge packages, and engineering artifacts.

Evidence and formula boundaries of Layered architecture of PETW Chapter 10 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Layered architecture of PETW." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 10 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

11 Knowledge base of each agent

11.1 Problem, concepts, and methods

11.1.1 Prerequisite status of knowledge preparation

Before undertaking a new task, each scientific agent should read the Wiki, RAG, review papers, Zotero collection, and existing results corresponding to its lane. Knowledge preparation is not a one-time attachment completed during construction, nor is it an organizing step after task execution; it is a continuously updated and on-demand-mounted research prerequisite. This prerequisite status has a methodological basis: the value of a scientific task depends on its underlying evidence; an Agent that executes a task without knowledge preparation will substitute model-internal knowledge for project-verified sources, producing output that cannot be traced back. PETW therefore gives sources priority over summaries: a source list is established before task execution, specific sources are cited during execution, and the reading scope and unread gaps are reported upon

return.

The operational importance of this discipline becomes clear when it is absent. If the Windfield Agent, when processing the radial profile of a moving downburst, does not first read the calibration conditions of the Holmes-Oliver model and the parameter optimization basis of the Xhelaj-Burlando model but instead generates a set of parameter values directly from the language model's internal knowledge, then the evidence identity of that output cannot be traced to a specific calibration dataset or published result. If the EVA Agent, when constructing a statistical population, does not first read the method comparisons for mechanism separation in the mixed wind climate literature but instead directly adopts a method, then the rationale for method selection will lack literature support. In both cases, the output may be formally correct but has a break in the evidence chain.

11.1.2 Five categories of knowledge resources and their provenance chains

PETW knowledge resources fall into five categories, with each category organized along the provenance chain from identity verification to project evidence.

Zotero literature management provides the authoritative layer for source verification. Each literature entry contains standard bibliographic information (title, authors, journal, year, DOI) as well as PETW-specific annotations (relevant Agent, full-text reading status, key numerical extractions and their qualifying conditions). In the final-state report as of 2026/7/19, the active bundle of the PETW LitWiki contained 368 registry records (341 direct records and 27 canonical pointers), of which 334 were `substantive_close_reading_verified`, 7 were `blocked_or_missing_full_text`, the direct semantic verification rate was 97.9%, and there were no pending or first-pass records. These numbers describe the coverage status of the knowledge layer and do not constitute a scientific quality score.

Wiki/RAG provides the claim-level evidence layer. The LitWiki organizes specific claims, methods, and values from the literature into Wiki pages, with each page containing a proposition description, source citation, applicable conditions, and known limitations. The RAG system indexes this information into searchable nodes and edges, enabling Agents to find existing knowledge relevant to the current task through semantic queries. The governance of the Wiki is transactional and role-isolated: source packets enter an isolation lane, the compiler produces candidate records, an independent reviewer profile re-reads the same batch of sources without inheriting the compiler's conclusions, and then the controlled registry executes the apply. Each record carries status labels such as `substantive_close_reading_verified` or `blocked_or_missing_full_text`, and commit markers bind the JSONL registry, SQLite mirror, backend registry, and governance manifest into a single verifiable bundle. Wiki publication and retrieval availability represent the engineering traceability of the knowledge layer and do not equate to scientific approval.

Review papers provide a continuously accessible academic synthesis. PETW has produced formal bilingual reviews or survey documents in six scientific lanes (Events, Windfield, EVA, WindTunnel, Structural, DesignLoad). These reviews are the highest-level expression of knowledge preparation for each lane, organizing scattered claims by topic and method into coherent academic narratives. Collected knowledge stores information from web pages, WeChat, and other external channels as a supplementary layer beyond literature sources. Project state and result reports store PETW's own work records and results as the project evidence layer.

The five resource categories follow a provenance chain: Zotero identity verification precedes Wiki claim extraction, review synthesis precedes Agent invocation, and project record updates

precede downstream task use. This sequence is not a linear pipeline but a resource tree traversed on demand during each task preparation.

11.1.3 Distinction between knowledge coverage and scientific sufficiency

Source counts, file sizes, retrieval pass rates, and full-text reading rates describe knowledge infrastructure coverage. Scientific sufficiency depends on whether these sources cover the current variables, events, scales, and comparison objects. A lane with a large amount of literature may still lack observations or verification under target conditions; an interface with very few records may still be clearly defined by highly relevant sources.

The research question should therefore define the completion criterion for knowledge preparation. Conducting event classification requires multi-source observations and mechanism literature; conducting EVA requires quantity endpoints, samples, and observation processes; conducting CFD requires source field variables, boundaries, grids, and pressure verification. A unified "knowledge base complete" label cannot substitute for task-specific coverage judgment.

Evidence and formula boundaries of Knowledge base of each agent Chapter 11 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Knowledge base of each agent." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 11 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

12 Usage and coordination of agents

12.1 Problem, concepts, and methods

12.1.1 Workflow for using a single agent

A single-Agent workflow begins with the Researcher's natural language task. After the Coordinator receives the task, it decomposes it into a structured task package containing the current scientific question, existing evidence status, new evidence to be produced, methods and tools to be used, required checking steps, completion conditions, stopping conditions, permitted interpretation range, and items requiring subsequent Researcher decision. The Coordinator selects the target Agent based on task content. The Agent reads sources relevant to the current task from its knowledge base, confirms existing evidence status, and then uses tools to execute the task within authorized scope.

After completing the work, the Agent generates a return package containing results, limitations, and decision requests. The return package is more structured than a free-text reply: numerical results are accompanied by units, source citations, and qualifying conditions; evidence status is annotated with formal labels; usage restrictions are expressed in explicit statements; and items requiring Researcher judgment are listed separately. The Coordinator aggregates the return packages, returns items requiring scientific judgment to the Researcher, and marks information that can be directly passed to downstream Agents as ready status.

Records at every stage make the results traceable. Task descriptions record problem definitions and boundary conditions. Execution records document the tools used by the Agent, sources read, and method configurations adopted. Return packages record result values, limitations, and decision requests. These records are stored in subproject directories and handoff files, forming a complete traceability chain from task initiation to result production.

12.1.2 Information flow among scientific agents

Information flows among the nine scientific agents along the scientific chain, following the causal order from events to design loads. This information flow is not an unconditional automatic transfer but a controlled transfer that must satisfy specific completeness requirements at each interface and can proceed only after the Researcher's cross-Agent use decision.

The Events-to-CM1/WRF interface transfers event identity, mechanism hypothesis, and observational context. This information enters the constraints of simulation configuration: the event's time window determines the simulation's time range; the organizational type of the parent storm influences the choice of simulation method (CM1 for mechanism prototype studies, WRF for event reconstruction); and observational data provide comparison benchmarks for simulation results. The interface from CM1/WRF to Windfield transfers three-dimensional time series of simulated wind fields and detailed parameters of simulation configuration. The Windfield Agent extracts engineering-relevant wind field features from these, including vertical profile shape, non-stationary characteristics, spatial coherence, and movement speed. The conversion from simulation grids to engineering wind field features involves coordinate system mapping, temporal sampling alignment, and profile fitting, among other specific operations, each of which introduces additional assumptions.

The Windfield interfaces transfer different forms of wind field feature packages. Transfer to EVA provides wind speed definitions, event independence information, and mechanism classification labels needed for extreme value statistical object construction. Transfer to CFD provides inlet conditions for building-scale transient simulations, involving scale matching issues from meteorological scale to building scale. Transfer to WindTunnel provides target velocity time history design references for accelerating-flow wind tunnel tests. The work in the three current directions progresses independently within their respective case conditions; data transfer under the same event conditions has not yet been achieved.

The CFD-to-Structural interface transfers transient surface pressure distributions or floor force time histories. Interface information must include inflow conditions (ABL profile parameters or non-stationary time histories), grid resolution (face element size and total count), temporal sampling (output frequency and total duration), reference pressure and reference density, solver settings (turbulence model and time advancement scheme), and evidence status labels. The interface from Structural to DesignLoad transfers response indicators (displacement, acceleration, internal forces, base shear, and overturning moment) and candidate correction factors, with each indicator accompanied by its dependent input load quality, structural model assumptions, and damping settings.

Every interface transfer includes an explicit declaration of usage restrictions. The upstream Agent annotates the applicability conditions and prohibited usage range when producing results. The downstream Agent must not automatically upgrade the evidence level or transfer numerical values across conditions upon receipt. For example, when CM1 simulation results under idealized cold-source conditions are transferred to Windfield, they carry the restriction "for mechanism research under idealized conditions only," and Windfield must not directly use those results for engineering analysis of specific real events. The write-out verification of the current PETWLoadPacket has not yet been closed; completing this verification and external independent solver cross-checking are among the tasks that need to be advanced in interface governance.

Table 4: Main interfaces among scientific agents and their transfer contents

Upstream agent	Downstream agent	Main content transferred at the interface
Events	CM1 / WRF	Event identity, mechanism hypothesis, observational context, time window
CM1 / WRF	Windfield	Simulated wind field time series, grid configuration, physical parameters, evidence status
Windfield	EVA	Wind speed definitions, event independence information, mechanism classification labels
Windfield	CFD	Inlet conditions, scale matching information, ABL profile parameters
Windfield	WindTunnel	Target velocity time history design reference, non-stationary feature parameters
CFD	Structural	Surface pressure distribution or floor force time histories, solver settings, temporal sampling
Structural	DesignLoad	Response indicators, candidate correction factors, input dependency conditions

12.1.3 Handoff semantics, fail-closed principle, and cross-lane consistency

A handoff is not merely a file notification; it is a usage-restricted transfer request for a named data object. The package specifies the producer, consumer, object and parent version, variables and units, coordinate and temporal semantics, method, evidence status, prohibited uses, and pending decisions. Sending only indicates arrival; acknowledgment only indicates readability; only after the downstream side completes field checking and the Researcher opens usage does the status change from received to admitted.

Interfaces are fail-closed by default. When event identity, units, time basis, coordinates, source version, or claim ceiling are missing, the downstream side must not guess or fill in default values. A failure receipt lists gaps, affected fields, preservable uses, and resubmission conditions. Timeout, no receipt, version mismatch, and status ambiguity are all treated as not admitted; computations and interpretations that depend on the gap remain closed.

When upstream and downstream agents disagree on variable definitions, evidence levels, or usage qualification, the Coordinator preserves both parties' opinions and organizes a comparison; the Researcher conducts arbitration. The decision record states the adopted definition, task scope, open interpretations, and artifacts requiring re-examination. The Coordinator does not substitute routing authority for scientific judgment; the receiver does not rewrite the upstream identity due to computational need. Disagreements that cannot be adjudicated remain parallel, and the interface stays closed.

Shared fields and usage-specific contracts jointly maintain cross-lane consistency. Event, variable, unit, coordinate, time window, version, and evidence status retain the same semantics; scale conversion and admission thresholds are specified separately for each interface. The Coordinator cross-checks the event, input version, and comparison axes of parallel packages. When upstream versions or Researcher decisions change, all consuming lanes enter impact review; unreviewed de-

rived products are marked as having stale dependencies. This status remains in effect until the relevant receivers complete named-field verification.

Evidence and formula boundaries of Usage and coordination of agents Chapter 12 retains only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Usage and coordination of agents." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 12 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

Part IV
Detailed Agent Descriptions and Results

13 PETW Coordinator

13.1 Evidence, results, and interpretation

13.1.1 Responsibilities and scientific positioning

PETW Coordinator is the coordination layer between the Researcher and nine scientific agents and fifteen support/collaboration agents. Its responsibilities are to decompose the Researcher's natural language scientific requirements into assignable work descriptions, select appropriate owner agents and support agents, arrange execution order, aggregate results from each lane, and return items requiring scientific judgment to the Researcher. The Coordinator does not hold conclusion authority over any scientific step, does not replace scientific agents in professional analysis, and does not substitute for the Researcher in scientific decisions. Its core function is to ensure that nine scientific agents answer the same PETW question rather than producing disconnected partial outputs: event classification labels must be consistent with EVA population definitions, CM1 output semantics must be consistent with Windfield feature definitions, and Structural response quantities must be consistent with DesignLoad candidate expressions. The Coordinator maintains this consistency through task packages, handoff rules, and status files. The coordination structure is an operational and methodological arrangement and does not establish any claim that scientific work has been completed.

13.1.2 Continuous interaction with the Researcher

Interaction between the Coordinator and the Researcher spans the entire life cycle of a task, covering all stages including task proposal, scope confirmation, agent selection, resource actions, execution supervision, result interpretation, writing review, and scientific conclusion formation. At task proposal, the Researcher states the scientific question, objectives, scope, and permissible discussion boundaries, and the Coordinator confirms understanding and translates the question into assignable work descriptions. During scope confirmation, the Coordinator lists the data, models, cases, comparison objects, and result forms involved in the task for the Researcher to confirm, and only proceeds to execution after confirmation. During agent selection, the Coordinator recommends an owner agent and support agents based on the scientific step and task nature; the final configuration is decided by the Researcher. Before remote computation, HPC job submission, facility actions, or experimental execution, the Coordinator aggregates the resource scope,

authorization scope, and stopping conditions and awaits the Researcher's explicit decision. When method disagreements, unclear data meaning, or unexpected result deviations arise during execution, the Coordinator records the disagreement and returns the decision request to the Researcher. During result interpretation, the Coordinator differentiates input preparation, computation results, observational evidence, model results, statistical inference, and engineering implications, helping the Researcher judge the evidence category and applicability boundary of each result layer by layer. During writing and external review, the Coordinator organizes sources, versions, and response suggestions, and the Researcher decides on text and response content. Scientific conclusions can only be made by the Researcher.

Project organization directly reflects the Researcher's structural decisions. The Step 5/6 merger decision of 2026/6/16 combined the original Step 5 design wind load derivation and the original code interface step into Step 5 unified under DesignLoad, eliminating Standardization as a separate lane. The scientific rationale is that the chain from wind, force, and response to design expression should be treated continuously, and the code interface cannot be separated from the wind, force, and response relationships under the non-stationary time-frequency expression; the core theoretical reference is the generalized wind loading chain of Kareem, Hu, Guo, and Kwon (2019)[Kareem et al., 2019]. The Coordinator is responsible for executing such decisions and updating the operational map, including Master State, each lane's workspace, and knowledge namespaces; the decisions themselves belong to the Researcher.

13.1.3 Current issues and future work

Areas where the Coordinator currently needs improvement include: cross-agent scientific variable definitions still depend on each lane's packet contracts and lack a project-wide unified variable registry, so that the semantics, units, and provenance of the same variable across lanes are not yet fully consistent; figure reuse depends on each lane's figure register, and cross-report harmonization still requires manual judgment; result comparison lacks a standardized comparison format, so that results from different models and different cases cannot yet be compared against the same condition matrix; the Researcher interface still relies primarily on direct conversation and report reading and lacks a structured decision queue view, so that items requiring judgment cannot yet be presented in a compact and checkable form. These improvements help reduce cross-lane disruptions and do not change the basic principle that the Researcher is the scientific decision-maker.

Evidence and formula boundaries of PETW Coordinator Chapter 13 preserves only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "PETW Coordinator." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 13 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

14 Events Agent

14.1 Scientific questions, knowledge, and Researcher decisions

14.1.1 Scientific purpose

Events Agent handles the event identification and reconstruction component of Step 0 in the PETW event classification and reconstruction pipeline. Its scientific purpose is not to directly provide mechanism ground truth but to organize heterogeneous meteorological records into auditable candidate evidence, specifically including multi-source event identification, event identity,

parent storm context, surface evidence, local mechanism hypothesis, and event evidence packages for downstream employ. In the literature, a downburst is defined as a convective phenomenon in which a localized strong downdraft from a thunderstorm diverges at the surface to form near-surface outflow[Fujita, 1985]; its spatial extent is far smaller than synoptic-scale processes, so event-level identification must rely on the joint interpretation of multi-platform observations rather than labels from a single source[Gaetano et al., 2014, Gunter and Schroeder, 2015]. High-resolution surface observation network studies have shown that the complete characterization of an event requires a combination of surface sequences, radar volume observations, and background environmental information[Iwashita and Kobayashi, 2019, Childs et al., 2021]. Based on this, the Events Agent divides its work units into five tiers: source record, evidence record, candidate state, reconstruction packet, and bounded handoff; each tier has an independent evidence gate, and no tier is automatically promoted because an upstream tier is complete. This component provides candidate inputs for the simulation value judgment of CM1 and WRF, provides upstream event classification evidence and uncertainty descriptions for EVA, and does not itself estimate return periods or produce any design quantities.

14.1.2 Knowledge base

The academic assets of Events comprise three parts: review archives, Wiki/RAG, and external literature libraries; these serve different roles. The Events Step 0 Review v2 Chinese edition totals 15 A4 pages, containing 55 source-bounded Events records and 1 Moore 2024 interface source, 56 bibliography keys and 2 vector schematic figures, with scientific status not_scientifically_approved. The function of this review is to define literature evidence and method boundaries, not to approve a classifier or event ground truth. The active Step 0 Wiki/RAG includes 488 records, of which 55 are approved source-bounded Step 0 evidence and 433 are unapproved; its design objective is to support evidence gap identification using source-bounded evidence, and retrieval results still require PI review (evidence status: engineering preparation).

The external literature library is clearly separated by permitted roles; the applicable scope and prohibited scope of each source are explicitly registered in the Events knowledge governance. Cook ASOS v2 is a thunderstorm wind classification dataset based on the Automated Surface Observing System; its development set includes 9,014 events, and the national analysis set exceeds 10 million events used for TNNDT/XTNNDT machine labels; it is permitted for employ in candidate discovery and source-derived weak label stratification but is not permitted for confirmed positive without independent external evidence, because ASOS records themselves do not include mechanism information. Sui v2 is based on GridRad v3.1 hourly volume observations with a minimum grid resolution of 1 km, providing parent storm context for four Texas cities; it is permitted for use in parent-storm context and radar-led candidate discovery but is not permitted for local-mechanism truth determination, because GridRad's temporal resolution and spatial smoothing prevent it from capturing transient divergence features at convective scales. GridRad-Severe is a rapid candidate ranking tool for severe convective cases with a starting height of approximately 0.5 km; it is permitted for rapid candidate ranking but not for 10 m divergence estimation or confirmed downburst labels, because its lowest height is far above the characteristic height of near-surface outflow. NEXRAD Level II (WSR-88D raw radar data) provides beam-geometry-aware radial velocity and polarimetric evidence; its key limitation is that the lowest-elevation beam at distant stations can be several kilometers above the surface, making near-surface divergence structures not directly observable. The Storm Events database provides surface report recovery, but its reported wind

speed values typically come from the nearest routine station or estimates rather than independent measurements. IEM provides additional surface data and cross-platform retrieval of historical observations. MRMS provides spatiotemporal context of storm evolution through multi-source reflectivity compositing at approximately 2-minute temporal resolution and 1 km spatial resolution. SWDI (Severe Weather Data Inventory) provides mesocyclone and TVS (tornado vortex signature) detection records for rotation signature contamination screening, preventing rotation-dominated events from being misattributed to downbursts.

Literature evidence indicates event classification and mechanism attribution methodologically depend on hierarchical combinations of source labels, morphological features, and auxiliary information[Cook, 2023a, 2024a]; systematic differences between different labeling systems (such as ASOS automatic labels and manual Storm Events labels) are among the main sources of classification uncertainty. The locality of thunderstorm outflow makes station network coverage a hard constraint on detectability[Gunter and Schroeder, 2015]: when the horizontal scale of a microburst is less than 4 km and the nearest station spacing exceeds 28 km, the event may be completely unrecorded by surface stations; even when recorded, a single-station time series is insufficient to differentiate between downburst outflow and competing mechanisms such as gust front leading-edge vortices. The above assets collectively constitute the established portion of event classification literature and classification tiers; no single source provides complete mechanism ground truth.

14.1.3 Researcher participation and decisions

In the Events stage, the Researcher's PI intervention centers on scientific boundary decisions rather than approving individual records one by one. PI decided to stop the BESSIE case re-run and package design unless a new explicit decision reopens it, and stipulated that BESSIE maintains its positioning as an observational convective outflow reference without changing any source labels. PI determined that the M0/M1 879-row data can only be used as candidate records for engineering preparation and must not be treated as 879 independent events; the three anchor events OKEM, BESSIE, and ERICK maintain provisional OBS status; further evidence recovery for M4 requires separate authorization. Regarding thresholds and definitions, PI decided that definitions for event detectability, candidate ranking, positive evidence, hard negative examples, and event grouping must first undergo definition registry review (S0); when traceable thresholds are missing, the field is left empty rather than guessed. PI also decided that the direct entry point for Events employs the Lark channel, the automation level is limited to A0 through A2, A3 is open only for explicitly approved specific execution scopes, and A4 closed-loop or hardware autonomy remains deferred. These decisions help maintain evidence tier separation: scope approval does not equal scientific approval, engineering completion does not equal mechanism establishment, and any result must pass the corresponding evidence gate before entering downstream.

14.2 Methods, results, and figures

14.2.1 Workflow and interactions

The Events evidence cycle begins with source records. Source records preserve the 19 columns of the source table, native integer knots, and derived unit conversions (mph and m/s), and generate the provisional OBS tier. Evidence records then separately store source-derived labels, derived screening fields, independent evidence, mechanism hypotheses, and promotion status. Candidate state and reconstruction packet then aggregate by case, and finally a bounded handoff transfers to downstream. The M0/M1 stage generated 879 OBS, 879 evidence, and 879 narrative records for 879 source rows; 242 EPISODE_IDs and 140 multi-row episodes explicitly preserved the multiplicity

of source episodes, with a maximum multiplicity of 42; event_id remains null (evidence status: engineering preparation). A total of 12 recovery tests passed, and two deterministic re-runs were byte-identical. The 879 rows are source records, not 879 independent events: truth_status = not_final; event independence, mechanism, Path C, EVA, CM1 validity, and confidence all remain fail closed.

Multi-source reconstruction integrates Mesonet surface sequences, Storm Events reports, NEXRAD Level II raw radar data, MRMS multi-source reflectivity composites, GLM lightning activity, and environmental profiles (soundings or reanalysis data) into an event reconstruction package. The combination principle is that each source in the reconstruction package preserves its native time, space, units, and instrument characteristics; associations between different sources are established through time windows and spatial distances rather than through data fusion that eliminates differences. The acquisition manifest for the BESSIE case includes 217 files and 26 compact manifest entries; hash verification PASS; parent verification passed. The acquisition scope covers several hours of Mesonet surface data before and after the event, the complete volume scan sequence from three NEXRAD Level II stations, MRMS reflectivity grids at approximately 2-minute intervals, per-event GLM lightning records, and the OUN sounding.

Detectability calculation and mechanism status are kept strictly separate. Detectability depends on observation geometry: the AGL height of the radar beam above the event location determines whether the radar can sense near-surface divergence structures; station network distance determines whether surface stations can record the peak wind speed within the footprint; and record availability (such as data quality flags and missing data) constrains the actual information content of each source. Mechanism attribution belongs to the interpretation layer and depends on the joint analysis of all available evidence; evidence from any single source is insufficient to determine the mechanism. The competing mechanism hypotheses include four: propagating convective outflow (gust front passage), characterized by a linear leading edge and uniform temperature drop; downburst or microburst, characterized by localized radial divergence and steep temperature drop; rotation contamination, where tangential velocity from a mesocyclone or TVS superimposes on the outflow; and propagating multi-feature process, where multiple convective features influence the same station during approximately the same period. All four hypotheses are retained in parallel without exclusion, because currently available evidence is insufficient to make a definitive determination among them.

The Events-to-CM1 interface provides conditional input. The event reconstruction package provides CM1 with the event time window, location and station geometry, observed wind speed and direction with their measurement definitions, candidate storm morphology (such as isolated cell, bow echo, or linear system), availability and quality assessment of radar and reanalysis data, the applicability rationale for CM1 simulation, and data gap inventory. CM1 is responsible for judging simulation worthiness; Events does not promote events with only ASOS records and no independent evidence directly to CM1-valid input. The interface from Events to EVA provides only upstream evidence: the event's classification labels, quality and coverage of source labels, confidence in competing mechanisms, spatiotemporal coverage of observations, and station exposure openness determinations. Event independence judgment (whether different phases of the same parent storm system should be treated as independent events), sample membership, tail distribution estimation, and design quantity boundaries are all the responsibility of EVA; Events does not prejudge these statistical questions. The current automation level is primarily at A0 through A2: A0 covers planning and schema definition, A1 covers read-only ingestion and metadata extraction, and A2

covers supervised processing, QA, and evidence packaging.

14.2.2 Completed work and main results

Recovery and staging of 879 historical candidate records. The M0/M1 stage performed a complete recovery and staging of 879 historical candidate records. The source table preserves 19 original columns, with wind speeds stored as native integer knots alongside derived mph and m/s conversion values; timestamps preserve the literal CST-6 to UTC semantic correspondence. Each source row generates three independent records: 879 provisional OBS records (preserving raw observational quantities and source identification), 879 evidence records (separately storing source-derived labels, derived screening fields, independent evidence, and mechanism hypotheses), and 879 narrative records (preserving the original text and source path of event narratives). Among the 879 rows, 242 EPISODE_IDs were identified, of which 140 are multi-row episodes; the maximum multiplicity is 42 (i.e., a single episode includes up to 42 source rows); event_id remains null across all records. A total of 12 recovery tests passed, and two deterministic re-runs were byte-identical. The 879 rows are source records, not 879 independent events: truth_status = not_final; event independence, mechanism, Path C, EVA, CM1 validity, and confidence all remain fail closed.

M3 bounded anchor recovery. The M3 bounded anchor recovery retrieved 35 files and 873,948 source bytes (upper limit 5,000,000 passed), used for data recovery and event information structure description. Three anchor events maintain provisional OBS status: the OKEM event peak time 2019-07-06T18:50:00Z, wmax 31.8 m/s; the BESSIE event 2019-07-04T02:10:00Z, wmax 34.9 m/s; and the ERICK event 2018-05-30T01:15:00Z, wmax 35.4 m/s. The selection criteria for the three anchors are high peak wind speed, good multi-source evidence availability, and value as conditional input for subsequent CM1 simulation. All promotion counters (new events, mechanism labels, Path C, EVA-ready, CM1-valid, WRF-valid, design-wind) remain at zero (evidence status: engineering preparation).

BESSIE multi-source observational reconstruction. The BESSIE observational reconstruction (2019-07-04) is the most complete observational result to date. The BESSIE observation package includes 217 acquisition files and 26 compact manifest entries; hash verification PASS. The BESS station (Oklahoma Mesonet) recorded WMAX 34.9 m/s at 02:10 UTC; the quantity definition for this value is

$$W_{\max}(t_0) = \max_{k=1,\ldots,N} v_k, \tag{3}$$

where v_k is the k-th 3-second observation (in m/s) within the 5-minute Mesonet period $[t_0, t_0 + 300\,\mathrm{s})$, N is the number of 3-second observations within that period, and the measurement height is 10 m. The 34.9 m/s obtained from Equation (3) is a bounded result within the observational domain; it is not equivalent to a recovered continuous 3-second time series. This value does not establish a unique mechanism, nor does it constitute a completed event catalog. The 68 kt record in the Storm Events database shares the same source as BESS Mesonet.

The same surface sequence shows a temperature drop of 5.2 °C and a pressure rise of 1.29 hPa within 10 minutes, with precipitation beginning at 02:15 (Table 5). The surface transition amplitudes are defined by endpoint differences:

$$\Delta T = T(t_2) - T(t_1), \qquad \Delta p = p(t_2) - p(t_1), \tag{4}$$

where T is temperature (°C), p is pressure (hPa), and taking t_1 = 02:05 UTC, t_2 = 02:15 UTC gives $\Delta T = -5.2\,^{\circ}\mathrm{C}$, $\Delta p = +1.29\,\mathrm{hPa}$ from Table 5 (evidence status: bounded observational

result). The transition sequence shows peak wind speed occurred during the rapid temperature drop phase and preceded the onset of precipitation; the 02:15 precipitation onset was accompanied by a 1.29 hPa pressure jump and rapid wind speed decay; by 02:20 wind speed had dropped to 13.6 m/s. The entire transition process completed within approximately 15 to 20 minutes, consistent with the time scale of convective outflow in the literature. The surface transition and convective association are supported; propagating convective features are supported only at the event-family scale (evidence status: candidate interpretation). Downburst-specific reconstruction remains NOT_ESTABLISHED; BESSIE is not used as a confirmed downburst mechanism. The station exposure and sensor maintenance adjudication for the BESS station on the event day remains open.

UTC	Temperature (°C)	Pressure (hPa)	Mean wind speed (m/s)	WMAX (m/s)	Precipitation (mm)	Interpre
02:00	26.9	951.28	14.4	19.8	0.0	Pre-tran
02:05	25.0	951.65	15.9	26.4	0.0	Pre-prec
02:10	22.9	951.62	20.5	34.9	0.0	Measure
02:15	19.8	952.94	14.5	18.6	3.56	Precipit
02:20	18.7	952.38	9.1	13.6	7.62	Post-tra

Table 5: BESS station observed surface sequence (2019-07-04). WMAX is the maximum of 3-second observations within a 5-minute Mesonet period at 10 m height and is not a recovered continuous 3-second time series; the station exposure and maintenance adjudication on the event day remains open. Source: BESSIE observational reconstruction package non-radar analysis (Events integrated program report, 2026-08-12); evidence status: bounded observational result.

Storm-scale evidence and detectability. Storm-scale evidence is summarized in Table 6. Three NEXRAD Level II stations (KFDR, KTLX, KVNX) provided 9 scans covering the BESSIE event period. The velocity gate heights above BESS are approximately 1.568 km to 3.031 km AGL, all above the boundary layer, and cannot directly sense near-surface divergence. 9/9 raw along-radial local gradients are negative, but 1 scan is a dealiasing artifact; dealiased values have nonzero sensitivity. Radial gradient is a 20 km local fit proxy, not complete horizontal divergence. This evidence supports beam-height-aware storm background analysis and does not establish near-surface divergence proof or classification exclusion.

At 02:10, the MRMS core was offset approximately 9.1 km east and 2.22 km north relative to BESS; the centroid velocity within the tracking window was 31.32 km/h; the bess_hsr_raw_value at 02:10 is a missing sentinel value and is not used as low reflectivity. Three-station intense convective core passage; maximum reflectivity within the nearest scan at 50 km is approximately 62 to 64 dBZ, representing storm background rather than local downburst ground truth. GLM recorded 40 flashes within 25 km and 44 flashes within 50 km in the 02:10 five-minute bin, indicating only active convection with low mechanism specificity. The nearest non-BESS station is approximately 28.27 km away; response is delayed and spatially inhomogeneous; the station network is too sparse to obtain a core-scale footprint, which is especially limiting for microburst/macroburst extent judgment. The OUN sounding is 148.3 km from the event and 2.17 h prior; MUCAPE 2188.9 J/kg, MLCAPE 1860.7 J/kg, DCAPE 728.0 J/kg; the presence of DCAPE supports regional-scale downdraft convective potential, but the spatiotemporal distance of the sounding means it can only characterize the regional environment, not the event's local conditions (evidence status: bounded observational result).

Job 3388245 conditional simulation diagnostics. The CM1 output of Job 3388245 is a source-traceable/display-only conditional simulation result: 241 NetCDF files, 3600 s integration, 5 spatial snapshots at 0 s/900 s/1800 s/2700 s/3600 s, and 60 s statistical sequence. Model-internal diagnos-

Evidence item	Observed value	Evidence status and limitations
MRMS core offset	02:10 maximum offset approximately 9.1 km east, 2.22 km north; centroid velocity within tracking window 31.32 km/h	SUPPORTED; bess_hsr_raw_value at 02:10 is a missing sentinel value
Level II reflectivity	Three-station intense convective core passage; maximum within nearest scan at 50 km approximately 62 to 64 dBZ	SUPPORTED; storm background
Raw radial gradient screening	9/9 raw along-radial local gradients negative; 1 scan is dealiasing artifact	PARTIAL; 20 km local derivative is proxy; beam height approximately 1.568 to 3.031 km AGL
GLM lightning	02:10 five-minute bin: 40 flashes within 25 km, 44 within 50 km	SUPPORTED; indicates active convection only
Adjacent station response	Nearest non-BESS station approximately 28.27 km; response delayed and spatially inhomogeneous	LIMITED; station network too sparse
OUN sounding	148.3 km from event, 2.17 h prior; MUCAPE 2188.9, MLCAPE 1860.7, DCAPE 728.0 J/kg	REGIONAL CONTEXT

Table 6: BESSIE storm-scale evidence and detectability. Source: BESSIE observational reconstruction package radar and non-radar analysis (Events integrated program report, 2026-08-12); evidence status: bounded observational result; no individual item establishes mechanism attribution.

tic peaks appear in Table 7. Maximum vertical velocity wmax 22.4418 m/s at 2340 s, maximum precipitation rate pratemax 0.00936125 $\mathrm{kg\,m^{-2}\,s^{-1}}$ at 2400 s, maximum downdraft velocity wmin −11.4658 m/s at 2520 s, and maximum three-dimensional divergence divmax 0.00118535 $\mathrm{s^{-1}}$ at 2580 s. The diagnostic time series shows the typical successive development of ascent, precipitation, descent, and divergence, but the model-internal sequence is not equivalent to the temporal evolution of the observed event; the retained 2700 s lowest height of approximately 2.10 km AGL does not establish a touchdown downdraft core. The s10 maximum of 2.98344 m/s is an instantaneous near-surface diagnostic, distinct from the lowest model level cumulative sws and observational WMAX. The Level-1 candidate status is NOT_SUPPORTED_UNDER_FROZEN_GATE (evidence status: candidate model diagnostic evidence).

Taken together, current evidence supports the following: the measured gust and convective outflow transition of BESSIE are supported; propagating convective features are supported at the event-family scale; downburst-specific mechanism reconstruction is NOT_ESTABLISHED; no EVT identity, mechanism ground truth, Path C, EVA-ready status, validated CM1, validated downburst, $\hat{V}_{3s}$, design wind, or design load has been created.

14.2.3 Figures

Figure 1 depicts the BESS Mesonet surface transition sequence. Figure 2 shows the station network geometry at 02:10 UTC. Figure 3 shows the storm-scale background from MRMS and GLM. Figure 4 shows the three-station Level II reflectivity and radial velocity displays. Figures 5 and 6 show the temporal diagnostics and 2700 s vertical cross-section from Job 3388245. Figure 7 shows five discrete Moore-style diagnostic center snapshots. Figure 8 shows the BESSIE v2 storm tracking and surface observation overlay. Figure 9 shows the Events work cycle.

Diagnostic quantity	Peak value	Occurrence time	Meaning and limitations
wmax	22.4418 m/s	2340 s	Model-internal maximum vertical velocity
pratemax	0.00936125 $\mathrm{kg\,m^{-2}\,s^{-1}}$	2400 s	Model-internal maximum precipitation rate diagnostic
wmin	−11.4658 m/s	2520 s	Model-internal downdraft diagnostic; retained spatial cadence did not sample all peak morphologies
divmax	0.00118535 $\mathrm{s^{-1}}$	2580 s	Model-internal maximum three-dimensional divergence; not observed surface divergence
s10 maximum	2.98344 m/s	60 s statistical sequence	Instantaneous near-surface diagnostic; distinct from cumulative sws and observational WMAX

Table 7: Job 3388245 model-internal temporal diagnostic peaks (241 NetCDF files, 3600 s integration, 5 spatial snapshots, 60 s statistics). Source: Job 3388245 results display package v2 temporal diagnostics JSON (Events integrated program report, 2026-08-12); evidence status: candidate model diagnostic evidence; Level-1 candidate status is NOT_SUPPORTED_UNDER_FROZEN_GATE.

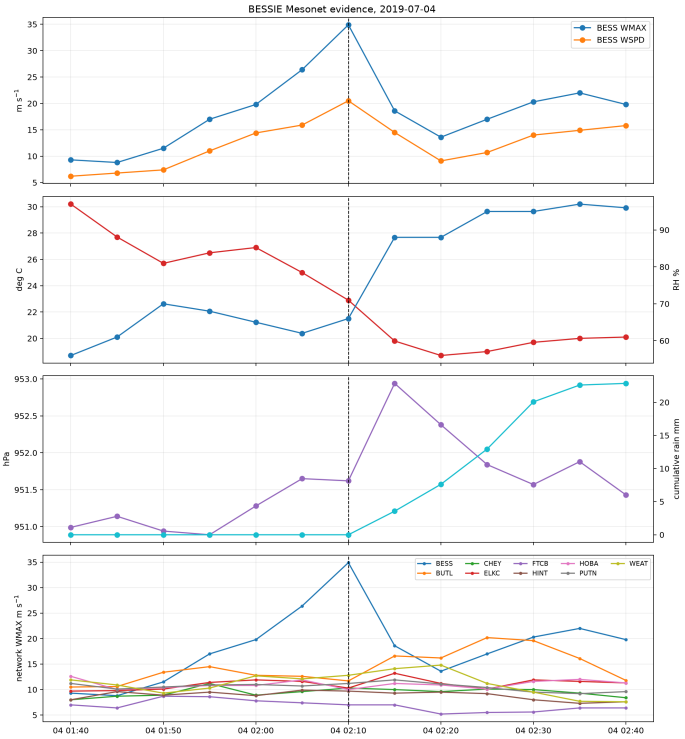


Figure 1: BESS Mesonet surface transition sequence (2019-07-04, 02:00 to 02:20 UTC): observed changes in temperature, pressure, wind speed, and WMAX, including the WMAX peak at 02:10 UTC. Source: PETW BESSIE observational reconstruction package (original figure); evidence status: bounded observational result; supports measured transition but not a unique local mechanism.

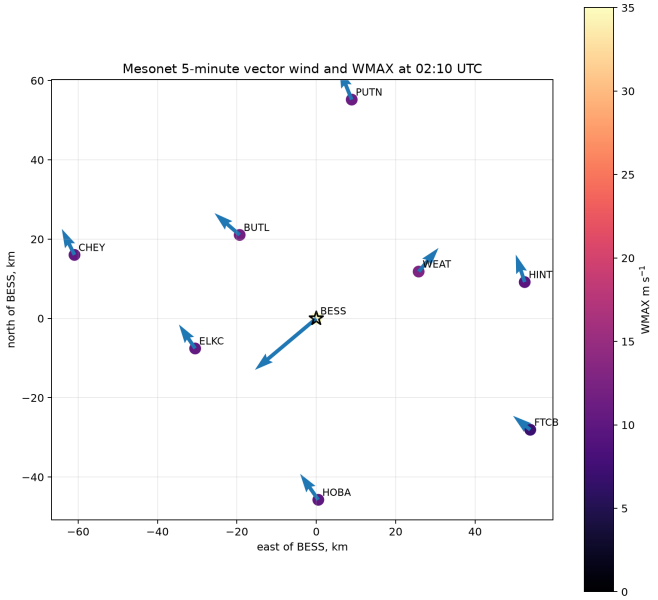


Figure 2: Station network geometry of BESS and adjacent stations at 02:10 UTC. Source: PETW BESSIE observational reconstruction package (original figure); evidence status: bounded observational result; nearest non-BESS station approximately 28.27 km away; sparse station network does not constitute a core-scale footprint.

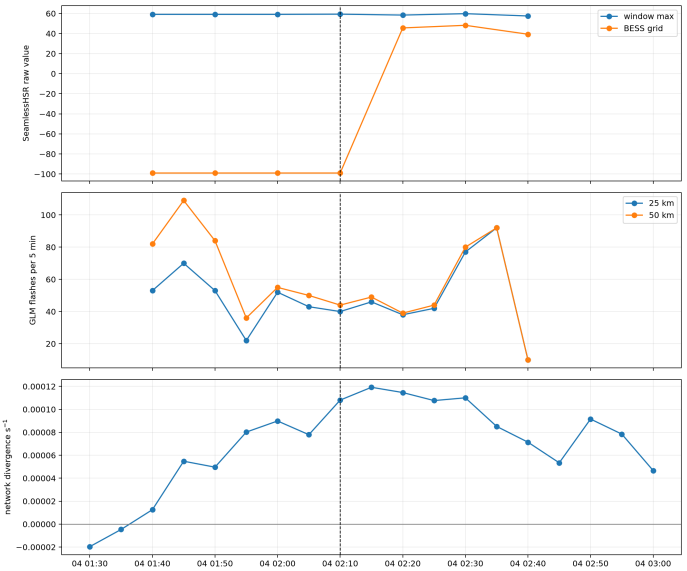


Figure 3: MRMS and GLM network proxy map: storm core offset and lightning activity background near 02:10 UTC. Source: PETW BESSIE observational reconstruction package (original figure); evidence status: bounded observational result; supports active convection and propagation background but not mechanism ground truth.

14.3 Limitations, diagnostics, and interfaces

14.3.1 Current scientific limitations and future research

14.3.2 Evidence association, detectability, and downstream qualification

The central method of Events is to preserve differences between sources and then determine what associations those differences permit. Surface record timestamps, radar volume scan effective times, MRMS grid times, GLM event times, and environmental profile times have different sampling support. The reconstruction package employs a unified UTC axis to establish association windows but does not interpolate all data into an apparently synchronous single sequence. Spatially, station positions, radar beam paths, grid cells, lightning location errors, and sounding representative ranges are likewise preserved. Temporal proximity and spatial proximity can only establish candidate

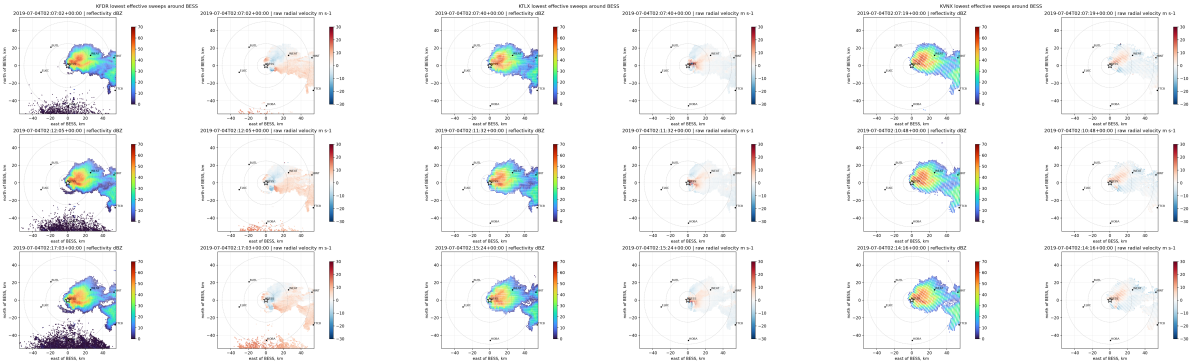


Figure 4: KFDR, KTLX, and KVNX three-station NEXRAD Level II reflectivity and radial velocity displays. Source: PETW BESSIE observational reconstruction package (original figure); evidence status: bounded observational result; beam heights approximately 1.568 to 3.031 km AGL; beam-height-aware radar background that does not constitute near-surface divergence proof; the three-station background does not eliminate single-Doppler and beam-height limitations.

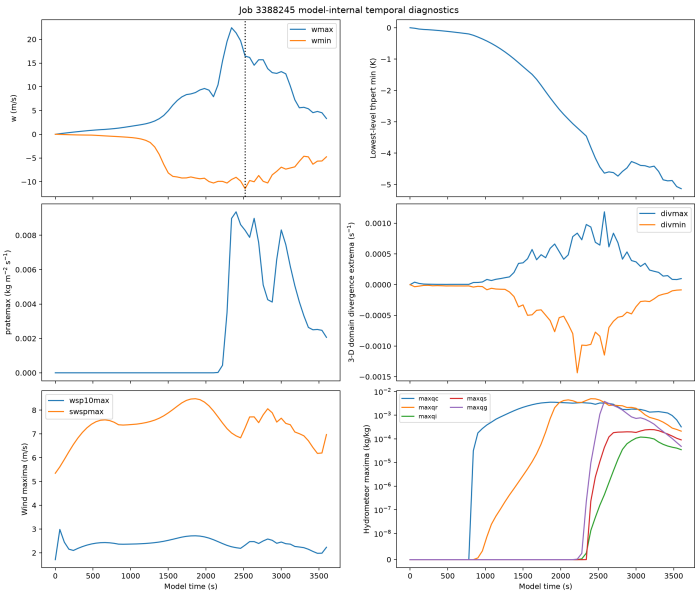


Figure 5: Job 3388245 model-internal temporal diagnostics from 0 to 3600 s (60 s statistical interval). Source: PETW Job 3388245 results display package v2 (original figure); evidence status: candidate model diagnostic evidence; model-internal sequence that does not constitute event reconstruction.

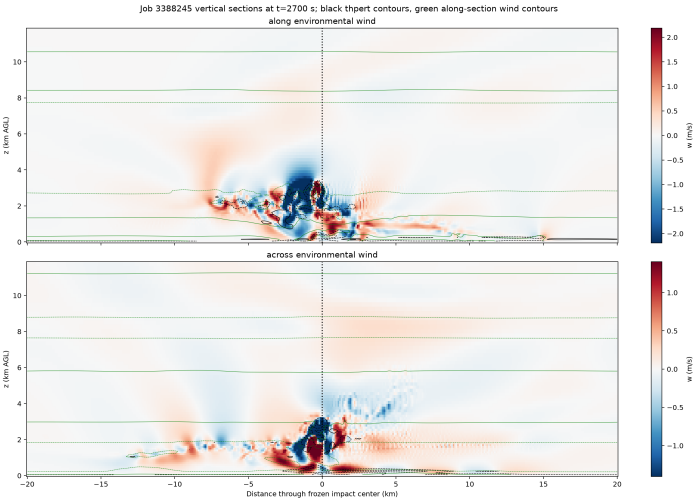


Figure 6: Job 3388245 vertical cross-section at 2700 s. Source: PETW Job 3388245 results display package v2 (original figure); evidence status: candidate model diagnostic evidence; retained 2700 s lowest height of approximately 2.10 km AGL does not establish a touchdown downdraft core.

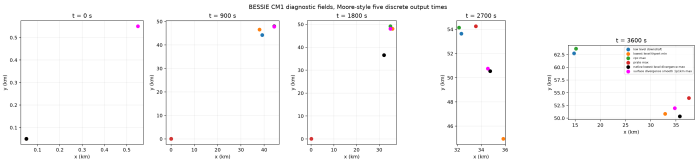


Figure 7: Job 3388245 five discrete Moore-style diagnostic center snapshots at 0/900/1800/2700/3600 s. Source: PETW Job 3388245 results display package v2 (original figure); evidence status: candidate model diagnostic evidence; discrete snapshots are not a continuous trajectory.

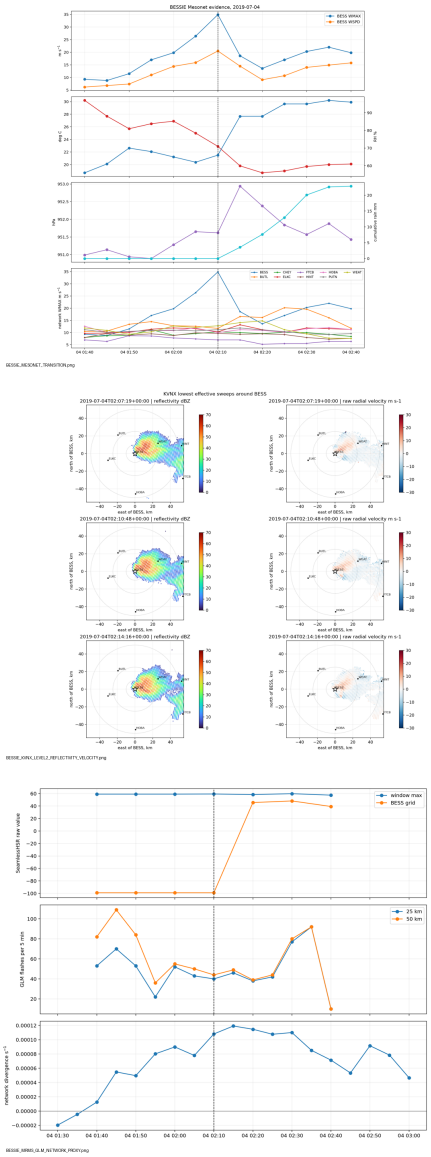


Figure 8: BESSIE v2 storm tracking and surface observation overlay. Source: PETW BESSIE observational reconstruction package v2 (original figure); evidence status: bounded observational result; supports propagating convective background but does not establish mechanism ground truth.

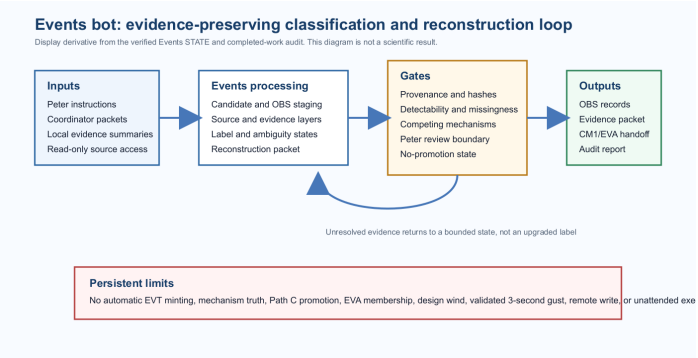


Figure 9: Events work cycle: source record, evidence record, candidate state, reconstruction packet, bounded handoff, and Researcher approval boundary. Source: Events integrated program report (flowchart); evidence status: generated input.

associations and cannot alone establish the same physical process. Each association must record the source pair, time difference, horizontal distance, vertical representative height, availability flag, and competing interpretation, avoiding the rewriting of "co-occurrence within the same window" as causal relationship during post-processing.

The measurement chain determines the identity of a station peak. The BESS station 34.9 m/s corresponds to the Oklahoma Mesonet BESS station, 10 m measurement height, 5-minute reporting period, and the maximum of 3-second observations within that reporting period. This meaning differs from a continuous 3-second moving average series, an event spatial maximum, a radar-retrieved wind, or a model grid peak. Its temporal localization accuracy is constrained by the reporting period; its spatial support is constrained to a single station; the exposure and maintenance status on the event day remains to be adjudicated. Thus, this observation can support "the BESS station recorded a strong wind transition" and, combined with temperature, pressure, and precipitation changes, can support "the observation is compatible with convective outflow passage," but it cannot alone support outflow scale, center location, movement speed, or a unique local mechanism.

Radar detectability must first be constrained by beam geometry. Let the radar antenna elevation be z_r (m MSL), the event surface elevation be z_s (m MSL), and the beam center elevation at a given radar range and elevation angle be z_b (m MSL); then the beam center height above ground at the event location is defined as

$$h_b = z_b - z_s, \tag{5}$$

where h_b is in m AGL. This height determines the atmospheric layer represented by the radial velocity evidence. The velocity gates of the three BESSIE stations are approximately 1.568 to 3.031 km AGL; thus, along-radial local gradients can describe only the storm background above the near-surface layer. Negative radial gradients are also affected by the radar line-of-sight direction, velocity dealiasing, local fitting window, and three-dimensional flow projection. Even if multiple scans have the same sign, they cannot be converted into surface two-dimensional divergence. Retaining the screening results of 9/9 raw scans and the one dealiasing artifact is intended to show diagnostic reproducibility and sensitivity rather than to employ counting to vote on the mechanism.

Station network detectability can be expressed by the relationship between the nearest effective station distance and the candidate footprint scale, but no approved general footprint threshold currently exists. If the candidate core influences only an area smaller than the station spacing,

then the absence of a comparable peak at adjacent stations cannot serve as hard negative evidence; if the station is located outside the core, then it records only the outflow margin. The nearest non-BESS station is approximately 28.27 km from BESS, far greater than the local scale commonly associated with microburst-scale events in the literature; thus, the delayed and inhomogeneous response at adjacent stations can neither reconstruct the core footprint nor exclude local strong winds. Under such conditions, Events reports only coverage limitation and keeps footprint, event independence, and mechanism status open.

Parent storm evidence and local mechanism evidence require stratification. MRMS core offset and centroid motion describe storm organization and propagation; GLM counts describe active convection; maximum reflectivity describes intense echo background; regional sounding describes instability and downdraft potential. These sources can collectively strengthen the judgment that "the event occurred within an active convective environment," but they do not have the same mechanism specificity. The surface temperature drop, pressure rise, wind direction shift, and precipitation transition increase the consistency of the convective outflow interpretation, but propagating gust fronts, local downbursts, rotation superposition, and multi-feature propagation may still produce partially similar signals. The reconstruction package thus preserves three categories of fields: supporting items, refuting items, and unobserved items, avoiding the misinterpretation of missing data as counterevidence.

The qualification judgment toward CM1 does not substitute wind speed ranking for scientific rationale. A candidate package must explain the mechanism hypothesis testable by idealized simulation, the environmental information available as boundary or initial conditions, the quantities available for comparison with observations, and the quantities that cannot be verified because of observation geometry. The sounding is 148.3 km from the event and 2.17 h prior; it is suitable only as a regional environmental clue and cannot serve as event-local profile ground truth. The qualification toward EVA is stricter because source records, episodes, and independent events are different statistical units. The M0/M1 879 rows include 242 source EPISODE_IDs and 140 multi-row episodes and have not yet formed an independent event set usable for extreme value analysis. Events cannot decide population membership for EVA based on row counts, peak rankings, or source labels.

Events processing follows the fail-closed principle. When data are missing, temporal semantics are unclear, unit conversions cannot be traced back, station identity conflicts arise, duplicate episodes are unresolved, mechanism evidence comes from only a single source, or promotion thresholds lack registered sources, records remain at the current tier. Deterministic re-run checking verifies whether the same input and version can produce the same structured products; it does not verify whether the event interpretation is correct. Hash checking confirms that files have not changed; it does not confirm that the values in the files represent the true mechanism. After passing these engineering gates, the Researcher must still read the evidence matrix, competing interpretations, and undetectable portions before deciding whether to form a truth-seed, whether to enter blind evaluation, or whether to maintain ambiguous status.

Closure of event identity also requires distinguishing multiple source records from the same parent storm from independent surface events. Source EPISODE_ID reflects the organizational structure of the database and does not necessarily correspond to a single physical outflow; reports from adjacent stations during similar periods may belong to the same propagating system; multiple rows from the same station may describe different phases of one process. Independence judgment

requires the joint consideration of parent storm track, time interval, spatial continuity, and local signal rather than de-duplication by a fixed number of hours alone. Events can prepare candidate groupings and sensitivity branches, but once grouping rules influence EVA event rates and tails, they must be reviewed jointly by EVA and the Researcher. Retaining null event_id currently helps avoid prematurely solidifying the data table row structure into a statistical population.

Quality control comprises record, association, and interpretation levels. Record-level checking covers units, missing sentinels, time formats, station metadata, and obvious instrument flags; association-level checking covers whether different sources correspond within the permitted spatiotemporal support; interpretation-level checking covers whether supporting items have mechanism specificity and whether competing interpretations are fairly preserved. The MRMS missing sentinel value cannot be interpreted as low reflectivity; the dealiasing artifact cannot be interpreted as a true velocity gradient; the Storm Events 68 kt record shares the same source as BESS Mesonet and cannot serve as a second independent wind speed evidence. Identifying these dependencies reduces the apparent number of evidence items but increases the effectiveness of the reconstruction package for downstream judgment.

The function of the blind evaluation stage is to separate candidate generation from mechanism adjudication. Candidate ranking can employ source-derived weak labels, wind speed intensity, and multi-source availability to allocate reading resources, but truth-seed determination should examine evidence without exposing ranking scores or expected categories. Blind evaluation still cannot overcome the hard limitations of observation geometry; thus, its output permits supported, opposed, or indeterminate. Retaining the complete packet for indeterminate cases permits assessment of the selection effect caused by station network and radar configurations rather than forcing every candidate into a binary category. This design enables subsequent EVA to explicitly handle membership and detection-process uncertainty.

Current limitations are as follows: no systematic verification chain for multiple event types yet exists; event independence rules, mechanism attribution, and EVA population still require research; single-Doppler geometry and beam height make low-level divergence ground truth unreachable and also prevent classification exclusion; sparse station networks prevent closure of the core-scale footprint for intense events; event-level Mesonet quality control adjudication, station exposure and sensor maintenance review, raw radar and satellite data interpretation, and formal association rules all remain open; the retained snapshots of Job 3388245 are discrete output observations, not a continuous trajectory. Future work advances in tiers S0 through S4. S0 freezes the definition registries for detectability, candidate ranking, and event grouping; S1 conducts read-only detectability census and Cook v2 candidate extraction; S2 performs Texas four-city overlap and stratified candidate ranking; S3 performs deep reconstruction and truth-seed determination; S4 freezes blind evaluation. All stages are planned work that does not start automatically; cases with insufficient evidence remain ambiguous. BESSIE maintains its observational convective outflow reference status; Job 3388245 is not modified or resubmitted. This study does not establish EVT identity, mechanism ground truth, Path C, verified wind, or any design quantities; any further promotion requires new evidence and PI scientific decision.

When Events hands off to downstream, it must also preserve the observational qualification of negative evidence. The absence of a signal recorded by a given source may result from insufficient coverage, sampling interval, quality control deletion, beam height, or event core offset rather than necessarily indicating that the physical signal does not exist. Only when the source genuinely

has detection capability at the target time, location, and height, and the quality control chain is complete, can non-detection serve as a limited counterevidence. The adjacent stations and high-beam radar of BESSIE do not meet the complete detection conditions for the local core; thus, non-detection fields remain coverage-limited. This treatment ensures that supporting evidence, refuting evidence, and unobservable evidence maintain different identities in the packet and also prevents the candidate ranking system from miswriting data richness as mechanism confidence.

Evidence and formula boundaries of Events Agent Chapter 14 preserves only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "Events Agent." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 14 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

15 CM1 Agent

15.1 Scientific questions, knowledge, and Researcher decisions

15.1.1 Scientific purpose

CM1 Agent handles PETW Step 1 mechanism-oriented idealized and conditional simulation: conducting downburst mechanism-oriented simulation with the CM1 cloud model, recovering Moore and Schumacher benchmark constraints, performing three-dimensional wind field diagnostics, and providing input support for Windfield and CFD. CM1 and WRF form a complementary division of labor: CM1 addresses mechanisms and idealized conditions, while WRF addresses real event reconstruction. Literature evidence indicates high-resolution three-dimensional cloud models can simulate thunderstorms that produce downbursts and their outflow structures[Orf et al., 2012, Saville et al., 2014]; idealized three-dimensional numerical simulation of downburst wind fields is a common pathway on the wind engineering side for characterizing this mechanism[Mason et al., 2010]; the cooling-source method provides a reference for outflow parameterization[Vermeire et al., 2011]; and the combination of CM1 and CFD has been used in research transitioning from storm scale to building scale[Dang et al., 2023]. The scientific objects of this component include benchmark reproduction, three-dimensional wind field diagnostics, input semantics for Windfield and CFD, and Level-1 candidate mechanism screening; CM1 does not bear responsibilities for code interfaces, design quantities, or engineering verification. The value of benchmark reproduction is to bind every simulation to traceable input identity, protocol, and output contracts, so that mechanism judgments are built on the same reviewable conditions[Wang et al., 2021]. Each CM1 run fixes archive-member identity, sounding hash, namelist, executable version, MPI decomposition, output writer, and postprocessing version; the output side differentiates scheduler completion, CM1 application completion, model-internal diagnostics, Level-1 label, formal three-dimensional validation, $\hat{V}_{3s}$ readiness, and engineering employ.

15.1.2 Knowledge base

The academic assets of CM1 comprise two review families and two governed Wiki/RAG namespaces. Family 1 is the Step 1 CM1 downburst modeling review; the Chinese v2 totals 29 pages with current candidate status pending_peter_scientific_review and scientific approval not_scientifically_approved; its 43-page English counterpart version completed 45/45 citation cross-checks, two deterministic builds, and 43-page visual inspection with PASS. Family 2 is the CM1/WRF downburst to high-rise building aerodynamic loads review; the Chinese v2.1 is a CFD-focused revision totaling

26 pages with status PASS_FOR_PETER_INTERNAL_REVIEW and scientific approval not_scientifically_approved; the 34-page English counterpart has been accepted by PI as the current internal bilingual review baseline. The Chinese package preserves 57 sources from 68 frozen sources and includes 1 boundary independent experiment entry VE-001; VE-001 does not verify moving downbursts, floor forces, base bending moments, torsion, or other aspect ratios. Namespace cm1_to_cfd39_topic contains 39 canonical sources, 54 Wiki pages, 100 chunks, 100 claims, and 24 relations; retrieval evaluation 47/47 PASS. Namespace step1_cm1 contains 45 source records; source-semantic verification 45/45; historical retrieval evaluation 10/10 PASS; retrieval contract is complete_semantic_source_fidelity. Compilation, retrieval, and archival completion does not equal scientific approval, nor does it equal benchmark verification.

15.1.3 Researcher participation and decisions

PI decides in the CM1 stage which events and mechanism hypotheses enter simulation, and the acceptance boundary for existing 40-case evidence. The current Moore reconstruction status is ACCEPT_IMPROVED_NOT_EXACT, meaning accepted as improved but not exact reproduction evidence, with paper_97_membership remaining unknown; this is PI explicit definition of the evidence level and is not an endorsement of exact reproduction. PI decided that 40-case Level-1 labels are to be used only as model-internal two-dimensional screening evidence and must not be treated as observations, paper-pass labels, or verified gusts; the sws_max 26 m/s stratification axis is retained only as a Moore-style intensity and composition proxy. PI decided that Level-2 dense source selection and CFD handoff are gated by review; the CFD handoff status is PENDING_PETER_REVIEW. For D420, PI decided to preserve the existing determination of NOT_CONSISTENT_CONTROLLED_COMPARISON; historical M0/M1 results do not serve as controlled comparison evidence; the matched-drive three-arm remains a frozen static package with no new submissions executed. For all-102 production, PI decided that before the source height order issue is resolved, blocked cases are not to be sorted, modified, clamped, or relabeled. These decisions collectively maintain the A0 to A2 supervised automation boundary; scope decisions do not establish scientific approval; any result entering the next stage requires new evidence and decisions.

15.2 Methods, results, and figures

15.2.1 Workflow and interactions

The auditable loop of CM1 is: input packet enters routing and identity gate, passes through versioned run package, runtime and output QA, diagnostics and Level-1 determination, forms a source or handoff package, and reaches the PI approval boundary. The frozen Moore protocol is PETW_MOORE_UNIFORM_AGL_V1 (Table 9): CM1 r20.3, $800 \times 800 \times 84$ grid, 7200 s integration, 60 s output interval, output_filetype = 3, 48 MPI ranks, 6×8 decomposition (evidence status: generated input). The input height conversion contract is

$$z_i = \mathrm{HGHT}_i - \mathrm{HGHT}_{\mathrm{surface}}, \tag{6}$$

where z_i is the height of the i-th layer (m AGL), HGHT_i is the original elevation (m MSL), and $\mathrm{HGHT}_{\mathrm{surface}}$ is the surface elevation (m MSL). The surface AGL is set to 0; pressure and profile fields are preserved during conversion. The time protocol is 0 s start, 2160 s checkpoint, 3600 s midpoint, 5400 s checkpoint, 7200 s endpoint, 60 s output cadence. The output contract differentiates sws from diagnostic u10/v10: sws is the lowest model level cumulative envelope,

u10/v10 are diagnostic near-surface winds; neither is a verified 10 m 3 s gust, and $\hat{V}_{3s}$ is not available.

On the interface side, Events provides CM1 with parent storm mode (such as isolated thunderstorm, supercell, or linear system), local mechanism hypothesis and its supporting and refuting evidence, station network and radar detectability assessment, surface signal time series and measurement definitions, sounding availability and spatiotemporal representativeness assessment, and explicit enumeration of ambiguity reasons. CM1 returns case identity (including archive member ID and sounding hash), frozen protocol identifier and input identity, runtime identity (CM1 version, MPI configuration, execution platform), output manifest (NetCDF file list, spatial snapshot times, and statistical cadence), model-internal mechanism diagnostics (temporal and spatial characteristics of updraft development, precipitation formation, downdraft triggering, and divergent outflow stages), near-surface response quantities (sws, u10/v10, CPC), footprint extent and duration, trajectory temporal evolution, and uncertainty description (including initial field sensitivity, microphysics scheme dependence, and grid resolution limitations). Model initialization cannot override observational evidence; CM1 feedback cannot by itself promote EVT, Path C, or validated event.

The handoff from CM1 to Windfield includes variable semantics (distinguishing the different physical meanings of sws cumulative envelope, s10 instantaneous diagnostic, and u10/v10 diagnostic near-surface wind), temporal cadence (time resolution limitations of 60 s statistical interval and 900 s spatial snapshot interval), spatial support (minimum resolvable scale determined by grid spacing), coordinate system (AGL height and horizontal Cartesian coordinates), source identity, and finite-value checks. sws and diagnostic u10/v10 cannot be directly called engineering gusts or $\hat{V}_{3s}$; Windfield must preserve their model-internal semantics when receiving these variables. The handoff from CM1 to CFD is more detailed: the explicitly selected dense source package includes complete three-dimensional fields at each time step; source times are determined by the clock mapping contract (e.g., D420 from 2120 to 2380 s); the position of native variables on the staggered grid and centering methods are verified by the interface pilot; the CM1 initial field provides starting conditions for the CFD domain; boundary driving provides time-varying input for each face of the CFD domain. CFD holds final responsibility for case definition (closure selection, grid design), mesh generation, boundary dictionary, solver configuration, and runtime QA. The CM1 Level-1 label is not CFD execution authorization; the CFD handoff status is PENDING_PETER_REVIEW. The downstream claim boundary is: Level-1 screening SUPPORTED (model-internal two-dimensional evidence), CFD handoff PENDING_PETER_REVIEW (source semantics priority), design wind/load NOT_PROMOTED (not established). ComputeOps holds connections, staging, scheduler, monitoring, and terminal recovery, but since 2026/8/2, CM1 submits directly and maintains its own TACC jobs; ComputeOps defaults to registration only.

15.2.2 Completed work and main results

40-case unified results. The Level-1 distribution of the existing 40 protocol-consistent reconstructed production cases is: VALID_DOWNBURST_CANDIDATE_LEVEL1 24 cases, WEAK_NEAR_SURFACE_OUTFLOW 9 cases, COMPLETED_BUT_SCIENTIFICALLY_INVALID 3 cases, BOUNDARY_TRUNCATED 2 cases, and STRONG_OUTFLOW_CANDIDATE_NEEDS_3D 2 cases. The cross-tabulation by the sws_max 26 m/s proxy stratification appears in Table 8 (evidence status: candidate model diagnostic evidence). The 24/9/3/2/2 distribution is a registry-derived model-internal Level-1 screening count; the determination basis for each category is as

follows. VALID_DOWNBURST_CANDIDATE_LEVEL1 requires that the outflow at the lowest model level exhibits a clear radial divergence pattern in the two-dimensional plane, cold pool development is complete, near-surface wind speed response is significant, and its duration is consistent with the time scale of idealized downburst outflow. WEAK_NEAR_SURFACE_OUTFLOW indicates outflow exists but its intensity does not reach the candidate threshold or its features are insufficiently clear. COMPLETED_BUT_SCIENTIFICALLY_INVALID indicates that the computation completed normally but the result is physically invalid (e.g., the outflow did not reach the lowest model level or the cold pool was not established). BOUNDARY_TRUNCATED indicates that the outflow reached the computational domain boundary and was truncated, affecting the completeness of diagnostic quantities. STRONG_OUTFLOW_CANDIDATE_NEEDS_3D indicates that two-dimensional diagnostics show a strong outflow candidate but three-dimensional mechanism judgment is needed for confirmation. All five category labels are model-internal two-dimensional screening results: they are not observational verification, not paper-pass labels, and not verified gust values. The 26 m/s stratification axis is a Moore-style intensity and composition proxy; its selection basis is the stratification standard used in the Moore series of works in the literature; it is not $\hat{V}_{3s}$, not a verified 10 m 3 s gust, and not a paper-membership decision. Among the 35 valid cases remaining after excluding 3 SCIENTIFICALLY_INVALID and 2 BOUNDARY_TRUNCATED cases, 12 satisfy sws_max $\geq$ 26 m/s. The overall benchmark status is ACCEPT_IMPROVED_NOT_EXACT, with paper_97_membership = unknown (evidence status: PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION). Spatial and metric comparisons with literature figures remain within the ACCEPT_IMPROVED_NOT_EXACT boundary across multiple diagnostic quantities, but exact reproduction of the results from the 97 cases in the literature is not claimed.

The CM1 native wind field components u, v, w are defined on a staggered grid. The lowest model level horizontal wind speed resultant is

$$|\mathbf{U}_h| = \sqrt{u^2 + v^2}, \tag{7}$$

where u, v are horizontal velocity components on the CM1 staggered grid (in m/s). The horizontal divergence is defined as

$$\nabla_h \cdot \mathbf{u}_h = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}, \tag{8}$$

in units of s^{-1}, used for model-internal divergence diagnostics and distinct from observed surface divergence. The sws in the output contract is the lowest model level cumulative envelope,

$$\mathrm{sws}(t) = \max_{\tau \leq t} |\mathbf{U}_h|(\tau), \tag{9}$$

which is a source-defined temporally cumulative quantity that takes the historical maximum at each time step; it differs from the instantaneous near-surface diagnostic s10 and also differs from observational WMAX. Classification thresholds must be accompanied by source and applicability conditions: the sws_max 26 m/s stratification is valid only within its source registry and 40-case screening context and does not establish any verification quantity.

Benchmark input conversion and literature comparison. The benchmark work chain has three components: input conversion (sounding to CM1 input), temporal development (complete integration process within the 0 to 7200 s protocol window), and literature comparison. Input

Level-1 label	sws_max ≥ 26 m/s	sws_max < 26 m/s	Total
VALID_DOWNBURST_CANDIDATE_LEVEL1	10	14	24
WEAK_NEAR_SURFACE_OUTFLOW	1	8	9
COMPLETED_BUT_SCIENTIFICALLY_INVALID	1	2	3
BOUNDARY_TRUNCATED	2	0	2
STRONG_OUTFLOW_CANDIDATE_NEEDS_3D	1	1	2
Total	15	25	40

Table 8: Moore 40-case Level-1 distribution and sws_max 26 m/s proxy cross-tabulation. Source: TABLE_40_DISTRIBUTION.csv and TABLE_40_CROSSTAB.csv (CM1 integrated program report, 2026-08-12); evidence status: candidate model diagnostic evidence; labels are model-internal two-dimensional Level-1 screening counts; the 26 m/s axis is a Moore-style intensity and composition proxy, not $\hat{V}_{3s}$, not a verified 10 m 3 s gust.

conversion takes the sounding and archive identity through the MSL to AGL conversion of Equation (6) and then connects to the namelist and CM1 executable. The temporal development process includes the complete model-internal evolution chain of convective triggering, updraft establishment, precipitation formation, downdraft development, near-surface divergent outflow, and cold pool expansion. Spatial and metric comparisons with literature figures remain within the ACCEPT_IMPROVED_NOT_EXACT boundary; cold pool width and grid RMSE metrics at 2160 s and 2220 s provide quantitative differences between uniform and B-prime cases. The Schumacher technical benchmark construction, MPI configuration, Thompson microphysics scheme initiation, NetCDF output format, and 240×240 extension and 480×480 output diagnostic evidence have been established; the complete scientific benchmark remains NOT_ESTABLISHED (evidence status: engineering preparation).

all-102 identity audit. The complete denominator is 102 unique archive identities; 102/102 identity unique. Existing production comprises 40 cases; the remaining 62 comprise 61 release-ready new inputs and 1 SOURCE_HEIGHT_ORDER_BLOCKED case. The blocked case 20190630_2100_42.870_−91.890 has an inconsistent original height order; after conversion, 0.00 m AGL is followed by −1.76 m AGL. This case has not undergone any sorting, deletion, clamping, interpolation, or relabeling; it remains blocked until the source height order issue is resolved (evidence status: engineering preparation). Job 3396402 remains at LIVE_EXECUTION with scientific_result = false; this is an operational status indicator and does not represent completion of 102 scientific cases.

BESSIE conditional CM1 output. Job 3388245 employs ZC100 conditional output: 241 NetCDF files, 48 MPI ranks, spatial output times 0/900/1800/2700/3600 s, statistical cadence 60 s. The model-internal sequence exhibits the typical development process of ascent, precipitation, descent, local divergence, and cold pool diagnostics. Peak wmax 22.4418 m/s (2340 s), wmin −11.4658 m/s (2520 s), divmax 0.00118535 s^{-1} (2580 s); temporally, the sequence shows successive ascent followed by precipitation, then descent, and finally divergence (evidence status: candidate model diagnostic evidence). Spatial snapshots at five discrete times 0/900/1800/2700/3600 s provide full-field three-dimensional variables, including three velocity components, temperature perturbation, water vapor mixing ratio, and hydrometeor categories. The 0 s snapshot reflects the initial field state; the 900 s snapshot captures early convective triggering; the 1800 s and 2700 s snapshots cover the development stage of the downdraft; and the 3600 s snapshot approaches the end of integration. However, the 900 s interval between snapshots means that the spatial structure of key transitions such as convective triggering, downdraft touchdown, and divergent outflow formation is not continuously captured, unlike the denser temporal sampling provided by

the 60 s cadence temporal diagnostic sequence. The fact that the retained 2700 s lowest height is approximately 2.10 km AGL indicates at this snapshot time the model-internal downdraft core had not yet reached the near surface and does not establish a touchdown downdraft core. Level-1 remains NOT_SUPPORTED_UNDER_FROZEN_GATE. The item-by-item determination under the frozen gate is: downburst-specific reconstruction NOT_ESTABLISHED (requires three-dimensional mechanism judgment and observational comparison), VALID_DOWNBURST_CANDIDATE_LEVEL1 NOT_SUPPORTED (no promotion under frozen gate), Path C NOT_PROMOTED (complete scientific path from CM1 to CFD not established), validated 10 m wind NONE (model-internal diagnostics are not equivalent to verified near-surface wind), $\hat{V}_{3s}$ NONE (the 3 s gust quantity does not exist in CM1 output definitions).

D420 CM1-to-CFD evidence. The primary case for the CM1-to-CFD interface is D420. CM1 source production job 3379790 completed; source-package manifest 1026/1026 PASS, cleanroom test 20/20 PASS, assembly QA 261/261 PASS, frame-flux QA 261/261 PASS. The single-frame velocity volume interface pilot verified native staggered u/v/w, centered representation, coordinate destaggering, and repeat ownership; this pilot is interface verification, not multi-time propagation results; cfd_handoff_allowed = false.

Historical M0 (Job 3384063) employs laminar closure, 200 × 200 × 29 grid (100 m horizontal spacing), 1,160,000 cells, 20 km × 20 km × 3.038 km domain, ground slip boundary condition, 0 to 260 s continuous integration, 1 s write interval. M1 (Job 3386652) uses kOmegaSST URANS closure, 400 × 400 × 100 grid (10 m horizontal spacing), 16,000,000 cells, 4 km × 4 km × 1 km domain, noSlip ground plus wall function, hot restart from $t = 40$ s with output at only four selected times 50/60/70/80 s (Table 10).

The item-by-item audit of boundary faces and flux contracts reveals direct conflicts in six dimensions. Bottom wall treatment: M0 employs ground slip, M1 uses noSlip plus wall function; the direct conflict makes it impossible to attribute any near-bottom differences to a single factor of grid or closure. Closure model: the CM1 source package does not fix the CFD-side closure choice; M0 selected laminar while M1 selected URANS kOmegaSST, constituting a direct closure conflict. Computational domain: M0 uses a 20 km × 20 km large domain covering the complete outflow development range; M1 uses a 4 km × 4 km small domain covering only the core region; the five-fold difference in domain size confounds boundary effects. Initial and restart state: M0 cold-starts from $t = 0$ using a non-uniform velocity field; M1 hot-restarts from $t = 40$ s using the original zero-velocity field plus restart evidence; restart alignment is unverified. Outer boundary fallback: M0 uses uniform zero velocity and uniform zero pressure on five faces; M1 uses internalField fallback on five faces; boundary effects are mutually confounded. Spatial support: M0 horizontal spacing of 100 m corresponds to a minimum resolvable feature scale one order of magnitude different from M1's 10 m spacing. The simultaneous inconsistency across these six dimensions means that any output difference between M0 and M1 cannot be attributed to a single factor; thus, the scientific conclusion of the historical comparison is NOT_CONSISTENT_CONTROLLED_COMPARISON. Job 3393320 produced bounded M0/M1 exploratory metrics without new solver execution and does not change the above determination.

The clock mapping is source 2120 s to 2380 s to CFD 0 s to 260 s, comprising 261 records at 1 s intervals; for coordinates, the vertical coordinate is converted per Equation (6), with the D420 CFD relative time $t_{\mathrm{CFD}} = t_{\mathrm{CM1}} - 2120$ s. The matched-drive three-arm is a frozen static package: D420_MC100_LAMINAR is a 40 × 40 × 10, 16,000-cell coarse-grid laminar reference;

D420_MF010_LAMINAR and D420_MF010_KOMEGASST_URANS are both $400 \times 400 \times 100$, 16,000,000-cell configurations providing fine-grid comparison under laminar and URANS closure respectively. The objective of the three-arm design is to provide controlled comparison along the grid and closure axes, but all three arms have not yet received new submissions (evidence status: engineering preparation).

Field	Value	Semantics
protocol	PETW_ MOORE_ UNIFORM_ AGL_V1	Protocol identifier
CM1	r20.3	Version
grid	$800 \times 800 \times 84$	Grid cells
duration	7200	s
cadence	60	s
output_filetype	3	CM1 output type
MPI	48	ranks
decomposition	6×8	MPI layout

Table 9: Frozen Moore protocol (PETW_MOORE_UNIFORM_AGL_V1). Source: TABLE_BENCHMARK_ INPUT_CONTRACT.csv (CM1 integrated program report, 2026-08-12); evidence status: generated input; the frozen protocol does not establish paper exact reproduction or scientific completion.

case	closure	mesh	cells	domain	time and wall
D420_M0	laminar	200×200×29, 100 m	1,160,000	20×20×3.038 km	0 to 260 s; slip
D420_M1	kOmegaSST URANS	400×400×100, 10 m	16,000,000	4×4×1 km	from t=40 s; noSlip
D420_MC100	laminar	40×40×10, 100 m	16,000	4×4×1 km	package only
D420_MF010_L	laminar	400×400×100, 10 m	16,000,000	4×4×1 km	package only
D420_MF010_K	kOmegaSST URANS	400×400×100, 10 m	16,000,000	4×4×1 km	package only

Table 10: D420 case identity and comparison conditions. Source: D420 T6 case identity table (CM1 integrated program report, 2026-08-12); historical M0/M1 are inconsistent in wall treatment, closure, domain, initial/restart state, outer-boundary fallback, spatial support, and write interval; comparison status is NOT_CONSISTENT_ CONTROLLED_COMPARISON; package-only rows are generated input, not solver results.

15.2.3 Figures

Figure 10 depicts a Level-1 near-surface divergent outflow case display. Figures 11 and 12 show benchmark spatial comparison and metric comparison. Figures 13 and 14 show the 40-case Level-1 distribution and stratified comparison. Figure 15 shows the all-102 status display. Figures 16 through 18 show the benchmark protocol workflow, input conversion contract, and time sequence. Figure 19 shows the CM1-side temporal diagnostics of Job 3388245. Figures 20 and 21 show the D420 M0 spatial structure and wind field time series. Figure 22 shows the Schumacher technical benchmark domain configuration. Figure 23 shows the Moore 40-case contact overview.

15.3 Limitations, diagnostics, and interfaces

15.3.1 Current scientific limitations and future research

15.3.2 Diagnostic tiers, comparison design, and interface closure

The interpretation of CM1 results must begin with run identity. A scheduler reporting success indicates only that the batch system completed the task; application completion further requires the

Variable	Definition or semantics	Unit	Boundary
u, v, w	CM1 native staggered grid velocity components	m/s	Model-internal
$\lvert\mathbf{U}_h\rvert$	Lowest model level horizontal wind speed resultant, Equation (7)	m/s	Model-internal
sws	Lowest model level cumulative envelope, Equation (9)	m/s	Source-defined cumulative quantity
u10/v10	Diagnostic near-surface wind	m/s	Not a verified 10 m 3 s gust
$\nabla_h \cdot \mathbf{u}_h$	Horizontal divergence, Equation (8)	s^{-1}	Model-internal
CPC	Cold pool velocity	m/s	Model-internal diagnostic
$\hat{V}_{3s}$	Not-established 3 s gust quantity	Not applicable	NOT_ESTABLISHED
z_i	Height conversion, Equation (6)	m AGL	Protocol contract

Table 11: Principal variables and semantics. Source: CM1 output contract and D420 tables (CM1 integrated program report, 2026-08-12); the variable semantic mapping does not prove propagation verification or any verified gust.

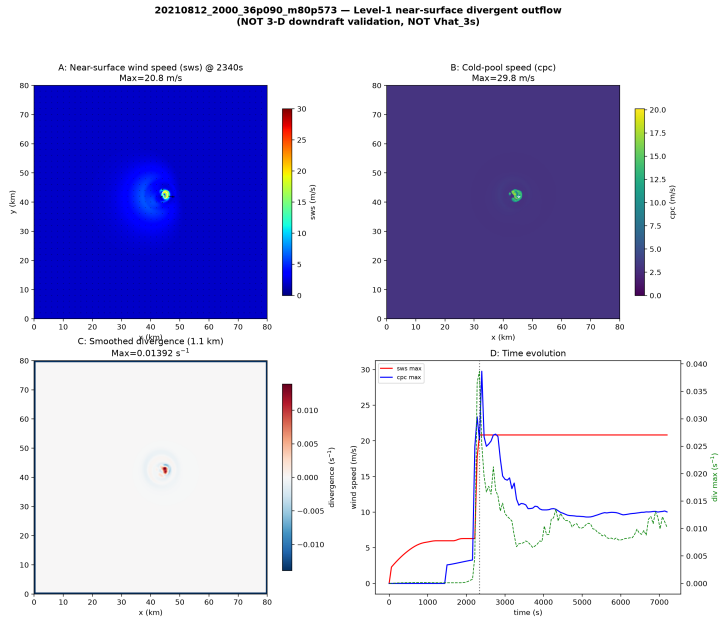


Figure 10: Level-1 near-surface divergent outflow case display (case 20210812_2000_36p090_m80p573): four-panel display of 2340 s near-surface sws, cold pool velocity CPC, 1.1 km smoothed divergence, and temporal evolution. Source: PETW Moore Level-1 40-event package (original figure); evidence status: candidate model diagnostic evidence; supports only bounded two-dimensional model-internal Level-1 evidence for this case; not a literature benchmark case, not three-dimensional verification, and not $\hat{V}_{3s}$.

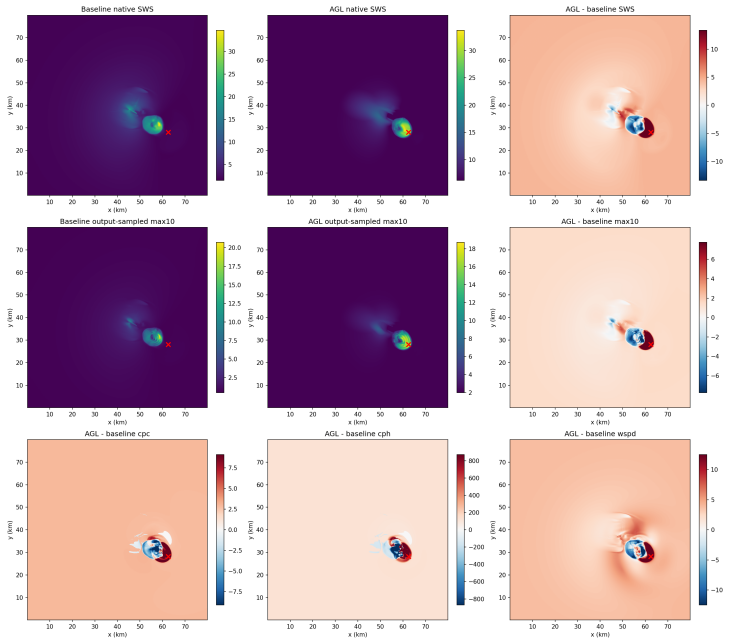


Figure 11: Benchmark spatial comparison (three-by-three panel): baseline native SWS, AGL native SWS, AGL minus baseline, output sampling max10, and CPC/CPH/WSPD differences, with red literature author point markers. Source: PETW Moore Figure5 case synthesis report (original figure); evidence status: candidate model diagnostic evidence; displays evidence but does not establish exact Moore 97 membership.

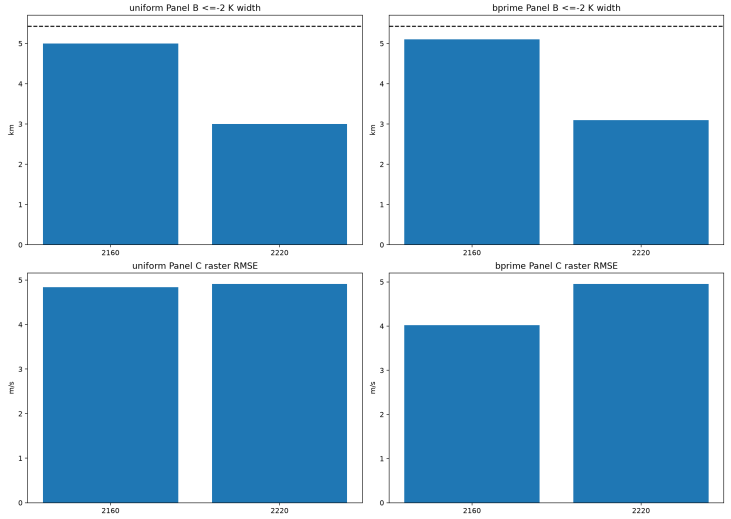


Figure 12: 2160 s and 2220 s metric comparison (four-panel): cold pool width and grid RMSE for uniform and B-prime cases. Source: PETW Moore Figure5 case synthesis report (original figure); evidence status: candidate model diagnostic evidence; comparison remains within the ACCEPT_IMPROVED_NOT_EXACT boundary.

model normal termination flag, expected output count, and NetCDF readability to be jointly confirmed. After passing the run gate, variables are still only model-internal output. Level-1 screening employs lowest model level planes, cumulative envelopes, cold pool, and divergence diagnostics to identify candidate morphologies; Level-2 requires examining the downdraft core, pressure perturbation, cold pool boundary, radial outflow, and temporal continuity in dense three-dimensional output; observational comparison further requires measurement height, averaging window, spatial support, and event time-phase compatibility. The three tiers cannot be replaced by a single completion flag.

The 24/9/3/2/2 counts of the 40-case come from the registry, not from re-judgment of classification results based on the body text. The registry fixes the identity, protocol, diagnostic status, and Level-1 label of each case in the same row, making the denominator and exclusion

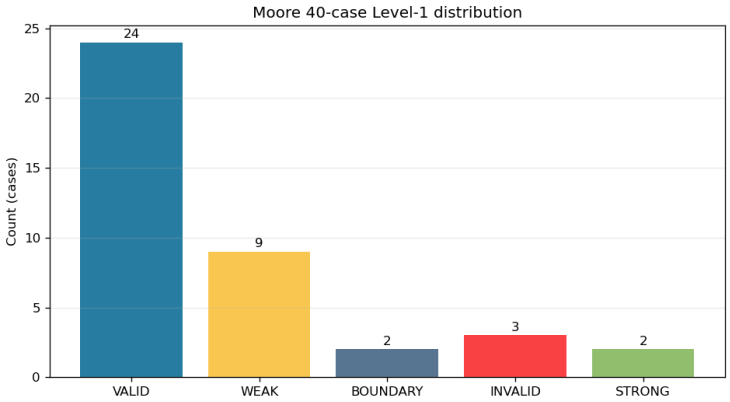


Figure 13: Moore 40-case Level-1 distribution: five-category registry counts (VALID 24, WEAK 9, BOUNDARY 2, INVALID 3, STRONG 2). Source: TABLE_40_DISTRIBUTION.csv and dual-axis registry (CM1 integrated program report, 2026-08-12); evidence status: candidate model diagnostic evidence; labels are not observations, not paper-pass labels, not verified gusts.

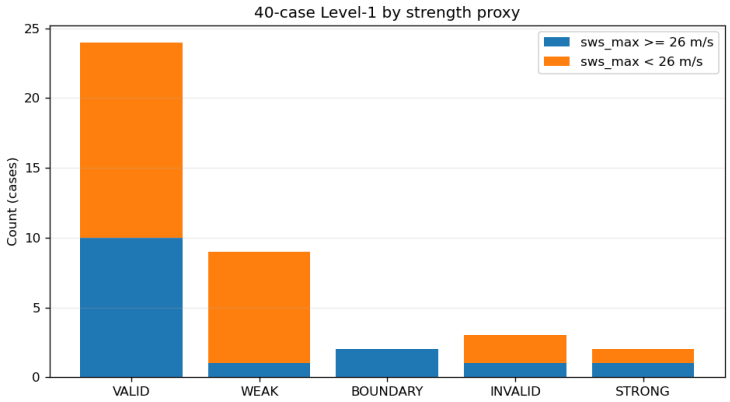


Figure 14: 40-case classification stacked comparison by sws_max 26 m/s proxy stratification. Source: TABLE_40_CROSSTAB.csv and dual-axis registry (CM1 integrated program report, 2026-08-12); evidence status: candidate model diagnostic evidence; the 26 m/s axis is a Moore-style intensity proxy, not $\hat{V}_{3s}$.

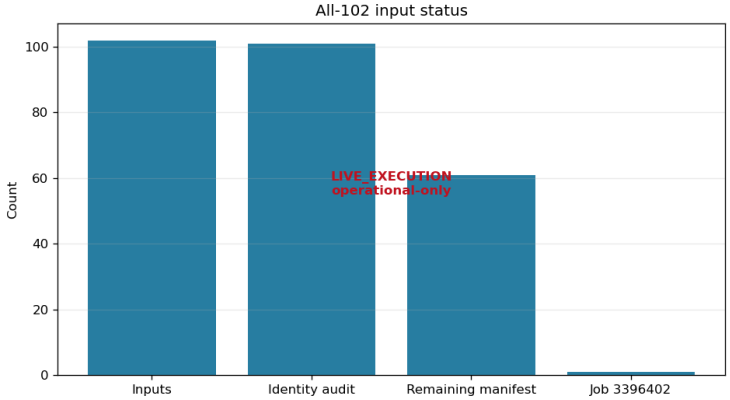


Figure 15: all-102 status display: Inputs 102, Identity audit 101, Remaining manifest 61, Job 3396402 is LIVE_EXECUTION and operational-only. Source: TABLE_ALL102_STATUS.json (CM1 integrated program report, 2026-08-12); evidence status: generated input; does not represent completion of 102 scientific cases.

Figure 16: Benchmark protocol workflow: sounding and archive identity, MSL to AGL conversion, namelist and CM1, raw output, Level-1 QA and diagnostics. Source: CM1 main results report v4 (protocol figure); evidence status: generated input; records only the process and evidence boundaries and does not prove solver completion or scientific verification.

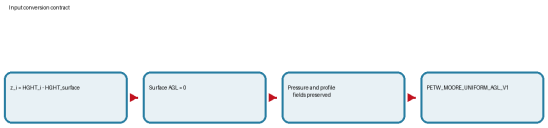


Figure 17: Input conversion contract: $z_i = \mathrm{HGHT}_i - \mathrm{HGHT}_{\mathrm{surface}}$, surface AGL set to 0, pressure and profile fields preserved, protocol identifier PETW_MOORE_UNIFORM_AGL_V1. Source: CM1 main results report v4 (protocol figure); evidence status: generated input; input identity does not prove paper membership or scientific completion.

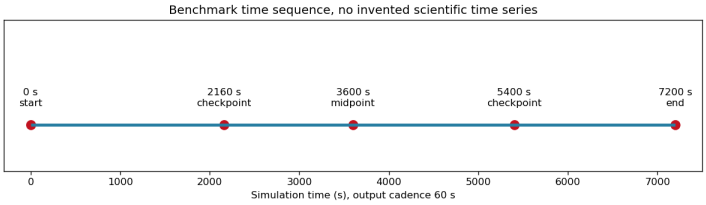


Figure 18: Benchmark protocol time sequence: 0 s start, 2160 s checkpoint, 3600 s midpoint, 5400 s checkpoint, 7200 s endpoint, 60 s output interval. Source: CM1 main results report v4 (protocol figure); evidence status: generated input; checkpoint figure, not a newly created scientific time series.

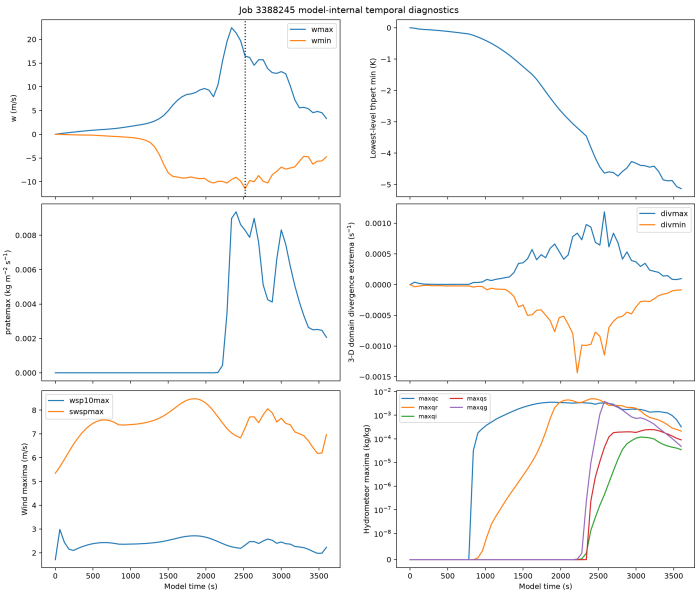


Figure 19: Job 3388245 temporal diagnostics from 0 to 3600 s, with source markers at 2340/2400/2520/2580 s (CM1 BESSIE reference diagnostics, from the same source as the figure in Section 14, Figure 5). Source: PETW Job 3388245 results display package v2 (original figure); evidence status: candidate model diagnostic evidence; does not establish event reconstruction, three-dimensional verification, or verified gust.

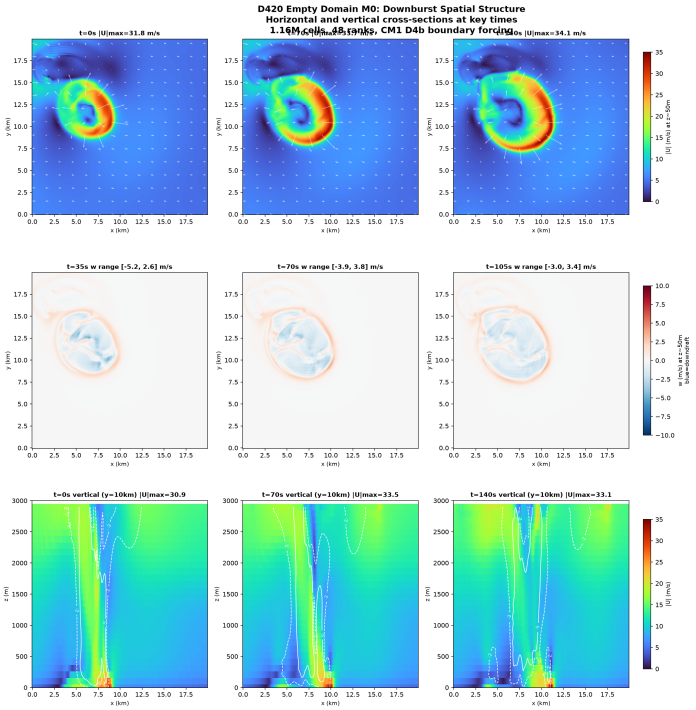


Figure 20: D420 M0 spatial structure (three rows): horizontal $|\mathbf{U}|$ with vectors, horizontal w, $y = 10\,\mathrm{km}$ vertical cross-section, $z = 50\,\mathrm{m}$, times $t = 0/35/70/105/140\,\mathrm{s}$. Source: PETW CFD automation output (original figure); evidence status: candidate model diagnostic evidence; post-processing display of existing solver output; does not constitute controlled verification and is not design wind.

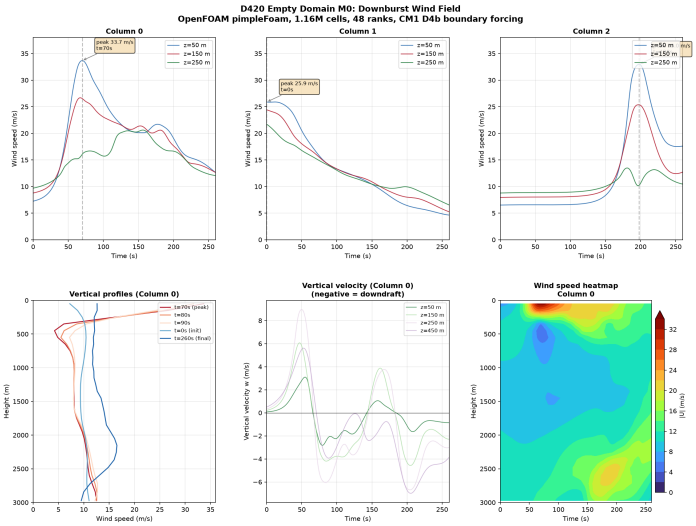


Figure 21: D420 M0 wind field time series (three-column velocity time series, Column 0 vertical profiles, Column 0 w and heatmap), 0 to 260 s, $z = 50/150/250/450\,\mathrm{m}$. Source: PETW CFD automation output (original figure); evidence status: candidate model diagnostic evidence; Column 2 peak annotations are partially occluded; not verified gust.

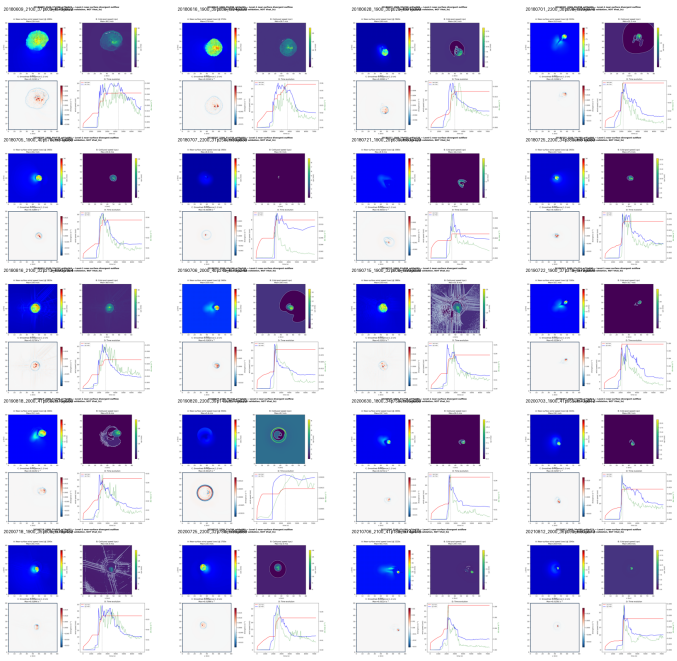


Figure 22: Schumacher technical benchmark domain configuration and Thompson microphysics scheme initiation. Source: CM1 integrated program report (original figure); evidence status: engineering preparation; complete scientific benchmark remains NOT_ESTABLISHED.

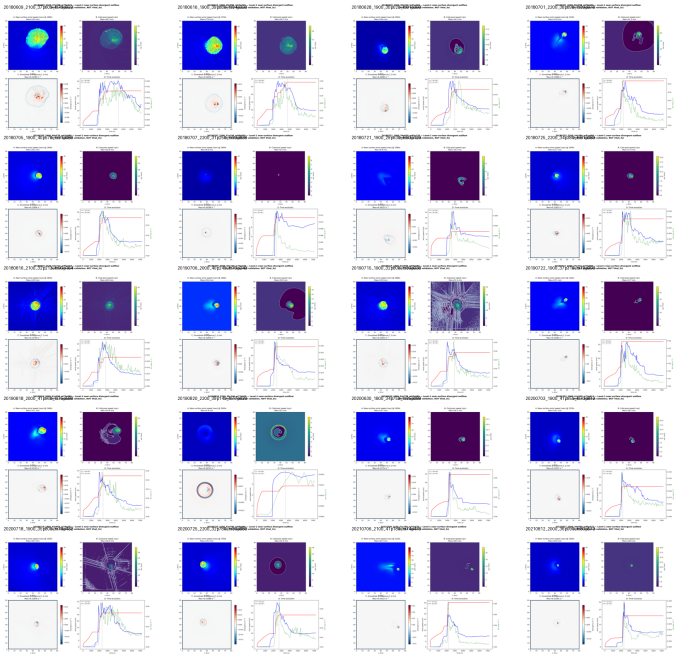


Figure 23: Moore 40-case contact overview. Source: CM1 integrated program report (navigation figure); evidence status: generated input; the contact overview is not an independent scientific result.

items reviewable. The 24 VALID_DOWNBURST_CANDIDATE_LEVEL1 indicate only candidates under frozen two-dimensional rules; the 9 WEAK_NEAR_SURFACE_OUTFLOW indicate weaker near-surface outflow evidence; the 3 COMPLETED_BUT_SCIENTIFICALLY_INVALID indicate runs completed but failed for scientific employ; the 2 BOUNDARY_TRUNCATED indicate outflow truncated by the boundary; and the 2 STRONG_OUTFLOW_CANDIDATE_NEEDS_3D indicate three-dimensional checking is needed. This distribution is registry-derived model-internal Level-1 screening. No category is an observational label, paper membership label, mechanism ground truth, or engineering inflow qualification; this constraint also specifies the interpretation ceiling that subsequent comparisons cannot exceed.

The distinction between cumulative and instantaneous quantities directly affects screening. The sws in Equation (9) preserves the historical maximum horizontal speed at each grid point up to time t; therefore, its spatial map may jointly include peaks that occurred at different times. This map is appropriate for showing the cumulative extent of influence but cannot be interpreted as the wind field existing simultaneously at a single instant. s10 or u10/v10 are model-diagnosed instantaneous or near-surface quantities, but their height definition, surface layer treatment, and temporal sampling still differ from the observational 10 m, 3 s quantity. If downstream needs trajectories, movement speed, or instantaneous CFD boundaries, three-dimensional source frames with explicit timestamps must be used; temporal evolution cannot be recovered from cumulative envelopes.

Spatial resolution constrains interpretable structures. The u, v, w on the staggered grid must be stored at native variable positions; before computing resultant speed, divergence, or transferring to CFD, an explicit centering method must be adopted. Different centering orders may alter peaks and positions in strong-gradient regions; thus, the interface pilot jointly checks coordinates, array shape, finite values, boundary face direction, and repeat ownership. A single-frame pilot proves that data structure and transformations can be executed; it does not prove that multi-frame propagation closes regarding mass, momentum, or pressure. The D420 1026/1026 manifest, 20/20 cleanroom, 261/261 assembly, and 261/261 frame-flux QA are source package and interface engineering evidence; cfd_handoff_allowed remains false.

The temporal interface likewise requires maintaining two clock systems. D420 employs

$$t_{\mathrm{CFD}} = t_{\mathrm{CM1}} - t_0, \tag{10}$$

where t_{CM1} is the CM1 simulation time (s), $t_0 = 2120$ s is the selected source window start, and t_{CFD} is the CFD relative time (s). This mapping corresponds the 261 source times from 2120 to 2380 s to 0 to 260 s without changing the physical identity of source frames. Clock alignment indicates only the index relationship between two file sequences; it does not prove that the CFD domain can receive time-varying boundaries without reflection, nor that the selected window includes the complete downburst life cycle.

The comparison failure of historical M0 and M1 comes from multiple axes changing jointly rather than from either case lacking output. The wall condition changed from slip to noSlip plus wall function, the closure changed from laminar to kOmegaSST URANS, the horizontal domain shrank from 20 km to 4 km, the grid spacing changed from 100 m to 10 m, the initial and restart states differ, the outer boundary fallback differs, and the write times are also inconsistent. Output differences may come from any combination of these factors; thus, NOT_CONSISTENT_CONTROLLED_COMPARISON is a judgment about attribution qualification. Retaining M0

and M1 figures still has value: they show the spatial structure and temporal organization of existing output, but the figures cannot be used to estimate grid effects, closure effects, or wall effects.

The matched-drive three-arm separates comparison into two axes through variable control. MC100 and MF010_LAMINAR maintain laminar closure while changing the grid; MF010_LAMINAR and MF010_KOMEGASST_URANS maintain the fine grid while changing the closure. The three arms also must share source frames, domain, boundaries, initial state, clock, output frequency, and postprocessing version to discuss grid and closure effects separately. Currently, all three arms are frozen static packages with no new submissions executed; thus, there are no reportable three-arm differences, convergence trends, or preferred closures. The Researcher's execution authorization must specify the computational scope, stopping conditions, and result usage; after computation is complete, an independent scientific comparison gate is still required.

The Moore benchmark status reflects the combined limitations of input, evolution, and comparison. The MSL to AGL conversion fixes the height semantics; the frozen protocol fixes the version, grid, duration, and output cadence; the 2160 s and 2220 s comparisons provide model-internal differences such as cold pool width and grid RMSE. At the same time, paper_97_membership remains unknown, and the target literature members cannot be closed case by case against current archive identities. Current evidence can support that the protocol-based reconstruction is more consistent than earlier paths and can show differences between uniform and B-prime at specific times and metrics, but it cannot support case-by-case exact consistency with the paper. The formal status remains PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION, corresponding to the ACCEPT_IMPROVED_NOT_EXACT boundary.

The BESSIE conditional output displays the model-internal process chain, but there is no closed loop between its initialization and the observed event sufficient to verify the mechanism. The 60 s statistical sequence can locate the succession of wmax, precipitation, wmin, and divmax, but the 900 s spatial snapshots cannot continuously track the downdraft core touchdown. The 2700 s snapshot preserves the lowest downdraft core at approximately 2.10 km AGL; no direct comparison exists between this and the surface 34.9 m/s observation at the same dimension, same time phase, and same spatial support. This case is thus suitable as conditional diagnostics and output semantic exercise but not as observational reconstruction or verified inflow.

The input management of all-102 maintains denominator transparency. Of 102 unique archive identities, 40 are existing production cases, 61 new inputs are release-ready, and 1 case is blocked because of source height order anomaly. The blocked case has -1.76 m AGL following 0.00 m AGL at the surface; automatic sorting, truncation, or interpolation would alter the correspondence between original layers and thermodynamic variables. Maintaining the original input unchanged and stopping the run protects source semantics. Even if all remaining 61 inputs are run to completion, new results still require case-by-case run QA, Level-1 screening, three-dimensional qualification, and source membership cross-checking; job completion counts cannot be directly written as scientific sample counts.

Handoff to Windfield closes when variables, time, coordinates, and support scales can be fully interpreted. Windfield can receive cumulative footprints, instantaneous fields, or vertical profiles, but each must be separately annotated; they cannot be combined into an unverified unified event wind field. The closure condition for handoff to CFD adds boundary face, flux, centering, clock, and driving consistency. Any interface to EVA further requires observational definitions and event independence; CM1 model-internal peaks do not have direct statistical membership qualification.

The Coordinator can track these gates; final usage is still decided by the Researcher based on scientific evidence.

Figure 24 summarizes the CM1 diagnostic ladder from run evidence to downstream qualification. Each tier in the ladder adds new checking objects, and passing a lower tier can only be a necessary condition for a higher tier. Run completion does not imply Level-1 candidate; Level-1 candidate does not imply three-dimensional mechanism; three-dimensional model diagnostics do not imply observational consistency; observational consistency does not by itself authorize engineering boundary input. This structure avoids covering distinct scientific questions with a single validated label.

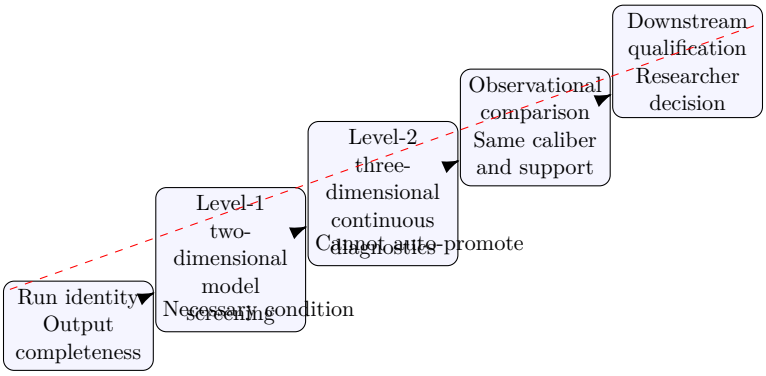


Figure 24: CM1 diagnostic ladder and promotion boundaries. The figure is a project-internal diagnostic method diagram and does not indicate that any case has reached observational verification or engineering use level.

Three-dimensional mechanism checking requires jointly observing dynamical, thermodynamical, and hydrometeor structures. The vertical continuity of the downdraft core should be compatible with negative buoyancy or precipitation drag background; near-surface divergence should correspond temporally with the cold pool boundary and pressure perturbation; the azimuthal structure of radial outflow should exclude uni-directional high-speed bands caused by translational background wind. Seeing high wind speeds at only the lowest level does not demonstrate that they were formed by downdraft core touchdown; seeing strong negative vertical velocities at only upper levels does not demonstrate that surface outflow has been established. Dense output must also cover sufficient periods before, during, and after the process occurs; otherwise, "not seen" may be merely a result of the write cadence. The value of Level-2 is to establish the common temporal and spatial relationships among these quantities, not to add more peaks.

A quantity crosswalk is the first requirement for observational comparison. Grid-point wind, grid-average wind, lowest model level wind, diagnostic 10 m wind, and cumulative sws in the model each have different spatial filtering, surface treatment, and temporal support; station-average wind, 3-second observation maximum, and five-minute reported value in observations are also different. Comparable items must specify height endpoints, averaging algorithm, sampling interval, spatial representativeness, and terrain exposure. If a same-caliber conversion cannot be established, weaker features such as process sequence, duration, or spatial position can be compared, but quantity comparison must be marked as unavailable. This rule applies equally to radar reflectivity and model hydrometeors; similar variable names do not equate to the same observation operator.

Uncertainty arises from at least initial profiles, microphysics schemes, grid resolution, domain boundaries, triggering methods, and output sampling. The current 40-case registry primarily supports lateral screening within the frozen protocol and cannot isolate the above factors. If sensitivity studies are conducted in the future, each comparison should change only one pre-registered axis while keeping archive identity, conversion scripts, model version, domain, output

cadence, and postprocessing consistent. Multi-factor simultaneous variation can only be used to explore candidate regions and should not be written as single-factor causal conclusions. Sensitivity results must also be linked to classification stability, judging whether a case's Level-1 label holds or changes under reasonable configuration variation rather than comparing only peak magnitudes.

Boundary-truncated cases require separate treatment because cumulative envelopes and maximum winds may continue to be constrained by the computational domain after the outflow reaches the boundary. Merging such cases with domain-complete cases would mix domain design error into the mechanism distribution. Scientifically invalid cases also cannot enter the valid denominator merely because the solver completed. Table 8 jointly preserves both categories of exclusions, making the relationship among 40 run identities, 35 cases not excluded by these two issues, and the counts of each label transparent. STRONG_OUTFLOW_CANDIDATE_NEEDS_3D remains in the Level-1 distribution because it indicates two-dimensional evidence is insufficient rather than that three-dimensional confirmation has been completed.

Checking per-face array shapes alone cannot address quality conservation at the CFD interface. Time-varying boundaries require checking normal velocity signs, area weights, inlet and outlet net flux, pressure or velocity boundary combinations, duplicate assignment at edge corners, and mass imbalance caused by source-domain to target-domain scale cropping. Frame-flux QA provides a structured entry point for these checks, but its determinations are valid only within the defined mapping contract. After CFD solving, it is also necessary to check the propagation, boundary reflection, and numerical dissipation of the driving field in the target domain before discussing whether source field features are preserved. CM1 does not decide closure and wall model for CFD, nor does it claim propagation has been verified because the source package is complete.

When handing off to Windfield, it is also necessary to differentiate descriptive features from generatable fields. Footprint scale, peak radius, movement path, and time envelope can serve as descriptive candidates, but if the original output cadence is insufficient or the mechanism is not yet closed, these features cannot be used to synthesize high-frequency inlet. Spectra, coherence, and turbulence structure also cannot be automatically completed from CM1's coarse temporal snapshots. If Windfield employs analytical or stochastic models to supplement scales, it must preserve the model source and new assumptions so that simulation output and generated components remain distinguishable in the packet. This interface principle prevents an engineering-readable data package from being misinterpreted as a complete cross-scale wind field.

The Researcher's next decisions should center on evidence increments rather than run counts. For all-102, the decision may be whether to resolve blocked inputs and whether to approve the release-ready denominator; for Level-2, cases can be selected based on Level-1 categories and source output availability that best test classification boundaries; for D420, the decision may be whether to execute the matched-drive three-arm and the permitted computational resources; for observational comparison, the decision may be what same-caliber quantities Events must supplement. Each decision should fix expected criteria and stopping conditions before execution to prevent changing the success definition after results appear.

Current limitations include: mechanism judgment from idealized simulation remains two-dimensional model-internal evidence, and formal three-dimensional verification is not available; paper 97 membership is unknown; 40-case labels are limited to model-internal two-dimensional screening and cannot be extrapolated to observational ground truth or paper membership; cases with inconsistent source height order cannot constitute valid input; BESSIE conditional output

cannot be written as event reconstruction; D420 historical M0/M1 comparison cannot constitute a controlled comparison, and the matched-drive three-arm has not received new submissions. Future work includes all-102 production decisions, extension of Level-1 screening to three-dimensional mechanism judgment, establishment of systematic comparison with observations, Level-2 dense source selection, CFD matched drive execution authorization, and semantic handoff closure with Windfield and EVA. All are planned work requiring independent approval and new evidence. This study does not establish $\hat{V}_{3s}$, design wind, design load, or any code recommendation; CFD and design employ remain gated; any further promotion requires PI scientific decision.

Inter-run comparison also requires fixing the observation operator of postprocessing. Two cases, even if they share grid, duration, and microphysics scheme, produce incomparable peaks and footprints if one employs cumulative sws while the other uses single-time horizontal wind, or if divergence smoothing scales and centering orders differ. The registry therefore must jointly store native variables, derived formulas, temporal aggregation, spatial smoothing, boundary exclusion, and finite-value rules. When the postprocessing version changes, existing Level-1 labels cannot be silently inherited but should be replayed on the same source output, and whether the labels are stable should be recorded. This requirement separates physical differences from observation operator differences, ensuring that 24/9/3/2/2 always preserves its original meaning of registry-derived model-internal Level-1 screening.

Selection of Level-2 cases should not rely solely on descending peak rank. A more informative combination covers Level-1 category boundaries, boundary truncation risk, cold pool completeness, three-dimensional output density, and source identity verifiability, thereby testing under what conditions two-dimensional labels hold or change. After candidates enter three-dimensional checking, the original Level-1 status should be preserved and an independent Level-2 determination field added, avoiding overwriting historical screening records with subsequent interpretation. If three-dimensional output does not cover the downdraft core development or near-surface divergence formation period, the conclusion remains insufficient temporal sampling rather than interpreting missing frames as mechanism non-existence.

Each three-dimensional checking result should also write back a source-frame manifest listing the times, variables, vertical layers, and spatial cropping extent used. This permits distinguishing two terminal states: diagnostics did not find the target structure versus source output did not cover the target structure at all; it also lets Windfield or CFD inherit the same sampling limitations when receiving candidate packages. Cross-sections or animations without manifest correspondence can serve only as display materials and cannot change the case gate.

Evidence and formula boundaries of CM1 Agent Chapter 15 preserves only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of "CM1 Agent." Content without physical equations belongs to the research method, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 15 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations, or scientific conclusions approved by PI

16 WRF Agent

16.1 Scientific Questions, Knowledge, and Researcher Decisions

16.1.1 Scientific Purpose

The WRF Agent is responsible for the real-event environment and mesoscale convective reconstruction work in Step 1 of the PETW scientific chain. Unlike CM1's mechanism-oriented idealized simulations, WRF targets specific historical events by using publicly available analysis data as forcing to reconstruct the environmental conditions, mesoscale organizational structure, and near-surface wind field during the development of convective systems. This study must answer three interrelated questions along the WRF route: whether the parent convective system of a given event can be reproduced in the model; how the model's domain, grid, physics options, and forcing choices influence reconstruction quality; and which quantities from the model output can enter Windfield, CFD, or other downstream analyses. The value of WRF lies not in producing mechanism-idealized fields but in converting the observational context at event scale into diagnosable and comparable model evidence. The evidence status of all current WRF output is candidate model diagnostic evidence, the scientific status is PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION, and the PI gate remains pending_peter_scientific_review.

16.1.2 Knowledge Base

The WRF Agent's knowledge layer comprises four components. The first is WRF operational RAG, which stores procedural knowledge of model runs, namelist configuration, restart, and output processing, covering engineering experience with compilation, decomposition, scheduling, and post-processing scripts. The second is the shared-governance LitWiki namespace `wrf_mesoscale_modeling`, used to organize literature claims related to mesoscale convection and near-surface strong winds; the scientific approval status of this namespace is `not_approved`, meaning its content may be used for retrieval and methodological background but does not establish approval of any model results or event interpretations. The third is the Childs-centered v3/v3.1 review assets, which center on downburst climatology and near-surface winds of mesoscale convective systems, providing scientific context for the reconstruction target. The fourth is the body of work accumulated by the Schumacher 2023 reconstruction itself, including target article definitions, input verification records, and comparison evidence. Together these resources support benchmark interpretation and event environment analysis, but consistency and retrieval availability of the knowledge layer do not imply that scientific conclusions are established.

16.1.3 Researcher Participation and Decisions

Researcher decisions on the WRF route occur at four points. First, event and target selection: which historical event to employ as the reconstruction target, which publication to use as the comparison benchmark, and the intended purpose of the reconstruction are all determined by the Researcher; the currently selected event is the 2020-06-09 Akron macroburst in Colorado, USA, and the comparison benchmark is the gravity wave breaking simulation study published by Schumacher, Childs, and Adams-Selin (2023) in *Monthly Weather Review*. Second, configuration boundaries: the domain nesting scheme, resolution, vertical levels, physics options, forcing source, simulation duration, and output variable list are pre-computation decisions; the Researcher must confirm that these choices adequately support the target comparison while specifying which configuration items remain consistent with the literature and which are adjusted because of available resources. Third, result interpretation: when quantitative differences exist between the reconstruc-

tion and the literature, which differences can be explained by initial conditions, lateral boundaries, or model version, and which point to mechanism-level inconsistencies, requires the Researcher to make judgments based on all verified evidence rather than having the agent attribute causes independently. Fourth, downstream use: whether reconstruction output can enter Windfield feature extraction, and whether it can form candidate input for CFD, require the Researcher to decide after comparison evidence is complete. Current results do not authorize any downstream use.

16.2 Methods, Results, and Figures

16.2.1 Workflow and Interactions

The upstream of the WRF route is Events. The event identity, observational evidence, and mechanism hypothesis provided by Events determine the reconstruction target and comparison focus. ComputeOps and RemoteOps provide computational resources, scheduling, and operational support within the scope authorized by the Researcher; Literature and LitWiki provide literature context for configuration choices and comparison interpretation. WRF output has two downstream interfaces: one enters Windfield, forming candidate sources for profiles, footprints, and near-surface wind field features; the other serves as meteorological-side evidence for the CM1/WRF to tall-building load review, connecting with CFD boundary transfer studies. Output organization follows a fixed hierarchy: model raw output, post-processed diagnostic quantities, and comparison quantities against literature, each recorded separately and annotated with its own evidence level without mixing into a single evidence class. Figure 25 shows the evidence chain organization from literature definition, input verification, run gates, segmented execution, to post-processing acceptance, with engineering diagnostics and scientific results preserved in separate layers. Figure 26 shows the verification vector definition for the reconstruction, listing comparison objects across four levels: initial-state comparison, run reproduction, output diagnostics, and literature comparison. Figure 27 presents the transfer gate structure from WRF to downstream agents; each gate includes three fields: input identity, comparison conditions, and known limitations, and gate opening requires item-by-item Researcher determination.

16.2.2 Schumacher 2023 Reconstruction Configuration

The target article for the reconstruction is “Intense Surface Winds from Gravity Wave Breaking in Simulations of a Destructive Macroburst” by Schumacher, Childs, and Adams-Selin (2023), published in *Monthly Weather Review* (DOI: `10.1175/MWR-D-22-0103.1`); the target event is the 2020-06-09 destructive macroburst near Akron, Colorado. The reconstruction employs WRF-ARW 4.7 (the target article uses 4.0.3) with triple two-way nesting: d01 at 4 km resolution, 748 × 600 grid; d02 at 800 m, 1001 × 701; d03 at 266.667 m, 901 × 931, with 80 vertical levels, model top at 5 kPa, time step 20 s, parent ratio 1/5/3, feedback=1. Physics options remain consistent with the target article: Thompson microphysics (mp=8), MYJ boundary layer (pbl=2), Noah land surface (sfc=2), RRTM/RRTMG radiation (ra_lw=ra_sw=4), and no cumulus parameterization. Initial time is 2020-06-09 00:00 UTC; d03 is brought online at 03:30 UTC; the canonical window is 03:30 to 06:45 UTC.

The three online nested domains total 159,146,560 three-dimensional grid cells. d03 covers approximately 240 × 248 km, and d02 approximately 800 × 560 km. Figure 28 shows the spatial relationship of the three-domain nesting configuration.

The main simulation line uses 64 MPI ranks with 8×8 decomposition, io_form_restart=102, and d01/d02/d03 output intervals of 60/15/1 min respectively. Initial-state verification shows that the shape difference between the author’s Dryad wrfinput and the reconstruction wrfinput for

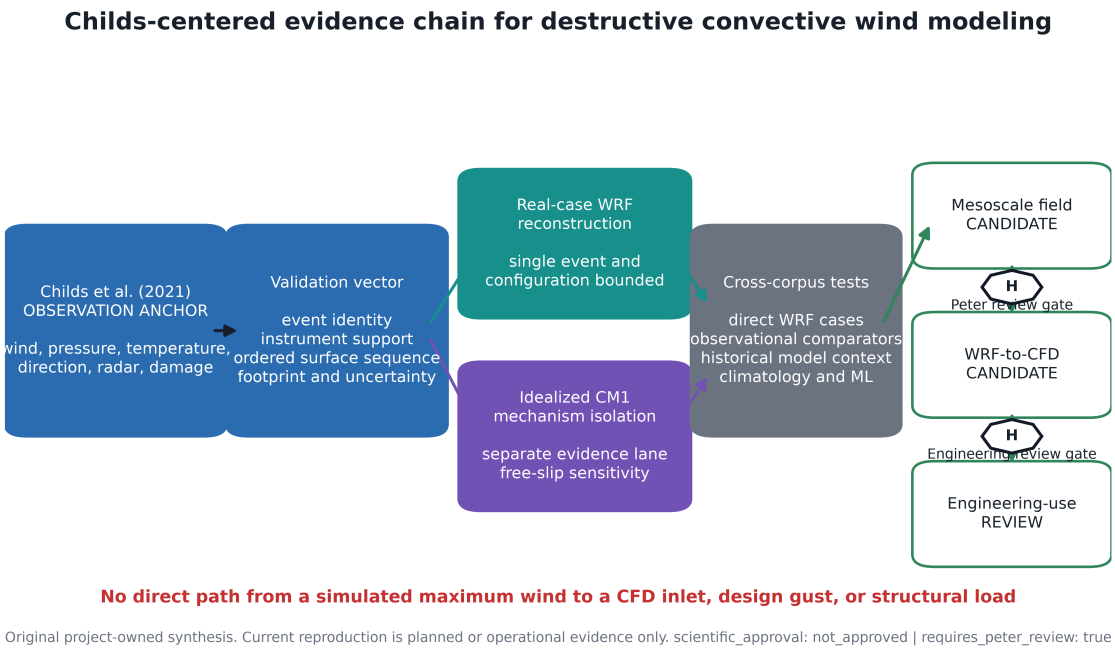


Figure 25: Evidence chain organization for the WRF reconstruction. From literature definition, input verification, run gates, segmented execution, to post-processing acceptance, engineering diagnostics and scientific results are recorded in separate layers. Source: PETW WRF integration process report (2026-08-12).

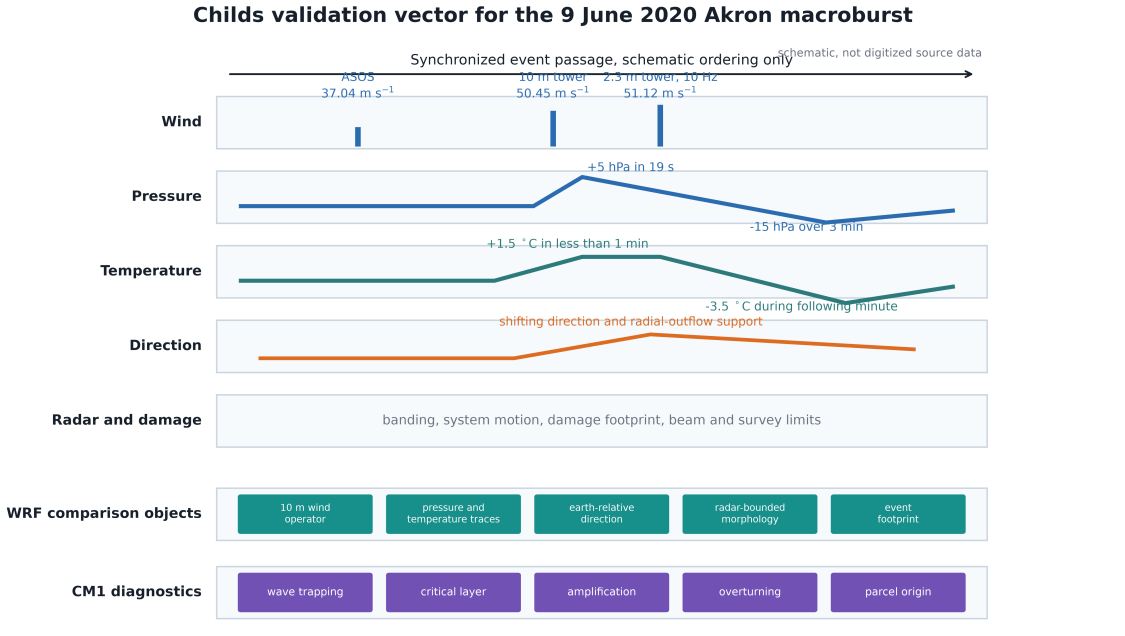


Figure 26: Verification vector for the WRF reconstruction. Initial-state verification, run reproduction, output diagnostics, and literature comparison define comparison quantities, expected precision, and determination criteria at four separate levels. Source: PETW WRF integration process report (2026-08-12).

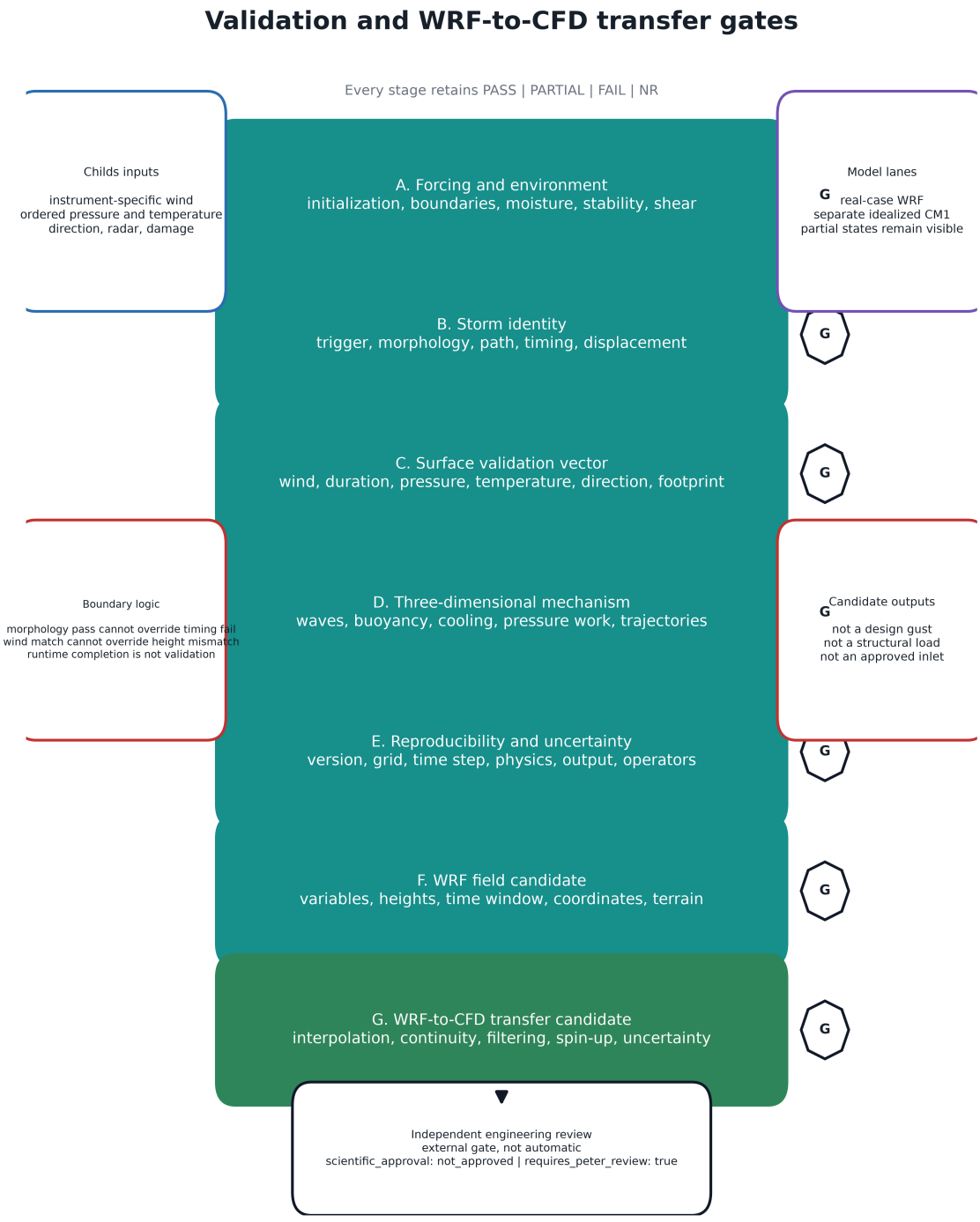


Figure 27: Transfer gate structure from WRF to downstream agents. Each gate contains input identity, comparison conditions, and known limitations; gate opening requires Researcher determination. All transfer gates currently remain unopened. Source: PETW WRF integration process report (2026-08-12).

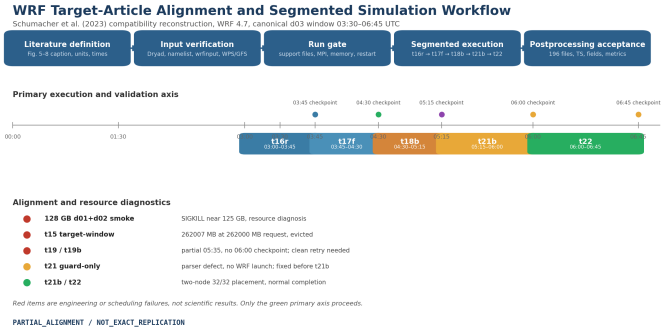


Figure 28: WRF three-domain nesting configuration. d01 (4 km), d02 (800 m), and d03 (266.667 m) are centered on the Akron event area in Colorado; d03 is brought online at 03:30 UTC. Source: PETW WRF Schumacher reconstruction report (2026-08-12).

all 192 common variables is 0. Table 12 presents a comparison of key configuration parameters between the reconstruction and the target article.

Table 12: Comparison of key WRF reconstruction and target article configuration parameters

Configuration item	Target article	This study reconstruction	Source of difference
WRF version	4.0.3	4.7	Available version
d01 resolution	4 km	4 km	Consistent
d02 resolution	800 m	800 m	Consistent
d03 resolution	266.667 m	266.667 m	Consistent
Vertical levels	80	80	Consistent
Model top	5 kPa	5 kPa	Consistent
Microphysics	Thompson (mp=8)	Thompson (mp=8)	Consistent
Boundary layer	MYJ (pbl=2)	MYJ (pbl=2)	Consistent
Land surface	Noah (sfc=2)	Noah (sfc=2)	Consistent
Radiation	RRTM/RRTMG	RRTM/RRTMG	Consistent
Cumulus parameterization	None	None	Consistent
Forcing	ERA5/GFS (not disclosed in detail)	Public forcing reconstruction	Nominally compatible
wrfbdy_d01	Original file not disclosed	Public forcing reconstruction	Not byte-equivalent

16.2.3 Reconstruction Results and Literature Comparison

Results are compared following the target article's figure definitions. Under the Fig. 5 definition (d03 1 km AGL reflectivity and cumulative maximum 10 m wind bands since 03:30 UTC, contour thresholds 30 and 40 m/s), the reconstruction at 05:45 UTC shows a reflectivity peak of 59.89 dBZ, WSPD10MAX peak of 28.83 m/s, and no area exceeding 30 m/s; at 06:15 UTC, reflectivity peak 64.24 dBZ, WSPD10MAX peak 33.62 m/s, area exceeding 30 m/s approximately 96.10 km^2, and still no area exceeding 40 m/s; at 06:45 UTC, values are 60.18 dBZ, 33.62 m/s, and 99.31 km^2 respectively. The target article reports wind bands exceeding 30 m/s and multiple localized areas exceeding 45 m/s. Figure 30 shows the time-segmented comparison panels for reflectivity and WSPD10MAX.

In the Fig. 6 06:11 UTC plan-view diagnostics, the 1 km reflectivity peak is 65.27 dBZ, 3 km vertical velocity ranges from −12.90 to +14.70 m/s, 0.5 km vertical velocity ranges from −10.08 to +9.95 m/s, approximate MSLP (lowest mass level hypsometric approximation) ranges from 1001.34 to 1011.11 hPa, WSPD10MAX peak is 32.48 m/s, grid points exceeding 30 m/s number 135, and those exceeding 40 m/s number 0. Figure 31 shows a side-by-side comparison of the literature and

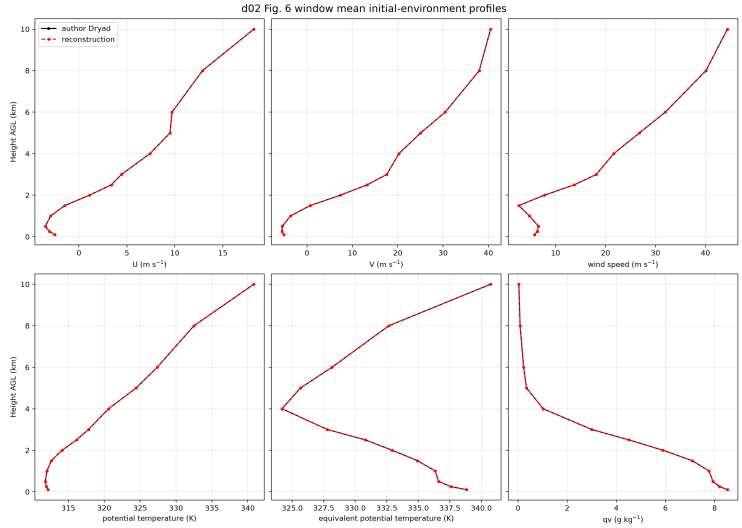


Figure 29: Event environment profiles (d02 event area). The initial environment is approximately rounding-equivalent to the author's Dryad data. Terrain RMSE is approximately 0.149 m; 10 m to 3000 m mean bulk shear is 23.14458 vs. 23.14463 m/s; 1000 to 3000 m mean θ gradient is 2.89225 vs. 2.89228 K/km. Source: PETW WRF Schumacher reconstruction report (2026-08-12).

reconstruction.

In the Fig. 7 vertical cross-section diagnostics, the v wind component ranges from −33.09 to +40.58 m/s; the low-level negative v wind column of approximately −45 m/s reported in the article is not recovered; the cross-section maximum wind speed is approximately 41.04 m/s, located between approximately 1 and 7.6 km MSL, which is not equivalent to near-surface wind speed. Figure 32 shows the vertical cross-section comparison.

The Fig. 8 four-fixed-point comparison is the most direct quantitative comparison. The four point mapping names are nwyu, swph, cph, and sedg; the mapping is inferred from spatial order and passage timing and represents a high-confidence mapping rather than an article-specified station name correspondence. Table 13 and Figure 33 show the peak comparison: the reconstruction four-point mean peak is 19.33 m/s versus the author's Dryad value of 42.52 m/s, giving a peak ratio of 0.468, systematically low by approximately 50%. The four-point mean temperature evolution correlation is 0.859; the reconstruction four-point mean pressure fluctuation amplitude is 9.04 hPa versus the author's 16.34 hPa; the four-point mean surface temperature change is 4.97 versus 6.89 °C; the wind mean correlation is only 0.329. Across the entire d03 domain, the maximum instantaneous wind speed is 33.4128 m/s and the WSPD10MAX maximum is 33.6211 m/s; none of the 196 canonical timestamps reaches 35 m/s.

Table 13: Schumacher 2023 reconstruction four-fixed-point peak comparison. Dryad peaks and article-annotated peaks are from the author's public data.

Point (mapping)	Article-annotated peak (m/s)	Dryad peak (m/s)	Reconstruction peak (m/s)	Difference (m/s)
1 (nwyu)	46.50	46.413	21.167	−25.333
2 (swph)	36.25	37.660	20.088	−16.162
3 (cph)	40.72	43.990	15.231	−25.489
4 (sedg)	42.52	42.025	20.829	−21.691
Mean	41.50	42.52	19.33	−22.169

Combining these comparisons, the reconstruction does produce strong deep convection (1 km reflectivity approximately 60 to 65 dBZ, clear ascending and descending motion) and partial near-surface wind bands exceeding 30 m/s, but does not organize the 40 m/s magnitude near-surface wind core shown in the article. Four-point wind peaks are systematically low by approximately

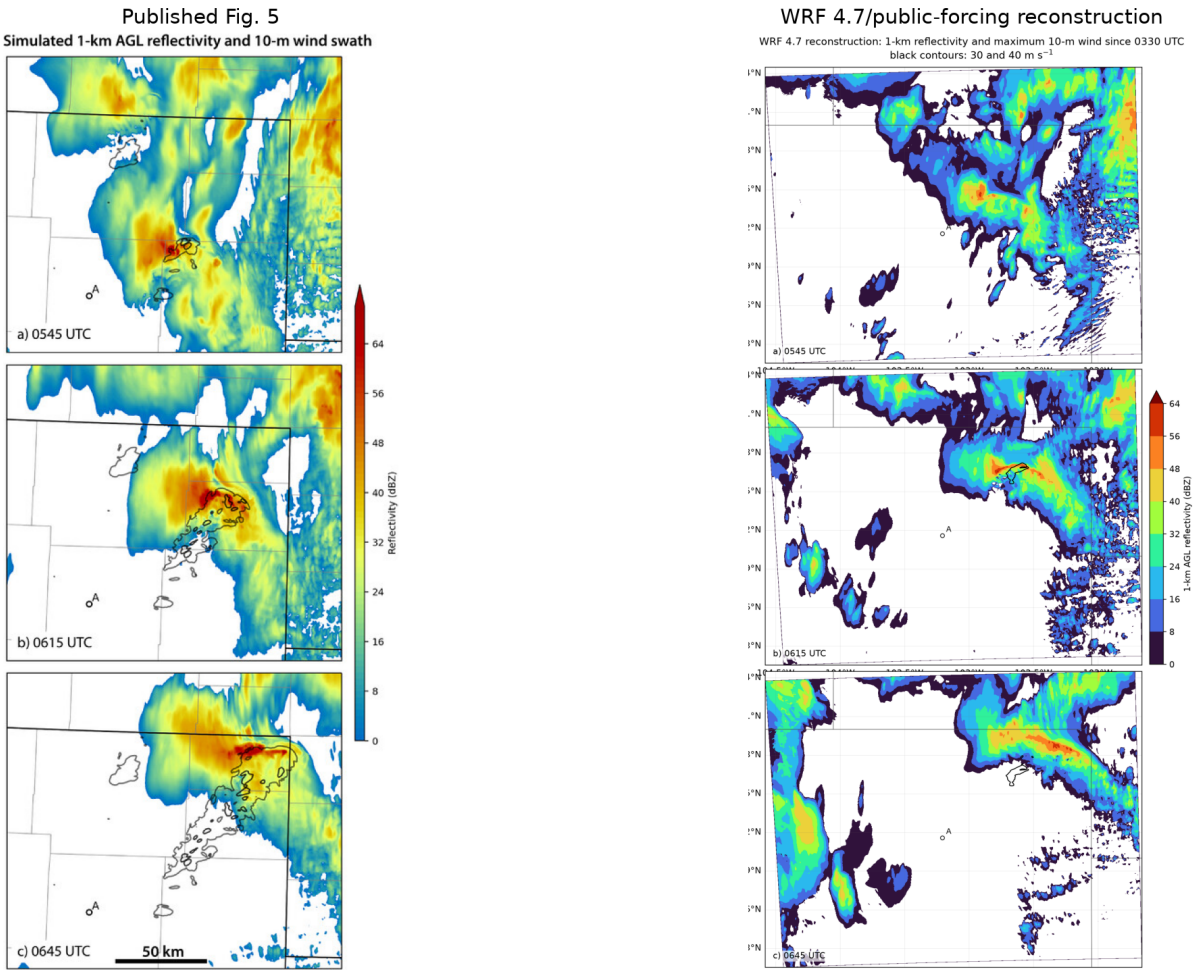


Figure 30: Reflectivity and WSPD10MAX time-segmented comparison under the Fig. 5 definition. The reconstruction produces strong reflectivity convection, but WSPD10MAX peaks are systematically lower than those in the article. Source: PETW WRF Schumacher reconstruction report (2026-08-12); status PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION.

50%, temperature evolution maintains moderate correlation (0.859), while wind maintains low correlation (0.329), indicating that the reconstruction captures the basic thermal evolution features but does not correctly reproduce the dynamic extreme processes. The domain-wide WSPD10MAX temporal evolution (Figure 34) remains consistently below 35 m/s throughout the canonical window, forming a significant gap with the article's multiple localized wind areas exceeding 45 m/s. Comparing the two sets of initial-state environment profiles (Figure 29), which are nearly point-by-point equivalent, permits initial environment differences to be ruled out as the primary cause.

Current evidence rules out initial state, grid, vertical levels, physics options, FDDA, feedback, time step, d03 online initialization, and output sampling. Explanations that cannot be uniquely determined from current evidence include lateral boundary evolution, parent domain trajectory, storm phase, low-level shear, and the primary cause within gravity wave breaking. The author did not disclose the original wrfbdy_d01 or the specific GFS cycle/lead; the reconstruction lateral boundary is a public forcing reconstruction, which is nominally compatible but not byte-equivalent. The magnitude of the domain-wide WSPD10MAX maximum can be summarized by the following statistic:

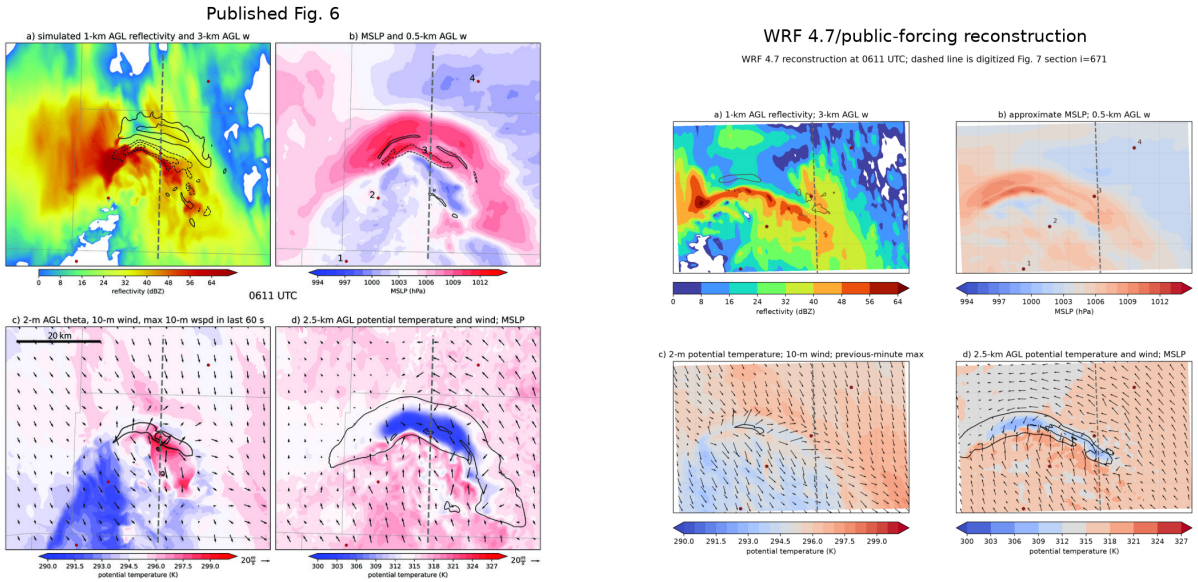


Figure 31: Side-by-side comparison of literature and reconstruction for Fig. 5 through Fig. 8. The reconstruction maintains strong reflectivity convection but does not produce the 40 m/s contour or 45 m/s localized wind areas shown in the article. Source: PETW WRF Schumacher reconstruction report (2026-08-12).

$$R_{\text{peak}} = \frac{\max(\text{WSPD10MAX}_{\text{recon}})}{\max(\text{WSPD10MAX}_{\text{Dryad}})} = \frac{33.62}{46.41} \approx 0.724, \tag{11}$$

Here, $\text{WSPD10MAX}_{\text{recon}}$ is the cumulative maximum 10 m wind within the reconstruction d03 domain (units m/s) and $\text{WSPD10MAX}_{\text{Dryad}}$ is the corresponding same-definition quantity from the author's Dryad data. This ratio indicates the degree of deficit at the domain maximum level but does not directly indicate the more severe 0.468 deficit at the fixed-point level. The formal scientific status is PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION.

16.3 Limitations, Diagnostics, and Interfaces

16.3.1 Current Scientific Limitations and Future Research

16.3.2 Hierarchy of Reconstruction Diagnostics and Error Attribution

Comparison objects for the WRF reconstruction comprise input consistency, process consistency, morphological consistency, and quantitative consistency. Input consistency checks common variable array shapes, grids, vertical levels, physics options, and initial-state profiles, and only indicates the two systems enter integration from approximately the same starting point. Process consistency checks whether the storm forms, propagates, and produces gravity waves and near-surface strong winds during a similar period. Morphological consistency checks the spatial organization of reflectivity bands, downdraft regions, pressure perturbations, and strong wind footprints. Quantitative consistency checks fixed-point peaks, domain maxima, area thresholds, and temporal correlations. The current high correspondence in initial state and major configuration, partial recovery of strong deep convection morphology, and clearly insufficient spatial organization and peaks of near-surface strong winds mean that near-equivalent input cannot replace precise result replication.

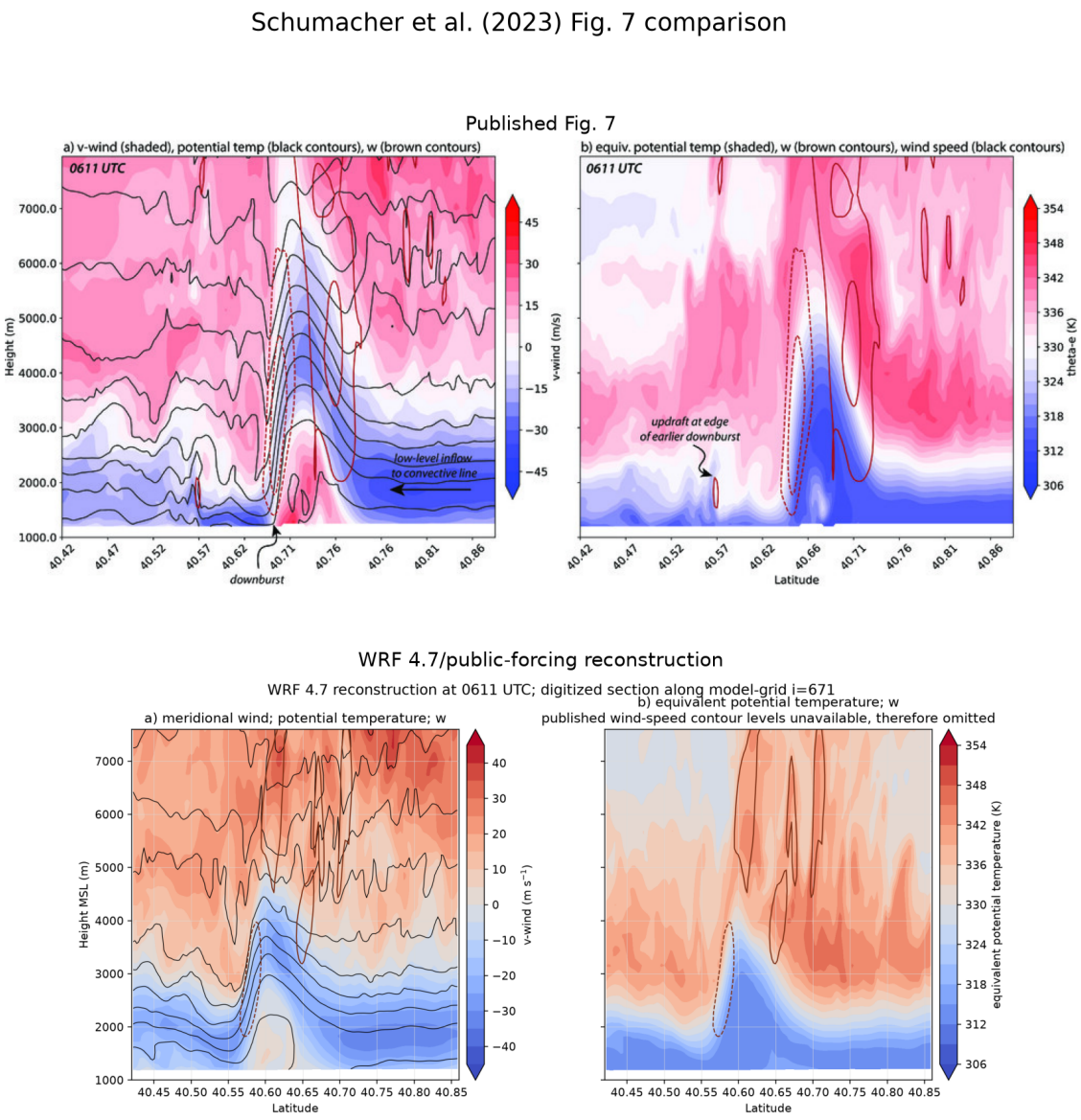


Figure 32: Vertical cross-section comparison for Fig. 7. The v wind component in the reconstruction does not recover the low-level negative v column of the −45 m/s magnitude reported in the article; the cross-section maximum wind speed is located between 1 and 7.6 km MSL. Source: PETW WRF Schumacher reconstruction report (2026-08-12).

The four-point comparison is more stringent than the domain maximum ratio because fixed points jointly constrain spatial position and passage timing. Domain maxima can occur at different locations and times, requiring only that some grid point in the model domain reaches the corresponding magnitude; fixed-point series require storm path, structural scale, and propagation speed to be jointly correct. The reconstruction domain peak ratio $R_{\text{peak}} \approx 0.724$ while the four-point mean peak ratio is 0.468; the difference suggests that the model not only underestimates intensity but may also fail to organize the strongest wind core onto the target path. Temperature correlation being higher than wind correlation indicates larger-scale thermal passage is more easily recovered than localized dynamic extremes. This interpretation is a bounded diagnostic and does not uniquely point to any forcing, boundary, or version factor.

Threshold area provides a morphological diagnostic rather than event verification. The area exceeding 30 m/s appears at 06:15 and 06:45 UTC, while the area exceeding 40 m/s remains zero, indicating that the reconstruction produces a limited strong wind band but does not recover the high-wind-speed core shown in the target article. Area calculations depend on the d03 grid, cumu-

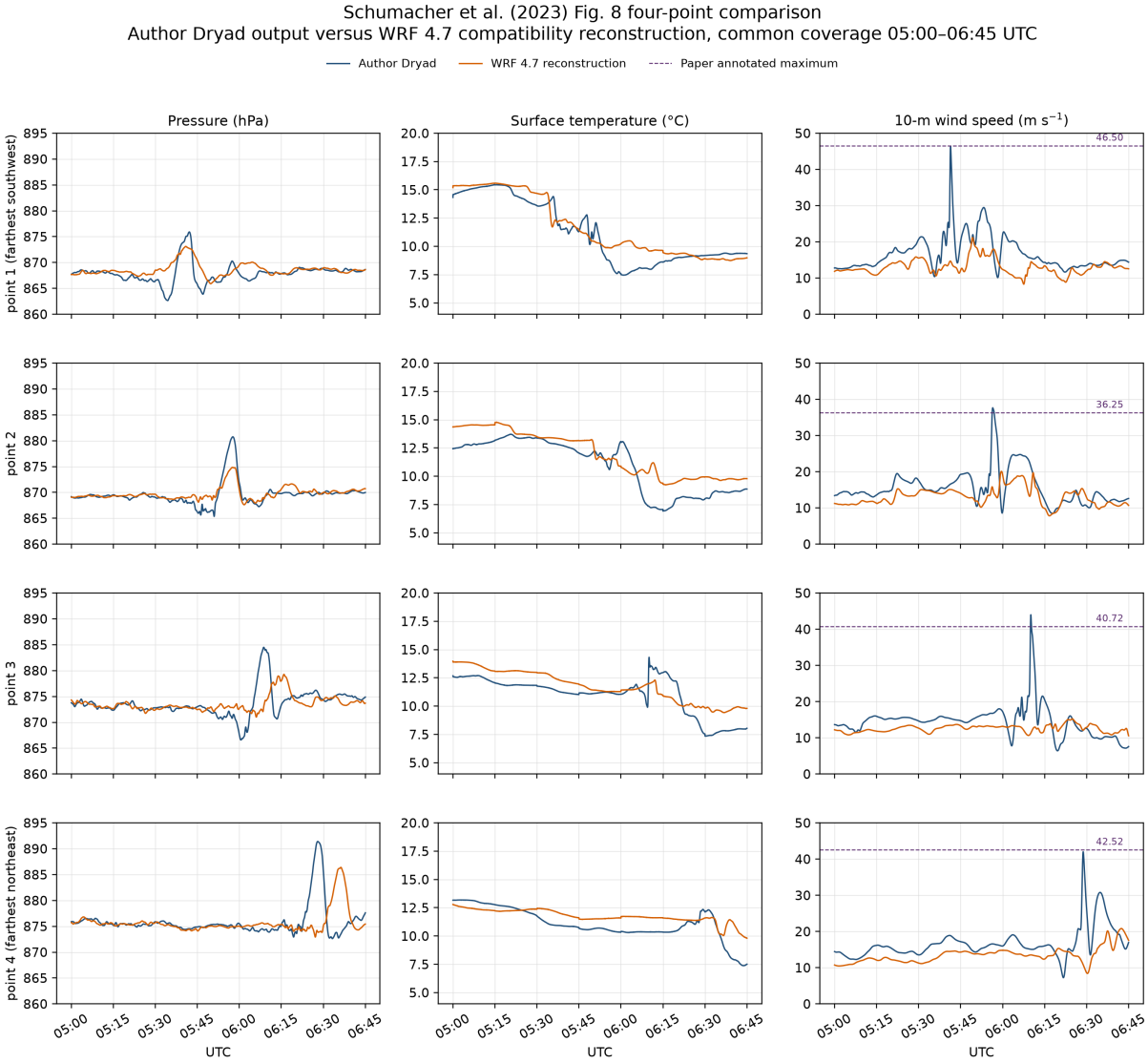


Figure 33: Schumacher 2023 reconstruction Fig. 8 four-point comparison. Author Dryad and reconstruction use native fixed-point time series; the reconstruction significantly underestimates wind speed peaks while temperature evolution is relatively close. Source: PETW WRF Schumacher reconstruction report (2026-08-12); status PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION.

lative maximum definition, and canonical window, and cannot be directly compared with different resolutions, different cumulative starting points, or instantaneous wind fields. WSPD10MAX also has a time-cumulative nature; contours at a given time include grid-point peaks from earlier times and are not equivalent to instantaneous 10 m winds at that moment. Interpreting Figure 30 requires separating the temporal semantics of instantaneous reflectivity from cumulative strong wind bands.

Vertical cross-sections are used to examine whether the dynamic structure is vertically coherent. The high-speed region between approximately 1 and 7.6 km MSL in the reconstruction cannot replace a near-surface wind core; the low-level negative v wind column does not reach the magnitude shown in the target article, nor does it form a path consistent with fixed-point peaks. If the cross-section location deviates from the actual storm core, intensity differences may include sampling error; if the storm phase shifts, comparing cross-sections at the same UTC time also mixes in temporal error. Cross-section diagnostics thus require simultaneous recording of the cross-section line, time, terrain height, variable levels, and target structure, and cannot compare

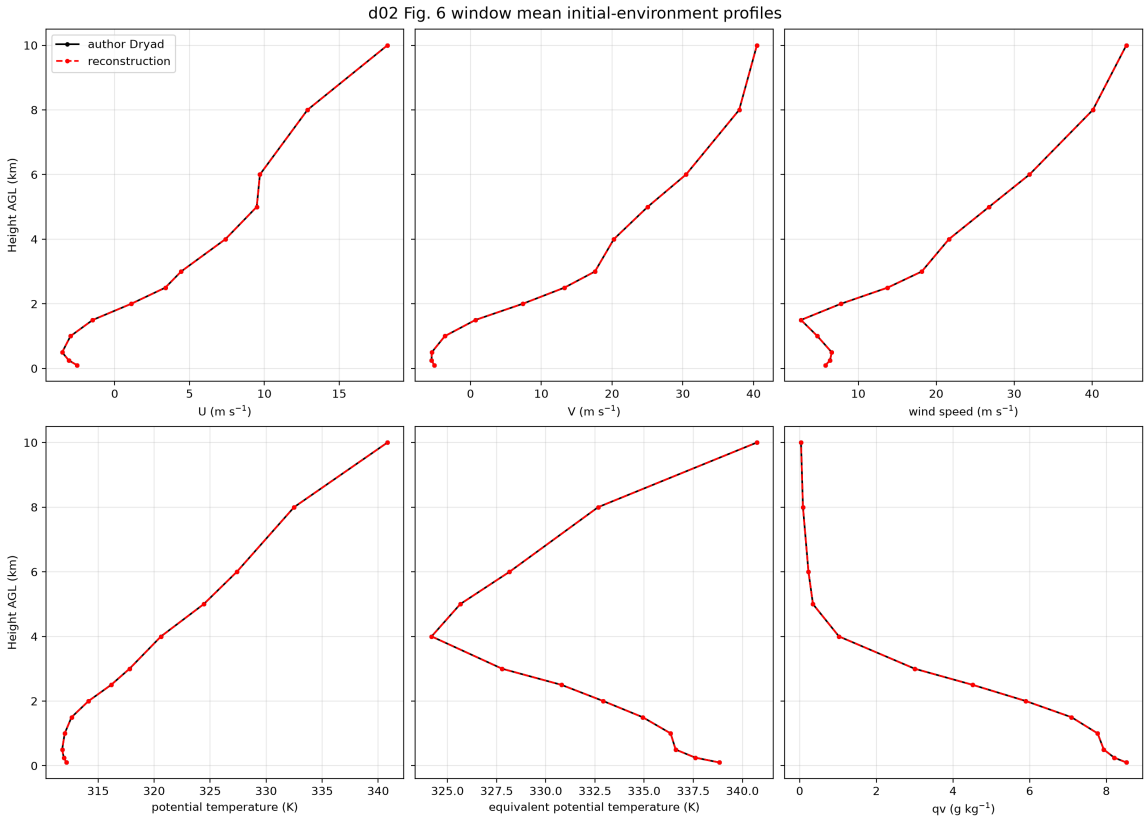


Figure 34: d03 domain-wide WSPD10MAX temporal evolution. Peaks remain consistently below 35 m/s throughout the canonical window of 03:30 to 06:45 UTC. Source: PETW WRF Schumacher reconstruction report (2026-08-12).

only color scale extremes.

Pressure and thermal diagnostics are similarly limited by derived algorithm constraints. The current MSLP employs a lowest mass level hypsometric approximation, and θ_e uses the Bolton approximation; both are used for internal structure identification within the reconstruction, not as point-by-point replicas of the author's native diagnostics. Pressure perturbation amplitudes and temperature changes can be compared trend-wise with Dryad series using the same definition, but algorithmic differences must be retained in the error budget. Four-point mean pressure fluctuation and temperature change are weaker than the target in the same direction as the wind speed bias, but this is insufficient to prove a single cold pool intensity error because storm path and sampling timing also jointly change these quantities.

The scientific importance of the lateral boundary issue arises from the continuous information input in the triple nesting. The initial field of d03 can be approximately consistent, but the d01 lateral boundaries and parent-domain convection evolving with time influence the storm position, low-level shear, gravity wave phase, and cold pool environment in d02 and d03. The author's original wrfbdy_d01 and complete parent-domain wrfout are not public, making the current forcing only nominally compatible rather than byte-equivalent. The 192 common variables with matching shapes in the shared wrfinput rule out obvious structural errors but cannot rule out lateral boundary values, forcing cycle, lead time, or data processing differences. This gap is an explanatory boundary and should not be rewritten as a confirmed primary cause.

The model version change from 4.0.3 to 4.7 is also retained as an independent sensitivity axis. Even with identical namelist options, physics scheme implementations, default values, numerical algorithms, and I/O details may change across versions. No controlled dual-version comparison under the same forcing and computing environment has been conducted, so the version contribution

cannot be quantified. A reasonable sensitivity design should hold domain, forcing, initial state, physics options, time step, nesting time, and post-processing all unchanged, changing only the model version, and then compare storm tracks, fixed-point peaks, pressure-temperature series, and dynamic cross-sections. Until such a comparison is completed, version differences can only be listed as secondary sensitivity.

Time window truncation influences peak conclusions and process conclusions differently. The canonical reconstruction ends at 06:45 UTC, while the target data extends to 07:15 UTC; the article-annotated key peaks occur before 06:45, so existing peak deficit comparisons remain valid. The missing final 30 minutes means that decay, secondary development, and late-stage strong winds cannot be completely compared. In the single-00Z extended diagnostics, late-stage high-speed values have not been closed with continuous downburst jet, near-surface divergent outflow, synchronous mesohigh, and thermal structure, and are thus retained as late-stage strong wind diagnostics. A single variable crossing a numerical value cannot replace a multi-variable mechanism chain.

The downstream gates shown in Figure 27 require converting model diagnostics into employ-specific qualifications. If Windfield receives WRF output, it must obtain the native grid, coordinates, terrain, vertical levels, time cadence, instantaneous and cumulative variable definitions, and limitations relative to observational comparisons. If CFD receives candidate boundaries, it additionally needs to address spatial clipping from meteorological grid to engineering domain, mass conservation, turbulence scale gaps, and time-varying boundary treatment. The current failure to precisely recover the strong wind core means that using this output as an event-specific engineering input would inherit unclosed intensity and path biases. Keeping the transfer gates closed does not deny the environmental diagnostic value of WRF but prevents model evidence from being elevated to verified engineering input.

Reconstruction comparison also requires maintaining three time axes. The data time axis records the UTC of forcing, observations, and author-public data; the model time axis records integration seconds from the initial time; the phenomenon time axis records relative stages of storm genesis, propagation, strong wind passage, and decay. Point-by-point comparison at the same UTC is appropriate for verifying event timing; comparison at the same phenomenal stage is suitable for analyzing phase shifts; the two cannot replace each other. If the storm arrives earlier or later in the reconstruction, directly comparing the same UTC attributes timing difference to intensity and position differences; if alignment is only by peak time, absolute forecast sequence information is lost. The WRF packet should therefore jointly preserve original time, relative integration time, diagnostic stage labels, and alignment method, so that fixed-point correlations, domain maxima, and vertical cross-sections can each be traced back to their respective comparison conditions.

Spatial comparison is similarly divided into grid consistency, geographic consistency, and structural consistency. Grid consistency checks projection, latitude-longitude, terrain, and variable staggering positions; geographic consistency requires compared quantities at the same location or with explicit regridding operators; structural consistency compares the relative configuration among reflectivity cores, pressure perturbations, downdraft regions, and strong wind bands. The high resolution of d03 does not by itself bring structural consistency because storm path or phase shifts can still place target fixed points outside the core. Domain maxima can verify whether the model produces a given magnitude but weaken position constraints; fixed-point series strengthen

position and timing constraints but may be highly sensitive to small path shifts. Only by juxtaposing both diagnostic types can intensity deficit, path deviation, and sampling misalignment be distinguished; current evidence is still insufficient to uniquely decompose these three contributions.

Near-surface wind variable observation operators must also enter the comparison contract. Instantaneous U10, cumulative WSPD10MAX, lowest mass level wind, and cross-section high-speed regions have different heights, time aggregation, and spatial support. Cumulative quantities preserve prior peaks at each grid point, suitable for describing event window footprints, but not suitable for synchronous causal interpretation with instantaneous reflectivity at a given time. Lowest mass level wind is affected by terrain-following coordinates and first-level height and cannot replace diagnostic 10 m wind. Fixed-point Dryad series and reconstruction series enter peak ratio and correlation calculations only when variable, time, and spatial mapping are consistent. Any post-processing update should record variable definition, cumulative starting point, interpolation method, missing-data treatment, and version hash, preventing identically named figures from masking changes in the observation operator.

Lateral boundary and version sensitivity should employ controlled branches rather than multivariable simultaneous adjustments. One branch replaces only the forcing cycle or lateral boundary source; another branch changes only the WRF version; all other domain, physics options, online nesting time, time step, output cadence, and post-processing remain fixed. Each branch must compare storm tracks, four-point time series, threshold areas, cross-section structure, and pressure-temperature evolution, rather than selecting only the single peak closest to the article. If multiple configurations change jointly, even if output comes closer to the target, it can only serve as an exploratory reconstruction and cannot attribute to any single factor. Pre-registered comparison matrices and stopping conditions can avoid adjusting success definitions after seeing results.

If the candidate package to Windfield is opened in the future, it must carry the current PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION status rather than transmitting only the wind field array. Intensity bias, fixed-point path inconsistency, lateral boundary non-byte-equivalence, incomplete final period, and derived algorithm differences all constitute limitations that downstream must inherit. Windfield can exercise variable semantics, coordinate conversion, and descriptive footprint within these boundaries, but cannot name resampled or randomly supplemented fields as having repaired the event reconstruction. The gate to CFD additionally requires proof that boundary clipping, mass flux, turbulence scale, and clock mapping are controllable. The most prudent current employ remains reconstruction method diagnostics and bounded literature comparison.

Run reproducibility also requires distinguishing restart continuity from scientific continuity. A restart file being readable and continuing integration only indicates the model state is continuous at the file interface; between segments, cumulative variable starting points, output times, nesting feedback, physics tendencies, and diagnostic scripts must also be verified for continuity. If cumulative variables such as WSPD10MAX are reset or double-accumulated after restart, the window footprint and peak timing change. Acceptance records must compare state variables, cumulative variables, and time indices before and after segment boundaries and preserve per-segment input-output hashes. Only when these checks pass can the segmented run serve as an engineering-continuous sequence of the same reconstruction, which still does not elevate scientific consistency.

Missing data and non-comparable items should explicitly enter the comparison vector rather than being silently removed from summary statistics. Four-point comparison requires recording

each point's common time window, effective sample count, interpolation, and temporal alignment; vertical cross-sections require recording common heights and cross-section lines; threshold areas require recording the effective domain and mask. If a comparison quantity exists only in the author data or the reconstruction, its status is one-sided available; if the two sides employ different algorithms that cannot be recomputed, the status is definition-incompatible. A complete verification vector jointly includes consistent, deviated, non-comparable, and unobserved result categories, preventing overall correlation or mean peak ratio from masking evidence gaps.

Scientific status updates require satisfying pre-registered evidence increments. A new forcing branch that only increases the domain maximum without improving fixed-point paths, pressure-temperature evolution, and vertical dynamic structure is insufficient to change NOT_EXACT_REPLICATION. Conversely, improvement at a single fixed point also cannot represent recovery of the entire strong wind band. Status review should examine input identity, temporal-spatial organization, multiple independent diagnostics, and remaining alternative explanations, and then the Researcher decides whether to maintain, tighten, or elevate the boundary. No such new controlled branch has been completed currently, and the PARTIAL_ALIGNMENT status remains unchanged.

PARTIAL_ALIGNMENT indicates some diagnostics are compatible under specified conditions but do not jointly cover intensity, path, timing, and dynamic structure. NOT_EXACT_REPLICATION indicates that the reconstruction cannot replace the target event realization and cannot use similar initial states or configuration to prove unrecovered processes. The former preserves bounded diagnostic value; the latter closes event equivalence and exact replication expressions.

This status directly constrains downstream. Windfield can examine variable semantics, coordinate conversion, and descriptive features but cannot name resampled fields as a replicated event wind field. CFD cannot employ it as an event-specific production input; EVA cannot convert model peaks to observational samples; Structural and DesignLoad cannot form loads or design endpoints along this chain. Sensitivity branches can only add diagnostic evidence and cannot open employs before Researcher review.

All comparison figures must still preserve source, variable definition, time window, and generation script version.

Scientific limitations of the reconstruction results center on six aspects. First, the uncertainty of lateral boundaries and parent-domain evolution cannot be fully closed from public data; the author's original wrfbdy and complete wrfout are unavailable, making point-by-point comparison valid only for common variables. Second, model version differences (4.0.3 vs. 4.7) are retained as secondary sensitivity; this study does not quantitatively isolate their effects. Third, the Fig. 8 point name mapping (nwyu, swph, cph, sedg) is a high-confidence mapping inferred from spatial order and passage timing, not an article-specified station name correspondence. Fourth, the canonical reconstruction covers only up to 06:45 UTC while the article timeline extends to 07:15; the final 30 minutes are missing; article-annotated extremes all precede 06:45, so this gap does not change peak comparisons, but the complete tail evolution cannot be compared. Fifth, MSLP and θ_e are hypsometric approximation and Bolton approximation respectively, and are not used as point-by-point equivalents of the author's native diagnostics. Sixth, the single-00Z extended diagnostics show 37.749 m/s U10 and 45.274 m/s lowest mass level wind speed at 07:02 UTC, but this late-stage strong wind does not satisfy the complete evidence chain of continuous downburst jet, near-surface divergent outflow, synchronous mesohigh, and thermal structure, and is retained as a late-stage strong wind event, not upgraded to delayed downburst. The formal status remains

PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION throughout. Current results do not authorize forcing sensitivity, ICX/SPR continuation, WRF-to-CFD transfer, design wind, or design load employs. Future research must first close lateral boundary and model version sensitivity before discussing whether reconstruction output can constitute bounded input for Windfield or CFD.

Evidence and equation boundaries of the WRF Agent Chapter 16 preserves only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of the WRF Agent. Content not presenting physical equations belongs to the research methods, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 16 remains a source-limited synthesis and cannot support verified design wind, design load, code recommendation, or PI scientific conclusion.

17 Windfield Agent

17.1 Scientific Questions, Knowledge, and Researcher Decisions

17.1.1 Scientific Purpose

The Windfield Agent is responsible for the wind field characterization work in PETW Step 2: organizing wind fields from observations, CM1, WRF, or other sources into engineering-relevant wind field features and checking whether these features meet the scale, resolution, averaging time, and spatial support requirements of downstream employ. The principal questions are: for a given event's wind field, which vertical profiles, horizontal footprints, near-surface turbulence, gust ramp-up, time histories, and translating outflow geometric features can be stably identified; how these features convert between meteorological grids and engineering inputs; and whether unit, coordinate, time window, and provenance information is complete throughout feature extraction. Windfield does not own event classification, EVA return level, CFD validation, wind tunnel facility actions, or design load approval authority.

17.1.2 Knowledge Base

Windfield's knowledge layer is centered on the Windfield Wiki (`windfield_step2` namespace, 138 files including 120 paper pages, 6 claim pages, and 7 concept pages; 4 close-reading verified, 112 active candidate, 3 source-limited supplement, 1 PI supplemented by feature review and topical materials on vertical profiles, horizontal structure, non-stationarity, and turbulence. The REV5 bilingual review is the formal synthesis for this lane: Chinese 53 pages, English counterpart 76 pages, containing 15 original vector figures, 13 local tables, and 6 numbered equations; the English version employs 116 admitted bibliography keys and 332 citation commands. 21 internal source-figure pointers remain `not_copied` and `rights_uncleared_internal_only`. Retrieval consistency of the knowledge layer and active generation passage represent only internal knowledge layer status and do not indicate any model or event has received scientific approval.

17.1.3 Researcher Participation and Decisions

Researcher judgments at the Windfield stage center on three questions. First, what downstream needs: the wind field quantities entering EVA, WindTunnel, CFD, or Structural differ; EVA concerns standardized wind speed endpoints, WindTunnel concerns executable target time histories, CFD concerns inlet boundary spatiotemporal fields, and Structural concerns floor forces or pressure time histories; the Researcher must determine which downstream the current task serves and thereby determine the required feature set and accuracy requirements. Second, what resolution is sufficient: the scale gaps among meteorological grids, observation stations, and engineering surfaces

must be explained item by item, and the Researcher decides which scale conversions are feasible under current evidence and which must be explicitly marked as unresolved. Third, which features can be compared: similar features from different sources (observations, CM1, WRF, literature) are comparable only when time averaging, spatial support, and height endpoints are consistent, and the Researcher determines the eligibility of comparison objects.

17.2 Methods, Results, and Figures

17.2.1 Workflow and Interactions

Windfield's inputs come from Events, CM1, and WRF: Events provides event identity and mechanism hypotheses; CM1/WRF provide simulated wind fields; observations and literature provide profile and turbulence feature references. Outputs go in three directions: to EVA providing wind speed quantities and event-level features, to WindTunnel providing controlled intensity or wind field descriptions, and to CFD and Structural providing candidate inlet fields or feature packets. Windfield has developed from an empty profile to a research support lane with source/equation contract, model functions, JSON/NPZ packets, validator, integration smoke, reports, and cross-bot handoff. The 2026-08-12 fresh replay yields 70/70 unit tests PASS, integration smoke PASS, canonical packet validator PASS; the canonical NPZ includes 14 arrays (evidence status: engineering preparation). The model layer covers Dang radial closure, nine-model GUI transcription, Xhelaj translating horizontal mean field, Roncallo-Solari Method A/B, EPSD, SRM, coherence/cross-PSD, Moeini source anchor, and quasi-steady force/Newmark SDOF interface. Reports explicitly differentiate large-grid examples, canonical $3 \times 3 \times 4$ packet, integration smoke, and source anchors as four evidence identity classes.

17.2.2 Multi-scale Wind Field Components and Methods

Multi-scale component organization is the main thread of the Windfield synthesis. Wind fields are structured in six component categories: storm scale (thermodynamic and momentum forcing), outflow scale (footprint, vortex, and path), near-surface scale (profile and surface), temporal structure (mean, direction, and duration), stochastic structure (variance, spectra, and correlation), and interface description (multivariate, pressure, and response). Figure 35 shows the transfer relationships among the six component categories as a vector diagram; arrows represent description quantities that must be transferred and do not represent independently verified closed causal chains. Figure 36 shows the method family classification, covering the main technical routes from observational profile fitting to computational fluid dynamics inlet generation. Figure 37 presents the logical structure of forcing routes, distinguishing different evidence requirements between storm-scale forcing and engineering-scale forcing.

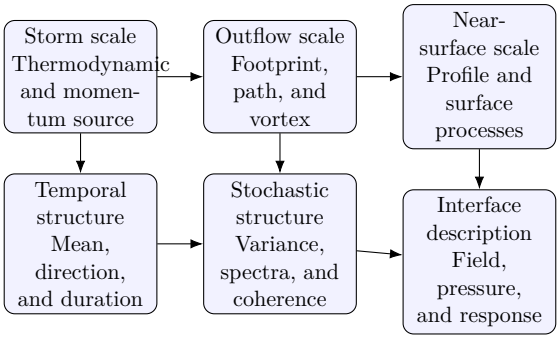


Figure 35: Windfield multi-scale components and description quantity transfer relationships. Arrows represent associations that packets must retain and do not represent inter-scale causal closure or verification completion. Original project method figure.

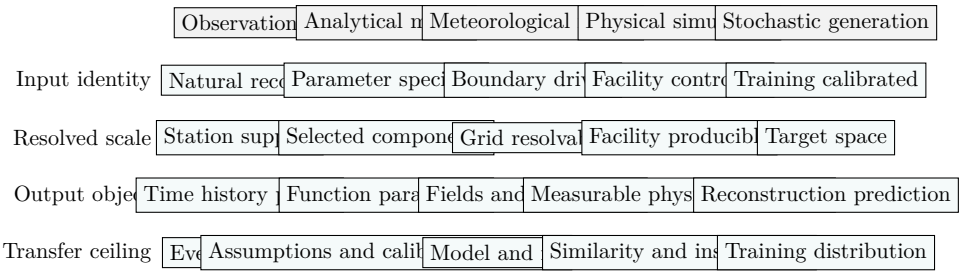


Figure 36: Windfield method family input identity, resolved scale, output object, and transfer ceiling. Evidence identities differ across columns; the figure does not provide a method superiority ranking. Original project method figure.

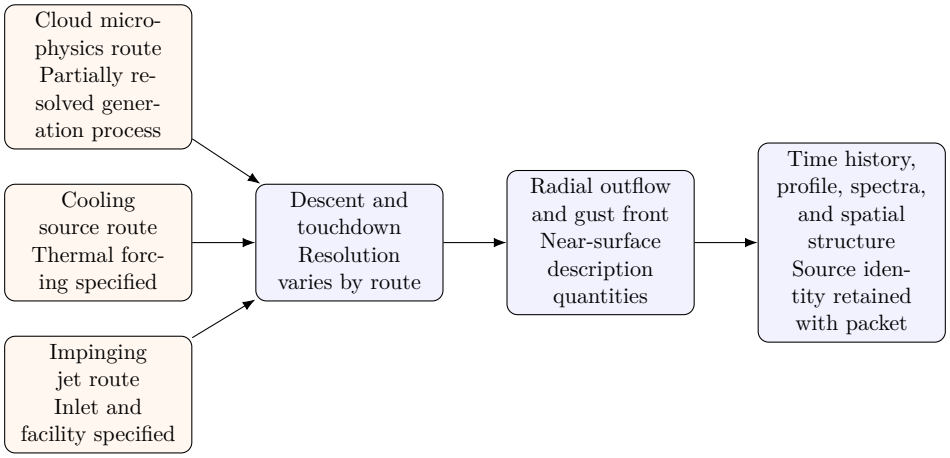


Figure 37: Separation of generation mechanism forcing and engineering forcing routes. Different routes can produce near-surface outflow descriptions, but the retained generation mechanism information differs; output morphological similarity cannot substitute for source equivalence. Original project method figure.

For engineering wind field description, the basic decomposition of non-stationary wind can be written as a three-component expression of instantaneous wind speed $U(z,t)$ at height z and time t:

$$U(z,t) = \bar{U}(z) + \tilde{U}(z,t) + u'(z,t), \tag{12}$$

where $\bar{U}(z)$ is the time-averaged component (units m/s), $\tilde{U}(z,t)$ is the slowly varying non-stationary component, and $u'(z,t)$ is the zero-mean stochastic turbulence component. Separation of the three components depends on the chosen averaging time window, filtering method, and stationarity assumption; different choices change the energy allocation between $\tilde{U}$ and u'. For thunderstorm outflows, the slowly varying component typically includes ramp and decay structures with durations ranging from tens of seconds to several minutes, far shorter than the quasi-stationary assumption time windows in synoptic-scale processes. Figure 38 shows a morphological comparison of common profile families under different assumptions. Figure 39 shows the correspondence between stationarity levels and time-frequency information retention, distinguishing the assumption implications of stationary assumed, piecewise stationary, and nonstationary retained levels.

Before downstream transfer, the spectral content of the wind field packet must match the frequency range of structural response. Under the quasi-steady assumption, the relationship between the base shear coefficient C_F and the reference wind speed U_{ref} is

$$F(t) = \frac{1}{2}\rho\, C_F\, A\, U_{\mathrm{ref}}^2(t), \tag{13}$$

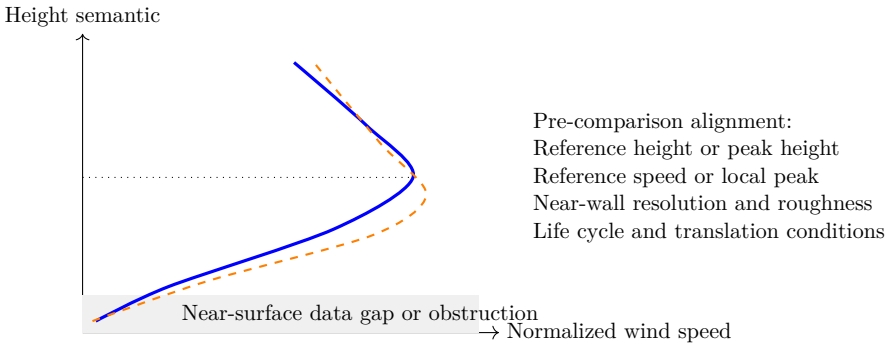


Figure 38: Vertical profile families and near-surface resolution boundary. Curves represent qualitative morphology only and do not represent universal profiles or determined parameters. Original project method figure.

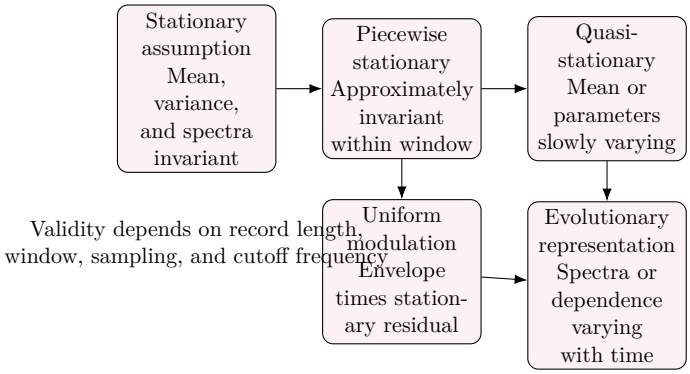


Figure 39: Hierarchy of stationarity and non-stationarity assumptions. Assumption type delimits the interpretation range of statistical quantities and does not indicate that any single type applies to all thunderstorm events. Original project method figure.

where ρ is air density (kg/m^3), A is the reference area (m^2), and C_F is the force coefficient related to cross-section and angle of attack (dimensionless). This equation assumes that wind speed variations are much slower than the adjustment time of the aerodynamic response; whether this holds during the ramp-up phase requires case-by-case assessment. Figure 40 shows a method comparison of time-frequency information retention. Figure 41 shows modeling levels of spatial coherence and known limitations.

	Time histor	Segmented sp	Evolutionar	Time-frequency distribution
Phase	Retained	Lost	Partial	Partial
Temporal localization	Retained	Window level	Retained	Retained
Total energy allocation	Partial	Retained	Retained	Retained
Instantaneous extremes	Retained	Lost	Indirect	Indirect

Figure 40: Differences in information retention among time-domain, segmented frequency-domain, evolutionary spectral, and time-frequency representations. Conversion between representations cannot automatically recover lost information. Original project method figure.

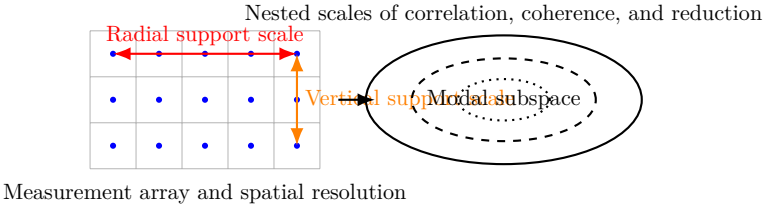


Figure 41: Spatial coherence, multivariate dependence, and array resolution. Correlation scale and reduced-order results depend on sensor layout and spatial support; single-point time histories cannot identify complete spatial coherence. Original project method figure.

The REV5 review defines a source/equation contract that records the source, valid domain, and known conflicts for each equation incorporated into the model. In the Dang radial closure, the relationship between the radial velocity field decay exponent and the Coles constant is

$$U_r(r, z) = U_{\max}(z)\, \exp\left[-\left(\frac{r - r_{\max}}{r_s}\right)^n\right], \tag{14}$$

Here, U_r is the radial velocity (m/s), r is the distance from center (m), z is height (m), $U_{\max}(z)$ is the maximum radial velocity at height z, $r_{\max}$ is the radius of maximum velocity (m), r_s is the scale parameter (m), and n is the decay shape exponent (dimensionless). Fixed constants such as the Coles constant in this closure are allowed to be overridden in local functions; the canonical runner does not expose these parameters, so the canonical packet is not affected by this path, but direct calls to local functions cannot claim strict source closure.

Table 14: Windfield evidence identity and permitted interpretations

Evidence identity	Content	Permitted interpretation	Prohibited interpretation
large-grid examples	Nine-model GUI transcription and demonstration	Method implementation and interface verification	Treating as a unified validated model family
canonical $3 \times 3 \times 4$ packet	14-array NPZ standard package	Field, unit, coordinate, shape, status, and provenance transfer exercise	Treating as measured event wind field
integration smoke	Cross-module connectivity test	Engineering readiness status	Treating as scientific result
source anchors	Source equations and constants	Source equation reference verification	Treating as having resolved all source conflicts

Table 15: Windfield method family input-output and transfer ceiling

Method family	Input	Output	Transfer ceiling	Known limitations
Observational profile fitting	Station time series, height	Parameterized profile	Within fitting domain	Limited height coverage
Idealized numerical	CM1/WRF fields	Feature fields	Grid resolution	Sub-grid processes missing
Analytical radial model	Parameter set	Velocity field	Parameter domain	Source conflicts not closed
Spectral/correlation model	Frequency, space	PSD/CSD	Model form	Non-stationarity applicability
Quasi-steady force interface	$U_{\mathrm{ref}}(t)$, C_F	$F(t)$	Time scale separation	Ramp phase uncertain

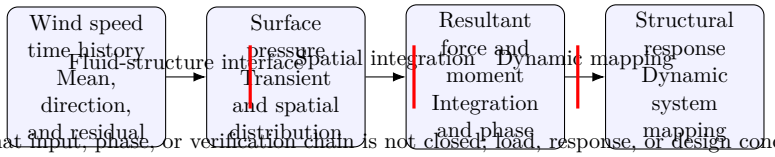


Figure 42: Information interface from wind speed to pressure, resultant force, and response. Each break point requires independent evidence closure; wind speed packet passage does not constitute structural use approval. Original project method figure.

Table 16: Source conflict and provenance risk retention list

Source/equation	Conflict content	Treatment	Risk level
Moeini Eq. 43 vs Dang Eq. 14	Parameter definition overlap in effective domain	Source labels retained separately	HIGH
Dang Coles override path	Local function allows overriding fixed constants	Canonical runner not exposed	HIGH
Roncallo-Solari A vs B	Methods A/B differ in terrain correction	Retained and compared separately	MEDIUM
EPSD non-stationarity assumption	Conflicts with SRM stationarity assumption	Layered treatment	MEDIUM

Table 17: Windfield to downstream agent interface input completeness status

Downstream agent	Required input	Current status	Missing items	Transfer condition
EVA	Standardized wind speed endpoint	Not established	Event packet	Researcher approval
WindTunnel	Target time history or intensity	Design layer ready	Event calibration	Researcher approval
CFD	Inlet boundary field	Not established	Real event packet	Researcher approval
Structural	Floor force/pressure time history	Engineering framework ready	CFD output	Researcher approval

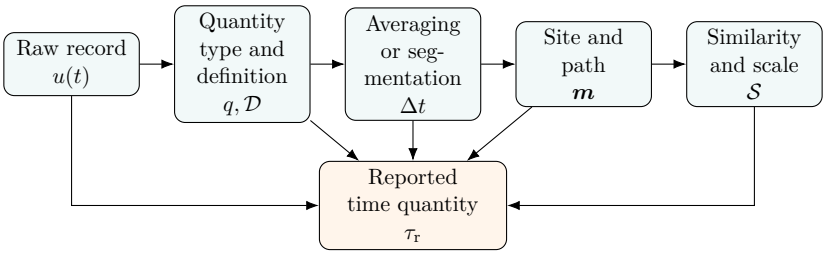


Figure 43: Conditional chain for reported time quantities. Raw record, quantity type, averaging algorithm, station path, and scale mapping jointly delimit reported values; unified units cannot substitute for condition verification. Original project method figure.

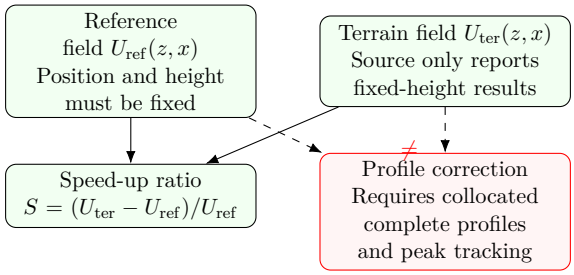


Figure 44: Distinction between terrain fixed-height speed-up ratio and profile correction. Without paired complete profiles, the speed-up ratio cannot substitute for peak height or gradient correction. Original project method figure.

Table 18: Windfield main symbols and unit definitions

Symbol	Definition	Units	Applicable domain
$U(z,t)$	Instantaneous wind speed	m/s	Height z, time t
$\bar{U}(z)$	Time-averaged wind speed	m/s	Within averaging window
$\tilde{U}(z,t)$	Slowly varying non-stationary component	m/s	After filtering
$u'(z,t)$	Zero-mean turbulence component	m/s	After detrending
$U_r(r,z)$	Radial velocity (Dang closure)	m/s	Within axisymmetric assumption
$r_{\max}$	Radius of maximum velocity	m	Within parameter domain
C_F	Force coefficient	Dimensionless	Cross-section and angle of attack dependent
ρ	Air density	$\mathrm{kg/m^3}$	Reference conditions

17.2.3 Current Scientific Limitations and Future Research

17.3 Limitations, Diagnostics, and Interfaces

17.3.1 Scale Transfer, Data Contract, and Comparability Gates

Windfield's primary task is not to apply a single model to all sources but to establish quantity comparability. A wind field quantity is defined by at minimum the physical variable, coordinate system, height datum, temporal support, spatial support, averaging or filtering algorithm, units, source identity, and uncertainty. Single-point time series from observation stations, lowest model level cumulative envelopes from CM1, gridded diagnostic winds from WRF, analytical radial models, and stochastic simulation fields all represent different objects even when all expressed in m/s. Only when definition endpoints are compatible do they enter the same comparison; otherwise the packet preserves parallel fields and the non-convertible reason. This gate can prevent spurious equivalence created by unit consistency.

Spatial scale transfer involves three distinct operations: resolution, interpolation, and supplementary modeling. Resolution computes footprints, maximum wind radii, gradients, or profiles on the native grid without adding information not present in the original data. Interpolation maps data at existing resolution to another coordinate or grid, with results still limited by original resolution and observation operator. Supplementary modeling employs spectral, coherence, or analytical closures to generate scales not resolved by the upstream, which must be tagged with the new model, parameter source, and random seed. Merging the three as "downscaling" hides whether information is rearranged or assumed to be generated. The Windfield packet thus must record the operation class for each array, allowing CFD, WindTunnel, or Structural to identify native and generated quantities.

Temporal scale transfer similarly cannot be automatically closed through resampling. WRF minute-level or CM1 sparse spatial snapshots cannot acquire high-frequency turbulence through interpolation to a denser time axis, and cumulative envelopes cannot recover instantaneous processes by inversion. If non-stationary decomposition is used, the averaging window and filter bandwidth determine the allocation between the slowly varying term and the stochastic term; if EPSD or SRM supplements high-frequency content, the spectral model must be compatible with event stage, spatial location, and measurement conditions. Generated time histories should jointly store the original slow-varying term, stochastic component, synthesized result, and frequency range, preventing downstream from interpreting stochastic supplements as direct upstream model output.

Equation (12)'s three-component expression has unique meaning only given a specified decomposition operator. For time histories containing fast ramps, fixed long-window averaging may

misassign ramp energy to turbulence, while overly short windows may remove low-frequency variations relevant to structural response. Reasonable diagnostics must compare multiple physically justified windows, check whether mean, variance, and spectra are stable across windows, and report boundary effects. If the time history length is insufficient to resolve the target low frequency, a complete PSD or coherence model should not be provided. The stationarity assumption is a method input, not a property automatically proven by a single fit.

Spatial coherence links the stochastic components of multiple points; its inputs include not only inter-point distance but also direction, frequency, mean wind direction, and local scale. When observational coverage is insufficient, empirical coherence parameters cannot be identified from single-point records; parameters migrated from boundary layer routine winds also cannot be assumed to apply to thunderstorm ramps by default. The canonical $3 \times 3 \times 4$ packet can verify array dimensions, coordinates, and the numerical interface of cross-PSD but lacks sufficient evidence for event coherence calibration. Before transfer to Structural, it must be stated whether coherence is source-measured, literature-adopted, model-assumed, or a sensitivity range.

The quasi-steady force Equation (13) maps wind speed input to force time history, but C_F may change with angle of attack, Reynolds number, unsteady separation, and acceleration. If the ramp time of $U_{\mathrm{ref}}(t)$ is comparable to vortex shedding or aerodynamic force adjustment time, using steady-state C_F would miss phase lag and overshoot. This interface is suitable as a reviewable baseline, not a substitute for WindTunnel or CFD. Windfield should transmit the height, area basis, force coefficient source, and applicable domain of U_{ref} together, enabling downstream to differentiate baseline estimation from dynamic aerodynamic evidence.

The risk description of the Dang radial closure illustrates that source contracts must enter code interfaces. In Equation (14), r_s, n, $r_{\max}$, and $U_{\max}$ jointly control peak position and radial decay; if local functions permit overriding source-fixed constants, output can remain numerically normal yet no longer represent the same source model. The canonical runner does not expose the override path, reducing the risk for the standard package, but direct calls to local functions can still change results. API hardening should make source-fixed parameters non-silently overridable, or record actual values and deviation reasons in output provenance. When source conflicts have not been adjudicated, Moeini and Dang routes should be run and reported separately rather than averaged into a compromise curve.

Roncallo-Solari Method A/B, EPSD, and SRM are retained in parallel to reflect different method objects. Reference heights, roughness, and directional conditions for terrain correction methods must be matched with the event packet; non-stationary spectral methods and stochastic simulations based on stationary processes have different assumptions. Method comparison should employ common input and common output metrics, record differences caused by method form, and avoid treating numerical proximity as physical verification. If two sources use the same symbol with different definitions, the packet should use namespaces or source tags and cannot rely on prose context to disambiguate.

Real event packet admission conditions are higher than the canonical validator. In addition to array shape, finite values, and units, event/run identity, mechanism status, coordinate datum, terrain, grid or station support, height, time window, sampling and filtering, original variable definition, QC, missing data, upstream claim limit, and permitted employ are also required. CM1 or WRF packets also must differentiate native, derived, and cumulative variables; observation packets need to preserve instrument and reporting period. When any critical field is missing, Windfield can

return a supplementation request or conduct an interface exercise that does not produce scientific conclusions, but cannot silently fill in default values and then generate a production packet.

The interface to EVA transmits only well-defined candidate wind speed endpoints and event-level features, without determining statistical sample membership. The interface to WindTunnel transmits target time histories or intensity descriptions, forming only design inputs that cannot be converted to facility instructions; target curves also require facility tracking and synchronized measurement. The interface to CFD requires complete boundary surfaces, time, coordinates, and scale supplementation notes, and upstream mechanism and intensity biases must be transmitted with the package. The interface to Structural can provide wind speed or quasi-steady force baselines, but real pressure and floor forces should still be provided by the corresponding aerodynamic route. The four directions have different employs, and a single validator PASS cannot jointly open all.

The Researcher's judgment centers on whether conversion is necessary, whether errors are acceptable, and whether downstream employ is matched. For the same source, a descriptive footprint serving literature comparison may qualify while a spatiotemporal field serving CFD inlet may not; a smooth ramp serving wind tunnel target design may qualify while a high-frequency time history serving real event reconstruction may not. Judgments should be based on use-specific input completeness and non-recoverable information. Windfield provides alternative models, sensitivities, and gaps, and the Researcher decides whether to continue supplementary evidence, limit use, or stop transfer.

17.3.2 Packet Observation Operators and Reproduction Boundaries

The engineering significance of the canonical packet lies in jointly freezing arrays and semantics. The fourteen arrays constitute a replayable object only when the common manifest records name, shape, dtype, units, coordinate axis order, time datum, missing-data convention, and generation version. JSON stores identity, methods, status, and limitations; NPZ stores numerical arrays; both must be bound by the same packet ID and hash. Validator PASS means these fields satisfy the current schema and arrays are readable; it does not check whether they represent a real event or whether parameters lie within the source model's valid domain. Integration smoke further indicates inter-module interfaces can transmit the same packet but still does not provide event calibration, observational consistency, or downstream qualification.

For each derived array, provenance should form a directed chain from upstream variables to output quantities. The chain records input file identity, variable selection, slicing range, coordinate transformation, interpolation kernel, filter, random seed, and output version. Resolution changes only the description form; interpolation changes sampling location; generative models add scales not present upstream; the three must employ different operation classes. If the final array mixes native slow-varying terms and stochastic supplements, the packet jointly stores both terms and the synthesis, and provides frequency band and units for each. Such decomposition permits downstream to recompute the synthesis process and prevents generated components from being miswritten as direct CM1, WRF, or observational output.

The coordinate contract differentiates at minimum horizontal projection coordinates, geographic coordinates, above-terrain height, and above-reference-plane height. Terrain-following levels of meteorological models, sensor heights of observation stations, wind tunnel model coordinates, and CFD engineering domains cannot be aligned solely by axis names x, y, z. Conversion requires fixed origin, axis direction, rotation angle, length ratio, terrain datum, and interpolation

support, with control points preserved before and after conversion. If the target domain exceeds the effective envelope of the source data, the extrapolated region should be left blank or explicitly marked; it cannot be filled with nearest-neighbor values and then claim source resolution. Boundary face normal direction and variable staggering positions must also be transmitted with the CFD candidate package.

A footprint is a derived geometry for a given quantity, time window, and threshold, not an event entity. Footprints from cumulative envelopes aggregate peak values from different times, suitable for describing the affected area within a window; footprints from instantaneous wind fields correspond to a single time and can be used for trajectory and propagation analysis. Even if the two have similar areas, they cannot be interchanged. If the threshold comes from model-internal screening, it is valid only within the corresponding source and protocol and cannot be converted to observational event boundaries. Maximum wind radius, center trajectory, and duration are also affected by center definition, smoothing scale, and domain boundary; the packet should preserve algorithm and sensitivity rather than only the final geometric quantity.

Vertical profile comparison first requires deciding on normalization endpoints. Normalizing by fixed reference height is appropriate for comparing a given engineering level; normalizing by local peak height is suitable for comparing morphology; normalizing by surface roughness or boundary layer scale introduces additional models. If the near-surface layer of observations or models has missing data, both peak height and near-wall gradient may be jointly unstable. Profile interpolation can map different height samples to a common coordinate but cannot add near-surface information. Windfield should report the common height range, interpolated layers, excluded extrapolation segments, and peak search boundaries, avoiding manufactured apparent consistency through normalization choices.

Non-stationary decomposition reproduction requires storing the filter kernel and endpoint treatment. Convolution, local regression, empirical mode, or segmented averaging can produce different slowly varying terms; recording the shorter-end endpoint padding method significantly influences the ramp onset and offset vicinity. Residual mean being close to zero is only one check of the algorithm result and does not prove that the residual is stationary or has the target spectrum. Diagnostics should also check variance, spectral peaks, time-local energy, and extreme value stability under window variations. If multiple reasonable windows yield different decompositions, the differences are retained as model-form sensitivity rather than taking the single smoothest curve as the true mean.

Spectral and coherence generation requires numerical conditions such as positive-definiteness and conjugate symmetry, as well as physical applicability domain. The cross-PSD matrix at each frequency should be consistent with point sequence and component definitions; if numerical correction changes matrix eigenvalues to ensure positive-definiteness, the correction magnitude must be recorded. Stochastic realizations should save the seed and generate multiple realizations to differentiate single-draw fluctuations from target spectra. Passing spectral matching cannot prove that ramp, extreme values, and spatial structure are jointly matched; time-domain envelope and frequency-domain energy must be verified separately. When event data are insufficient for coherence calibration, parameters can only serve as source model values or sensitivity ranges and cannot be termed event identification results.

The target time-history transfer to WindTunnel closes only at the design level. Windfield can output a target curve with endpoints, ramp shape, time axis, and tolerances, and state which

components derive from events, source models, or design choices. Whether the facility can track this curve, the actual Cobra wind speed deviation from target, and the synchronization of pressure and PIV belong to the WindTunnel measurement chain. Windfield does not produce actuator commands and does not claim facility input realization merely because the design file is readable. CFD boundary packages similarly need CFD-side completion of mass flux, boundary reflection, grid support, and propagation checks. Interface responsibility separation prevents a single candidate time history from gaining unreviewed common qualification across facility and numerical routes.

Transfer to EVA is limited more strictly to candidate endpoints. Cumulative sws, model U10, three-second observed maximum within a five-minute reporting period at a station, analytical model peak, and synthetic time history peak are different statistical objects. Windfield can establish a quantity crosswalk recording whether a traceable conversion model exists, the parameters required for conversion, and error sources; it cannot provide a unified `target_3s_gust` value when no conversion exists. Event identity and independence continue to be managed by Events and EVA. Even if the wind field packet is engineering-complete, if event membership and observational process are not closed, it cannot enter the EVA sample.

Source conflict disposition should produce auditable decision records. Records include the conflicting equations, symbol definitions, valid domains, code paths, current choice, choice rationale, and affected outputs. If adjudication is deferred, the two routes run separately on the same input, outputs employ different source tags, and differences are reported without averaging. API hardening for the Dang override path can be implemented by prohibiting silent overrides or forcing actual constants to be written; testing should cover default values, explicit deviations, and erroneous parameters. Code tests passing proves only contract execution, not which source model better fits real events.

Downstream admission should generate receipts separately by employ. The EVA receipt checks wind speed endpoints and event fields; the WindTunnel receipt checks target time history and facility items pending confirmation; the CFD receipt checks boundary field and coordinate flux; the Structural receipt checks pressure or force source and phase. One receipt's PASS should not modify other interface statuses. Receipts store check version, input hash, item-by-item results, human decisions, and claim ceiling, enabling subsequent updates to identify whether data change, schema change, or Researcher decision changed transfer qualification.

All current Windfield products are model output, interface verification, and candidate features, not validated wind fields. Three limitations determine next steps. First, no real Events/CM1 event packet exists; all packet exercises are synthetic or canonical, lacking source-bound inputs with complete event/run identity, mechanism confidence, coordinates, resolution, height, time window, terrain, units, and uncertainty. Second, no event-calibrated reduced downburst PSD exists; the connection between spectral models and real events has not been established. Third, source conflicts (Moeini Eq. 43, Dang Eq. 14, etc.) and Dang override risk API hardening are not complete; review has found that some public Dang local functions permit overriding source-fixed constants, and overriding the Coles constant within the valid domain can produce differences of up to 8.3655 m/s. Currently, production CFD inlet, wind tunnel actuator command, structural design load, or DesignLoad recommendation cannot be generated; any extreme value or load conclusion automatically derived from the packet is explicitly prohibited. The minimum next step is to receive a source-bound real event packet, complete source/equation conflict disposition and event-level calibration, and then discuss the possibility of issuing a candidate packet to EVA, CFD, or

Structural.

The first rehearsal of a real packet should select a case with relatively complete evidence fields and limited employ, first verifying whether source identity, coordinates, units, time, and missing data can enter the schema without loss, then running the minimum descriptive feature extraction. The rehearsal does not supplement high-frequency stochastic fields, does not produce CFD production inlet, does not estimate EVA endpoints, and does not generate facility instructions. Output includes admission receipt, variable crosswalk, operation-class list, validator record, and Researcher decision request. If critical quantities cannot be aligned, the terminal state remains unadmitted with an explanation of non-recoverable information. This minimum closed loop can verify whether the engineering framework genuinely accommodates source-bound data while not crossing scale transfer and scientific approval boundaries.

Publishing a packet also requires a version compatibility strategy. When the schema adds required fields, old packets cannot be automatically upgraded because the parser supplies defaults; a migration report should first be generated listing fields that can be losslessly moved, need recomputation, and cannot be recovered. If numerical arrays remain unchanged, metadata updates still need a new hash and parent version link. This prevents validator version upgrades from misrecording historical engineering artifacts as new scientific evidence and also keeps the quality meanings of 70/70 unit tests and real source admission distinct.

Evidence and equation boundaries of the Windfield Agent Chapter 17 preserves only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of the Windfield Agent. Content not presenting physical equations belongs to the research methods, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 17 remains a source-limited synthesis and cannot support verified design wind, design load, code recommendation, or PI scientific conclusion.

18 EVA Agent

18.1 Scientific Questions, Knowledge, and Researcher Decisions

18.1.1 Scientific Purpose

The EVA Agent executes Step 3 of the PETW pipeline and is responsible for converting event classification, observational or simulated output into well-defined engineering wind speed samples with uncertainty that can enter extreme-value analysis, while maintaining source and claim boundaries among mechanism separation, mixed wind climate, and design interfaces. The scientific objective is mechanism-aware statistical modeling and extreme-value analysis. Step 3 is not a single GEV or POT fit but rather converts Step 0 mechanism labels, confidence levels, and classification tiers, Step 1 detectability and simulation route evidence, Step 2 wind field features, resolution, and alternative model data packages, and Researcher-approved literature method cards into EVA-ready input tables, inclusion-exclusion rules, baseline-only EVA runner specifications, mechanism-aware sensitivity specifications, uncertainty propagation plans, and claim boundary descriptions. The output side faces DesignLoad and Manuscript, providing only candidate research interfaces and explicit non-transferable boundaries.

The research target is the standardized 3 s gust `target_3s_gust`. This quantity currently has only a definitional requirement and no PETW numerical value (evidence status: NOT_ESTABLISHED). The framework fixed six core distinctions at the planning stage: an observation station gust is not the true event maximum value; a classification label is not ground truth; a confirmed-only sample

is not the true population; a CM1 peak or surrogate feature is not the final `target_3s_gust`; a baseline long-term return level estimate is not a design endpoint conclusion; a sensitivity estimate is not a code recommendation. These distinctions dictate that every EVA output can only be named according to its evidence tier. All content remains not_scientifically_approved and does not form any design endpoint, design load, or code recommendation.

18.1.2 Knowledge Base

EVA's knowledge assets are bounded by controlled sources. The active literature namespace `eva_vhat3s` includes 154 files; the source-bounded candidate claims EVA-STEP3-C01 through EVA-STEP3-C08 are permitted only for research design and analysis preparation and are not permitted to serve as input basis for extreme-value analysis conclusions (evidence status: engineering preparation). The core synthesis asset is the 2026-08-12 source-bounded thunderstorm wind EVA literature review: Chinese version 21 pages, English counterpart 25 pages, both with 11 first-level sections, 47 subsections, 5 original method figures, 3 synthesis tables, and 41 unique citation keys; English QA confirms citation scope consistency with the Chinese, with no added or missing citation keys (evidence status: engineering preparation). Page differences arise from language typesetting and do not establish content inconsistency. The source recovery terminal state is 43 readable_full_source and 6 source_limited; 5 results-mode candidates were not accepted because of insufficient source bytes or checksum evidence; D5AF3D82 was excluded due to source traceability conditions. BESSIE and CM1 in the knowledge base have only conditional identity: BESSIE is an observational convective-outflow control, not a standardized `target_3s_gust` input; CM1 diagnostics do not equal engineering 10 m, 3 s gust.

18.1.3 Researcher Participation and Decisions

At this stage, the Researcher (Liang Hu / PI acts as the scientific decision maker. At the entry end, decisions are needed on: which mechanisms constitute the primary analysis population; how confirmed, probable, and possible labels enter EVA; whether mechanism-mixed modeling is acceptable; whether Bayesian hierarchical EVA is applicable or premature; how detection probability is defined; how `target_3s_gust` standardization is handled; and whether alternative model estimates of `target_3s_gust` can enter EVA and at which tier. At the exit end, decisions are needed on: which results are manuscript-ready versus exploratory; which results can be handed off to DesignLoad and which must not be transferred. As of 2026-08-12, the only item awaiting PI decision is whether to approve a small-scale, non-production EVA rehearsal that must not form design conclusions, once a qualifying real data package arrives. No approval of long-term return level extrapolation, design endpoint, or design-load elevation is currently required.

18.2 Methods, Results, and Equations

18.2.1 Main Methods

The method review follows the inference chain without pre-ranking any route.

18.2.2 Wind Speed Quantity Definition and Standardization

Peak duration, averaging algorithm, and instrument response determine the statistical identity of the recorded quantity[Kun and Ahsan, 2014, Lombardo and Zickar, 2019, Steinheuer and Friederichs, 2020]; measurement height, exposure, roughness, and directional sector delimit the standardization endpoint[Simiu et al., 2007, Giammanco et al., 2016]; spatial support differentiates station peaks, event-level network peaks, radar or dual-Doppler reconstructions, reanalysis gusts, and simulated winds[Mehner, 2020, Steinheuer and Friederichs, 2020, Krupar et al., 2016]. In

hurricane research, gust factors are limited by land or marine exposure, measurement height, and boundary layer assumptions and cannot serve as universal conversion quantities for thunderstorm winds[Vickery PI and Skerlj PI 2005]. Common units are not a sufficient condition for comparability; candidate comparable endpoints are formed only when temporal support, measurement chain, spatial support, and site endpoint are jointly fixed (Figure 45, Table 19).

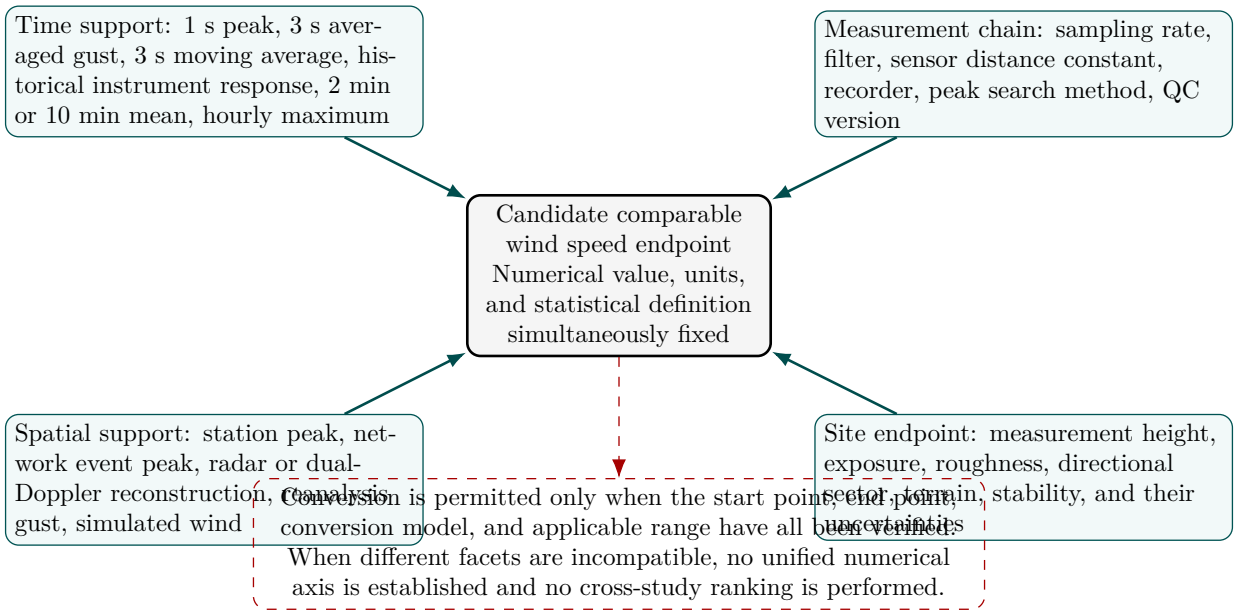


Figure 45: Wind speed definition and standardization comparison. Four non-numerical facets represent temporal support, measurement chain, spatial support, and site endpoint respectively; each facet does not constitute a unified numerical axis; conversion is possible only when the start point, end point, conversion model, and applicable range have all been verified. Drawn from verified literature synthesis; source-specific parameters are not directly transferred to PETW.

Table 19: Wind speed variables and comparability conditions (based on verified literature synthesis; source-specific parameters are not transferred to PETW)

Wind speed quantity	Peak window and averaging algorithm	Comparability condition	Prohibited direct conversion cases
1 s peak	Instantaneous or filtered peak	Requires verified instrument response and filtering	Not equal to 3 s average
3 s averaged gust	Non-overlapping block average	Differs from moving average	Cannot be interchanged with moving average
3 s moving avg max	WMO moving average	WMO recommended definition	Not equal to block average
Approx. 5 s historical response	Instrument equivalent filtering	Valid only within source instrument chain	Cannot be directly treated as 3 s
10 min mean	Fixed block	Meteorological standard reference quantity	Cannot be combined with gust
Hourly max gust	Maximum gust within 1 h	Gust window must be specified	Not equal to annual maximum
Reanalysis gust	Gridded diagnostic parameterization	Model level and grid conditions	Not equal to observed gust

18.2.3 Mechanism-aware Event Population and Observational Process

Mechanism classification can draw on thunder-lightning labels, meteorological variable rules, wind speed morphology rules, or automated classifiers[Twisdale and Vickery, 1992, Choi and Hidayat, 2002, Vallis et al., 2019, Chen and Lombardo, 2020]; classifier accuracy, precision, and recall describe label discrimination performance and are not physical measures of detection probability[Chen and Lombardo, 2020]. Missing data, QC peak removal, instrument replacement, and

station network density determine whether potential events can be recorded and retained[Cook, 2023b, Spassiani et al., 2023, Cechet and Sanabria, 2010]. Event independence requires declustering time, which has a different meaning from the classification evidence window[Palutikof et al., 1999, Brabson and Palutikof, 2000]; independence intervals in the literature, ranging from approximately 2 days, 4 days, 24 h, to 0.5 days to 4 h for thunderstorms, are valid only under source conditions[Cook, 1982, Lombardo and Ayyub, 2017, Choi and Tanurdjaja, 2002, De Gaetano et al., 2014]. Event rate, classification probability, detection probability, and conditional amplitude distribution are four statistical objects that must be kept separate; mixing them would cause systematic bias in tail estimates[Cook et al., 2003, Twisdale and Vickery, 1992, Cook, 2024b, Spassiani et al., 2023] (Figure 46).

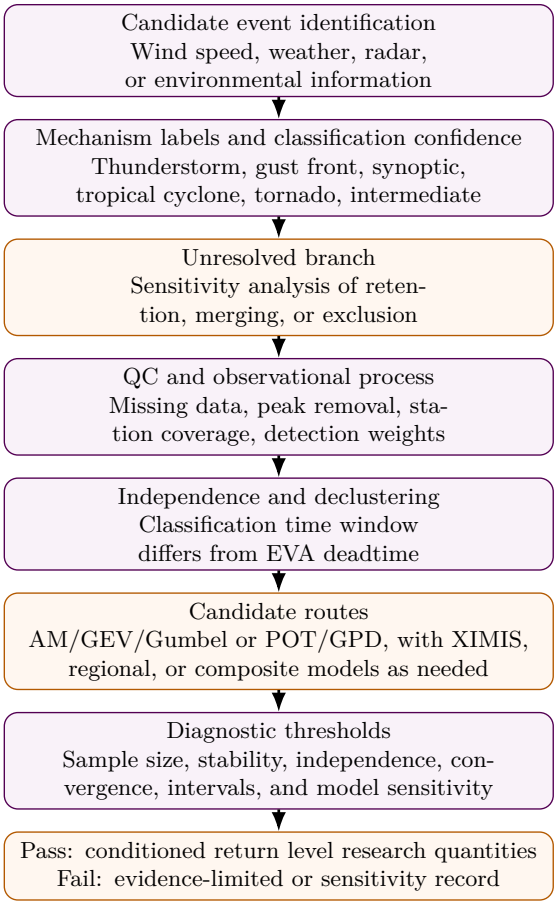
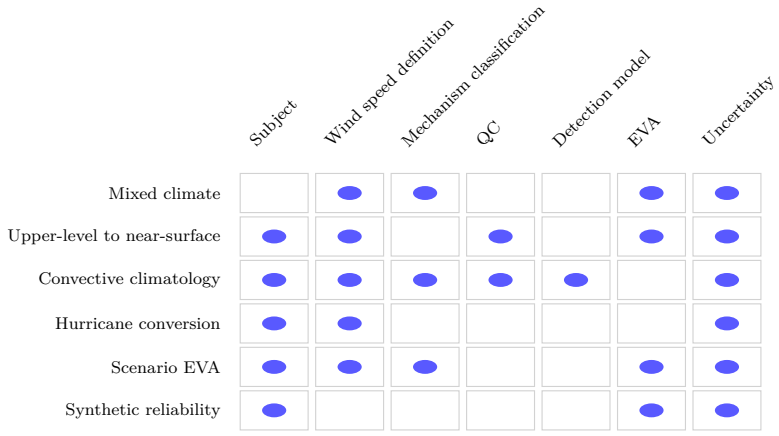


Figure 46: Method flowchart from event identification to EVA. Mechanism evidence and classification, observation and quality control, unresolved branches, independence and declustering, candidate routes, and diagnostic thresholds are connected sequentially. The classification evidence window differs from the declustering time; classification probability differs from detection probability. Drawn from verified literature synthesis; source-specific parameters are not directly transferred to PETW.

18.2.4 Candidate Extreme-value Routes

The annual maximum method with GEV or its special case Gumbel constructs samples from block maxima; block definition and shape parameter diagnostics determine its applicability[Palutikof et al., 1999]. POT/GPD employs threshold exceedance information and requires threshold stability and deadtime declustering diagnostics[Palutikof et al., 1999, Brabson and Palutikof, 2000, Mohr et al., 2017]. XIMIS starts from finite event rates and parent distributions to handle independent storms and multi-day maxima[Harris, 2009]. The Method of Independent Storms expands samples by defining storm boundaries through trough crossings[Cook, 1982]. Regional frequency analysis requires prior homogeneity and discordancy testing[Hong and Ye, 2014]. Directional analysis strat-

ifies by sector and reports per-sector sample sizes[Cook, 1982, Zhang and Chen, 2015]. Figure 47 shows the correspondence between method routes and evidence fields; the matrix only indicates whether a field is explicitly addressed and does not establish a research quality ranking.



• indicates that verified sources support this field; blank indicates not established by that source.
The matrix is used to identify interface gaps and does not constitute a research quality ranking or unified numerical scale.

Figure 47: Literature method and evidence field matrix. Rows represent method or evidence categories; columns represent research subject, definition endpoint, mechanism information, quality control, detection model, EVA route, and uncertainty coverage. Dots only indicate that the corresponding field is explicitly addressed and do not represent research quality ranking. Drawn from verified literature synthesis.

18.2.5 Conditional Synthesis for Mixed Wind Climates

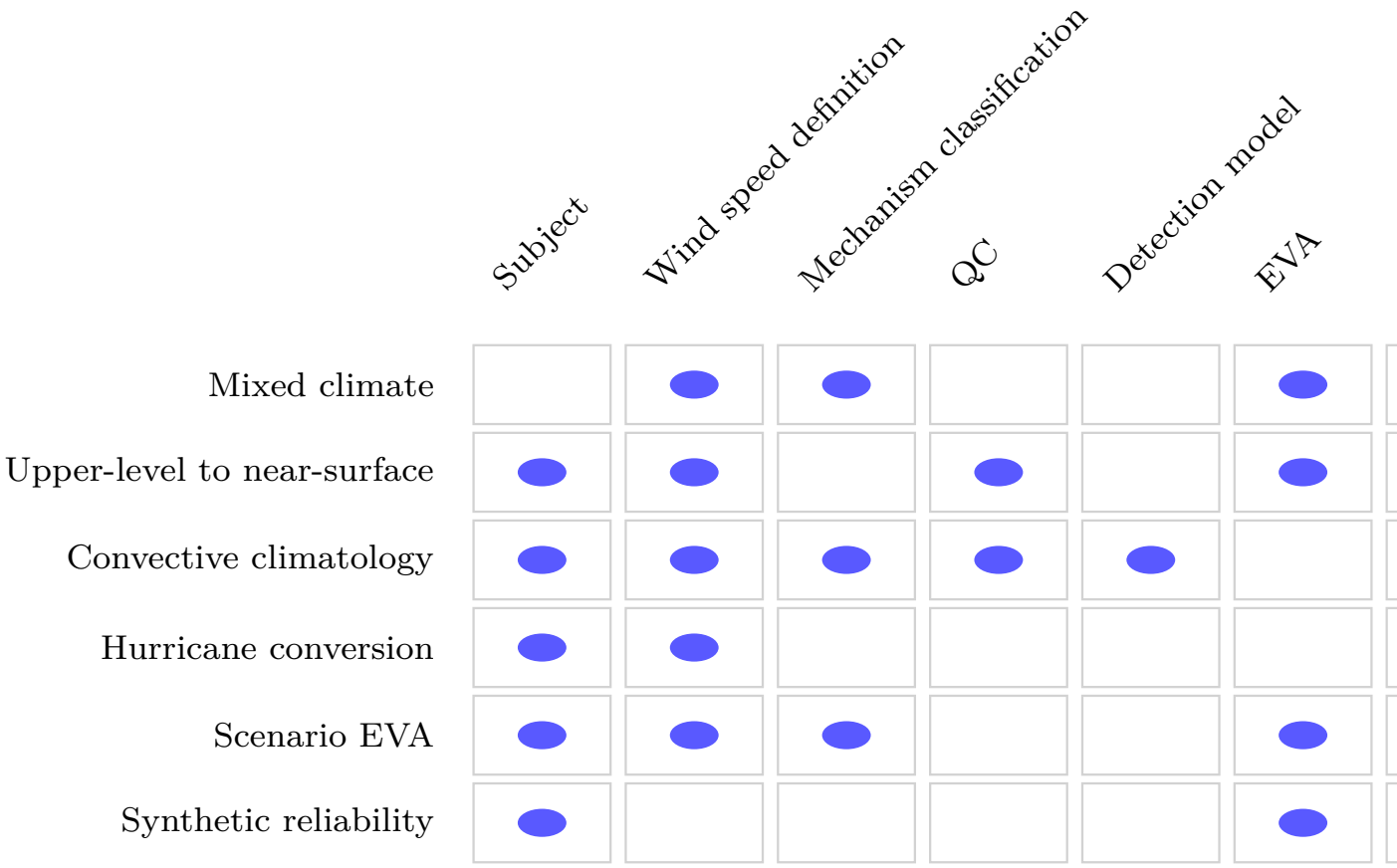
Conditional distributions precede unconditional mixed distributions: mixing is discussed only after category definition, occurrence rate, wind speed endpoint, and dependence assumptions are compatible[Gomes and Vickery, 1978, Cook, 2004]. Component tails, event occurrence rates, and composite long-term return level estimates must be modeled separately[Gomes and Vickery, 1978, Cook et al., 2003]; thunderstorm outflows, gust fronts, tropical cyclones, tornadoes, and background synoptic systems may have different event rates, durations, and conditional tails[Gomes and Vickery, 1978, Cook et al., 2003, Fan and Pang, 2019, Lawrence A. Twisdale et al., 2023]. Unresolved events should at minimum be retained under three sensitivity schemes: independent retention, conditional merging, and exclusion[Zhang et al., 2018, De Gaetano et al., 2014]. Figure 49 shows the structure and branches of conditional synthesis.

18.2.6 Six Categories of Uncertainty

The framework retains sampling, model form, measurement/conversion, membership, detection-process, and scenario uncertainties item by item (Table 20, Figure 50). Scenario uncertainty covers historical and future periods, global models, downscaling methods, and convective surrogate quantities; convective environment changes cannot be directly transcribed into future 3 s gust changes[Trapp et al., 2010, Wang et al., 2026, Wu et al., 2026]. Without joint dependence structure evidence, the six categories of intervals cannot be combined into a single total interval; when critical endpoints are unclear, diagnostics are unstable, or the model does not converge, long return period extrapolation is stopped[Cook, 2023c, Palutikof et al., 1999].

18.2.7 Equations

These equations are general forms of literature methods, used only for structural illustration and not for providing PETW numerical values. Let λ_m be the annual event rate for mechanism m (units yr^{-1}), whose value depends on event definition, observation period, and independence rules; let the conditional peak distribution given mechanism and observational conditions be denoted



• indicates that verified sources support this field; blank

The matrix is used to identify interface gaps and does n

Figure 48: Candidate extreme-value routes and evidence field coverage. Input requirements, applicable domains, and known limitations for GEV, POT/GPD, XIMIS, Method of Independent Storms, and regional frequency routes are separately annotated. Drawn from verified literature synthesis.

Table 20: Six categories of EVA uncertainty: action stages and retention rules (based on verified literature synthesis)

Uncertainty category	Action stage	Retention rule
sampling	Record length, event count, rare mechanism sample	Intervals increase with extrapolation distance; source numerical values cannot be transferred as PETW error bounds
model form	Threshold, block, distribution family, stationarity assumption	Route differences cannot be hidden within a single parameter standard error
measurement/conversion	Averaging time, instrument response, height, exposure conversion	Conversion errors cannot be assumed independent and identically distributed or mechanism-independent by default
membership	Classification error, intermediate, unresolved attribution	Not merged with detection-process; retained under sensitivity branches
detection-process	Station network density, reporting cadence, missing data, QC peak removal	Without an explicit detection model, only scenario analysis is performed; detection probability is not assigned
scenario	Time period, global model, downscaling, convective surrogate quantity	Not subsumed under model form; not transcribed into future gust changes

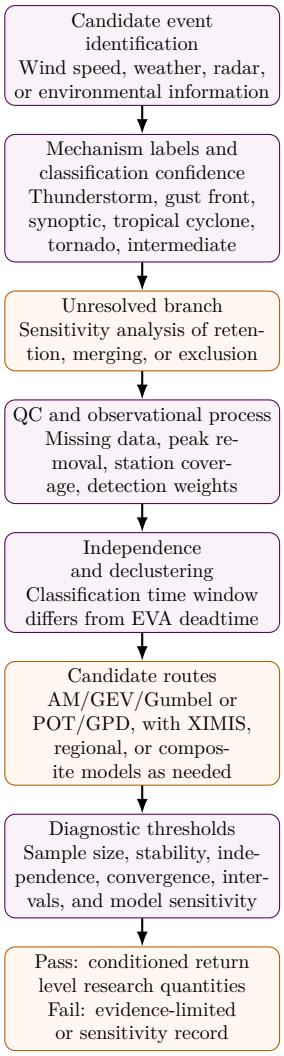


Figure 49: Conditional structure for mixed wind climates. Component definition, occurrence rate, conditional tail, and composite long-term return level estimates are modeled in separate layers. Unresolved events are retained under three sensitivity schemes. Drawn from verified literature synthesis.

$F_m(x)$ (dimensionless cumulative probability). Under the independent-mechanism assumption, the annual maximum distribution for a mixed wind climate can be written as the product of the component annual maximum distributions[Gomes and Vickery, 1978]:

$$F_X(x) = \prod_m F_m(x), \tag{15}$$

Here, the applicable conditions are that categories are separable, wind speed endpoints are compatible, and mechanisms are independent; directly fitting a single distribution to unseparated extreme-value samples may produce an apparent Fréchet characteristic, which is an artifact of mixed tails rather than an underlying physical process[Cook, 2004]. The generalized extreme value distribution (GEV) general form for the T-year return level estimate is

$$x_T = \mu + \frac{\sigma}{\xi}\left[\left(-\ln\left(1-\frac{1}{T}\right)\right)^{-\xi} - 1\right], \qquad \xi \neq 0, \tag{16}$$

Here, μ is the location parameter (m/s), $\sigma > 0$ is the scale parameter (m/s), ξ is the shape parameter (dimensionless), and T is the return period (yr). When $\xi \to 0$, the expression degenerates to the Gumbel form $x_T = \mu - \sigma \ln(-\ln(1-1/T))$. Applicable conditions are annual

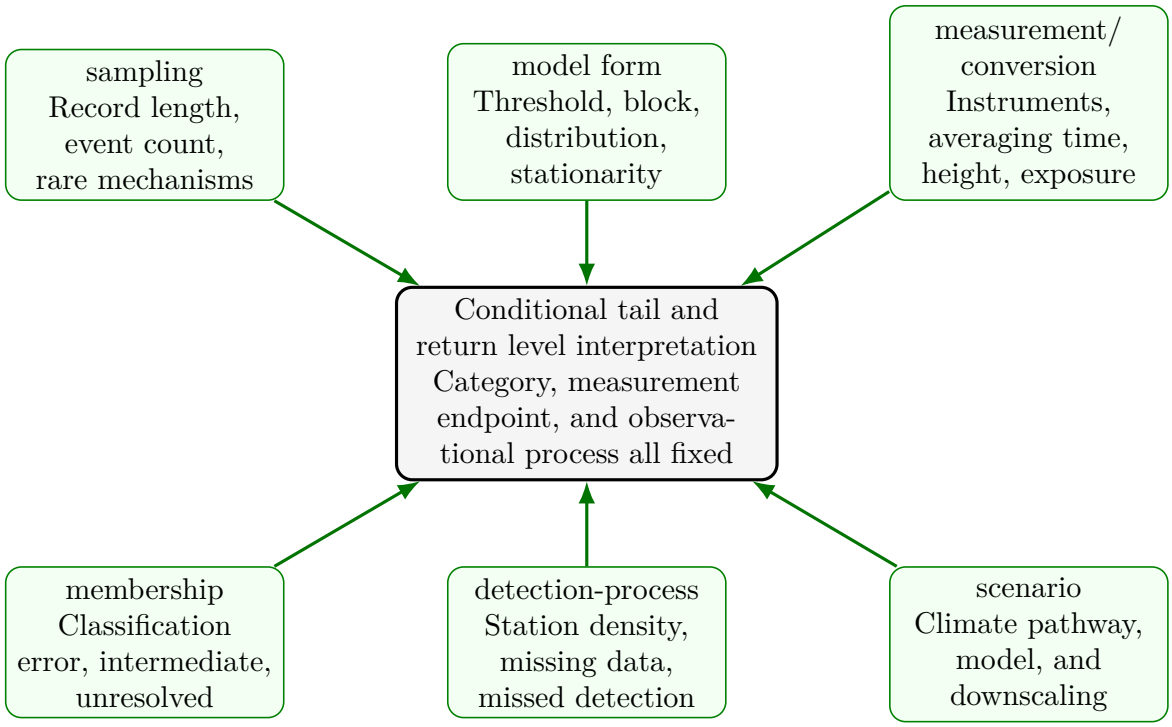


Figure 50: Propagation of six categories of uncertainty to conditional tail interpretation. Sampling, model form, measurement/conversion, membership, detection-process, and scenario are retained separately; membership is not merged with detection-process, and scenario is not subsumed under model form. Stopping condition: when critical components are unidentifiable, diagnostics are unstable, or the model does not converge, long return period values are not reported. Drawn from verified literature synthesis.

maximum or block maximum samples, well-defined block definition, and stable shape parameter diagnostics[Palutikof et al., 1999, Cook, 2023c]. The peaks-over-threshold (POT/GPD) route general form is

$$x_T = u + \frac{\sigma_u}{\xi}\left[(\lambda_u T)^{\xi} - 1\right], \qquad \xi \neq 0, \tag{17}$$

Here, u is the threshold (m/s), σ_u is the scale parameter (m/s), and λ_u is the annual exceedance rate (yr^{-1}). Applicable conditions are threshold stability, independence of threshold-exceeding events after deadtime declustering, and diagnosable exceedance rate[Brabson and Palutikof, 2000, Palutikof et al., 1999]. The above forms are used only within the meaning of the literature synthesis in this chapter (evidence status: literature-supported background).

18.2.8 Completed Work

As of 2026-08-12, the EVA Agent has completed the Step 3 concept and governance framework, covering wind speed quantity definition, mechanism classification, sample independence, threshold or block extreme-value routes, uncertainty propagation, mixed wind climate conditioning, and the interface boundary from EVA to DesignLoad. On the same date, the source-bounded thunderstorm wind EVA literature review, source recovery audit, and pilot-input ad-

mission checker were delivered. The admission receipt records assessed rows 0, admitted rows 0, production_eva_allowed=false. Therefore, no EVA fit was performed that day, no return level estimate was computed, and no design endpoint or design load was formed. This zero-input terminal state, stopped at the admission gate, is the correct state, not a failure or omission. Figure 51 shows the evidence chain and transfer boundaries: raw records pass through quantity definition, endpoint verification, mechanism classification, observational process, event independence, and conditional EVA to reach the candidate research interface; design endpoint, design load, Path C, and official guidance lie beyond additional approval boundaries. Figure 52 shows the admission gate determination structure and current terminal state.

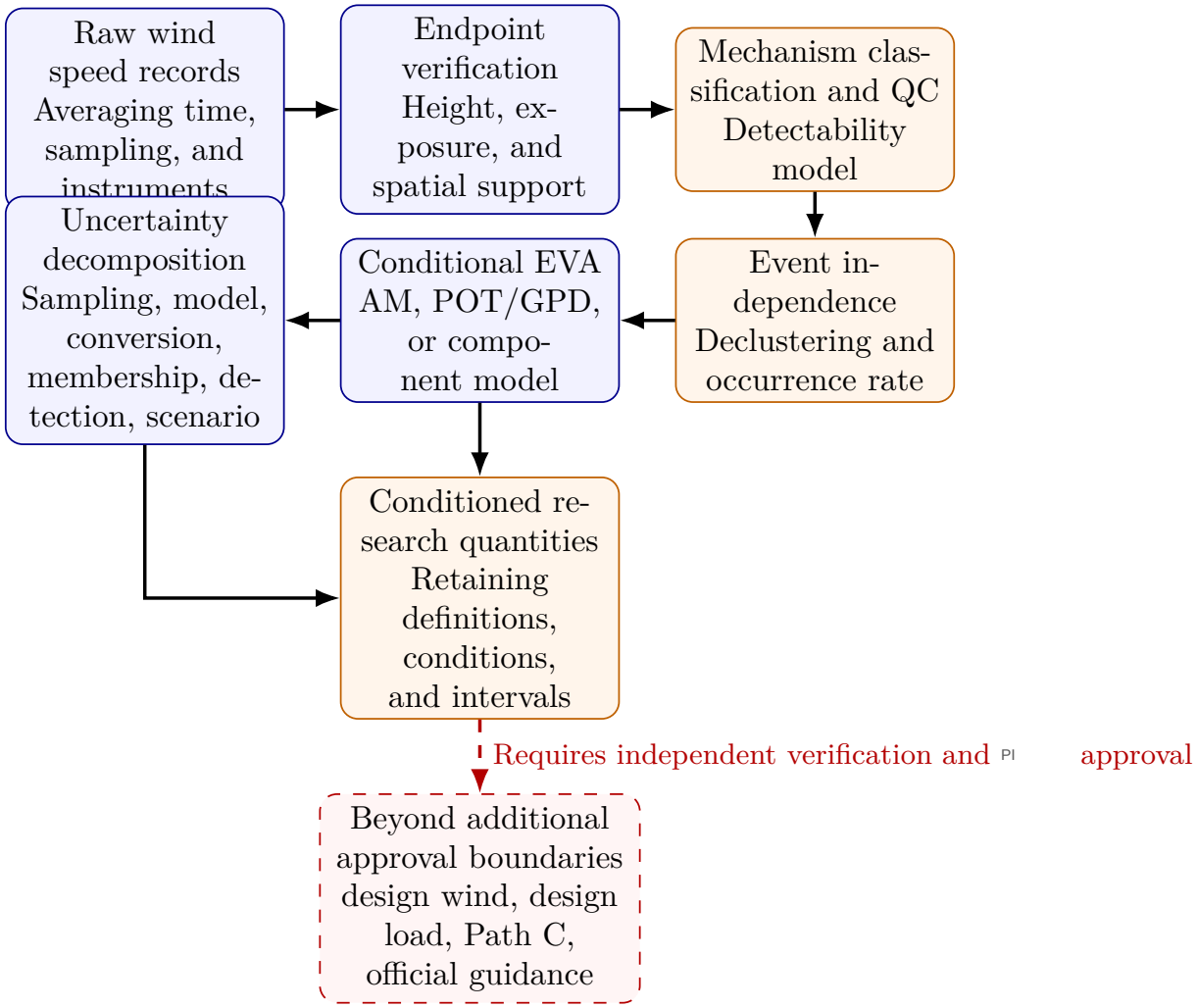


Figure 51: Evidence chain and PETW transfer boundaries. Raw records pass through quantity definition, endpoint verification, mechanism classification, observational process, event independence, and conditional EVA to reach the candidate research interface; design endpoint, design load, Path C, and official guidance lie beyond additional approval boundaries. Drawn from verified literature synthesis.

The safe output language for EVA is limited to baseline estimate, mechanism-conditioned estimate, classification-sensitivity estimate, detection-aware exploratory estimate, and candidate input to design-load synthesis; the employ of final design endpoint wind speed, verified target three-second gust quantity, ASCE-ready return level estimate, and code recommendation is prohibited.

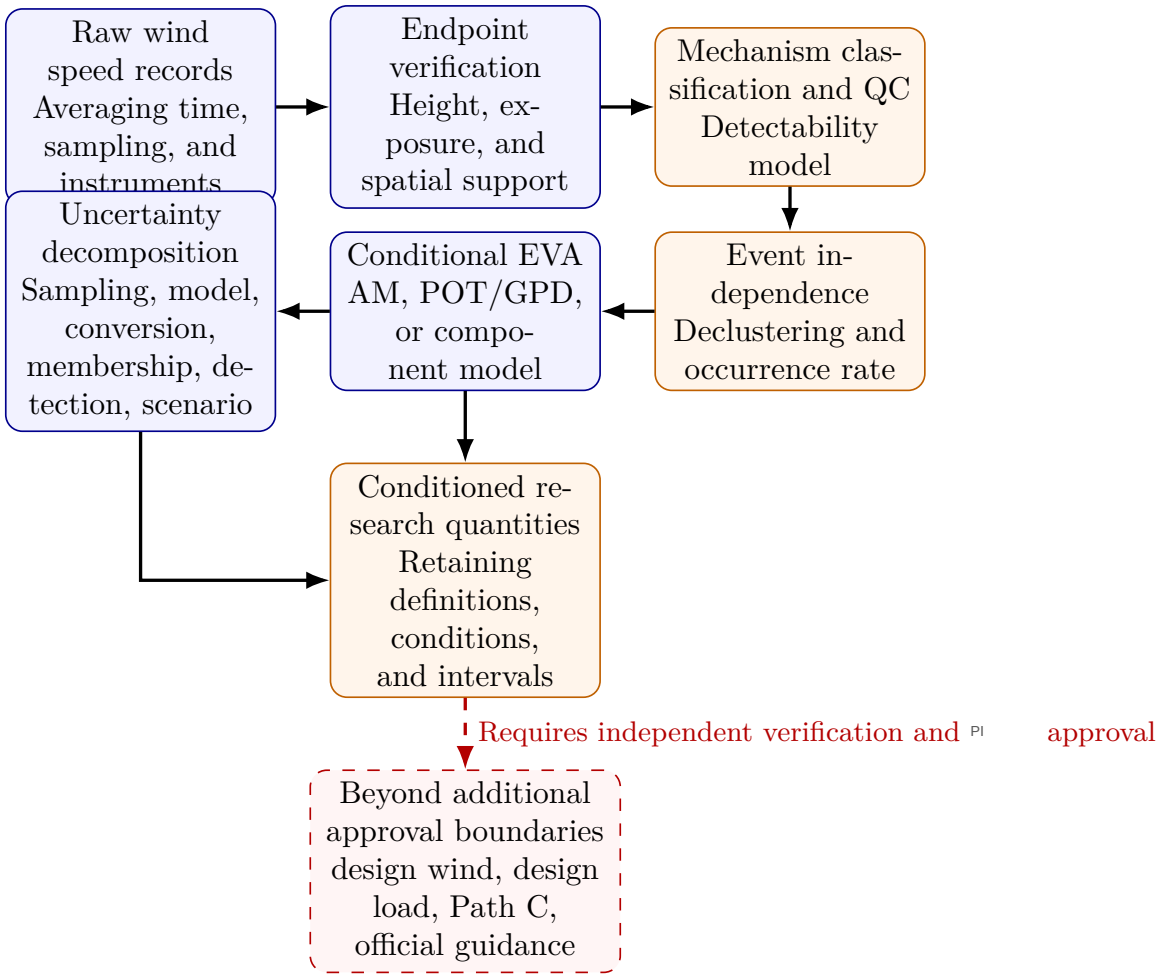


Figure 52: EVA admission gate determination structure and current terminal state. assessed rows = 0, admitted rows = 0, production_eva_allowed = false. The zero-input terminal state indicates that the admission gate correctly executed stop logic, not a system fault.

18.3 Evidence Audit, Interfaces, and Limitations

18.3.1 Current Scientific Limitations and Future Research

Current limitations are stated as facts. There is no formal PETW EVA sample, no return level estimate, and no design endpoint; the review supports only literature synthesis and cannot support PETW return level estimates, design endpoints, Path C, design loads, or official guidance (evidence status: NOT_ESTABLISHED). BESSIE remains an observational convective-outflow control, not a standardized `target_3s_gust` input; CM1 diagnostics do not equal engineering 10 m, 3 s gust. The source recovery terminal state preserves 6 source_limited entries; real Events, CM1, and Windfield data packages have not yet entered admission. The entry for future research is defined by the admission gate: only after receiving a real data package with stable event identity, mechanism confidence, native wind speed metadata, QC, `target_3s_gust` conversion, and uncertainty is the admission checker rerun; only after real rows pass admission and PI approves a bounded rehearsal may a small-scale EVA be attempted, and that rehearsal must not form design conclusions. Before that point, the correct action is to maintain the zero-input terminal state.

18.3.2 Admission Schema and Statistical Units

The minimum EVA input row is not a row from a source database, nor a model output grid peak, but a candidate statistical unit with stable event identity, quantity definition, and membership status. Each row must preserve event ID, parent storm ID, station or model-run identity, mecha-

nism label and its evidence tier, native wind speed quantity, measurement height, time averaging algorithm, reporting period, spatial support, quality control, exposure, missing data, event independence status, and provenance. Missing fields cannot be filled with default values. If the same parent storm influences multiple stations, station records can first be retained in parallel; event-level reduction rules must be separately registered to avoid counting station numbers as event numbers.

Observational process and event amplitude process must be kept separate. Whether a potential event can be captured by the station network depends on station density, instrument operating status, reporting cadence, quality control, and event footprint; the peak value recorded after capture is further affected by the sensor position relative to the core and instrument response. Without an identifiable detection model, EVA does not assign a numerical detection probability but instead establishes coverage scenarios and inclusion sensitivity. Classifier precision or recall also cannot replace detection probability because the former describes label discrimination of already-observed records while the latter describes the chance of potential events entering the recording system. If the two processes are merged, event rate and conditional tail shift jointly and are difficult to diagnose.

Rules for event independence must be frozen before tail fits are examined. Temporal declustering is only part of candidate rules; path continuity, spatial overlap, radar organization, and surface signals from the same parent storm also require joint consideration. Shorter deadtime may split a propagating system into multiple events; longer deadtime may merge independently independent convective cells. Time intervals from the literature can only serve as method candidates and cannot be directly adopted as PETW thresholds. The admission receipt should record the mapping between original source records and candidate event groupings, allowing different independence schemes to be replayed in sensitivity branches without modifying original records.

18.3.3 Endpoint Conversion and Model Route Diagnostics

The conversion contract for `target_3s_gust` must specify start-point quantity, end-point quantity, algorithm, and applicable domain. The BESS station 34.9 m/s already carries the semantics of station, 10 m height, five-minute reporting period, and three-second observed maximum within that period, but the event-day exposure and maintenance determination remain open, and this value is not the event spatial maximum. Model cumulative sws, diagnostic U10, lowest model level wind, or synthetic time history peak have no corresponding observation operator. Without a traceable conversion, EVA preserves the native quantity and marks endpoint not comparable rather than using an empirical gust factor to complete the target endpoint.

A uniform constant error after sample formation cannot represent endpoint conversion uncertainty. Conversion errors for different instruments, eras, exposures, and mechanisms may be heteroscedastic and may be correlated with wind speed intensity. If a qualifying conversion model is obtained in the future, it should be stored with parameter source, applicable range, covariates, and correlation structure, and propagated through repeated sampling or hierarchical representation into the conditional tail. Station representativeness and event spatial maximum differences beyond the conversion model remain independent uncertainties and do not disappear because the quantity value has been standardized.

The GEV route first checks block definition, completeness of each block, trends, and shape parameter stability. Only with long-term records and consistent endpoints do annual maxima have a common statistical identity. The POT route must jointly check the number of above-

threshold samples, parameter stability across thresholds, exceedance intervals, and declustering results. Thresholds cannot be selected solely to obtain more samples or smoother diagnostics. XIMIS, independent storm method, and regional frequency analysis correspond to different data organization and assumptions, and should not be simultaneously fit in the same rehearsal with the most favorable result then selected. Route selection is determined by data structure and pre-registered diagnostics.

For component composition in a mixed wind climate, the event rate and conditional amplitude distribution of each mechanism must first be established. If mechanism labels are probable or unresolved, weight propagation, independent retention, and boundary merging can be adopted as separate sensitivity branches, but any branch must maintain transparency of total record mapping. The confirmed-only branch typically has higher label purity but may miss events that are difficult to detect or have incomplete evidence; the relaxed branch increases coverage but also introduces membership uncertainty. The difference between the two should be reported as classification sensitivity, and neither route should be called the true population.

Regional pooling requires that endpoints, exposures, record periods, and wind climates across stations be comparable. Passing homogeneity tests does not prove that all stations share identical tails; it only indicates the specific test did not find differences sufficient to reject pooling. Directional analysis further reduces per-sector sample sizes and introduces wind direction measurement and sector boundary sensitivity. Without sufficient denominators, directional models may be less stable than non-directional models. EVA should jointly report effective sample counts for each mechanism, station, direction, and time period, preventing total sample size from masking local stratification sparsity.

18.3.4 Uncertainty Ledger and Stopping Rules

Sampling uncertainty is driven by finite event count and extrapolation distance; model-form uncertainty comes from distribution family, threshold, block, stationarity, and covariate choices. Measurement/conversion uncertainty belongs to the quantity chain; membership uncertainty belongs to event classification; detection-process uncertainty belongs to the observational entry mechanism; scenario uncertainty belongs to historical or future conditions. Six ledgers separately store inputs, assumptions, diagnostics, and sensitivity results. Without a joint dependence model, a seemingly complete total interval is not produced by sum of squares or simple addition. Separate reporting permits the Researcher to identify which gap dominates the current unavailable status.

Stopping rules precede model output. When endpoints are incompatible, event identity is unstable, independence cannot be defined, effective sample size is too small, threshold diagnostics are unstable, shape parameters change dramatically with reasonable choices, optimization does not converge, or the posterior is dominated by the prior, long-term extrapolation stops. After stopping, data audit, descriptive statistics, and diagnostic plots can still be saved, but return level estimates must not be reported. The ability of a model to return numerical values does not mean the statistical problem is identifiable. For rare mechanisms especially, it is necessary to differentiate computational convergence from information convergence, the latter requiring that the data genuinely constrain the tail.

The objective of a bounded rehearsal is solely to verify admission, data structure, and diagnostic workflow. If a small number of real rows pass the gate in the future, the rehearsal should pre-freeze input snapshot, endpoints, independence branches, candidate routes, and stopping conditions; output should employ exploratory or baseline language. It does not form a final sample,

does not provide return levels, does not establish $\hat{V}_{3s}$, and does not produce design wind. All numerical values are limited to rehearsal input and method verification and must not be transferred to DesignLoad as design endpoints.

The handoff receipt to DesignLoad must record item by item the mechanism population, wind speed endpoint, record period, event rate definition, model route, diagnostics, six categories of uncertainty, extrapolation range, and Researcher decision. Currently these fields have no real EVA results to fill; receipt status remains NOT_ESTABLISHED. Manuscript can report the method framework, zero-input gate, and gaps, but cannot employ literature general equations to fill PETW numerical results. This interface separates the existence of statistical methods from the existence of project endpoints.

Following future data arrival, the primary task remains auditing, not fitting. Events packets need first to resolve the mapping from source rows to independent events; Windfield packets need first to resolve endpoints and observation operators; station records need first to resolve exposure, instruments, and missing data. Only when these fields form a freezable snapshot is there qualification to discuss model rehearsal. Any upstream update should trigger a new admission receipt rather than reusing old status. Through this versioned entry, sample formation, model diagnostics, and scientific decisions can be tracked separately.

Data snapshots should include exclusion logs as well. Each excluded or deferred candidate row records the triggering rule, missing fields, whether remediation is possible, reviewer, and version, preventing retention of only the final inclusion table while losing the selection process. If subsequent supplementary evidence changes a row's status, a new decision record should be added while retaining old status without overwriting history. The exclusion log can both test whether selection correlates with wind speed magnitude or mechanism label and provide traceable input for detection-process and membership sensitivity.

Model diagnostics should preserve failed results and unstable branches. Threshold plots, parameter trajectories, residuals, probability plots, optimization status, and repeated fit results collectively indicate whether a route is usable; saving only one parameter table is insufficient. If different initial values or reasonable thresholds cause tail parameters to change significantly, the result is marked as model-form unstable. If sample size is too small and the shape parameter interval mainly reflects boundary or prior, the result is marked as weakly identified. Both states trigger stopping extrapolation but can guide the additional record length, event count, or endpoint information needed.

Subsequent research can include non-stationarity and climate scenarios only when historical observational processes are comparable. Station relocation, instrument updates, reporting regime changes, and urban exposure evolution may produce time-correlated record differences; if trend models do not handle these changes, observational system changes are mixed into climate signals. Convective surrogate quantities from future models also cannot directly substitute for surface three-second gusts. Scenario branches need separate explanation of global model, downscaling, surrogate quantity, and endpoint conversion chain; when any link is not closed, only scenario method or intermediate quantities are reported, not future design wind changes.

Researcher review materials should divide data qualification, model qualification, and employ qualification into three panels. Data qualification states which candidate rows passed admission; model qualification states which routes passed pre-registered diagnostics; use qualification states which outputs are permitted to enter Manuscript or the candidate DesignLoad interface. One

panel passing cannot compensate for another panel failing. Currently, the three panels are respectively zero rows admitted, no fits executed, and design transfer not permitted; thus maintaining NOT_ESTABLISHED is an information-complete terminal state.

Record period completeness should be audited segment by segment per station and quantity endpoint. Continuous year labels cannot replace period-by-period availability because maintenance downtime, communication interruptions, QC flags, and instrument replacement change observable exposure time. The annual maximum route must confirm that each block's coverage is sufficient to form a maximum; the POT route must calculate exceedance rates using effective exposure time. If gaps concentrate in the convective season or specific periods, simple scaling by annual missing-data ratio ignores selectivity. The audit should output effective observation duration, seasonal coverage, and gap reasons rather than only total years.

Covariate models must differentiate physical covariates from observational covariates. Season, environmental state, or large-scale climate indices may describe changes in event occurrence and amplitude; station network density, instrument type, and reporting regime describe the observational process. If both covariate types are mixed into the same location or scale parameter, statistical trends may lack physical interpretation. When sample size is insufficient, adding covariates should not be used to compensate for fitting inadequacy. Candidate covariates need source justification, temporal support, missing-data treatment, and pre-registered comparison, and identifiability of tail parameters and residual changes should be checked after inclusion.

The Bayesian hierarchical route applies only when hierarchical structure and prior information are defensible. Stations, regions, mechanisms, and directions can form partial pooling levels, but inter-level exchangeability cannot be assumed solely because names are the same. When priors come from external literature, wind speed endpoints, mechanisms, and site conditions must be confirmed as compatible; with weak samples, posteriors may mainly reproduce priors. Reporting should juxtapose prior predictive, posterior diagnostic, and sensitivity results, and clearly state data contributions. The current zero-input state does not support establishing or selecting any PETW hierarchical model.

Model checks must cover different risks of interpolation fit and tail extrapolation. Probability plots and residuals can check within-sample fit; the far end of the return-level curve is mainly controlled by shape parameter and model form. Bootstrap or posterior intervals propagate uncertainty only within the chosen model and do not cover omitted mechanisms, detection bias, or endpoint error. If multiple reasonable models are close within the sample range but diverge significantly in the far end, the divergence should be treated as model-form uncertainty rather than selecting the model with the narrowest interval.

If probability weights for sample members become available in the future, their calibration target and dependence structure must be stated. Per-event probabilities cannot be assumed mutually independent by default because the same classifier, station, or parent storm produces correlated errors. Simply treating probabilities as replicate weights may change effective sample size and event rate. A more prudent sensitivity approach is replaying under pre-registered membership scenarios or using models that can differentiate label error from event process. Without calibration evidence, probable and possible are retained as discrete branches and not assigned precise probabilities.

The relationship between spatial maximum and station maximum requires an independent observational model. When an event passes through the station network, the highest recorded station value depends on core path, station spacing, exposure, and instruments, rather than being

an unbiased observation of the true event spatial maximum. Adding station numbers may increase the chance of capturing extreme cores but also changes comparability across historical periods. If EVA employs network event maxima, it should record the available station set and coverage geometry for each event, and perform sensitivity analysis on station network evolution. Without such a model, the network maximum remains a quantity within observational support.

Endpoint audits also cover units and rounding. When sources store values in knots, mph, or m/s, native values, conversion formulas, and display precision should be retained to avoid multiple conversions of derived values. Historical integer reporting produces discretization and tied peaks; inclusion status near thresholds may be affected by rounding. Sensitivity should be handled within native resolution range rather than constructing spurious precision from display decimals. This rule also applies to event time and location; the center or endpoint of a reporting bin cannot be treated as the precise peak time.

The Manuscript EVA results template should be fixed before running. The template separately reports data snapshot, endpoints, sample formation, model route, diagnostics, sensitivity, stopping items, and permitted claims; vacant fields remain NOT_ESTABLISHED. Figure and table titles must differentiate literature method schematic, rehearsal diagnostic, and project result, and cannot label general GEV or POT curves as PETW estimates. Such a pre-established template reduces selective presentation after results appear and also enables the zero-input terminal state to be completely recorded.

Should future admission pass only a small number of cases, the appropriate output is a data qualification report and model feasibility assessment, not a forced attempt at tail parameters. Descriptive distributions must also avoid implying population representativeness and list unadmitted records and selection mechanisms. Only when record period, event rate, endpoints, independence, and diagnostics jointly meet pre-registered standards can more distant statistical questions be discussed. This order protects Step 3 from being pushed beyond evidence by the availability of computational tools.

Evidence and equation boundaries of the EVA Agent Chapter 18 preserves only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of the EVA Agent. Content not presenting physical equations belongs to the research methods, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 18 remains a source-limited synthesis and cannot support verified design wind, design load, code recommendation, or PI scientific conclusion.

19 WindTunnel Agent

19.1 Scientific Questions, Knowledge, and Researcher Decisions

19.1.1 Scientific Purpose

The WindTunnel Agent forms the accelerating-flow wind tunnel testing and validation lane for Step 4a in the PETW pipeline. The scientific question is how surface pressure, aerodynamic forces, separation and reattachment, vortex shedding, time-cell structure, pressure/PIV phase relationships, and quasi-steady deviation change under controlled accelerating or decelerating inflow for rectangular cross-sections, and whether this evidence can form a bounded experimental basis for CFD verification and subsequent Structural handoff. The target comparison objects are three aspect-ratio models with $B/D = 1$, 3, and 5 from the 60 mm series, where B is the cross-section width (mm) and $D = 60$ mm is the common reference depth. Scientific endpoints include wake

instability onset delay, aerodynamic behavior approaching steady-state flow within approximately five to six shedding cycles, time-cell structure, pressure distribution deviation relative to quasi-steady baselines, and whether pressure or integral quantity overshoot exists.

The agent is not the UF facility owner and does not have control authority over the wind tunnel, FFM (fan frequency modulator), DAQ, Scanivalve, Cobra, PIV camera, laser, traverse, force balance, trigger, or safety interlock; facility execution can only be carried out by qualified UF personnel following facility procedures and requires explicit approval from PI and UF. All content remains not_scientifically_approved; the formal experiment status remains DESIGN_ONLY_NOT_ACTIVE.

A velocity ramp represents only a controlled accelerating-flow mechanism and does not equal a complete downburst, thunderstorm outflow, or PETW storm-to-structure field. Track 1 accelerating-flow evidence and Track 2 meteorological downscaling evidence maintain different evidence levels and must not be mixed. Upstream inputs are PI scientific objectives, controlled intensity or wind field descriptions formed by Windfield/EVA, UF facility capability, and source-governed literature evidence; downstream outputs face CFD, Structural, and Manuscript, among which only verified pressure/force histories can enter Structural response packets, and WindTunnel does not directly generate design loads. Figure 53 shows the definition boundaries and dimensionless parameter schematic for nine inflow families.

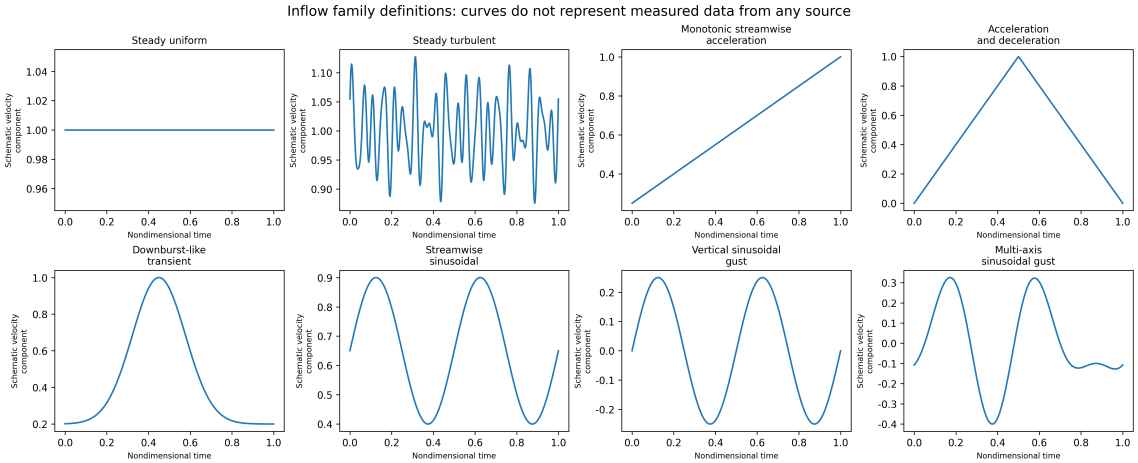


Figure 53: Nine inflow family definitions and dimensionless parameter schematic. Design figure, not experimental measurement. Different inflow families require independent definitions; schematic amplitudes or durations are not measured values.

19.1.2 Knowledge Base

Knowledge assets are organized in ten layers. RCC knowledge base entry, topic map, and method/interface knowledge cards constitute K01 through K06. WT-M1 accelerating-flow LitWiki (`wt_m1_accelerating_flow` namespace, 104 actual files) is K07. Graph/RAG mirror is K08; it is a lightweight JSONL mirror, not a vector database or approved scientific knowledge base, and all scientific answers preserve requires_peter_review=true. Zotero classified literature snapshot and citation-expanded readable literature map constitute K09 and K10.

At the evidence governance level, 55/55 source cards and WP1a/WP1b with engineering closure are complete: 52 non-quarantined sources have completed engineering freeze, 3 sources remain in quarantine; currently scientifically_admissible=0/55; engineering governance closure does not equal scientific approval of 55 sources (evidence status: engineering preparation). Principal review R07 is an all-themes data-driven review: Chinese 92 pages, English 109 pages; master sources 55,

unique citation keys 55, deep-extraction rows 1,642, data-driven figures 16, source figures 56; English QA passes 85/85 labels, 886/886 citation commands, 74/74 graphics. Four blockers identified by the independent scientific audit, including PIV source count, b018 inflow-family classification, b030 B/D coverage, and cross-facility overclaim, have all been remediated within source scope and re-verified; RESOLVED_VERIFIED indicates technical closure of the audit issue and does not indicate scientific approval. An additional 26-page bilingual Accelerated-Flow WindTunnel Design Review reviews model and PIV design figures. Figure 54 shows the coverage architecture of the 55 sources by aspect ratio, inflow family, and measurement channel.

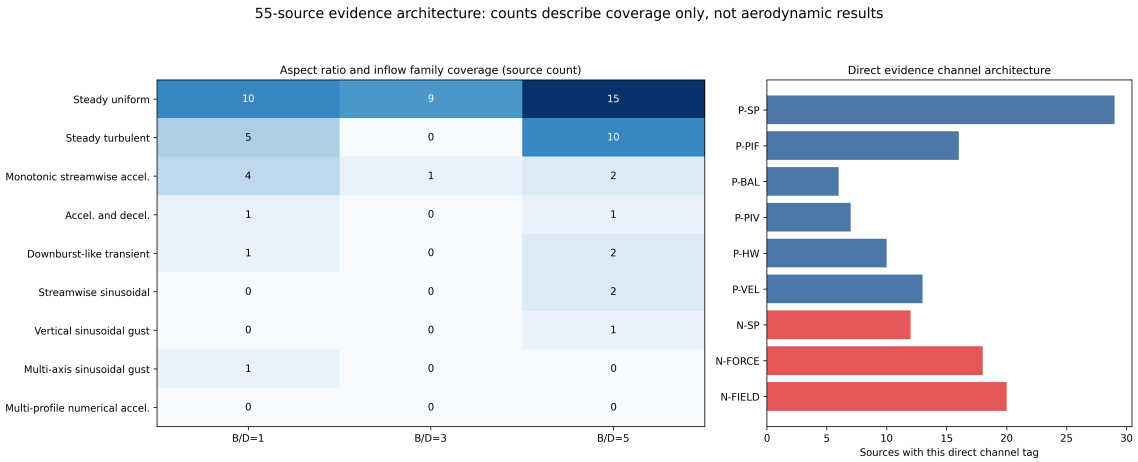


Figure 54: 55-source evidence architecture: aspect ratio, inflow family, and measurement channel coverage. Literature synthesis figure, not experimental measurement. Coverage is uneven across cross-sections, inflow families, and channels; counts do not constitute aerodynamic effect or source quality judgments.

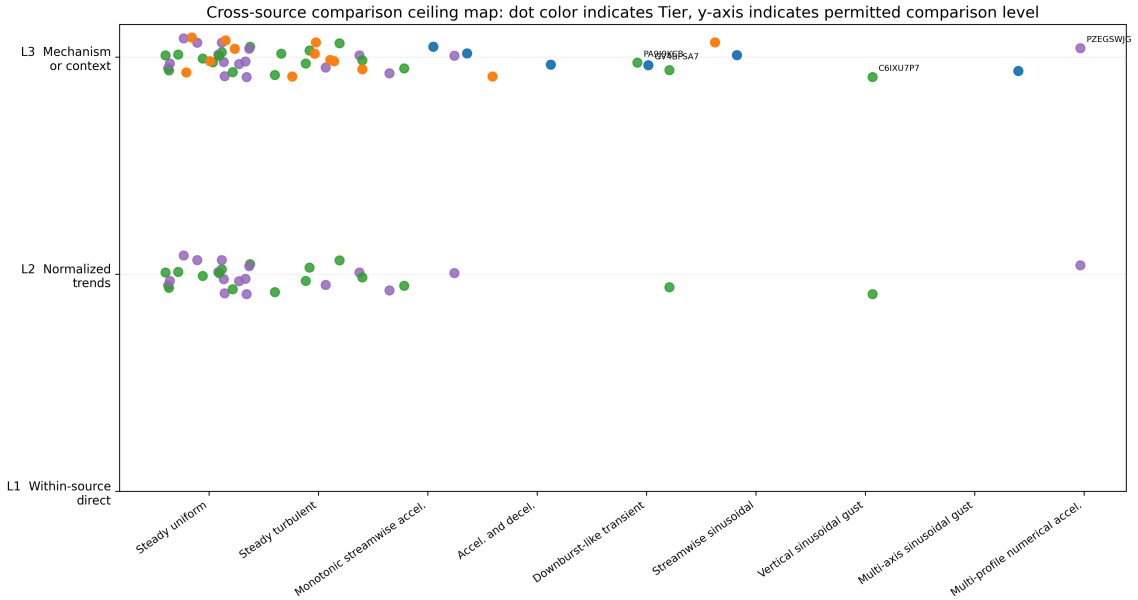


Figure 55: Comparison ceiling matrix for 55 sources. Comparison conditions, comparability, and non-comparability reasons are classified by source. Literature synthesis figure, not experimental measurement. Comparison ceilings do not constitute source quality ranking.

19.1.3 Researcher Participation and Decisions

The current design baseline reflects decisions by the Researcher (Liang Hu / PI On 2026-07-08, three cross-sections (60×60, 60×180, 60×300 mm, common reference depth $D = 60$ mm) and pressure 64-channel, 1.2 m tubing, PIV 300 Hz first-round requirements were confirmed, and transient pressure was confirmed to be reduced from full angle of attack to representative angles.

On 2026-07-28, six UF-compatible YM endpoints within the UF achievable range of approximately 2 to 15 m/s were confirmed, retaining source condition labels and respective a_p but not claiming preservation of source velocity ratio, source dimensional rise time, or source Reynolds number. Subsequently, local Gaussian smoothing parameters, Tier 1/2/3 stratification, pressure repeat convergence rules, PIV 3 to 5 repeats, and 400 Hz planned sampling rate were approved, and absolute time anchors solidified UP as PIV 6 to 18 s, common ramp start 8 s, target and pressure 0 to 20 s; for DOWN, strict inversion from the UP discrete array was required, with PIV window 21 to 33 s and descent start 22 s.

As of 2026-08-12, items awaiting PI decision include: whether to freeze v6.7 UP as the sole current candidate design baseline while maintaining DESIGN_MATERIAL_NOT_FACILITY_APPROVED; whether to continue keeping DOWN at DESIGN_ONLY_NOT_ACTIVE; whether to initiate claim-by-claim scientific-admissibility adjudication of the 55 sources; whether to issue a bounded RFI to UF and prepare a minimum pilot after receiving a written configuration response. Any formal experiment still requires UF facility procedures, safety approval, qualified operators, and specific operational approval, which are not within PI sole decision authority.

19.2 Experiment Design, Measurement, and Literature Evidence

19.2.1 Model and Tap Design

The three models are labeled M11 (60×60 mm, $B/D = 1$, 26 taps), M31 (60×180 mm, $B/D = 3$, 60 taps), and M51 (60×300 mm, $B/D = 5$, 62 taps), totaling 148 taps; models are installed per model-event and no single model exceeds the project requirement of 64 channels. Nominal span is 1200 mm; typical wall thickness is 0.250 in (6.35 mm); pressure tubing length is 1.2 m. M51, as the highest aspect ratio model, holds a central position in separation-reattachment mechanism research: $B/D = 5$ means that after inflow separation, there is sufficient surface length for reattachment to occur; how the reattachment position responds to Reynolds number and pressure gradient changes during the ramp phase is a core scientific question for the experiment design to answer. Table 21 shows the geometric parameters and tap allocation for the three models. Figure 56 shows the tap layout design schematic for M51.

Table 21: Three-model geometric parameters and tap allocation (design values; evidence status: engineering preparation)

Model	B (mm)	D (mm)	B/D	Tap count	Tap distribution characteristics
M11	60	60	1	26	Uniform distribution on all four faces
M31	180	60	3	60	Densified at leading edge and reattachment region
M51	300	60	5	62	Three-segment densification at leading edge, mid-section, and reattac

Tier 1 steady-state pressure baseline includes 69 model-angle combinations (M11 31, M31 13, M51 25), each at least 60 s with at least 3 repeats. Tier 2 transient pressure comprises 21 angle-of-attack values times 6 YM conditions; final repeat count is determined by phased convergence pilot. Tier 3 PIV core matrix consists of 13 values and 23 signed acquisition angles, selected by W_{TE} (trailing-edge wake window), W_{LE} (leading-edge separation window), and W_{W1+} (first reattachment mechanism window), with each dedicated window defaulting to 3 repeats and critical or unstable conditions up to 5. The steady-state layer also provides positive-negative angle pressure symmetry gate and instantaneous quasi-steady overshoot assessment.

19.2.2 Target Velocity Time Histories and YM Conditions

The local Gaussian smoothing model generates the six UP target time histories. Central acceleration and nominal rise time are defined as

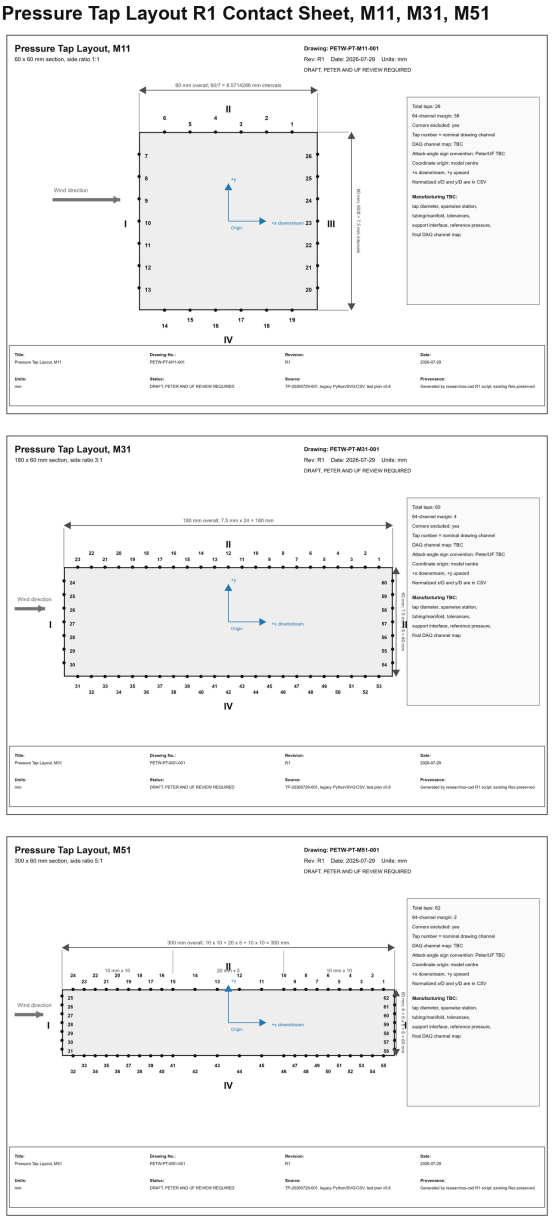


Figure 56: M51 ($B/D = 5$) model tap layout design schematic. 62 pressure taps are densified in three segments at leading edge, mid-section, and reattachment region. Design figure, not experimental measurement. Actual tap positions to be confirmed after model fabrication.

$$a_{\text{central}} = a_p \frac{U_t^2}{D}, \qquad T_{\text{linear}} = \frac{U_t - U_s}{a_{\text{central}}}, \tag{18}$$

Here, a_p is the dimensionless acceleration parameter (dimensionless), U_t is the target final speed (m/s), U_s is the starting speed (m/s), D is the reference depth 60 mm = 0.06 m, and T_{linear} is the nominal rise time without smoothing (s). Smoothing acts only on the slope transitions between plateau and ramp and between ramp and hold segments, preserving the central quasi-linear region and maintaining a_p traceability without producing command acceleration overshoot. Smoothing ratios are

$$\frac{\tau_s}{T_{\text{linear}}} = 0.20, \qquad \frac{\sigma_g}{\tau_s} = 0.35, \qquad T_{\text{rise}} = T_{\text{linear}} + 2\tau_s = 1.40\, T_{\text{linear}}, \tag{19}$$

where τ_s is the smoothing half-width (s), σ_g is the Gaussian kernel standard deviation (s), and T_{rise} is the actual rise time including smoothing (s). If aerodynamic force overshoot occurs, it cannot be simply attributed to target command spikes. Table 22 shows the current endpoint values for the six UP conditions. Figure 57 shows the target design $U(t)$ construction.

Table 22: v6.7 six UP target condition endpoints (native MATLAB target files; evidence status: generated input)

Condition	U_s (m/s)	U_t (m/s)	Pairing and notes
YM01	2.3	15.0	Shares $U_t = 15\,\text{m/s}$ and $a_p = 0.0012$ with YM03; compares initial speed
YM02	2.3	11.0	Shares $U_t = 11\,\text{m/s}$ with YM04; similar a_p
YM03	5.0	15.0	Paired with YM01
YM04	5.0	11.0	Paired with YM02
YM05	8.3	15.0	High initial speed inhibits zone expansion
YM06	11.0	15.0	High initial speed inhibits zone expansion

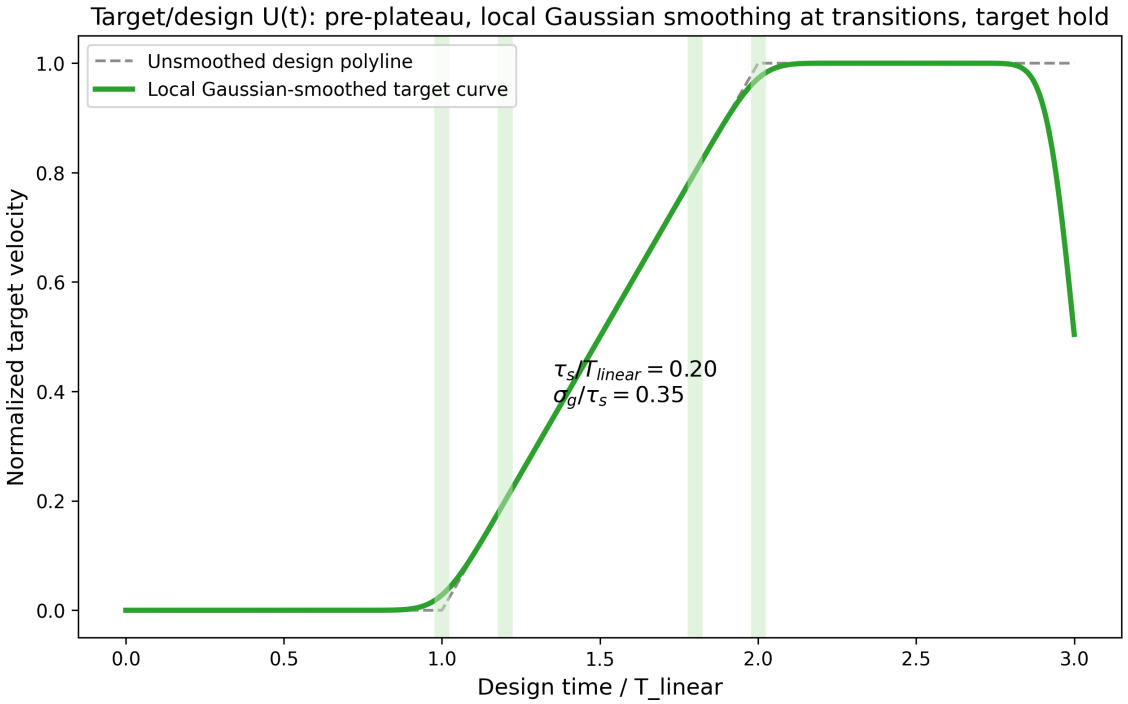


Figure 57: v6.7 UP target design velocity profiles. Design figure, not experimental measurement. Normalization conditions $\tau_s/T_{\text{linear}} = 0.20$, $\sigma_g/\tau_s = 0.35$; curves encode target design profiles and are not measured or verified facility inflow.

The UP main branch uses the following time axis: 0 to 6 s is the pressure continuous-acquisition pre-steady-state period where PIV does not save; 6 to 8 s is the PIV-saved 2 s U_s steady-state baseline; 8 s is the common ramp start for all six conditions; 8 to 18 s is ramp and U_t hold within the PIV window; 18 s PIV saving ends; 18 to 20 s pressure continues acquisition; 20 s target and pressure end.

The DOWN design branch spans 0 to 33 s with PIV window 21 to 33 s, common descent start at 22 s, where 21 to 22 s provides 1 s advance recording of U_t steady state. The strict symmetric construction of DOWN is

$$U_{\text{down}} = \text{flip}(U_{\text{up}}), \qquad a_{\text{down}} = -\text{flip}(a_{\text{up}}), \tag{20}$$

Here, flip denotes reversing the UP discrete sequence, and acceleration after reversal changes sign. Independent QA yields maximum absolute symmetry error of 0 for discrete velocity and

acceleration across all six conditions (evidence status: generated input). DOWN status is DESIGN_ONLY_NOT_ACTIVE. Native MATLAB packages are 6 UP files with 20,001 rows each, 6 DOWN files with 33,001 rows each, dt=0.001 s.

19.2.3 Instruments, Synchronization, and PIV

The planned PIV capacity boundary is

$$T_{\mathrm{PIV}} = \frac{N_{\mathrm{frames}}}{f_{\mathrm{PIV}}} = \frac{4800}{400} = 12 \text{ s}, \tag{21}$$

where $N_{\mathrm{frames}} = 4800$ is the planned frame count and $f_{\mathrm{PIV}} = 400$ Hz is the planned sampling rate. The distinction among frame, image-pair, and vector rate awaits UF confirmation. The nominal field of view is approximately 200×200 mm, with vertical position fixed at $y = 0$ to 200 mm and only horizontal traversal. Pressure, fixed Cobra, and PIV preserve their respective native sampling rates and are aligned via common hardware trigger and explicit clock relationship; different PIV windows come from different repeats and can only be termed local synchronization, event-conditioned, or phase-conditioned reconstruction, not full-domain instantaneous stitching. The fixed tunnel-global Cobra is the common $U(t)$ reference for all production runs; formal analysis must not employ the target curve in place of the Cobra measured curve. Figure 58 shows the PIV window plan and spatial relationship with models. Figure 59 shows measurement channel identity and non-merge rules.

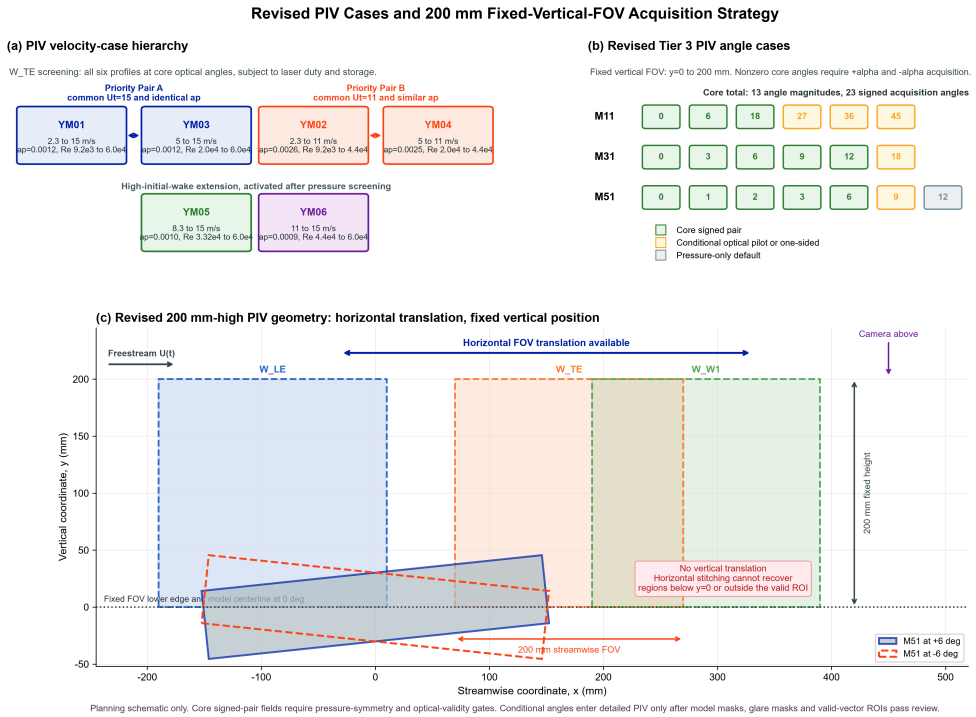


Figure 58: PIV window plan and spatial relationship with models. W_{TE} (trailing-edge wake), W_{LE} (leading-edge separation), and W_{W1+} (first reattachment mechanism) window types are defined by spatial position and scientific objective respectively. Design figure, not experimental measurement.

19.2.4 Separation-reattachment Mechanism and Time Cells

Figure 60 shows a schematic of separation-reattachment mechanisms for $B/D = 1$, 3, and 5 cross-sections. For $B/D = 1$, separation occurs at the four leading-edge corners, and the wake has no reattachment; for $B/D = 3$, post-separation flow may reattach at the rear section of the side face, forming a reattachment bubble; for $B/D = 5$, the reattachment position is far enough from the

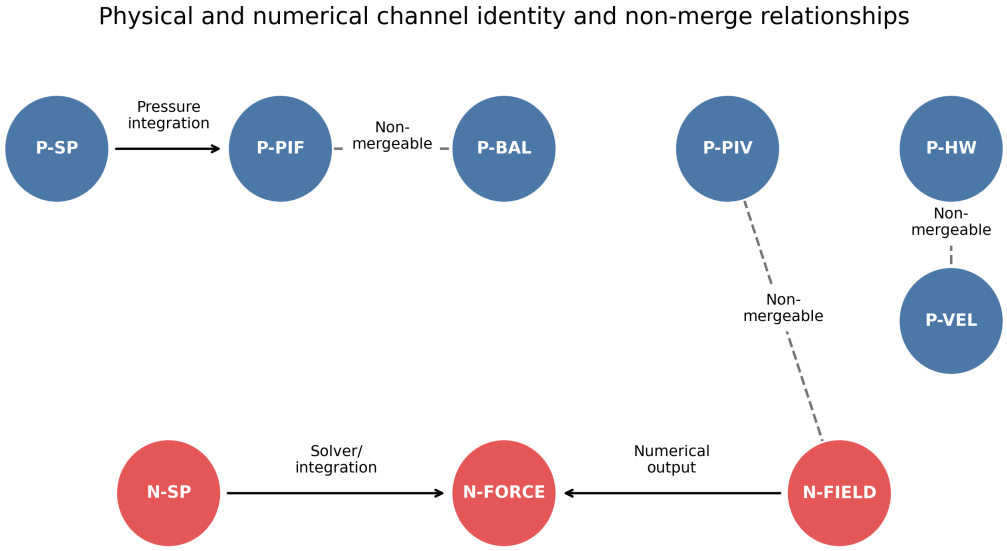


Figure 59: Measurement channel identity and non-equivalence relationships. Pressure-integrated forces, balance forces, various anemometry types, and numerical channels retain their respective identities; channel existence does not constitute comparability or evidence quality judgment. Design figure, not experimental measurement.

leading edge that pressure recovery in the reattachment region becomes the primary aerodynamic feature. Under accelerating inflow, the reattachment position may shift with changes in Reynolds number and pressure gradient during the ramp phase, and whether this shift produces observable pressure overshoot is a scientific question for the experiment design to answer.

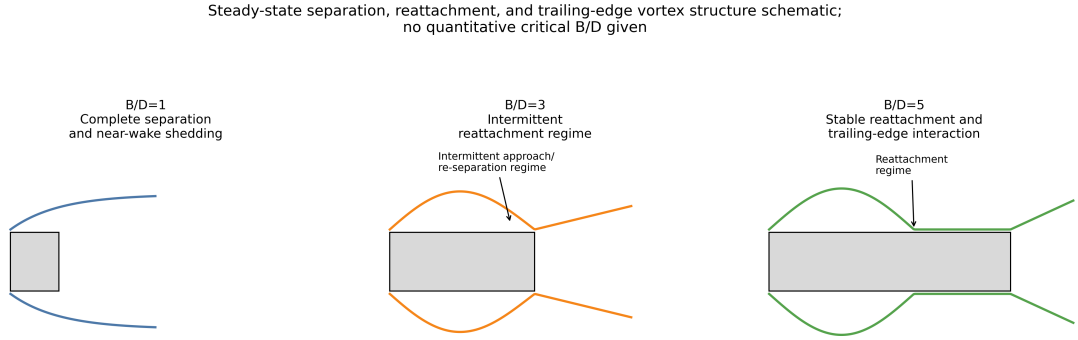


Figure 60: Separation-reattachment mechanism schematic for $B/D = 1$, 3, and 5. Design figure, not experimental measurement. Different aspect ratios require separate separation-reattachment narratives; the schematic does not provide quantitative critical ratios or Reynolds number thresholds.

The time cell is a type of discrete aerodynamic force jump structure reported in source literature that appears under ramp or pulsed inflow. Figure 61 shows a conceptual schematic of constant-frequency time cells: under conditions of approximately constant vortex shedding frequency, inflow acceleration causes the vortex circulation to increase within each shedding-frequency period, and when the intensity difference between new and old vortices exceeds a certain threshold, step-like jumps in force coefficients may occur. The existence of time cells, number of cycles, jump amplitude, and quantitative relationship with inflow acceleration are objects to be experimentally tested.

19.2.5 Literature Evidence

Asymmetry between accelerating-flow and decelerating-flow responses in source literature is a key reference for the experiment design. Figure 62 shows the maximum drag coefficient drop percentage

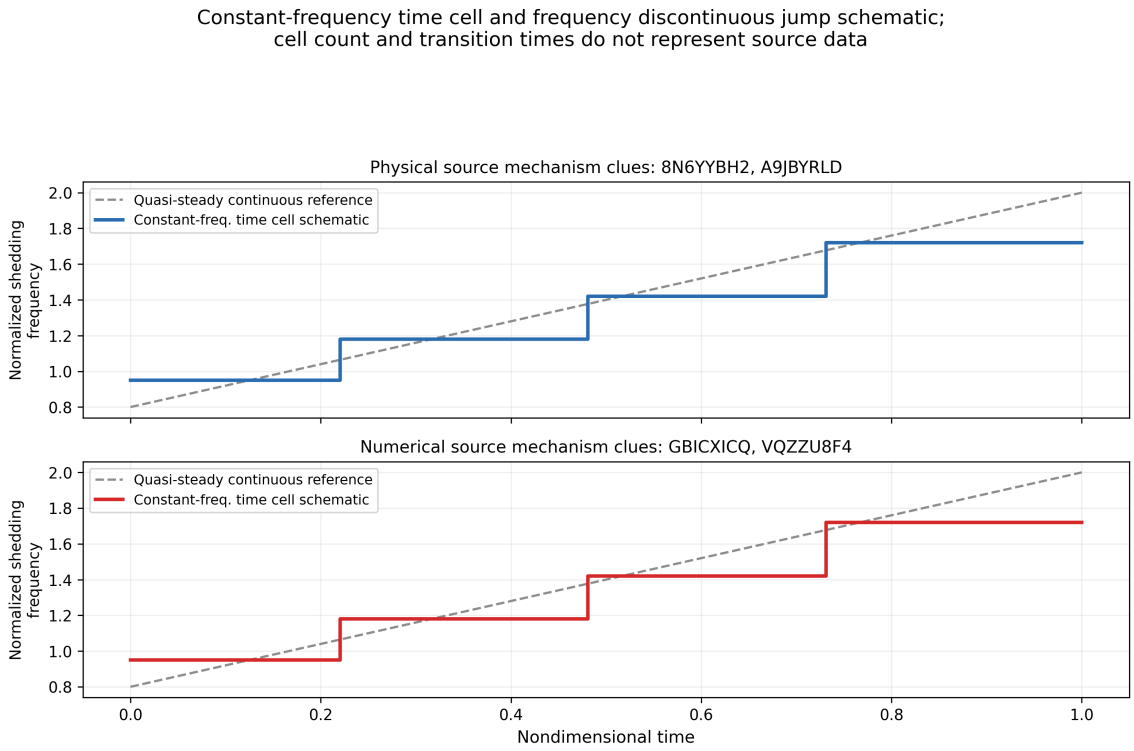


Figure 61: Constant-frequency time cell schematic. Design figure, not experimental measurement. Information derives from qualitative findings in source reports; the schematic does not predict cell count, duration, jump amplitude, or numerical verification.

comparison between acceleration and deceleration for GV4BFSA7 source-internal UF1 through UF4 conditions; this percentage is a source-internal reported value, not a universal correction factor. Figure 63 shows a synthesis comparison of drag and force coefficient response characteristics for $B/D = 1$ cross-sections reported across different sources. Figure 64 shows a literature synthesis of force overshoot and pressure distribution changes for $B/D \geq 3$ cross-sections. These literature data synthesis figures require license determination before external distribution; literature example results are not PETW experimental results.

19.3 Stage Gates, Data Products, and Limitations

19.3.1 G0 through G9 Stage Gates

The experiment program is controlled by ten stage gates from G0 through G9; each gate includes completion conditions and stopping conditions. G0 is literature and design closure; G1 is no-model bench pilot (confirming FFM tracking of target $U(t)$, Cobra synchronization, and DAQ channels); G2 is optical configuration confirmation (PIV camera, laser, and ROI in position); G3 is Tier 1 steady-state baseline (all 69 angle combinations' steady-state pressure completed); G4 is transient pilot (single-angle, single-YM pressure repeat convergence and initial PIV); G5 is Tier 2 transient main matrix start; G6 is Tier 2 convergence checkpoint (intermediate repeat count assessment); G7 is Tier 3 PIV main matrix start; G8 is data integrity audit; G9 is final data release and handoff to CFD/Structural. Currently G0 is partially complete (literature closure complete, facility closure not complete); G1 through G9 are all not started. When any gate's stopping condition is triggered, subsequent gates must not proceed. Table 23 shows the brief definition and current status of each gate.

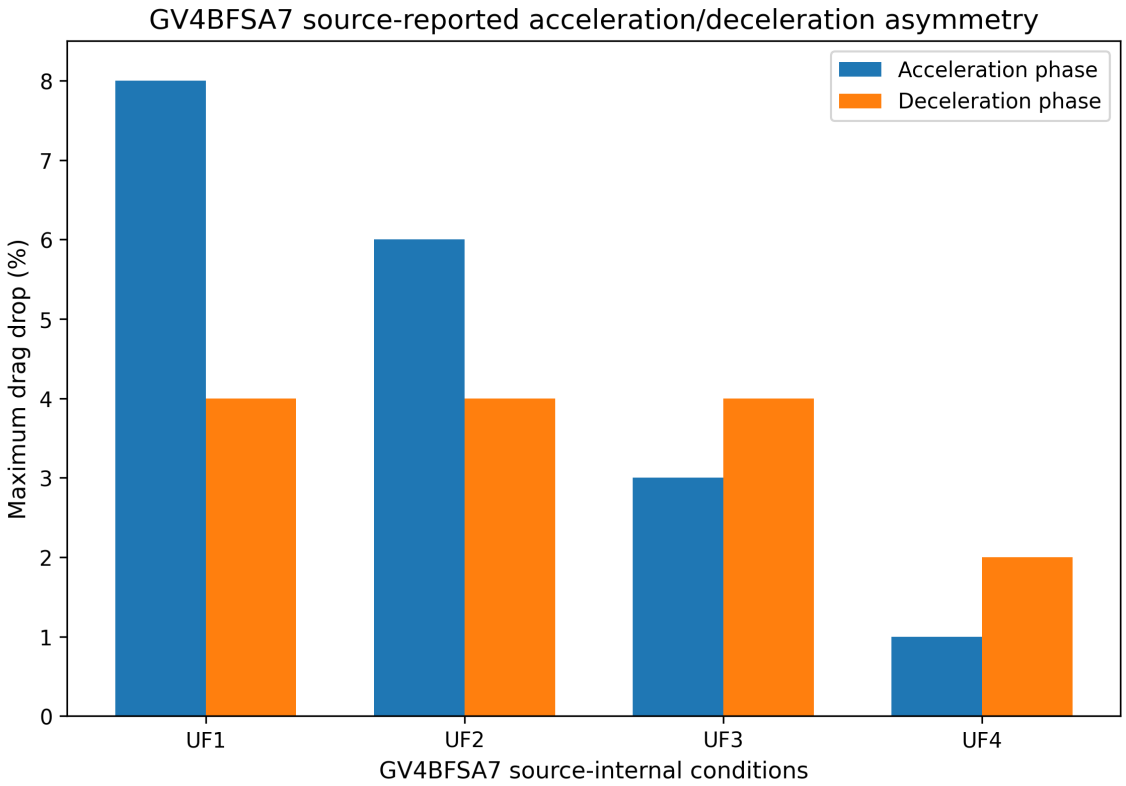


Figure 62: GV4BFSA7 source-internal UF1 through UF4 acceleration and deceleration maximum drag drop comparison. Literature synthesis figure, not experimental measurement. Percentages are source-internal reported values, not universal correction factors.

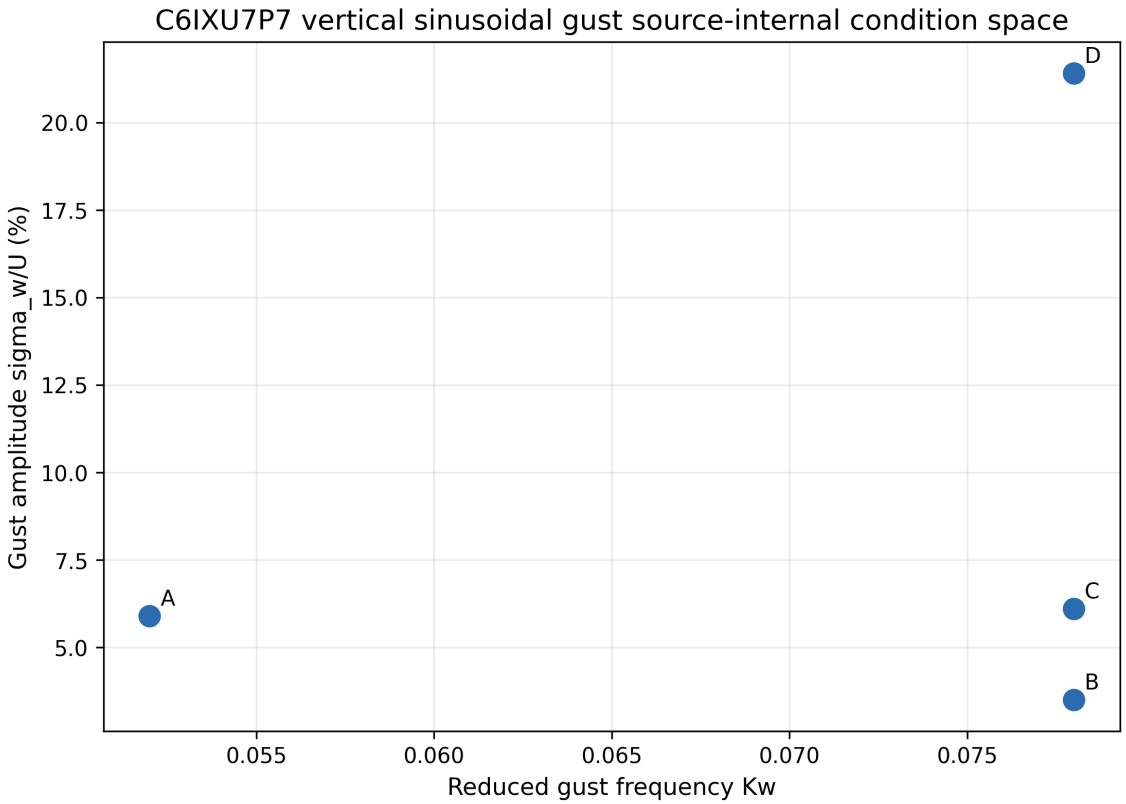


Figure 63: Drag and force coefficient response characteristic synthesis for $B/D = 1$ cross-sections across different sources. Literature synthesis figure, not experimental measurement. Comparison conditions differ across sources and do not constitute direct quantitative pooling.

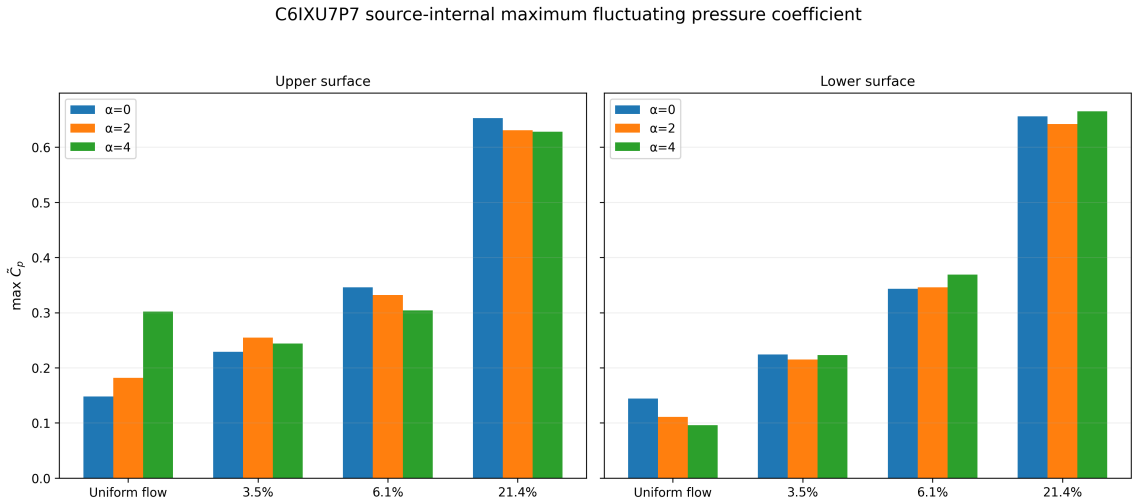


Figure 64: Literature synthesis of force overshoot and pressure distribution changes for $B/D \geq 3$ cross-sections. Literature synthesis figure, not experimental measurement. Overshoot magnitude and conditions vary by source and do not constitute a unified overshoot coefficient.

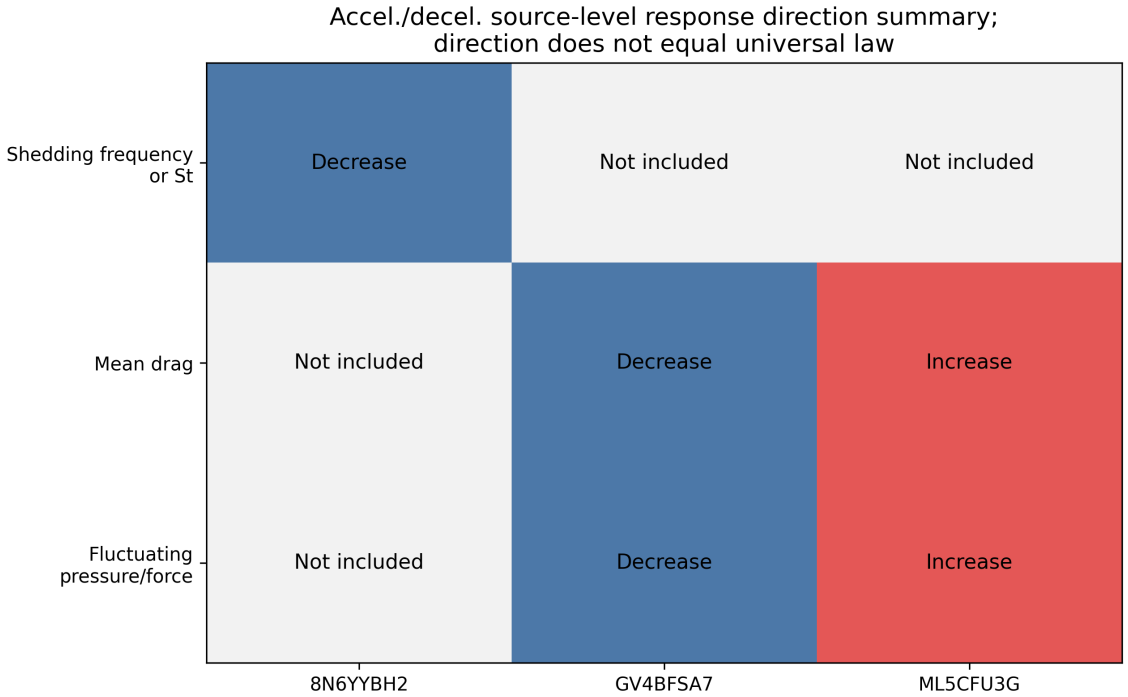


Figure 65: Accelerating and decelerating response direction comparison. Literature synthesis figure, not experimental measurement. Asymmetry of aerodynamic response between acceleration and deceleration varies by aspect ratio and acceleration magnitude.

Table 23: G0 through G9 stage gate definitions and current status

Gate	Name	Completion condition (brief)	Current status
G0	Literature and design closure	55-source engineering closure, v6.7 UP/DOWN delivery, model and tap design complete	partially complete
G1	No-model bench pilot	FFM tracking, Cobra synchronization, DAQ channel confirmation	not started
G2	Optical configuration	PIV camera, laser, ROI in position and aligned	not started
G3	Tier 1 steady-state baseline	69 angle combination steady-state pressure at least 3 repeats each	not started
G4	Transient pilot	Single-angle single-YM pressure repeat convergence, initial PIV	not started
G5	Tier 2 main matrix start	G4 convergence satisfies expansion conditions	not started
G6	Tier 2 convergence checkpoint	Intermediate repeat count assessment	not started
G7	Tier 3 PIV main matrix	G6 passed with optical expansion conditions	not started
G8	Data integrity audit	All raw data hash, metadata, provenance complete	not started
G9	Final release and handoff	Researcher-approved release to CFD/Structural	not started

19.3.2 Completed Work

Completed content supports no statement beyond engineering preparation and design material. On the experiment design side, deliverables include three cross-section and pressure/PIV stratified experiment design, target velocity time histories, time windows, angle-of-attack matrix, run-ID, metadata, synchronization and QA specifications; v5.8 complete experiment plan and v6.0 through v6.6 timing amendments; v6.7 UP/DOWN MATLAB target histories, whose native MATLAB generation and read-back pass, with independent QA showing velocity and acceleration symmetry errors of 0. On the literature and delivery side, 55-source evidence governance, principal review R07 bilingual manuscript, 26-page design review, and UF technical pre-fill matrix (540-row planning representation, not 540 executed tests) are complete. These assets prove that numerical implementation and file read-back are correct and design choices are traceable; they do not prove that the UF wind tunnel can track the target $U(t)$ or that experiments have been executed.

Table 24 defines the formal experiment-status boundaries. Target histories, literature-digitized data, and published-paper figures in the workspace are not new UF experimental measurements.

19.3.3 Verification Path and Planned Data Products

The verification path design logic is layered comparison. The first layer is internal consistency: steady-state baseline positive-negative angle symmetry, multi-repeat statistical convergence for the same condition, and PIV versus pressure-integrated force consistency checks. The second layer is bounded comparison with literature: comparing this study's steady-state pressure coefficients with literature-reported values within the condition range declared by the source, with comparison objects limited to subsets matching in B/D, Reynolds number, and angle of attack. The third layer is quantification of accelerating-flow versus steady-state deviation: at the same B/D and angle of attack, the time-dependent deviation of transient pressure coefficients relative to the steady-state baseline. The fourth layer is delivery of CFD verification data packages: pressure and PIV data confirmed for repeatability and resolution, delivered to the CFD Agent in a Researcher-approved format. Figure 66 shows the coverage and gaps of the planned 60 mm series experiment program

Table 24: WindTunnel formal experiment status boundaries (as of 2026-08-12)

Item	Current status	Meaning and boundary
Experiment plan	complete as internal candidate	Internal candidate baseline, not a UF-accepted matrix
Facility configuration	not confirmed	Configuration and controller interface awaiting UF written confirmation
Instruments and calibration	not closed	Scanivalve, Cobra, PIV, trigger, and tubing pending closure
Hardware execution	none	No facility, DAQ, PIV, or pressure hardware run
Raw experimental measurement	none	No newly acquired pressure, PIV, or Cobra data for this project
Processed experimental data	none	No newly processed results for this project

relative to the 55 sources.

Figure 67 depicts the facility execution gates as a dual serial connection of evidence and authority. Target files, model design, and source synthesis belong to project design evidence; facility configuration, risk assessment, operator qualification, and written operational approval belong to UF authority and safety conditions. If either branch is not closed, facility execution remains prohibited. This figure does not convert internal file completion into facility readiness status.

Figure 68 shows the traceable nodes in the planned measurement chain. Target $U(t)$ and fixed Cobra measured inflow are stored in parallel; pressure, PIV, and other channels enter time alignment via common trigger and independent native sampling rates before forming derived pressure coefficients, velocity fields, or integrated forces. If any clock, calibration, or tubing condition in the chain is not closed, derived quantities maintain planned status.

Planned data products comprise five categories: (1) steady-state pressure coefficient matrix (C_p versus angle of attack and B/D, with repeat mean and standard deviation); (2) transient pressure time histories ($C_p(t)$ versus YM condition and angle, with Cobra measured $U(t)$ as common time reference); (3) PIV velocity field sequences (instantaneous u, v fields versus YM condition and window position, with mask and spatial resolution metadata); (4) integrated force time histories (pressure-integrated $C_D(t)$, $C_L(t)$ versus YM condition and angle); (5) quasi-steady overshoot metrics (transient-to-steady-state force coefficient peak ratio, with confidence intervals). All of the above are planned data product definitions; as of 2026-08-12, no actual data exist (evidence status: engineering preparation).

19.3.4 Current Scientific Limitations and Future Research

Current limitations fall into three layers. The first layer is the distinction between design input and measurement: v6.7 target histories are generated input; formal experimental measurements are none. It cannot be claimed that UF has approved the experiment, the wind tunnel has realized the target $U(t)$, pressure/PIV/Cobra are synchronized, the pilot is complete, new aerodynamics measurements exist, CFD has been verified, or that a universal overshoot multiplier, design coefficient, or PETW design guidance exists.

The design matrix also must map scientific questions to observables. The steady-state baseline establishes angle-of-attack, aspect ratio, and pressure distribution references; transient pressure identifies time-dependent deviation relative to the baseline; PIV windows observe separation shear

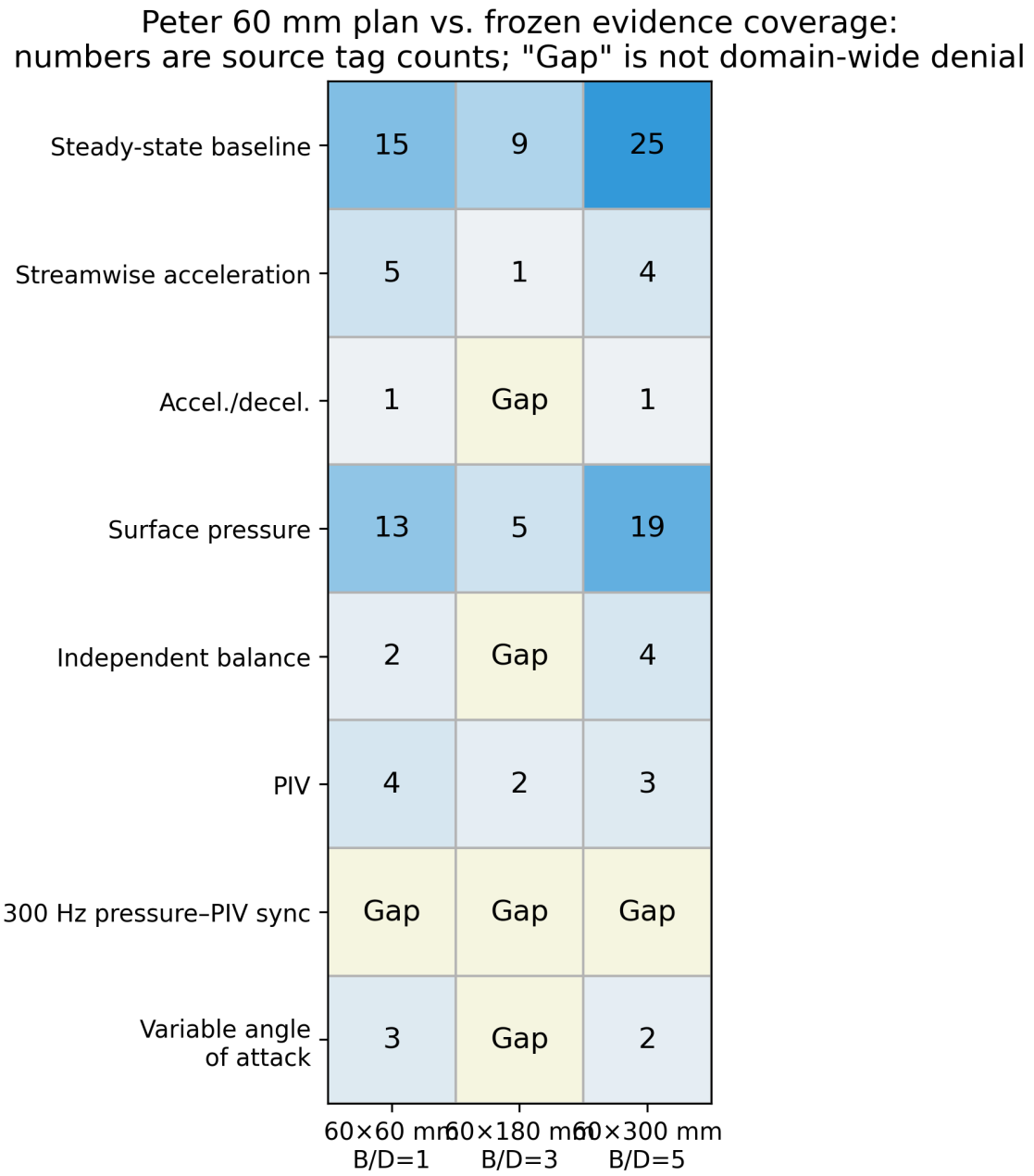


Figure 66: Coverage and gaps of the planned 60 mm series experiment program relative to 55 sources. Design figure, not experimental measurement. Blank cells do not constitute a claim of domain-wide literature gap or facility readiness.

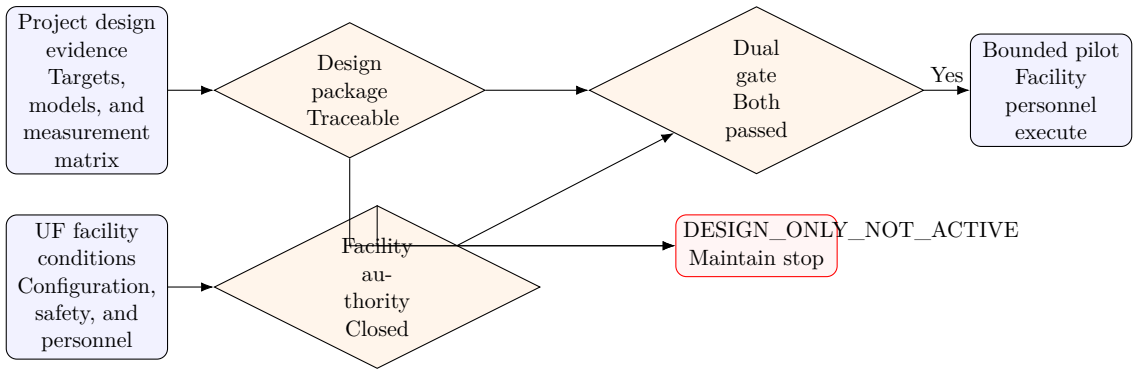


Figure 67: WindTunnel facility execution design evidence gate and UF authority safety gate. Currently the dual gate has not been simultaneously closed. Original project governance method figure; does not indicate that a pilot has been approved or executed.

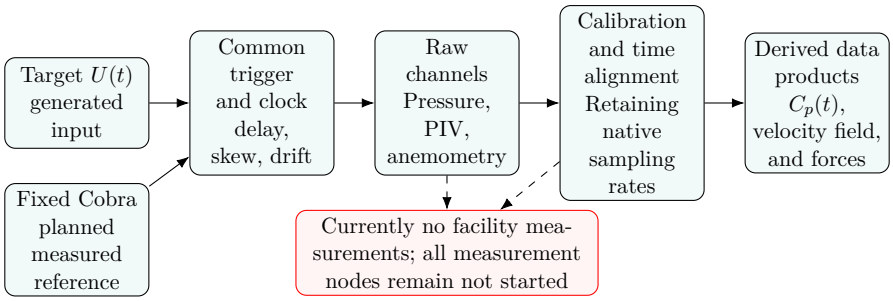


Figure 68: Planned measurement chain and raw channel identity. Target curve cannot substitute for Cobra measurement; channel existence does not indicate that synchronization, calibration, or data quality have been closed. Original project method figure.

layers, wakes, and candidate reattachment regions; fixed Cobra provides the common inflow time reference for every run. Different windows come from different repeats and cannot be stitched into a full-domain instantaneous field at the same moment. If pressure-integrated forces and balance forces are both available in the future, they should also maintain measurement chain identity; comparison is only valid after time, coordinates, calibration, and reference area are consistent.

Repeat strategy is determined by measurement stability rather than fixed count alone. Tier 1's minimum three repeats provide preliminary repeatability for the steady-state baseline; Tier 2's final repeat count must be jointly determined by convergence of peaks, time history morphology, and spatial pressure distribution in the pilot; Tier 3's three to five PIV repeats also must consider particle, illumination, mask, and correlation quality. If the mean appears stable but tails or phase are unstable, early stopping is not warranted. Stopping rules, augmentation rules, and maximum repeat count should be frozen after the pilot and stored with run ID, preventing repeat selection based on desired results.

Target tracking error must be decomposed into endpoint deviation, ramp shape deviation, time shift, local oscillation, and inter-repeat variation. A single RMSE may mask correct endpoints but incorrect central ramp timing, or may mask short-time controller oscillations. The no-model bench pilot should jointly compare endpoints, derivatives, rise time, and hold-segment stability between target and Cobra, and check FFM commands, actual wind speed, and channel clocks. Only when these quantities are within pre-registered tolerances can the target time history be termed a facility-trackable candidate, which still does not mean aerodynamic results with the model are valid.

Pressure tubing produces amplitude attenuation and phase shift on transient signals. The 1.2 m tubing length is only a design parameter; actual frequency response also depends on internal diameter, sensor, Scanivalve, connection volumes, and sampling configuration. Steady-state calibration cannot replace dynamic tubing characterization. If dynamic correction is conducted in the future, the transfer function source, valid frequency band, deconvolution regularization, and noise amplification checks should be stored. Pressure peaks without dynamic closure cannot be used to determine overshoot, nor to establish precise phase relationships with PIV vortex structures.

The PIV planned value of 400 Hz requires closure of facility definitions for frame, image-pair, and vector rate. The twelve-second capacity holds only when N_{frames} and f_{PIV} employ the same convention. Window field of view, spatial calibration, particle displacement, laser energy, mask, correlation peak quality, and dropped frames all limit usable vector sequences. Local windows can answer local separation, wake, or reattachment questions but cannot represent the full-domain flow field corresponding to model-wide pressure. Cross-window comparison requires event-conditioned or phase-conditioned reconstruction while retaining identity of different repeats.

Assessment of quasi-steady deviation should reference the steady-state baseline at the same cross-section, angle of attack, reference speed definition, and measurement chain. If Reynolds number, reference pressure, or angle definition is inconsistent between transient and steady state, deviation mixes baseline differences. Peak ratio also requires uncertainty and repeatability; it cannot simply take the maximum single point across all runs. Time-cell identification should require discrete jumps to show repeatable temporal organization across multiple channels or integral quantities, and controller and tubing artifacts should be checked. Design schematics do not preset cell count, period, or amplitude; actual evidence must come from future measurements.

CFD validation handoff requires raw data, processing scripts, geometry, boundaries, measurement uncertainty, and synchronization information to jointly form a packet. Providing only a single pressure curve or one PIV image is insufficient to verify a numerical model. Comparison should first fix cross-section, angle of attack, measured inflow time history, Reynolds number, coordinates, time zero, and observation operator between CFD and experiment before distinguishing pressure distribution, integrated force, flow field structure, and phase matching. The existence of experimental data does not by itself approve CFD; the Researcher must determine the verification scope based on employ and error.

The facility RFI should be limited to configuration facts and should not request UF to make scientific judgments on behalf of the project. The RFI must confirm FFM interface and executable waveform, Scanivalve actual rate and channel configuration, Cobra model and position, PIV frame/pair/vector definition, camera and laser capacity, common trigger, and safety procedures. Responses should form a versioned configuration receipt, compared item by item with the internal candidate plan. If responses change sampling rate, window, or channel limit, the design matrix should be updated with a new version rather than silently overwriting the v6.7 record.

Model fabrication acceptance requires storing CAD nominal geometry, actual dimensions, and tap coordinates separately. Common reference depth $D = 60\,\mathrm{mm}$ and nominal span $1200\,\mathrm{mm}$ are design values; after actual model completion, external shape, edges, wall thickness, surface, hole positions, tubing connections, and installation posture also must be measured. Tap numbering must have one-to-one mapping across CAD, Scanivalve channel, DAQ field, and post-processing integration order. Any missing hole, blockage, wrong connection, or coordinate deviation must enter run exclusion or uncertainty records and cannot be silently reordered in post-processing.

Angle-of-attack identity similarly requires joint definition of mechanical zero, model coordinates, and inflow direction. Positive-negative angle symmetry gates are meaningful only when the zero position and geometric symmetry can be verified. Installation eccentricity, blockage, support interference, or tunnel flow angularity can change surface pressure and wake; thus the steady-state baseline is first used to check systematic bias, not to directly seek transient effects. If symmetry fails, the zero position, channel, or facility cause should be located before entering Tier 2, and positive-negative angle averaging must not be used to mask problems.

The temporal support of reference pressure and velocity determines the identity of $C_p(t)$. If transient C_p employs time-varying Cobra dynamic pressure as the denominator, noise amplification at low-speed start, Cobra-to-pressure time shift, and spatial position differences must be handled; if a fixed target final-speed dynamic pressure is used, it represents a different normalization. Both definitions can be juxtaposed as diagnostics but cannot be mixed. Analysis plans should prespecify primary definition, auxiliary definition, and low-speed masking rules, and employ measured references rather than target files for formal calculations.

Data management must start from the raw level of each run. Run ID connects model, angle, YM, UP or DOWN, repeat, window, facility configuration, operator, calibration, trigger, and file hashes. Raw files are stored read-only; derived processing writes to independent version directories; all exclusion and reprocessing records are auditable. The G8 data integrity audit passes only after expected files, channels, time axes, metadata, and hashes are closed; figure generation or a single successful processing step cannot replace this audit.

Safety stop conditions are independent of scientific convergence. Controller tracking anomalies, interlocks, laser safety, model or support vibration, pressure system anomalies, and PIV particle and field-of-view issues can all stop a run even if the remaining matrix has scientific value. Facility personnel have on-site stop authority; project design must not override facility procedures. Whether data after stopping are usable for engineering diagnostics must be determined on a case-by-case basis for the actual anomaly period and channels and does not by itself enter the valid denominator because the run was started.

The interface between Track 1 and Track 2 can only be connected through explicit similarity and scaling contracts. The accelerating wind tunnel ramp controls inflow velocity time variation and cross-section aerodynamic response but does not resolve thunderstorm generation, descent core, surface radial expansion, or complete turbulence structure. Even if a target curve originates from Windfield, it only indicates a controlled intensity description was selected. If meteorological association is discussed in the future, it must be stated which dimensionless time, velocity, or geometric quantities are preserved and which change because of facility scope, with the differences treated as design assumptions; wind tunnel results must not be named as complete downburst validation.

The minimum pilot is intended only to assess facility executability and measurement chain closure. The no-model phase checks target tracking and clocks; the optical phase checks PIV capacity and vector quality; the synchronization phase checks Cobra, pressure, and PIV trigger relationships; the model phase then checks repeatability and initial aerodynamic response. Each phase enters only after the preceding phase meets completion conditions; failure returns to configuration modification and generates a new version. This order reduces consuming the full model matrix when basic measurement chain closure has not been achieved, and also maintains DESIGN_ONLY_NOT_ACTIVE until UF gives actual execution approval.

Blockage and similarity checks should be recalculated from UF test section actual dimensions before fabrication. Model span, cross-section, support, and angle-of-attack variation jointly determine effective blockage; the three B/D values cannot be assessed solely by common depth. If correction methods must be used in the future, source, applicable geometry, and uncertainty should be recorded; correction coefficients should not be preset without facility confirmation. Model-to-wall or model-to-floor end conditions also influence two-dimensionality and reattachment structure; the nominal span in design figures cannot replace actual installation boundary conditions.

Steady-state pressure integration requires specifying the control area, normal direction, missing-point treatment, and reference center for each tap. Densifying taps at leading edge or reattachment candidate regions improves local resolution, but integration weights must reflect surface geometry rather than tap density. Interpolation methods may smooth local suction peaks; thus direct piecewise integration and controlled interpolation should be compared and differences recorded. If critical taps fail, whether integrated forces are still usable must be assessed under pre-registered coverage rules, and complete curves must not be formed by neighbor copying.

Selected discrete angles limit the representativeness of the transient angle-of-attack matrix. Reducing from full angle-of-attack sweep concentrates pressure and PIV resources but means that overshoot, reattachment, or time-cell behavior at untested angles remains unknown. The steady-state layer can be used to select representative angles and identify symmetry but cannot prove that transient response varies monotonically with angle. If the pilot reveals new transition regions, matrix expansion requires new Researcher and facility approval while retaining the original plan and modification rationale.

Strict inversion of UP and DOWN discrete arrays proves only mathematical symmetry of the design time history and does not predict aerodynamic response symmetry. Facility controller, wind tunnel inertia, boundary layer development, pressure tubing, and flow memory can all cause measured inflow and aerodynamic forces to differ. Formal comparison should employ each Cobra-measured time history as the condition, checking endpoints, derivatives, and time alignment before discussing response differences. DOWN currently maintains DESIGN_ONLY_NOT_ACTIVE; no comparable facility data exist.

The uncertainty budget should differentiate geometry, inflow, pressure, PIV, synchronization, repeatability, and processing model components. Components may be correlated; for instance, a common Cobra error jointly influences multiple pressure coefficients, and clock deviation simultaneously affects pressure and PIV phase. Without dependence evidence, all components should not be combined into a single precise interval. At the planning stage, components and measurement methods are defined first; after the pilot, actual calibration and repeat data are used for updates, before determining which data products qualify for CFD comparison.

Data release must preserve license and source boundaries. Synthesis figures from the 55 sources, digitized data, and published paper figures each have their own license status and cannot be mixed into an undifferentiated public package with future UF raw data. Internal source figures, original project design figures, facility configuration files, and experimental measurements should be separately marked with source, rights, review status, and permitted employ. G9 handoff releases only data and metadata approved by Researcher and facility license, and does not default to external distribution because it is technically copyable.

The second layer concerns unclosed configuration limitations: the 1200 Hz project request conflicts with UF Scanivalve documentation maximum of 625 Hz; the 400 Hz stereo PIV frame/pair/vector rate, laser energy/duty, ROI, 4800-frame capacity, and download time are unconfirmed; Cobra model, position, frequency response, sampling rate, and common clock are not frozen; pressure/PIV/Cobra common trigger, delay, skew, and drift are not measured; model/tap geometry, tubing, calibration, reference pressure, and fabrication acceptance are not closed; the Yang and Mason digitized point count of 15,517 differs from the review build count of 15,518 by one; 3 quarantine sources remain isolated.

The third layer concerns evidence-level boundaries: Track 1 accelerating-flow evidence and Track 2 meteorological downscaling are different evidence levels; a velocity ramp does not equal a complete downburst. The entry for future research is determined by stopping conditions: no dynamic pilot with the model may be executed until any safety/interlock, pressure rate, PIV joint configuration, fixed Cobra, synchronization, or data-capacity gate is confirmed; no scientific validation or downstream design claim may be formed without pilot measurement and uncertainty closure. The minimum next step is for PI to freeze the v6.7 UP baseline, UF to designate an owner and provide written answers on joint configuration, and completion of a no-model or

optically simple bench/optical/synchronized pilot, followed by determining whether to enter the model pilot based on measured tracking, synchronization, data quality, capacity, and repeatability.

Evidence and equation boundaries of the WindTunnel Agent Chapter 19 preserves only content directly related to the scientific questions, methods, evidence, limitations, and interfaces of the WindTunnel Agent. Content not presenting physical equations belongs to the research methods, knowledge, or coordination layer and does not substitute formalized equations for missing physical evidence. Chapter 19 remains a source-limited synthesis and cannot support verified design wind, design load, code recommendation, or PI scientific conclusion.

20 CFD Agent

CFD Agent is the Step 4b execution unit in the PETW pipeline, responsible for converting meteorological-scale or idealized transient wind fields into building-scale flow, facade pressures and aerodynamic forces, and handing off force data packets with complete provenance and physical parameter semantics to the downstream Structural Agent. Upstream inputs come from CM1/WRF transient wind fields and Windfield wind field descriptions; CAD provides building geometry, ComputeOps supports computational execution, and WindTunnel provides independent facility comparison targets. This chapter reports the state of the agent as of 2026-08-12 in seven sections covering scientific purpose, knowledge base, Researcher involvement and decisions, workflow and interactions, completed work and results, key formulas, and current scientific limitations and future research. All results remain not_scientifically_approved, constrained by specific cases, grids, turbulence closure models, computational domains and executed time spans, and do not establish completed validation, generalized building aerodynamic conclusions, final solver selection, or completed long-duration HPC computation.

20.1 Scientific Questions, Knowledge, and Researcher Decisions

20.1.1 Scientific Purpose

The scientific objective of the CFD Agent is to extract aerodynamic loads across scales. Given three-dimensional transient downburst outputs produced by CM1/WRF or idealized transient wind fields, this study constructs a reproducible one-way multi-scale workflow: establishing boundary conditions and initial conditions for fine-scale CFD, resolving the wind field around buildings, extracting facade pressures and floor forces, and forming aerodynamic endpoints including base shear V_{base} (unit: N), two overturning moments M_x, M_y (unit: N·m) and torsional moment M_z (unit: N·m). The target building class defined at the review level is an isolated, rigid, enclosed, sharp-edged, square prismatic tall building.

This objective organizes seven stages into a continuous auditable chain: source field generation, variable contract and coordinate transformation, spatial and temporal interpolation, interface turbulence reconstruction, building-scale solving, pressure reference and normalization, and load integration. The source field generation stage extracts three-dimensional velocity fields $\mathbf{u}(x, y, z, t)$ (unit: m/s) and thermodynamic state from CM1 or WRF output; the variable contract specifies the physical quantities, units and coordinate definitions required for inlet, outlet, top and lateral boundary conditions of the CFD domain. Spatial interpolation is performed between meteorological grids and CFD grids; temporal interpolation converts the finite storage time steps of meteorological output to the finer time steps required by CFD. Interface turbulence reconstruction superimposes unresolved turbulence components from meteorological-scale output onto the mean field using synthetic methods. Building-scale solving completes the discretization and

time advancement of the Navier-Stokes equations within the computational domain. The pressure reference and normalization stage determines the semantics of p_{ref} (unit: Pa); load integration integrates surface pressures along facet normals into discrete floor forces. Any missing or inconsistent stage causes the downstream force data packet to lose traceable physical semantics.

Literature-side evidence delineates the nature of the input fields. Full-cloud models can produce downburst wind fields with asymmetric outflow heights and short-duration near-surface peaks[Orf et al., 2012]; circumferential analysis of the same simulation yields a peak-to-mean ratio of 1.936, conditioned on the three strongest wind instants and the radial range not dominated by adjacent downdrafts[Saville et al., 2014]; idealized three-dimensional downburst wind fields have independent numerical implementations[Mason et al., 2010]. Cold-source scaling studies show that peak radial wind speed at $t/T_0 = 4.0$ can be scaled to $u_{r,\max}/V_0 = 3.25$ with RMS deviation below 10%[Nguyen, 2021], where T_0 (unit: s) is the cold-source characteristic time and V_0 (unit: m/s) is the reference velocity. In comparisons between full-cloud CM1 and axisymmetric URANS releases, the position, magnitude and timing of peak radial wind speed agree within 5%, with $t_{\max}$ of 590 s and 600 s respectively, though the stated agreement is limited to approximately one minute before the peak to three minutes after the peak[Anabor et al., 2018]. These literature facts show that the input fields confronting CFD Agent require case-by-case retention of conditions regarding spatial asymmetry, peak scaling and event time window selection, and cannot be summarized by a single scalar.

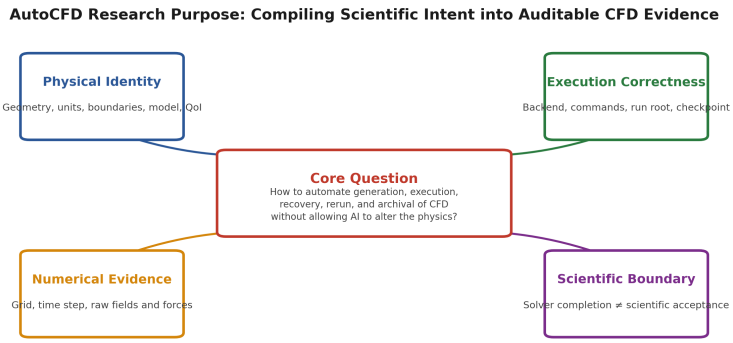


Figure 69: CFD Agent research purpose and cross-scale problem chain schematic. PETW original figure, conditioned on isolated rigid square prismatic target and transient wind field input.

20.1.2 Knowledge Base

The knowledge layer of CFD Agent consists of eight Wiki namespaces totaling 801 files, distributed as follows: cfd_automation 84, cfd_m1_general_cwe 132, cfd_m3_accelerating_flow 61, cfd_m4_downburst_cfd 289, cfd_validation 16, cfd_structural_handoff 67, cm1_to_cfd39_topic 54, and crc_docs 98. All 47/47 knowledge layer retrieval tests passed; their evidential significance is limited to retrieval and method evidence and does not establish support for any solver setting or load result. Review assets include the AutoCFD/Agent-CFD v9 review (35 pages in Chinese, 36 pages in English) and the AutoCFD/CM1-CFD program research summary (43 pages in Chinese, 65 pages in English, 25 figures).

The executable asset line comprises the AutoCFD/Agent-CFD automated workflow, Case011Up, Case02Up LayerForce50, DFG 2D-2 cylinder diagnostic cases, the CM1-D420 interface adapter, and the three-arm matched-drive plan. Knowledge assets are used to organize evidence and codify method boundaries, and do not replace case-by-case execution evidence; passing retrieval tests only shows that evidence can be located and cited, not that any numerical conclusion has been

validated.

20.1.3 Researcher Involvement and Decisions

The Researcher (Liang Hu / PI intervenes in the CFD stage at two positions: input-side assumptions and output-side identity. On the input side, decisions are needed on which transient events and mechanism hypotheses enter building-scale simulation, how target geometry is provided by CAD, whether the pressure reference p_{ref} (unit: Pa) uses a fixed event reference, time-varying reference or height reference, and how velocity and density reference values are recorded with the data packet. On the output side, decisions are needed on which pressure or force data packets may be handed off to Structural Agent and which physical parameters and provenance fields accompany the handoff.

Decisions formed to date include: the DFG case series executed as cylinder diagnostic cases to a uniform duration of 30 s; Case011Up and Case02Up LayerForce50 advanced as tall building cases to their respectively recorded execution times; D420 C0/M0/M1 matched-drive three-arm retained as a frozen static package, without a submitted computation and no new scientific output. The condition for data to enter Structural is that pressure and force data packets simultaneously carry pressure reference semantics, normalization denominator, geometric description, grid information and time synchronization information, and are handed off as candidate research inputs, not as design loads. The Researcher has not yet selected a final solver scheme, nor approved any case results for design use.

20.2 Methods, Results, and Formulas

20.2.1 Workflow and Interactions

The workflow uses a one-way multi-scale chain: CM1/WRF/Windfield provides transient wind fields or target profiles, which enter building-scale CFD through boundary and initial condition construction; after pressure field extraction, floor forces and base resultants are formed by facet integration, producing provenance-bearing force data packets that enter Structural. On the geometry side, CAD provides the target building exterior surface definition; automated geometry generation in computational wind engineering has reusable OpenFOAM workflows[Rodriguez et al., 2025], and the scope of OpenFOAM applications in computational wind engineering also has review evidence[Tiwari et al., 2024]. Cross-scale interfaces draw on downscaling schemes from mesoscale meteorological models to microscale CFD[Cao et al., 2011].

The denominator semantics of pressure coefficients and normalization determine whether the integrated forces and moments share a common scale. A fixed event reference uses $q = \frac{1}{2}\rho U_{\max}^2$ (unit: Pa) as denominator, where ρ (unit: kg/m^3) is air density and $U_{\max}$ (unit: m/s) is the event maximum wind speed; an instantaneous reference uses $q(t) = \frac{1}{2}\rho U(t)^2$ (unit: Pa) as denominator, where $U(t)$ is the time-varying reference wind speed. Pressure coefficients under the two references are not interchangeable[Li et al., 2025]; time-varying static pressure can change local C_p by approximately 0 to +0.8[Iida and Uematsu, 2019]. CFD Agent therefore retains dimensional pressure as the primary quantity and computes coefficients only when the denominator semantics are fully documented. Table 25 provides the definitions and applicable conditions for three pressure reference types.

The workflow also applies per-stage QA gates and the definition of an eight-level verification ladder: R1 and R2 constrain consistency of source field and interface variables, R3 constrains fine-scale velocity fields and turbulence spectra, R4 and R5 constrain facade mean pressures and fluctuating pressures, R6 constrains floor forces, R7 constrains base shear and overturning moments,

Table 25: CFD Agent pressure reference type definitions. Pressure coefficients under the three references are not interchangeable; dimensional pressure is the primary handoff quantity.

Reference type	Denominator definition	Applicable conditions
Fixed event reference	$q = \frac{1}{2}\rho U_{\max}^2$	Single maximum wind speed normalization for the entire event
Time-varying reference	$q(t) = \frac{1}{2}\rho U(t)^2$	Reference wind speed varies with time
Height reference	$q(z) = \frac{1}{2}\rho U(z)^2$	Reference wind speed varies with height

and R8 constrains torsional moment. Passing at a lower level cannot substitute for evidence at a higher level. Verification metrics and turbulence frequency limits for LES wind load assessment have been defined in the literature[Geleta and Bitsuamlak, 2022]. Table 26 presents the constraint objects and current status for each verification ladder level.

Table 26: CFD Agent eight-level verification ladder definition and current status. Passing at a lower level does not establish verification evidence at a higher level.

Level	Constraint object	Comparison quantity	Current status
R1	Source field variables	Inlet velocity, temperature, density	Partially checked for some ca
R2	Interface consistency	Coordinate mapping, time synchronization	Partially checked for some ca
R3	Fine-scale velocity	ABL profiles, turbulence spectra	Not closed
R4	Mean pressure coefficients	Surface mean C_p	Not closed
R5	Fluctuating pressures	Surface RMS C_p'	Not closed
R6	Floor forces	Along-height distribution and time history	Output available for some ca
R7	Base shear and moments	V_{base}, M_x, M_y	Not closed
R8	Torsional moment	M_z	Not closed

WindTunnel provides independent facility pressure or force measurements as comparison targets; ComputeOps provides computational resource execution support. The DFG 2D-2 cylinder cases serve a method diagnostic function; Case011Up and Case02Up LayerForce50 serve the tall building load chain function.

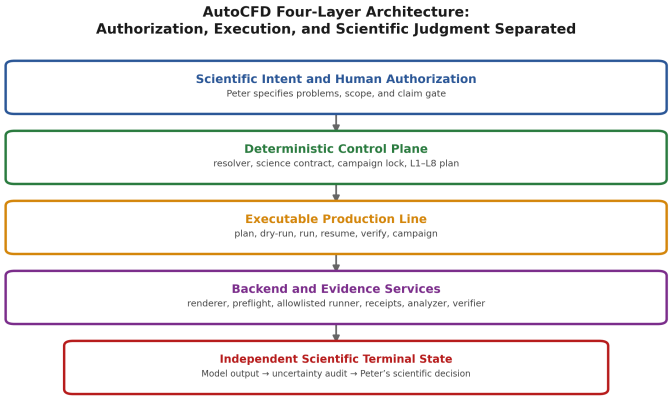


Figure 70: Four-layer architecture schematic of the CFD automated workflow. PETW original figure.

20.2.2 Completed Work and Results

As of 2026-08-12, completed work by CFD Agent encompasses three categories: DFG cylinder diagnostic cases, tall building cases, and historical interface packages. All results are constrained by their specific cases, grids, turbulence closure models, computational domains and executed time spans, and do not establish completed external validation or generalized building aerodynamic

conclusions. Table 27 presents the parameters and execution overview for each case.

Table 27: CFD Agent executed case overview. All results are constrained by their respective cases, grids and executed time spans.

Case	Geometry	Execution time	Primary output	Evidence status
DFG 2D-2	Cylinder	30 s	Drag coefficient time history	bounded
Case011Up Condor32	Tall prism	11 s	Flow field snapshots	bounded
Case02Up LF50	Tall prism	11 s	50-layer synchronized forces	bounded
Case011Up ABL	Empty domain (no building)	time 1000	ABL profiles	bounded
D420 C0/M0/M1	matched-drive	Not submitted	Frozen static package	engineering

All five DFG campaign cases were executed to 30 s. These cases use two-dimensional cylinder geometry for diagnosing the solver behavior in drag coefficient C_D (dimensionless) and lift coefficient C_L (dimensionless) time histories on a simple bluff body flow problem. The C_D time histories across five cases show sensitivity to spatial resolution, and mesh independence cannot be claimed. Lift amplitude differences across different grids show that lift-amplitude is sensitive to spatial resolution.

Case011Up Condor32 was executed to 11 s; this case targets a tall square prism to verify the pressure extraction workflow from inlet boundary to building surface. Case02Up LayerForce50 was executed to 11 s and generated 50-layer synchronized force data, using `rhoInf`=1 as the normalization reference density; consequently the synchronized force output carries normalization semantics rather than actual physical force or design load meaning. The 50-layer force profile of this case provides force component time histories in the along-wind, crosswind and vertical directions at each floor along the building height.

Case011Up medium empty ABL case job 3396039 was executed to time 1000; this case contains no building and is designed to diagnose the preservation of ABL profiles in the empty domain. The comparison between inlet profile and target profile (Figure 71) and the relative deviation distribution (Figure 72) show that the inlet conditions have not yet reached a closed acceptance ABL reference condition.

D420 C0/M0/M1 matched-drive three-arm remains a frozen static package, without new computations submitted and no new scientific output. The driving conditions of the historical D420 M0/M1 are inconsistent; this data set cannot serve as evidence for controlled comparison. Figure 78 presents the D420 M0 wind field spatial structure, shown only as content display of the frozen package.

Literature evidence provides comparison benchmark boundaries for these runs. For tall building LES, mean pressure R^2 can exceed 0.97 while base moment differences range from 10.5% to 13.0%[Thordal et al., 2020]; when mean pressure deviation is below 10%, RMS pressure deviation can reach 19%[Ricci et al., 2018]. Pressure errors do not necessarily attenuate after area integration; therefore surface mean pressure, surface fluctuating pressure, local extremes, floor forces, base shear, two overturning moments and torsional moment each require separate error reporting, and a preceding level indicator cannot be used to infer that a subsequent level has passed.

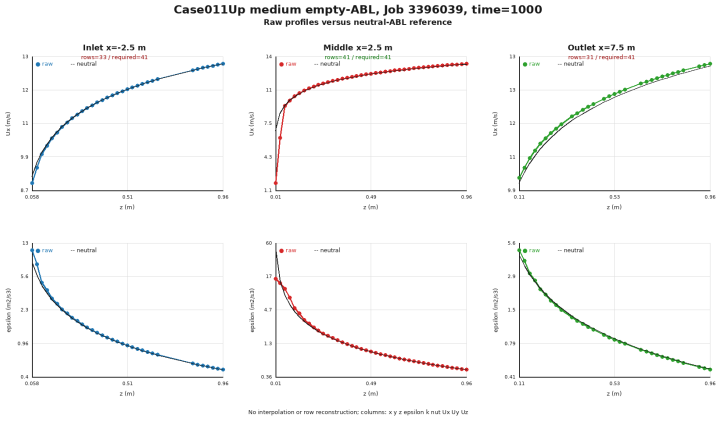


Figure 71: Medium empty ABL reference profile from job 3396039. PETW original figure, executed to time 1000; vertical axis is height (m), horizontal axis is wind speed (m/s).

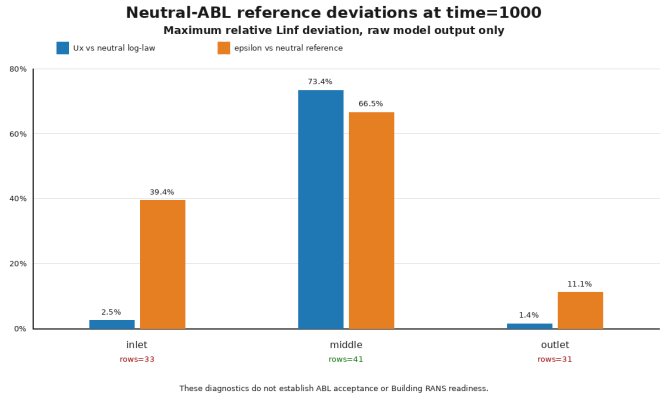


Figure 72: Deviation distribution relative to ABL target profile from job 3396039. PETW original figure, executed to time 1000.

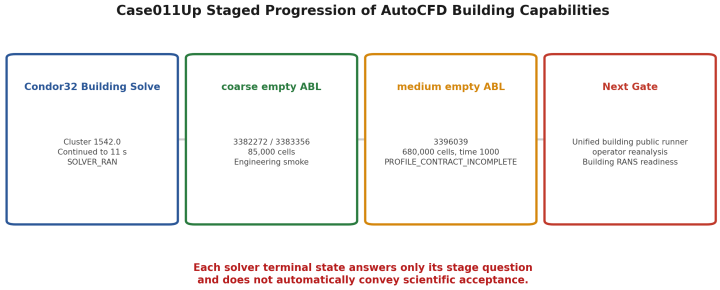


Figure 73: Case011Up case series progression schematic. PETW original figure, executed time span to 11 s.

20.2.3 Key Formulas

CFD Agent uses dimensional pressure $p(t)$ (unit: Pa) as the primary quantity. Let $p_{\mathrm{ref}}(t)$ (unit: Pa) be the pressure reference, which may be a fixed event reference, time-varying reference or height reference; $\hat{n}$ (dimensionless) be the facet outward normal unit vector; and dA (unit: m^2) be the facet area. The facet force is defined as

$$d\mathbf{F}(t) = -\big(p(t) - p_{\mathrm{ref}}(t)\big)\,\hat{n}\,dA \tag{22}$$

where $d\mathbf{F}(t)$ (unit: N) is the force vector produced by the net pressure on the facet, and the negative sign shows that positive pressure (above reference) produces an inward force against the outward normal. The floor force for the i-th facade band is the integral over the band area A_i (unit: m^2):

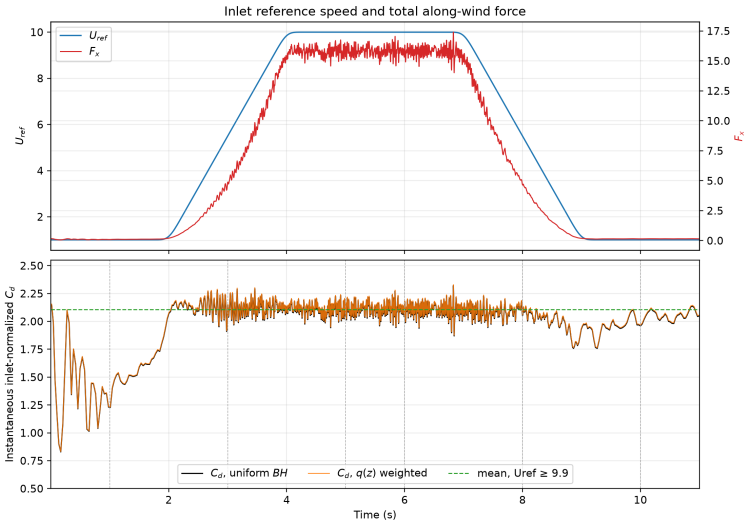


Figure 74: Drag coefficient C_D time history for the DFG 2D-2 cylinder case. PETW original figure, covering the 30 s executed time span across five cases.

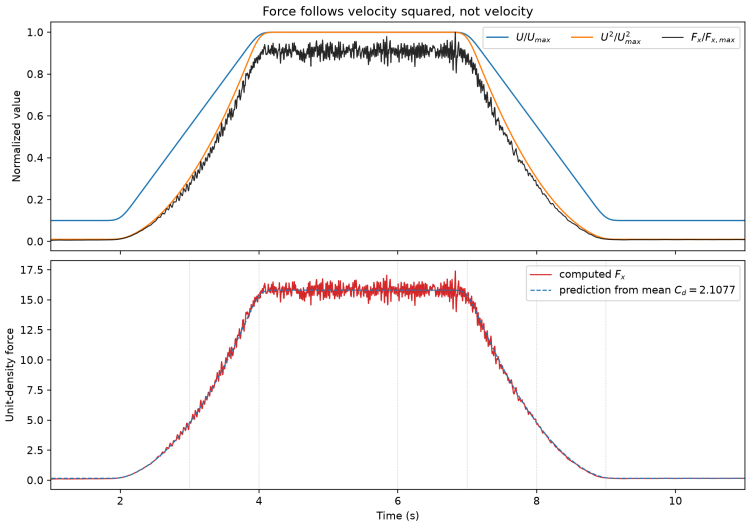


Figure 75: Force versus velocity squared linearity diagnostic. PETW original figure, DFG cylinder case.

$$\mathbf{F}_i(t) = \int_{A_i} d\mathbf{F}(t) \tag{23}$$

where $\mathbf{F}_i(t)$ (unit: N) is the resultant force on the i-th band at time t. The base shear is the sum of all facade band forces, currently discretized into 25 or 50 bands:

$$\mathbf{V}_{\text{base}}(t) = \sum_{i=1}^{N_{\text{band}}} \mathbf{F}_i(t) \tag{24}$$

where N_{band} is the number of facade bands. The base moment is the integral of the moment of facet forces about the reference point (base center), with $\mathbf{r}$ (unit: m) being the position vector from the facet to the reference point:

$$\mathbf{M}(t) = \int_A \mathbf{r} \times d\mathbf{F}(t) \tag{25}$$

In the right-hand building coordinate system (x along the nominal approach flow direction, y as the crosswind direction, z as the vertical direction, with origin at the base center), the discretized two overturning moments and torsional moment are

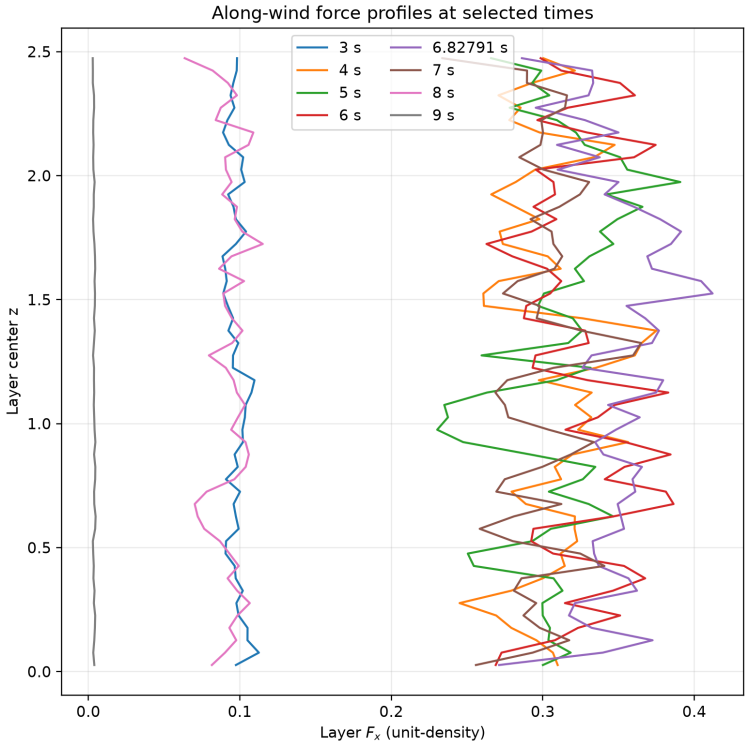


Figure 76: Along-height floor force profiles. PETW original figure, 50-layer synchronized force data from Case02Up LayerForce50.

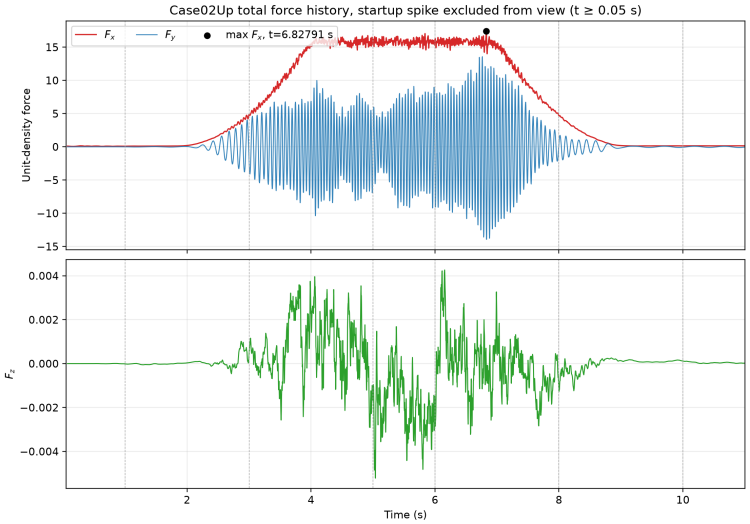


Figure 77: Total force time history. PETW original figure, Case02Up LayerForce50 executed time span to 11 s.

$$M_x(t) = -\sum_{i=1}^{N_{\text{band}}} F_{y,i}(t)\, z_i \tag{26}$$

$$M_y(t) = \sum_{i=1}^{N_{\text{band}}} F_{x,i}(t)\, z_i \tag{27}$$

$$M_z(t) = \sum_{i=1}^{N_{\text{band}}} \left(\Delta x_i\, F_{y,i}(t) - \Delta y_i\, F_{x,i}(t)\right) \tag{28}$$

where z_i (unit: m) is the height from the i-th band center to the base, Δx_i and Δy_i (unit: m) are the horizontal offsets from the i-th band center to the reference point, and $F_{x,i}$, $F_{y,i}$ (unit:

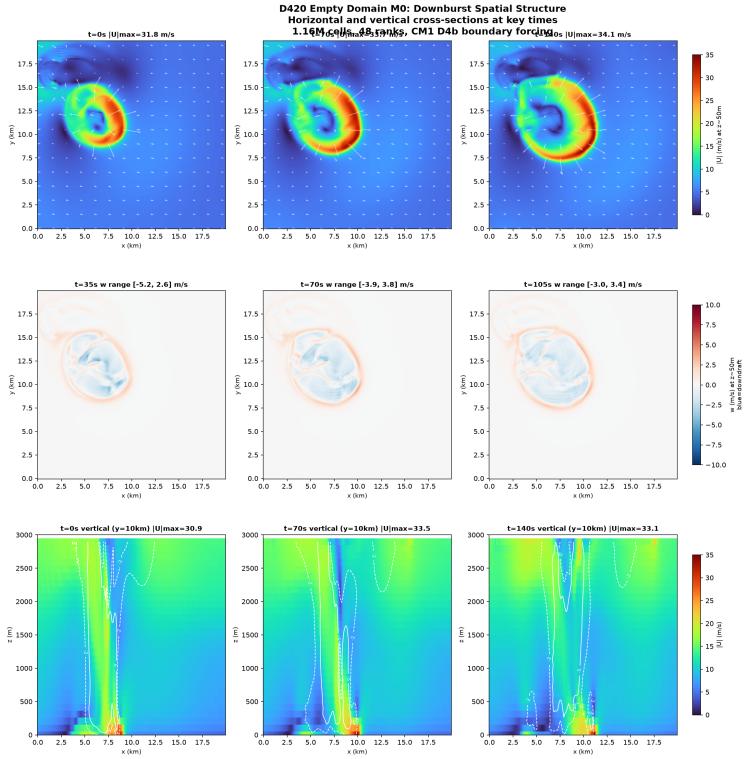


Figure 78: D420 M0 wind field spatial structure. PETW original figure, from the frozen static package with no new scientific output.

N) are the components of the i-th band force in the x and y directions. Pressure coefficients have comparable meaning only when the pressure reference, density, velocity reference and reference height are fully documented.

The force data packet must carry all variables listed in Table 28 when handed off to Structural. The data packet is handed off as a candidate research input, not as a design load.

Table 28: Force data packet handoff variable contract. The data packet is handed off to Structural Agent as a candidate research input.

Variable	Physical meaning	Unit	Source
$\mathbf{F}_i(t)$	i-th band floor force vector	N	CFD integration
$p_{\mathrm{ref}}(t)$	Pressure reference	Pa	CFD domain definition
ρ_{ref}	Reference density	kg/m^3	Input conditions
U_{ref}	Reference wind speed	m/s	Input conditions
z_i	Band center height	m	Geometry
Δt_{out}	Output time step	s	Solver settings

The formulas above close the computational path from surface pressures to target aerodynamic endpoints, but do not close the validation of those endpoints. Validation requires ascending level by level along the same data chain, retaining independent comparison targets and failure records at each level.

20.3 Evidence Audit, Interfaces, and Limitations

20.3.1 Refinement of Input Contracts, Numerical Diagnostics, and Evidence Transfer

CFD inputs are first decomposed into three categories of objects: source fields, transformation rules, and target domains. Source fields store velocity, thermodynamic state, surface conditions, temporal coordinates and spatial coordinates; transformation rules store interpolation, rotation,

resampling and unresolved turbulence processing; target domains store the boundary semantics of inlet, outlet, lateral, top, ground and building surfaces. The three categories must maintain independent version identities. Storing only the interpolated inlet file without preserving the source field would prevent subsequent personnel from determining whether anomalies originate from the upstream field or the interpolation process; storing only the source field without preserving the target domain definition would make it impossible to reconstruct the normal, tangential and height coordinates on boundary surfaces. The D420 data tables separately record source provenance, clock mapping, coordinate mapping, boundary surfaces and flux contracts precisely so that each category of error has an independent traceback location.

Temporal contracts must simultaneously distinguish source output instants, interpolation instants, solver internal time, and result archiving instants. Adjacent meteorological output files may be continuous in physical time, but file indices do not necessarily equate to a continuous clock; the CFD internal time step is further governed by stability and convergence controls, and archiving frequency should not be substituted. Consequently, each inlet field frame must retain both source time and target time, and result files must simultaneously retain solver time and archiving sequence number. For transient outflows, rapid changes near the peak, the rising flank and the decay segment carry different physical information; temporal interpolation that only ensures numerical continuity without checking the rate of change may smooth local peaks or shift load peaks. The current D420 package provides engineering preparation evidence for clock and coordinate mapping, but the driving conditions of the historical M0 and M1 are inconsistent, so differences between them cannot be attributed to a single numerical setting.

Spatial contracts include not only coordinate origin and axis directions, but also velocity component definitions, boundary surface outward normals, and building orientation relative to the approach flow. After horizontal components in meteorological coordinates are transferred into building coordinates, the rotation matrix must use the same convention as the geometric orientation. If the velocity on the inlet surface contains normal reverse-flow regions, their physical origin and boundary treatment must be documented rather than truncated without explanation. Vertical interpolation must also distinguish terrain-following height from geometric height; the first available meteorological level near the surface does not automatically equal the first CFD grid layer. The current interface diagrams express the geometric contracts of five boundary surfaces, the mapping from native coordinates to center coordinates, and the three-branch case identities separately, avoiding the merging of geometric consistency, variable consistency and case comparability into a single pass marker.

Inlet quality checks must maintain a hierarchy between the empty domain and the domain containing a building. The empty domain isolates the effects of inlet generation, ground treatment and numerical diffusion on profiles; the domain with the building superimposes blockage, separation and wake feedback. If the profile contract in the empty domain has not yet been closed, pressure differences in the building domain cannot be entirely explained by building aerodynamic mechanisms. Figures 79 through 82 therefore form a diagnostic chain: streamwise drift checks whether profiles change with propagation position, iteration evolution checks whether the variation is still controlled by the initialization process, resolved height checks which height ranges have sufficient grid support, and the building upstream comparison checks the actual approach flow before entering the building influence zone. The combined purpose of all four diagnostics is to expose gaps in the input contract, not to append validation labels to existing results.

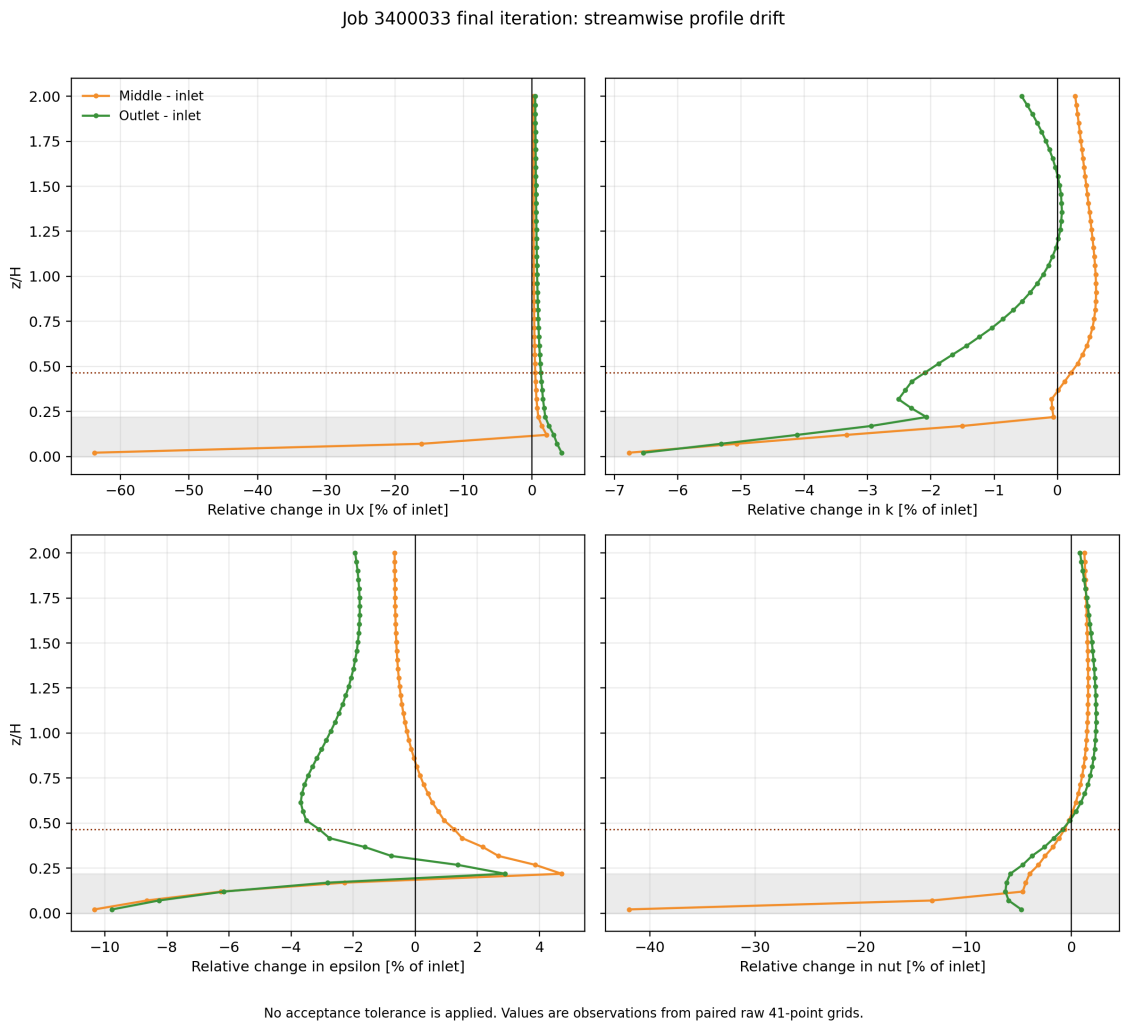


Figure 79: Empty domain ABL streamwise profile drift diagnostic. PETW original project figure; this figure is used to locate inlet condition preservation issues within the computational domain, and does not show that the solver has been validated.

Numerical strategy selection must be organized around the target output rather than a single flow field image. If the target is mean pressure, the time averaging window, inlet mean profile and grid representation of separation points are the primary checks; if the target is fluctuating pressure or floor force peaks, time resolution, unresolved turbulence, pressure sampling and preservation of spatial coherence become additional constraints; if the target is base moment and torsional moment, facade band division, facet normals, reference point and multi-face synchronization further affect integration results. This forms a progressive chain from velocity to pressure, from pressure to floor forces, and from floor forces to base effects. An apparently stable output at a preceding level only shows that the diagnostic at that level has not exposed obvious anomalies under current conditions, and cannot substitute for independent verification at the subsequent level.

Grid checks also need to be bound to target quantities. Local grid refinement may improve windward face pressure gradients or separation zone representation, but does not necessarily improve low-frequency wake structures; an overall increase in degrees of freedom also does not equate to the target load statistical quantity having achieved mesh independence. The current DFG cylinder cases show spatial sensitivity of lift amplitude, so they serve a numerical diagnostic function rather than solver certification. For tall prisms, subsequent grid studies should compare velocity profiles, mean pressure, fluctuating pressure, floor forces and base effects under the same inlet,

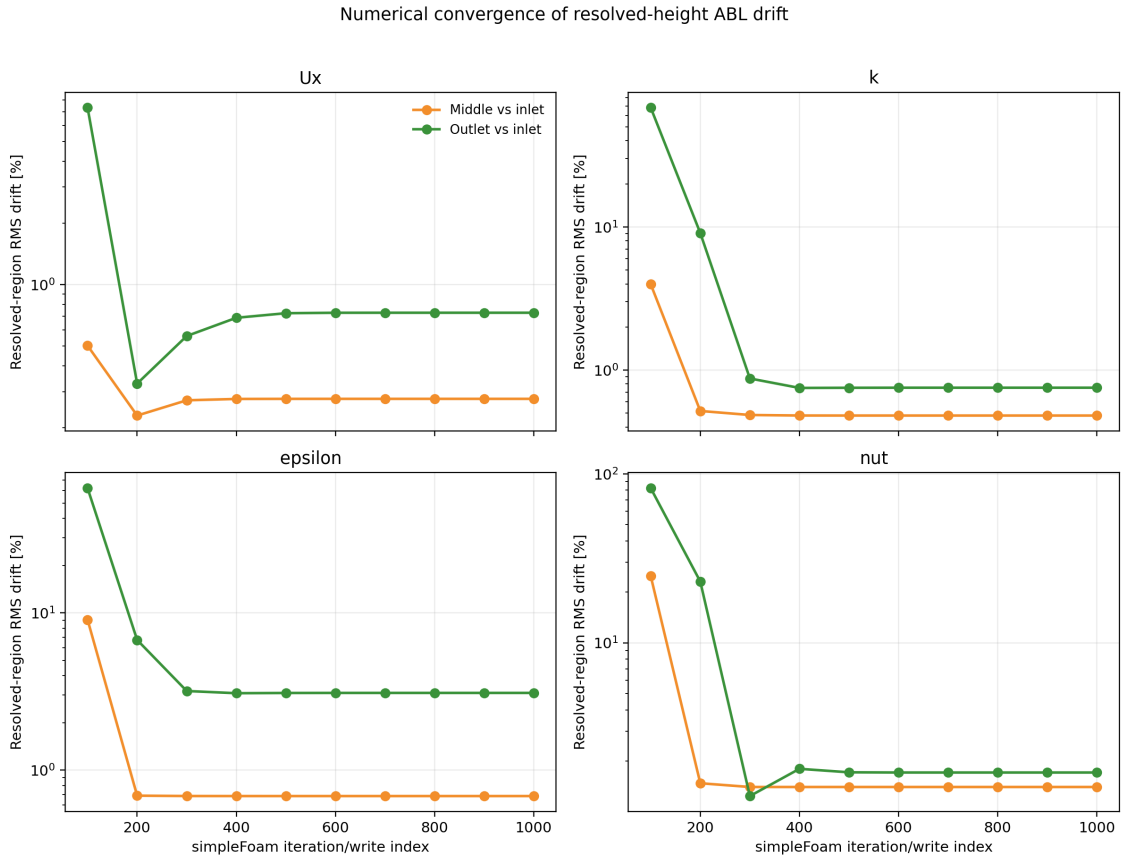


Figure 80: Empty domain ABL diagnostic quantity evolution with iteration advancement. PETW original project figure; the evolution trend corresponds only to the executed interval and does not support long-term statistical convergence assessment.

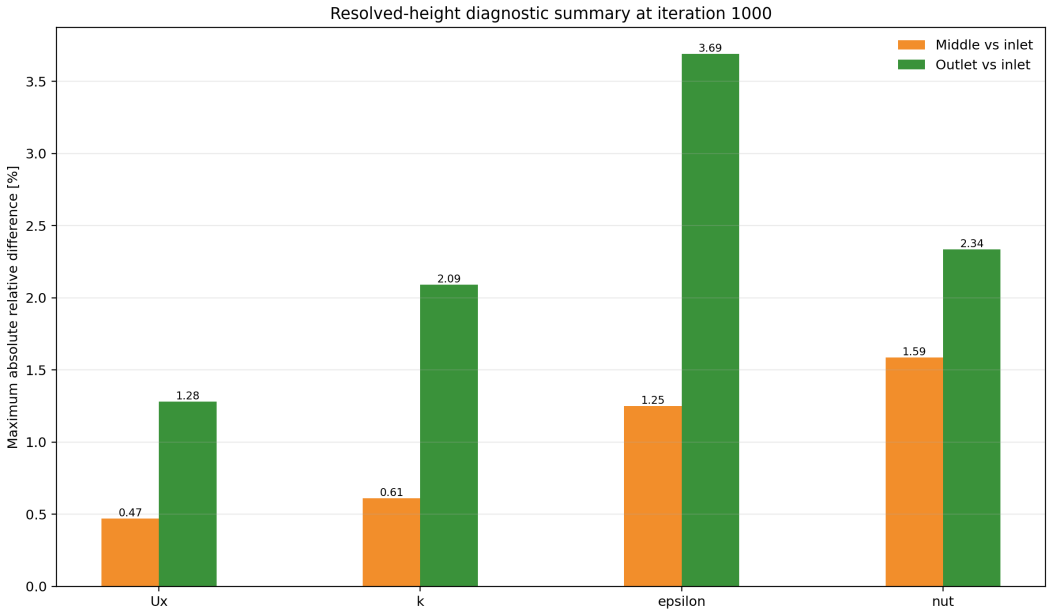


Figure 81: Empty domain ABL resolved height diagnostic summary. PETW original project figure; the resolved range is constrained by the current grid, domain and executed time span.

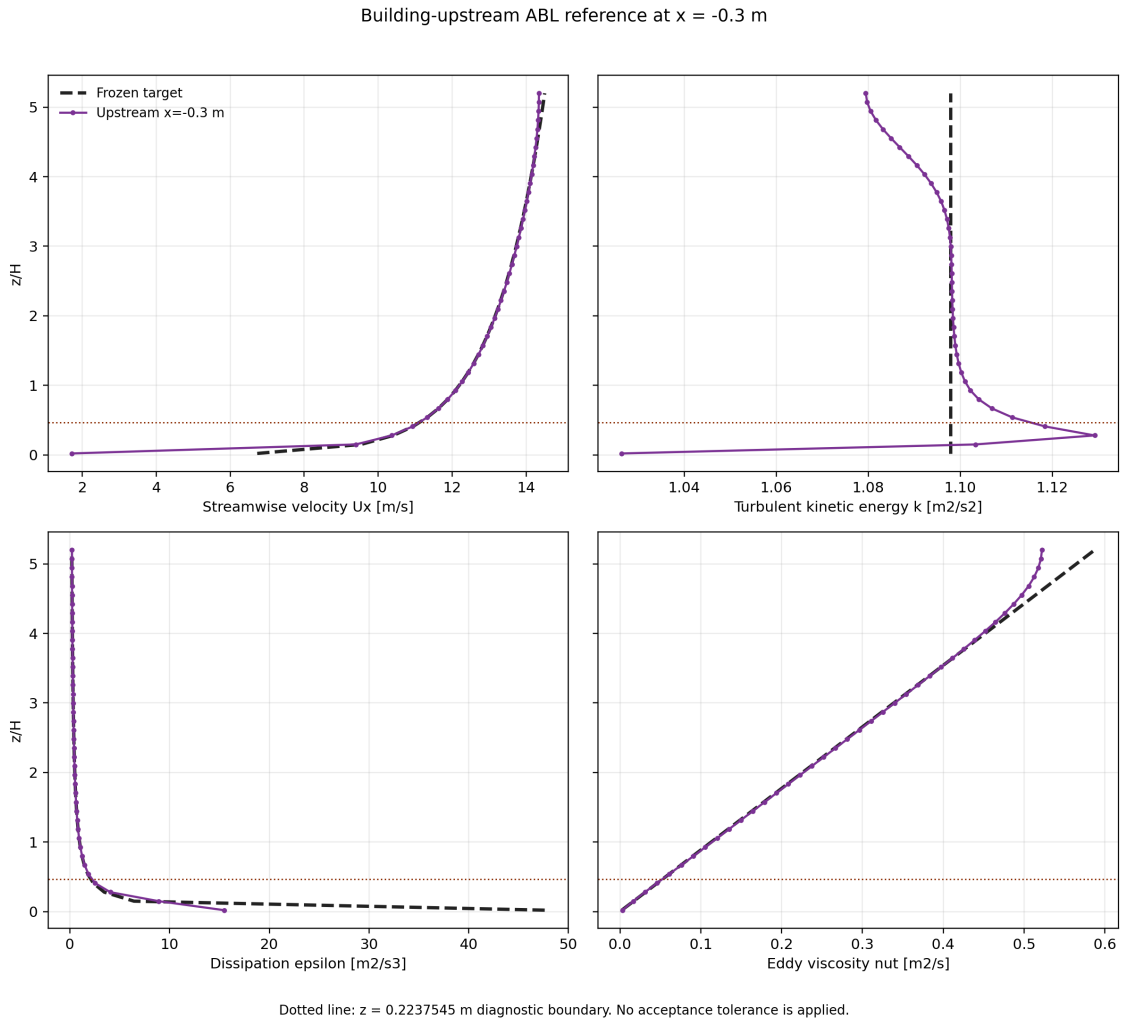


Figure 82: Building upstream location versus target profile comparison. PETW original project figure; this comparison is an inlet preservation check, not a comparison with wind tunnel measurements.

computational domain, turbulence closure and statistical window. Changing the grid while simultaneously changing the inlet or executed time span makes it impossible to attribute differences to the grid itself.

Turbulence closure and inlet turbulence likewise cannot be merged into a single setting. Inlet turbulence describes how upstream unresolved fluctuations enter the target domain; the closure model describes how unresolved scales within the target domain affect resolved motions. Both can alter surface pressure fluctuations, but their physical roles differ. Large-scale non-stationary changes provided by meteorological fields also cannot be treated as a substitute for inlet turbulence, because event envelopes, coherent gusts and small-scale random fluctuations occupy different temporal and spatial scales. Current documentation is sufficient to prescribe that these variables should be recorded separately, yet insufficient to select a unique closure scheme or demonstrate that a particular reconstruction method is applicable to all events.

Pressure processing must retain three parallel representations: original dimensional values, reference pressure, and pressure coefficients. Original dimensional pressure supports direct integration; reference pressure defines the baseline for net pressure; pressure coefficients enable comparison under specific denominator conditions. If only pressure coefficients are retained, the downstream cannot recover original pressures when the reference velocity or density definition changes; if only absolute pressures are retained, it becomes difficult to identify the normalization

basis across different cases. The synchronized forces from Case02Up use a normalization reference density; its output can only be interpreted under that semantic. Directly labeling it as physical force would exceed the declared evidence ceiling of the data packet.

The spatial organization of floor force data packets also affects usability for Structural. Each facade band must carry height range, representative height, area, facet set and force component coordinates. When the number of bands changes, the downstream should not treat the same index as the same physical height, but should remap based on height ranges. Total force can be verified by summing floor forces; base moment can be reconstructed from floor forces and lever arms; however, torsional moment, which depends on the horizontal offsets and tangential components of each face, requires more complete facet or sub-face information. Consequently, the minimum completeness of the data packet depends on the downstream target response, and cannot be assessed merely by whether the file is readable.

Interpretation of current results employs a three-layer evidence identity framework. The first layer is execution evidence, indicating that a specific case produced output under the recorded configuration and time span; the second layer is diagnostic evidence, indicating that the output has undergone profile, clock, coordinate, grid or integration consistency checks; the third layer is comparison evidence, requiring comparison against analytical solutions, independent solvers or controlled experiments under common definitions. DFG, Case011Up and Case02Up have provided first-layer and partial second-layer information, but have not yet formed a third-layer closed loop covering building pressures and overall loads. Execution success does not equal diagnostic pass, and diagnostic pass does not equal external validation.

The Researcher's handoff decision should be made per object. The velocity field being usable for diagnosing inlet preservation does not mean the pressure from the same case is usable for Structural; floor force file format passing does not mean the magnitudes carry physical force semantics; the data being complete for one response target does not mean another response target receives the same support. Handoff records must specify permitted uses, prohibited uses, case identity, time range, passed diagnostics and unclosed items. Only when this record accompanies the data packet can the downstream distinguish executable inputs from interpretable evidence.

The comparison path from WindTunnel to CFD remains a preparatory interface. Common geometry, common inlet time history, common pressure reference, common measurement point mapping and common statistical window all require item-by-item confirmation after the experiment is conducted. Wind tunnel design documents can provide the variable structure needed for future comparison, but cannot substitute for measured data. Existing empty domain and building cases from CFD can support refinement of interfaces and diagnostic scripts, but cannot be described as having already completed comparison with experiment. Long-duration HPC results likewise do not yet exist; the currently executed interval is insufficient to support conclusions about long-term statistical convergence and event tail behavior.

The CFD-to-Structural interface therefore retains a constrained candidate status. The core of the handoff is not an isolated total force curve, but a complete package containing source field, case configuration, geometry, coordinates, units, reference quantities, floor mapping, time axis, diagnostic status and prohibited inferences. Structural can use this package to check reading, resampling and response algorithms, but whether to use it for scientific response analysis remains a decision for the Researcher based on input quality. This interface discipline allows progress in numerical execution to be utilized by the downstream while preventing unclosed aerodynamic

evidence from being erroneously written as structural design input.

20.3.2 Interpretation Framework for Case-Level Diagnostics

The DFG cylinder, tall prism empty domain, tall prism with building, and D420 interface package each examine different problems. The cylinder case uses simplified geometry to isolate flow-around and discretization settings; the empty domain case isolates inlet profiles and ground boundary; the domain with building checks pressure and floor force extraction; and D420 checks the cross-scale contract from meteorological output to CFD input. These four categories of objects do not establish a single case series ranked by maturity. When presenting cylinder results, empty domain ABL and building floor forces together, the text must state the diagnostic object of each; the pass status of one case cannot be used to endorse another case.

In the DFG results, drag and lift serve different diagnostic roles. Drag primarily reflects mean and low-frequency effects in the approach flow direction; lift is more readily affected by wake shedding, lateral grid and time advancement. The current lift amplitude sensitivity to spatial resolution shows that its statistical quantities have not yet formed a stable comparison baseline. Even if the drag curve appears stable during a given time segment, it cannot be inferred that lift, surface pressure or tall building loads have achieved mesh independence. Diagnostics must retain the reference area, reference velocity, sampling interval and initial transient treatment for each coefficient.

The objective of the empty domain ABL is not to produce building loads, but to check whether the target profile is preserved after the inlet. Consistency at the inlet point does not represent consistency within the computational domain interior; consistency at the building upstream location also does not represent that the full height range is resolved. Streamwise drift, resolved height and iteration evolution provide complementary evidence: drift reveals spatial propagation error, resolved height reveals grid support range, and iteration evolution reveals whether the current output is still controlled by initialization. If the three diagnostics give different signals, the conflict should be retained and its cause identified, rather than selecting the most favorable one as the overall conclusion.

In the domain with building, pressures must first undergo surface mapping audit. The normals, areas and groupings of grid faces versus geometric faces determine the direction of pressure integration; if face grouping has omissions or duplications, systematic bias will appear in total force. Facade band division also requires checking that each facet enters exactly one height band, and when band boundaries do not align with structural floor boundaries, the mapping rules must be preserved. Total force reconstruction, sub-face force summation and facet pressure integration should be compared under the same reference pressure to avoid apparently unclosed differences caused by reference value changes.

Case02Up floor force output provides synchronized arrays and along-height diagnostics, but synchronization only shows that all floors share a time axis. It does not automatically show that each floor force's unit, reference density, band area and coordinate direction satisfy downstream physical interpretation requirements. Heat maps are suitable for identifying inter-floor anomalies, missing data or local abrupt changes; profiles are suitable for comparing along-height distributions; envelopes are suitable for checking extreme value ranges within the executed interval. All three types of plots are produced from the same data but answer different questions, and the number of plot types cannot be treated as independent verification evidence.

The choice of executed time span directly affects statistical quantities. The initial transient

may include inlet field establishment, pressure field adjustment or structural wake development; the statistical segment must be diagnostically separated from the transient. Current short-duration execution can support checks of the data pipeline and transient behavior, but cannot support long-term means, low-frequency spectra, extreme value stability or event tail statistics. Before extending the computation time span, the inlet contract and grid diagnostics should first be fixed; otherwise, a longer run would only accumulate more output on unclosed settings.

Convergence must distinguish algebraic convergence at each step, time advancement stability, statistical quantity stability, and grid convergence. Residual decrease belongs to the algebraic level; non-divergent time history belongs to the advancement level; mean or spectral stability with the statistical window belongs to the statistical level; target quantities converging across multiple grids belongs to the grid level. The four levels may fail simultaneously or pass partially. “Completed” in the report should explicitly indicate which level is meant, avoiding writing a normal job termination as complete convergence.

Boundary condition audit includes mass conservation and momentum semantics. Fluxes at inlet, outlet, lateral boundaries and top should be checked under the current discretization and compressibility assumptions; when the building and ground are solid walls, wall treatment and roughness semantics must also be confirmed. Transient inlets cause instantaneous flux to vary with time, so reasonableness cannot be judged from a single aggregate quantity. If outlet or lateral boundary feedback affects the building region, this needs to be identified through domain size or boundary sensitivity diagnostics rather than attributing all variation to the approach flow event.

The choice of pressure reference also affects cross-case comparison. A fixed reference is suitable for maintaining a uniform denominator within the same event; a time-varying reference can highlight local pressure changes relative to instantaneous approach flow; a height reference is used when profiles exhibit significant vertical variation. Numerical differences among the three coefficient types may originate from the denominator rather than the surface pressure itself. The report prioritizes dimensional pressure and presents the reference type alongside any coefficient usage precisely so that readers can distinguish physical changes from normalization changes.

Integration from pressure to overall effects is not information-neutral. Facet integration suppresses some local high-frequency variations while potentially retaining or amplifying common variations through spatial coherence; along-height moments further weight high-elevation contributions through lever arms. Small mean pressure errors do not guarantee small fluctuating pressure errors, and small local pressure errors do not guarantee small torsional moment errors. The verification ladder is graded by output object to require each level to have its own comparison quantity, rather than assuming that errors will naturally vanish after integration.

Source field uncertainty must be separated from CFD numerical uncertainty. The position, timing, resolution and physical parameterization of CM1 or WRF determine the large-scale structures entering CFD; interpolation, inlet reconstruction, grid and turbulence closure determine numerical transfer within the building domain. If output is inconsistent with future experiments, controlled branches are needed to determine whether the difference originates from the source field or CFD. A comparison that simultaneously changes the source field and solver settings cannot complete such attribution.

The significance of the D420 three-branch plan lies in establishing comparison under common driving conditions, while the existing historical M0 and M1 do not satisfy this condition. The frozen static package retains case identity, variable semantics and plans awaiting execution, and

can support checks of input consistency for future runs. It has not submitted new solves nor produced new output, so the text can only describe plans and interface preparation. Separating package-only rows from solver result rows is a necessary step in evidence identity management.

WindTunnel comparison requires commonality in measurement points and statistical definitions. CFD surface fields can be sampled at measurement point locations or integrated by pressure bands, but the two comparison types have different spatial support. Experimental pressure channels have sampling, tubing and synchronization characteristics; CFD output has time step, write-out interval and filtering characteristics. Future comparison should first align the time basis and spatial support, then compute mean, fluctuation and time history metrics. Currently no measured pressures or PIV are available to conduct this work; existing design drawings only support establishing a comparison schema.

Data reproducibility also requires layered storage of processing scripts and raw output. Raw solver files record case results; extraction scripts define variables and regions; analysis scripts generate diagnostic quantities; plotting scripts handle presentation. Any change at any layer should produce a new identity rather than overwriting upstream artifacts. This approach allows determination of whether a change in a figure originates from a new solve, re-extraction, or merely a plotting modification. A complete reproducibility path still does not elevate scientific status, but it makes subsequent independent verification possible.

The Researcher at the CFD stage must separately decide case purpose, allowable input range, resource execution, diagnostic acceptance and downstream handoff. Resource execution decides whether a given plan is launched, which is not equivalent to accepting its scientific assumptions; diagnostic acceptance decides which types of analysis the output is usable for, which is not equivalent to approving external validation; downstream handoff decides whether a given data packet may serve as a candidate input, which is not equivalent to approving structural or design use. Recording these decisions separately prevents a single operational authorization from being misread as scientific approval.

The ordering of near-term work should aim to reduce non-attributable differences. First close the empty domain ABL and coordinate-time contracts; then fix the grid and pressure reference for building cases; subsequently extend the executed time span and check statistical stability; and finally, once WindTunnel data are available, establish comparison under common conditions. If upstream source field identity conflicts still exist, retain multiple controlled branches rather than merging before the inlet. This ordering ensures each incremental piece of evidence has a clear scope of interpretation.

The figures, tables and equations in this chapter collectively define a case-bounded evidence package. Figures display diagnostic objects; tables store parameters, ladders and interfaces; equations specify variable relationships from pressure to force and moment. None of the three categories can independently establish validation; only when they are mutually consistent under a common case identity and compared with independent evidence can they potentially support higher-level judgments. The current state remains at case-limited model output and interface preparation, with no completed long-duration HPC results.

For external readers, the most likely misreading is interpreting “output exists” as “validation exists.” This chapter maintains the necessary interpretive conditions by repeating the constraints of case, grid, closure, domain and time near each figure and table, and by distinguishing package-only, diagnostic output and solver results. These constraints are not appended disclaimers but

part of the definition of numerical results. If any one of them changes, the original result no longer belongs to the same comparison object, and a new case identity and corresponding diagnostics must be established.

20.3.3 Current Scientific Limitations and Future Research

Current evidence is constrained by several unclosed conditions. The synchronized forces from Case02Up LayerForce50 use `rhoInf`=1, representing synchronized force output under normalization semantics rather than physical forces or design loads. The DFG campaign exhibits lift-amplitude spatial sensitivity, and mesh independence cannot be claimed. The Case011Up empty ABL profile contract is not closed; the inlet profiles from this case cannot yet serve as accepted ABL reference conditions. The driving conditions of the historical D420 M0/M1 are inconsistent; this data set cannot serve as controlled comparison. In the end-to-end chain, source field variable contracts, interface turbulence reconstruction, grid convergence and error propagation from pressure to moments have not been simultaneously closed on the same downburst event; therefore, the results of any single case can only be interpreted as source-constrained candidate diagnostic evidence. The Researcher has not yet selected a final solver scheme; long-duration HPC computation has not been completed; the executed time spans are insufficient to assess statistical convergence or long-term behavior of time integration errors.

Future research requires completing the following specific items: closing the Case011Up empty ABL profile contract, establishing quantitative diagnostics for lift-amplitude spatial sensitivity, unifying `rhoInf` and pressure reference semantics across cases, and level-by-level verification of floor forces, base shear, overturning moments and torsional moment with independent comparison targets. Until these conditions are completed, CFD output retains its candidate research input status, and force data packets entering Structural Agent continue to carry pressure reference semantics and normalization denominator documentation.

Evidence and Formula Boundaries of CFD Agent Chapter 20 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "CFD Agent." Content without physical equations belongs to the research methods, knowledge or coordination layer, and formalized formulas are not used to substitute for missing physical evidence. Chapter 20 remains a source-constrained synthesis and cannot support validated design wind, design loads, code recommendations, or scientific conclusions approved by PI

21 Structural Agent

Structural Agent is the Step 4c execution unit in the PETW pipeline, responsible for converting upstream pressure or floor force time histories into structural displacement, acceleration, internal force and overall response, and handing off response metrics and evidence boundaries to Design-Load in a controlled manner. Upstream inputs are pressure and force data packets from CFD or WindTunnel, and analytical load models; MATLAB supports computation and plotting. This chapter reports the state of the agent as of 2026-08-12 in seven sections covering scientific purpose, knowledge base, Researcher involvement and decisions, workflow and interactions, completed work and results, key formulas, and current scientific limitations and future research. All results remain not_scientifically_approved and are model-internal benchmark output; they do not establish validated full-building response, design response, design guidance, code recommendations, or η value adoption conclusions.

21.1 Scientific Questions, Knowledge, and Researcher Decisions

21.1.1 Scientific Purpose

The scientific objective of the Structural Agent is to assess structural dynamic response under mechanism-specific wind fields. This Agent does not itself generate wind fields, nor does it adopt design loads; it converts Windfield, CFD, WindTunnel or analytical loads into structural dynamic response, response metrics, uncertainty evidence and controlled handoff to DesignLoad. Response metrics include displacement $x(t)$ (unit: m), velocity $\dot{x}(t)$ (unit: m/s), acceleration $\ddot{x}(t)$ (unit: m/s^2) time histories and their peak and RMS values, background and resonant decomposition within the evolutionary power spectral density (EPSD) framework, non-stationary extreme value distribution and peak factor g_{ns} (dimensionless), generalized gust-front factor G_{GFF} (dimensionless), and response-specific equivalent static wind load (ESWL) candidate quantities (unit: N).

The literature background delineates comparable benchmarks. The deterministic-stochastic hybrid downburst model[Chen and Letchford, 2004a] and the empirical downburst profile model[Holmes and Oliver, 2000] form the method basis of the analytical load chain; the Chen and Letchford [2004b] parametric study of CAARC building along-wind response provides the comparison target for current reproduction; the comparison of spectral methods and time-domain methods for thunderstorm outflow structural response[Solari and De Gaetano, 2018] delineates the position of method divergence. These method evidence items support constructing benchmarks and comparison metrics, but response assessment itself must be executed on the structural models and parameter combinations of this study.

21.1.2 Knowledge Base

The knowledge layer of Structural Agent centers on the Step4c Wiki namespace, totaling 413 files, including 192 Zotero metadata identities, 146 detailed readings, 146 accepted semantic records, 14 out-of-scope pending items, 22 metadata-only items, and 10 blocked items. The Researcher has approved this knowledge layer only for evidence-status publication and for Claims 1 through 6 literature-supported background and methods-governance purposes, and has not approved it as a basis for scientific results, design guidance, η values or code recommendations of this study. The review asset is the Step4c review, 45 pages each in Chinese and English, covering 146 unique sources and 458 source-binding occurrences.

Program assets include the 46-layer shear chain, 18-element Euler-Bernoulli beam, Chen-Letchford non-stationary load model, PETWLoadPacket/v1 data packet, Newmark-β solver, EPSD pathway, non-stationary extremes module, G-GFF module, and response-specific ESWL output. Knowledge layer and program layer remain separated; program output does not automatically enter the literature knowledge layer; accepted records in the literature knowledge layer also do not directly serve as program acceptance targets.

21.1.3 Researcher Involvement and Decisions

The Researcher's (Liang Hu / PI decisions at this stage are concentrated in three categories. The first is benchmark selection: using v0.12 CAARC beam-18 as the current shared scientific benchmark, with displacement reproduction confirmed by the Researcher; d_0=2900 m is not a value explicitly given in the literature but was identified from the main peak at 150 to 170 s and secondary peak at 330 to 350 s in Chen and Letchford [2004b] Figures 12 through 14; this value places the model main peak at approximately 159 s (evidence status: candidate interpretation). The second is reproduction discipline: literature reproduction does not use post-hoc correction factors, with differences directly reported and root causes sought; after the internal consistency

audit of velocity and acceleration, the Researcher decided that the literature displacement column and incoherent acceleration can serve as external anchor points, while the velocity and acceleration under Davenport and full coherence are not used as program acceptance targets. The third is usage boundary: only evidence-status publication and limited use for Claims 1 through 6 are approved; scientific result, design guidance, η values and code recommendations are not approved.

The conditions for results to enter DesignLoad include an end-to-end closed loop with real upstream PETW load packets and cross-verification with an external independent solver; both are currently unmet. Candidate interpretations of method comparisons still await the Researcher's final decision.

21.2 Methods, Results, and Formulas

21.2.1 Workflow and Interactions

The program is organized as callable closures, with the chain proceeding from structural configuration and load source through adapter verification into four methods: TDA (time-domain analysis), EPSD (evolutionary power spectral density), extremes (non-stationary extremes) and G-GFF (generalized gust-front factor), producing DownburstResponseResult/v1 result packages, array files, figures and QA records. Structural configuration automatically identifies model type from YAML, supporting both the 46-layer shear chain and the 18-element Euler-Bernoulli beam.

The load model implements the Wood vertical profile, Holmes-Oliver time function[Holmes and Oliver, 2000], Kaimal spectrum, Davenport or constant coherence, and single-index and Deodatis double-index spectral representations. The Wood vertical profile describes the mean wind speed distribution as a function of height z (unit: m) with $U(z)$ (unit: m/s); the Holmes-Oliver time function describes the temporal evolution of wind speed in a transient downburst event with $U(t)$ (unit: m/s) as a function of time t (unit: s), with characteristic parameter d_0 (unit: m) controlling event duration and spatial scale. The Kaimal spectrum provides turbulent velocity power spectral density as $S_u(f)$ (unit: $\mathrm{m^2/s^2/Hz}$), where f (unit: Hz) is frequency. Ensemble statistics use regular trajectories and 64-seed extreme trajectories, avoiding the use of single random realization statistics as verification gates.

PETWLoadPacket/v1 serves as a source-neutral interface; upon reading, it checks time, height, shape, SI units and provenance metadata. Current data packet primarily supports TDA; the EPSD and extremes pathways require uniform modulation decomposition information, and requesting G-GFF automatically enables EPSD and extremes in the pipeline. Output can be read back in JSON, CSV, NPZ and NPY formats. Force data packets from WindTunnel and CFD enter through the same interface; DesignLoad receives response evidence, metrics and ESWL candidate quantities while retaining structural, load, target response and verification scope information.

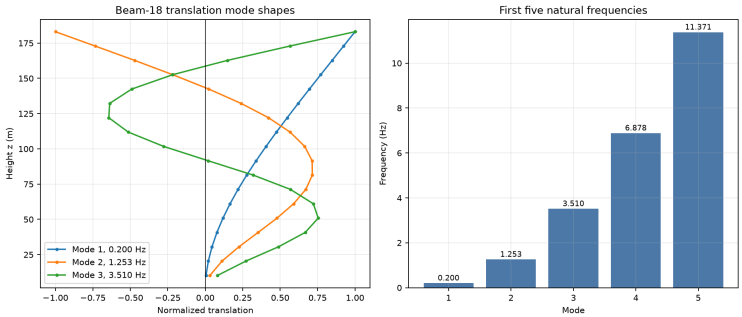


Figure 83: 18-element Euler-Bernoulli beam structure and mode shape schematic. PETW original figure, total height approximately 183 m, fundamental frequency 0.20 Hz, uniform modal damping 1.5%.

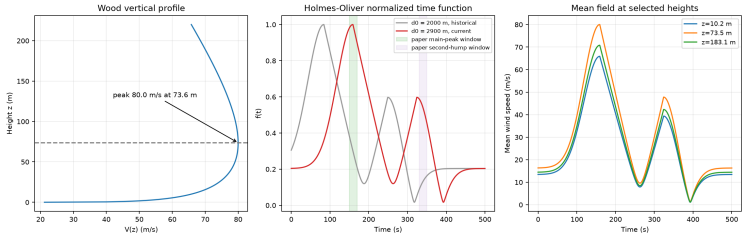


Figure 84: Time-varying mean wind field composed of the Wood profile and Holmes-Oliver time function, d_0=2900 m. PETW original figure.

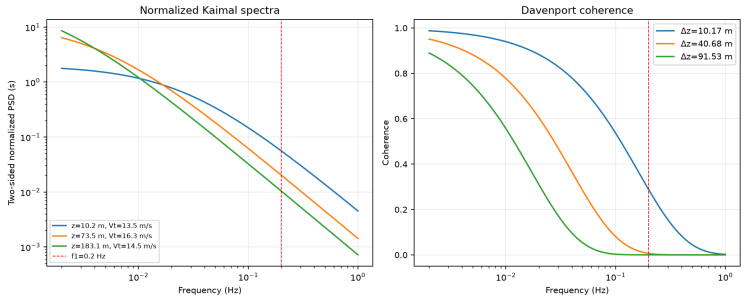


Figure 85: Kaimal spectrum with three coherence cases: incoherent, Davenport, and 0.99 constant coherence. PETW original figure.

21.2.2 Completed Work and Results

The program evolved through versioning to the current v0.15 callable closure; on-site re-verification on 2026-08-12 passed all 6/6 verification gates. Table 29 presents the structural model parameters adopted for the current scientific benchmark v0.12.

Table 29: v0.12 benchmark structural model parameters. The model type is an 18-element Euler-Bernoulli beam; the parameter combination constrains the applicability of all results.

Parameter	Symbol	Value	Unit
Total height	H	≈183	m
Moment of inertia	I	1215	m^4
Mass density	ρ_s	160	kg/m^3
Fundamental frequency	f_1	0.20	Hz
Uniform modal damping ratio	ζ	1.5	%
Equivalent elastic modulus	E	≈26.5	GPa
Holmes-Oliver d_0	d_0	2900	m
Time step	Δt	0.1	s

The 16-seed mean maximum displacement differences relative to Chen and Letchford [2004b] Table 2 for three coherence cases are +1.7%, −2.7% and −4.3%, without post-hoc correction factors (Table 30).

Table 30: v0.12 benchmark maximum displacement compared with Chen and Letchford [2004b] Table 2. Displacement unit is m; conditions are CAARC beam-18, d_0=2900 m, 1.5% uniform modal damping and 16-seed ensemble mean; evidence status: candidate benchmark verification.

Coherence case	v0.12 16-seed mean	Table 2	Difference
Incoherent	1.068	1.05	+1.7%
Davenport	1.265	1.30	−2.7%
Fully coherent	1.809	1.89	−4.3%

The method comparison under the Davenport case yields results from four pathways, summarized in Table 31. TDA 64-seed ensemble maximum displacement mean is 1.271 m (ensemble standard deviation 0.449 m). The EPSD-Poisson pathway yields 1.393 m, 9.6% above TDA. Using

only the deterministic mean load yields 0.848 m, 33.3% below TDA, indicating that the transient mean pulse at the current structural fundamental frequency has a dynamic amplification factor of 0.9995; this analytical mean pulse is approximately quasi-static for the current tall structure. The classical stationary peak factor pathway yields 1.887 m, 48.5% above TDA, indicating that directly applying stationary assumptions to a non-stationary event overestimates the response.

Table 31: Comparison of maximum displacement across four method pathways under the Davenport case. Conditions are CAARC beam-18, d_0=2900 m, 1.5% uniform modal damping; evidence status: candidate methodology comparison, limited to this benchmark and parameter combination.

Method pathway	Maximum displacement (m)	Relative to TDA	Evidence status
TDA 64-seed mean	1.271	Baseline	candidate
EPSD-Poisson	1.393	+9.6%	candidate
Deterministic mean load	0.848	−33.3%	candidate
Stationary peak factor	1.887	+48.5%	candidate

Accompanying outputs include EPSD peak standard deviation of 0.3764 m (occurring at approximately 159.7 s), non-stationary peak factor $g_{\mathrm{ns}} = 1.456$ (dimensionless) and stationary factor $g_{\mathrm{s}} = 2.774$ (dimensionless). The resonant variance fraction at peak time is approximately 0.198, indicating that the background component dominates under the current parameter combination.

The v0.15 Davenport benchmark output is $G_{\mathrm{GFF}} = 1.64985$ (dimensionless), with mean pulse component 1.00211, fluctuating component 1.64637, and B/R variant 1.65997. G_{GFF} and ESWL are bound to the roof displacement target and cannot be transferred to base bending moment, member internal forces or cladding pressures; the current value is a method verification output, not a design coefficient. The technical volume contains 8 calculation cases, 30 equation environments and 13 result figures; numerical verification confirms consistency among v0.12 CSV and result JSON, EPSD peak, G-GFF product and 18-node ESWL reconstruction.

The v0.13 internal consistency audit yields an incoherent first-mode acceleration difference of +3.8%, which can serve as a limited external anchor point; the velocity and acceleration under Davenport and fully coherent conditions cannot be simultaneously reproduced through published model specifications and therefore are not used as program acceptance targets.

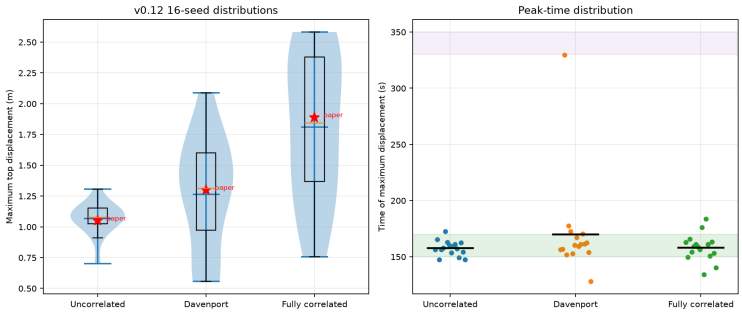


Figure 86: v0.12 displacement ensemble and main peak timing for three coherence cases. PETW original figure, comparison target is Chen and Letchford [2004b] Table 2.

21.2.3 Key Formulas

The equation of motion for the multi-degree-of-freedom structure applies the mass matrix $\mathbf{M}$ (unit: kg), damping matrix $\mathbf{C}$ (unit: N·s/m) and stiffness matrix $\mathbf{K}$ (unit: N/m) to the displacement vector $\mathbf{x}(t)$ (unit: m), balanced by the time-varying load vector $\mathbf{f}(t)$ (unit: N):

$$\mathbf{M}\,\ddot{\mathbf{x}}(t) + \mathbf{C}\,\dot{\mathbf{x}}(t) + \mathbf{K}\,\mathbf{x}(t) = \mathbf{f}(t) \tag{29}$$

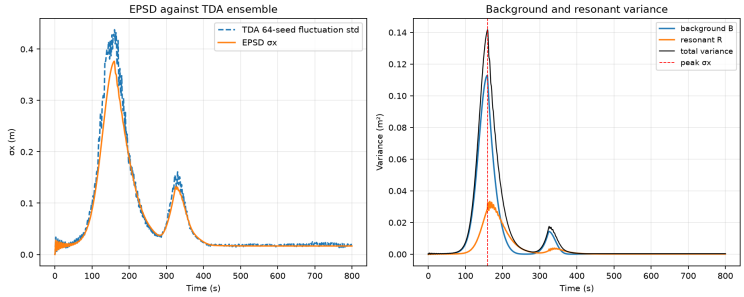


Figure 87: Background and resonant decomposition of the evolutionary spectral pathway, with peak occurring at approximately 159.7 s. PETW original figure.

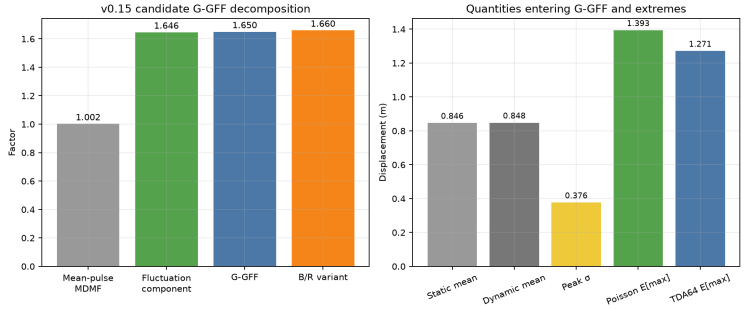


Figure 88: G-GFF mean pulse component and fluctuating component decomposition. PETW original figure, Davenport benchmark, $G_{\mathrm{GFF}} = 1.64985$.

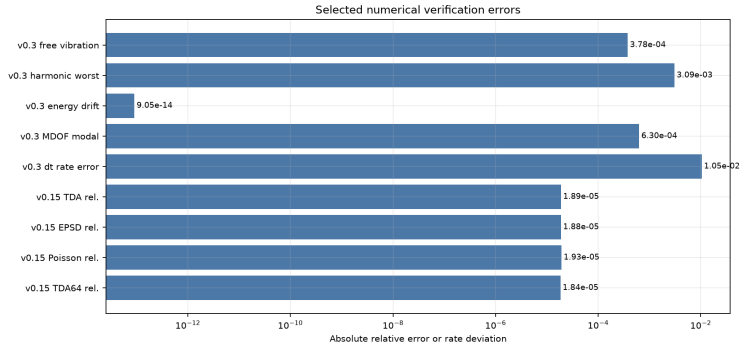


Figure 89: Verification gate errors compared with tolerances. PETW original figure, v0.15 on-site re-verification passed all 6/6 gates.

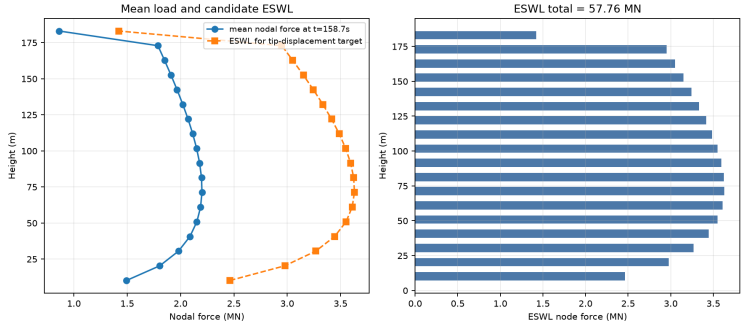


Figure 90: Floor force profile and ESWL nodal force reconstruction bound to roof displacement target. PETW original figure, 18-node beam.

where $\dot{\mathbf{x}}(t)$ (unit: m/s) is the velocity vector and $\ddot{\mathbf{x}}(t)$ (unit: m/s^2) is the acceleration vector. Modal transformation projects physical coordinates to mode shape space:

$$\mathbf{x}(t) = \Phi\,\mathbf{q}(t), \qquad m_j\,\ddot{q}_j(t) + c_j\,\dot{q}_j(t) + k_j\,q_j(t) = \phi_j^{\top}\mathbf{f}(t) \tag{30}$$

where Φ (dimensionless) is the modal shape matrix, $\mathbf{q}(t)$ is the generalized coordinate vector, $m_j = \phi_j^{\top}\mathbf{M}\phi_j$ (unit: kg) is the j-th modal mass, c_j (unit: N·s/m) and k_j (unit: N/m) are defined analogously, and ϕ_j is the j-th mode shape vector. Time-domain integration uses the general form of Newmark-β recursion:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t\,\dot{\mathbf{x}}_n + \tfrac{\Delta t^2}{2}\left[(1-2\beta)\,\ddot{\mathbf{x}}_n + 2\beta\,\ddot{\mathbf{x}}_{n+1}\right] \tag{31}$$

$$\dot{\mathbf{x}}_{n+1} = \dot{\mathbf{x}}_n + \Delta t\left[(1-\gamma)\,\ddot{\mathbf{x}}_n + \gamma\,\ddot{\mathbf{x}}_{n+1}\right] \tag{32}$$

where Δt (unit: s) is the time step, β (dimensionless) and γ (dimensionless) are Newmark parameters, and subscripts n and $n+1$ denote the current step and next step respectively. The program adopts the constant average acceleration method ($\beta = 1/4$, $\gamma = 1/2$), which is unconditionally stable for linear systems, with a time step of 0.1 s.

The evolutionary spectral pathway represents the non-stationary response as a modulated stationary process, with the evolutionary power spectral density as

$$S_{xx}(t,\omega) = |\mu(t,\omega)|^2\,S_{xx}(\omega) \tag{33}$$

where $S_{xx}(t,\omega)$ (unit: m^2/s^2/Hz) is the time-varying spectral density, $\mu(t,\omega)$ (dimensionless) is the modulation function, $S_{xx}(\omega)$ (unit: m^2/s^2/Hz) is the base stationary spectral density, and ω (unit: rad/s) is the angular frequency. Evolutionary spectral modeling of thunderstorm outflow has a corresponding framework in the literature[Roncallo and Solari, 2020]. Non-stationary extremes through the time-varying upcrossing rate $\nu_x(x,t)$ (unit: 1/s) and Poisson model yield the conceptual form of the event maximum distribution:

$$P\left(\max_{0\le t\le T} x(t) \le x\right) \approx \exp\left(-\int_0^T \nu_x(x,t)\,dt\right) \tag{34}$$

where T (unit: s) is the event duration. Methods for estimating time-varying upcrossing rates and non-stationary peak factors have been specifically studied in the literature[Huang and Gu, 2019, Li et al., 2023].

The Davenport exponential coherence model defines the spatial coherence function between two heights z_1 and z_2 (unit: m):

$$\gamma(f,\Delta z) = \exp\left(-\frac{C_z\,f\,|z_1 - z_2|}{\bar{U}}\right) \tag{35}$$

where C_z (dimensionless) is the vertical decay coefficient, f (unit: Hz) is frequency, and $\bar{U}$ (unit: m/s) is the mean wind speed at the two heights. The incoherent case sets $\gamma = 0$; the constant coherence case sets $\gamma = 0.99$; the choice among the three cases affects ensemble response statistics and serves as an independent variable in the method comparison.

The generalized gust-front factor is implemented according to Kwon and Kareem [2013]

Eq. (8), organizing peak load effects into a product structure of mean pulse component G_{mean} (dimensionless) and fluctuating component G_{fluc} (dimensionless)[Kwon and Kareem, 2009]:

$$G_{\text{GFF}} = G_{\text{mean}} \times G_{\text{fluc}} \tag{36}$$

This quantity is bound to a specific target response (currently roof displacement) and cannot be transferred to base bending moment, member internal forces or cladding pressures. The current Davenport benchmark $G_{\text{GFF}} = 1.64985$, with $G_{\text{mean}} = 1.00211$ and $G_{\text{fluc}} = 1.64637$; all are model-internal method verification output, not design coefficients.

21.3 Evidence Audit, Interfaces, and Limitations

21.3.1 Load Identity, Response Targets, and Verification Levels

The first substantive task of Structural is not solving the equation of motion but confirming the physical identity of the input loads. Surface pressure, facade band force, floor resultant, generalized force and equivalent static wind load are objects at different levels. Surface pressure retains local spatial distribution and can support different integrations for cladding and overall loads; facade band forces have already undergone one spatial compression; floor resultants further depend on bandwidth and floor mapping; generalized forces additionally introduce mode shape projection; ESWL is further bound to a specific response target for inversion. If the input object has already lost local spatial information, Structural should not recover nonexistent details through interpolation, nor apply the roof displacement equivalent load to member pressure or torsional response.

Input data packets must treat numerical arrays and semantic metadata as jointly required content. Numerical arrays contain time, spatial position and force components; metadata contain source lane, case identity, coordinate system, units, reference quantities, normalization method, sampling interval, missing data treatment and permitted uses. Consistent array shape proves only that the reading interface is executable, not that physical quantities are consistent. For example, two files may both be represented as time-by-floor matrices, but one may be dimensional force while the other may be solver output normalized by reference density. If the metadata do not declare this difference, the structural response can still be computed numerically but loses interpretable physical scale.

Temporal processing must distinguish original sampling, resampling and integration time step. The upstream load sampling interval determines the resolvable frequency range; the structural solver time step determines integration accuracy; the output interval determines the saved response detail. The three may differ, but any resampling must document the filtering and interpolation rules. For non-stationary events, a uniform time step is not merely programmatic convenience but relates to peak timing, envelope variation and transient frequency content. If the upstream time axis has interruptions or duplications, direct infilling would alter the upcrossing rate and peak statistics; therefore, data issues must first be reported, with the Researcher deciding whether to permit constrained repair.

Spatial mapping must preserve the relationship between load application heights and structural degrees of freedom. CFD facade band centers, WindTunnel measurement points or pressure bands, and analytical model continuous height functions all need to be mapped to structural nodes or elements. The mapping must conserve the target quantity: total force conservation when checking force sums; overturning response additionally requires checking the moment about the base;

torsion requires preserving in-plane eccentricity. Checking only total force may mask along-height distribution errors, while checking only nodal values may violate overall equilibrium. The current 18-element beam and 46-layer shear chain use different discretization objects; their load mapping rules cannot directly be interchanged by node index.

Structural model identity is jointly determined by mass, stiffness, damping, boundary conditions and degrees of freedom. The same fundamental frequency does not guarantee that two models have the same mode shapes, generalized mass or higher-mode contributions. The CAARC beam-18 benchmark provides a model-internal comparison target under a specific set of parameters. It can support checks of the program's reproduction of known displacement anchor points, but cannot represent all tall buildings. The model parameter table, mode shape figures and response figures must be read together; any standalone displacement value does not carry sufficient structural identity information.

Damping treatment also requires preserving its source and form. Current benchmark adopts uniform modal damping ratio; the damping matrix in the linear equation of motion is constructed from this assumption. For the combined input of non-stationary pulse and random fluctuation, damping simultaneously affects transient decay, resonant bandwidth and peak statistics. Changing the damping ratio changes the response, but this parametric sensitivity is not equivalent to determining the actual structural damping. Subsequent real building analysis must audit the source, applicable amplitude range and uncertainty of damping together with the structural model.

The time-domain pathway and the EPSD pathway answer different diagnostic questions. The time-domain ensemble directly produces random realizations and can check individual time histories, ensemble dispersion and the feasibility of nonlinear extension; the EPSD pathway computes second-order statistics under prescribed modulation, spectral and coherence assumptions and can decompose background and resonant contributions. When the two pathways produce similar results, this only indicates method consistency under the current model and statistical definition; when results disagree, the modulation separability, spectral discretization, coherence representation, ensemble size and peak model need to be checked. Differences between methods cannot directly be interpreted as physical mechanism differences.

The non-stationary extremes module organizes time-varying variance, time-varying derivative statistics and upcrossing rate into a candidate distribution for the event maximum. This method relies on process approximation, Gaussianity and upcrossing event treatment; applicability must be assessed jointly with the specific response. The process characteristics of roof displacement differ from those of acceleration, base bending moment or local internal force; a peak factor from one response cannot be transplanted to another response. Current difference between g_{ns} and g_{s} belongs to the model-internal method comparison of the Davenport benchmark, and cannot be interpreted as all thunderstorm wind responses having the same ratio.

The G-GFF pathway further compresses mean pulse and fluctuating effects into a response ratio. The compressed form facilitates comparison but also conceals time history shape, phase and spatial distribution. Accordingly, each G-GFF output must carry target response, baseline definition, structural model, load model and statistical interval. The current decomposition under the roof displacement target cannot automatically serve the base bending moment, nor can it become a cross-structure universal coefficient. When DesignLoad receives this result, it should receive the complete response identity rather than only a dimensionless number.

The purpose of ESWL is to construct a static load representation for a given target response.

It requires the static load to reproduce the dynamic peak or candidate effect under the selected response functional, and does not require reproducing the original time history response across all degrees of freedom. A roof displacement ESWL may be equivalent in roof displacement yet not equivalent in base shear, local internal force or acceleration. If multiple response targets are needed, ESWL should be constructed separately for each and their differences compared; the equivalence under one target cannot substitute for verification under another.

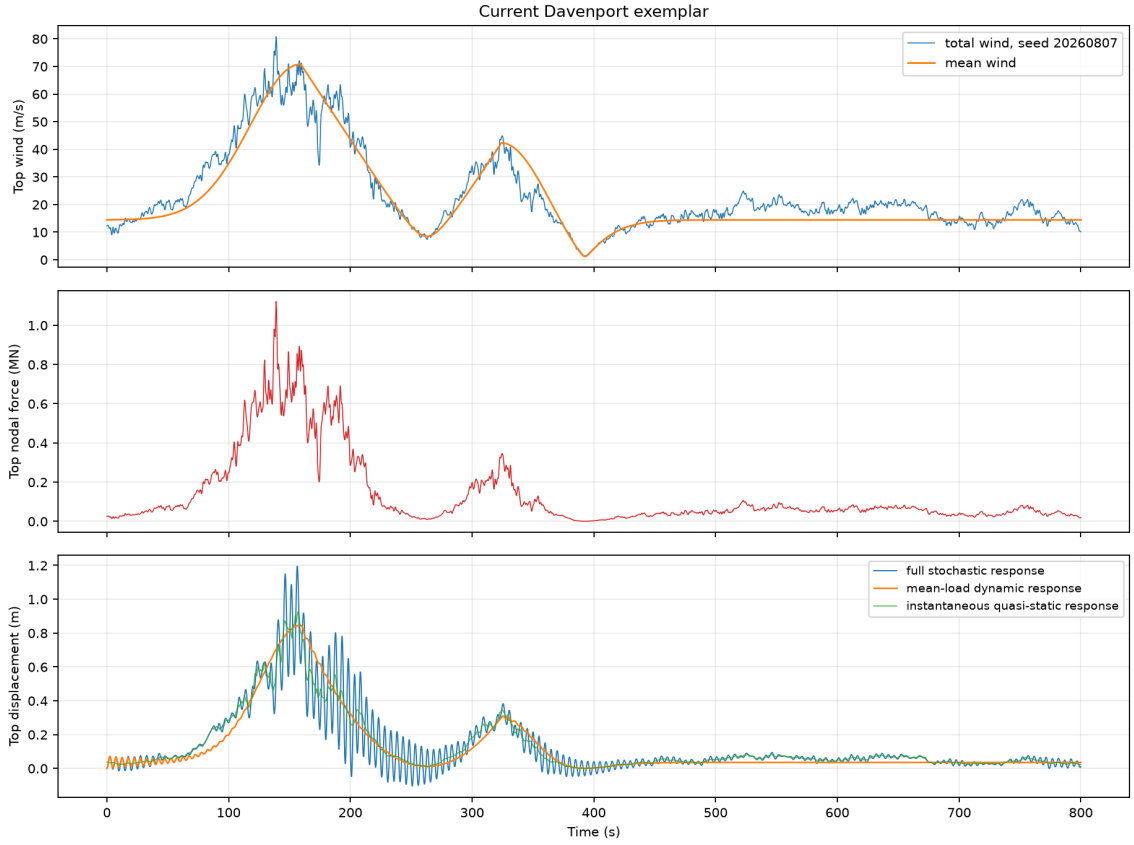


Figure 91: Exemplar wind, force and structural response time histories from the benchmark. PETW original project figure; each time history corresponds only to the current load model and structural parameter combination.

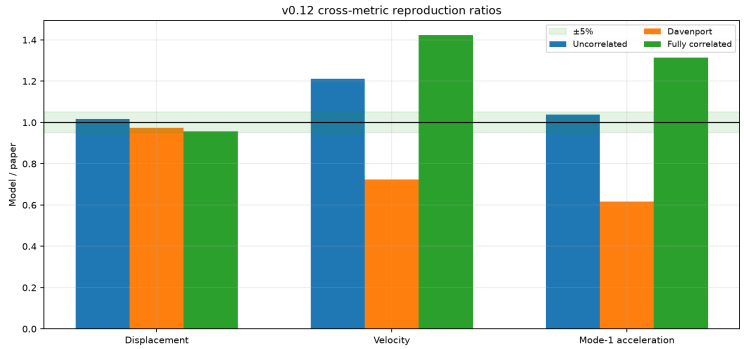


Figure 92: v0.12 ratio diagnostics between different response metrics. PETW original project figure; ratios are used for identifying metric dependence and do not support cross-response transfer.

Figures 91 through 94 provide supplementary displays of the hierarchy from load time histories, cross-metric differences to extreme value diagnostics. Exemplar time histories are used to check the temporal relationships among input, force and response; cross-metric figures are used to expose different proportions of displacement, velocity and acceleration within the same model; the original comparison figure is used to preserve the external anchor point context of the benchmark; the extreme value figure is used to check the candidate extreme value module. The four categories of

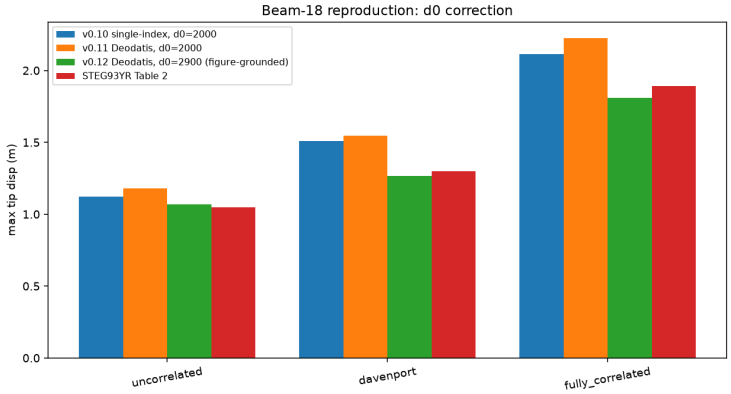


Figure 93: v0.12 original benchmark comparison figure. PETW original project figure; comparison identity is retained as model-internal benchmark.

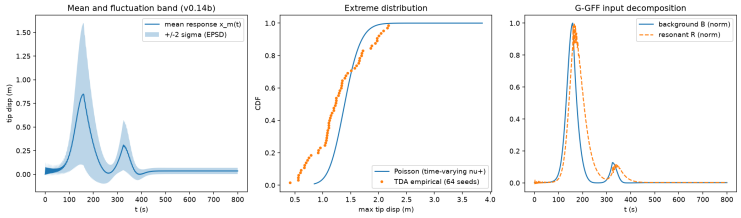


Figure 94: v0.14b non-stationary extremes pathway model-internal diagnostic. PETW original project figure; this does not show that real building extreme responses have been validated.

figures cannot be merged into a single unified validation because they constrain different objects.

Current benchmark validation should distinguish four levels: mathematical implementation, internal consistency, literature anchor point and real input. Mathematical implementation checks matrix assembly, integration recursion and output reconstruction; internal consistency checks whether time-domain, frequency-domain or independent reconstruction agree under common definitions; the literature anchor point compares model output with published benchmark specific responses; the real input level requires input from upstream governed data packets with independent response evidence. Existing work covers partial content of the first three; the fourth level has not yet been established. Writing 6/6 program gate passes as structural response validation would conflate program checking with scientific evidence.

Ensemble statistics must simultaneously report central tendency, dispersion and sample identity. The 16-seed and 64-seed trajectories serve different purposes: the former for the current literature anchor point comparison, the latter for method pathway ensemble statistics. Increasing the number of random seeds can improve statistical quantity estimates under the current model, but cannot reduce uncertainty from structural parameters, load models and upstream inputs. A single random realization also cannot substitute for ensemble results, as its peak and timing may be significantly affected by random phase.

Boundary conditions and initial conditions are part of the response identity. Fixed base, linear elasticity and small deformation assumptions constrain the current model; initial rest conditions constrain how the transient response develops from the event onset. If real event input already has wind-induced response before the recording starts, rest initial values may produce a startup transient inconsistent with the physical process. Subsequent data packets need to provide a sufficient lead-in period or an explicit initialization strategy to avoid incorporating artificial startup transients into event peak values.

Current work primarily covers along-wind overall response. Crosswind and torsion require ad-

ditional aerodynamic force components, spatial coherence and structural degrees of freedom; local member response requires higher spatial resolution pressure input; nonlinearity requires material or geometric models and corresponding integration and verification. The existing along-wind benchmark cannot be extrapolated to these objects. Explicitly listing the missing directions preserves the evidence boundary better than using a blanket "structural response" label.

The Structural-to-DesignLoad handoff should be completed through a response evidence package. The package must store structural model version, load data packet identity, target response, time-domain or EPSD method, ensemble settings, peak definition, result units, passed checks, unclosed items and prohibited uses. DesignLoad can use this information to organize candidate expressions, but must not rewrite model-internal benchmarks as design responses. The Researcher's handoff decisions should be recorded separately for each target response; permitting the roof displacement method comparison does not mean that base bending moment, member internal force or comfort metrics simultaneously receive handoff qualification.

The priority target for future real-input runs is not to pursue more output figures, but to close a minimal end-to-end object: select a single, source-identified upstream force data packet, complete unit and coordinate verification, establish a load-to-node conservation mapping, verify the same linear model with two independent implementations, and store a complete evidence package for one clearly defined response target. After this minimal object is completed, extension to additional directions, models and parameters becomes appropriate. Current documentation supports specifying this pathway but does not support claiming that end-to-end response has been completed.

21.3.2 Per-Item Interpretation of Model-Internal Benchmark

The current CAARC beam-18 results must be understood as bounded checks of program pathways. The benchmark fixes geometry, mass, stiffness, damping, mean wind model, time function, turbulence spectrum and coherence model as a single combination, and compares displacement anchor points within that combination. It does not use actual pressures or floor forces from PETW upstream events, and therefore cannot answer what response a specific thunderstorm wind event produces on a real building. The scientific role of the benchmark is to separate program errors, method differences and input uncertainty into layers, rather than to provide design response samples.

The identity of d_0 requires particular attention. This parameter originates from a candidate interpretation of the main peak and secondary peak timing in literature figures, not a calibrated value directly reported in the literature. The choice of d_0 affects the timing and duration of the mean wind pulse, and consequently affects the overlap between structural response and random envelope. The current value placing the model main peak consistent with the identified timing only shows that this candidate interpretation has usability within the current benchmark; it cannot be written as a universal event-scale parameter. If subsequent work changes d_0, the time function, peak timing and response changes should all be reported simultaneously.

The three coherence cases are used to isolate spatial correlation assumptions. Incoherent, Davenport and constant coherence represent, respectively, different vertical correlation structures; they are not three measurement results of the coherence of a real event. The 16-seed displacement mean being close to the literature table supports the current program's benchmark consistency on a specific displacement metric; the inability to simultaneously satisfy the same external anchor point for velocity and acceleration shows that the literature specification or implementation basis

still has unclosed parts. The report retains this inconsistency because it is more consistent with reproduction discipline than making all metrics appear to match through post-hoc correction.

Displacement, velocity and acceleration assign different weights to frequency content. Displacement is more influenced by low-frequency and quasi-static components; velocity and acceleration progressively emphasize higher frequency components. Accordingly, displacement anchor point agreement does not guarantee velocity or acceleration agreement. The v0.13 internal consistency audit provides only a limited external anchor point for the specified incoherent acceleration; other coherence cases still cannot serve as acceptance targets. This difference also shows that Structural handoff decisions must be made per response quantity rather than for the entire program.

The 64-seed time-domain ensemble mean and dispersion jointly describe the current stochastic model. The ensemble mean can be compared with the EPSD candidate peak; the ensemble standard deviation reveals the influence of random phase on event peaks. The difference between the EPSD-Poisson pathway and TDA may originate from peak models, modulation assumptions and numerical discretization, and cannot be simplified to one method being more accurate. The deterministic mean pathway being below the time-domain ensemble shows that stochastic fluctuations contribute to roof displacement under the current combination; the magnitude of this contribution remains bound to the current spectrum, coherence and structural model.

The stationary peak factor pathway being above the non-stationary pathway shows that directly applying stationary duration and peak assumptions changes the response estimate of the current benchmark. The comparison target here is computational methods, not real event observations. The non-stationary pathway similarly depends on the selected envelope and upcrossing rate approximation, and cannot be called more realistic simply because its result is lower than the stationary method. Independent event response evidence is needed to evaluate the physical performance of both methods.

The background and resonant decomposition provides model-internal diagnostics for response mechanisms. The resonant variance fraction at peak time reflects the combined effect of the current structural frequency, damping, spectrum and modulation, and cannot be generalized as a general ratio for tall building thunderstorm wind response. If the fundamental frequency, damping or spatial coherence changes, the background-resonant allocation may change. Figures should be read together with the parameter table to prevent readers from treating the decomposition ratio as a structural-class constant.

The G-GFF mean pulse and fluctuating decomposition provides compressed method metrics. The product form requires both components to use a common baseline, and the output target is fixed as roof displacement. The difference between the B/R variant and the main output belongs to a definition difference within the current model. Even if numerical reconstruction is consistent, this only proves that formulas, code and result files agree under common definitions, and does not support using them as design coefficients or cross-response parameters.

The ESWL reconstruction checks whether the target response can be reproduced by static nodal forces. The consistency of the 18-node reconstruction is internal closure evidence, depending on the mode shapes, flexibility and target response used. It does not prove that the real static load distribution has been identified, nor does it prove that the same nodal forces can reproduce other responses. When ESWL and the original floor forces are presented together, it should be clear that the former is a response-equivalent representation and the latter is an upstream aerodynamic force input; their physical identities differ.

The multi-modal multi-degree-of-freedom pathway is used to check the extensibility of higher modes and response recovery. Whether higher modes are important depends on the spatial distribution, frequency content and target response of the input load. Current project figures demonstrate that the program can organize this pathway, but the lack of real upstream data and independent cross-verification means that multi-modal thunderstorm wind response cannot be claimed as validated. Prioritizing closure of the minimal object for a single response remains a more auditable approach than simultaneously extending all modes and directions.

Structural uncertainty includes at least model form, parameters, damping, boundary conditions and linearity assumptions. Load uncertainty includes upstream field, aerodynamic force conversion, spatial coherence and time sampling. Method uncertainty includes time-domain ensemble, EPSD, extreme value model and ESWL construction. The three categories of uncertainty should be recorded separately; increasing random seeds cannot substitute for structural or load uncertainty, and averaging multiple method results cannot eliminate method divergence.

Interface QA should be executed once before and once after computation. Pre-computation checks examine time axis, units, shape, coordinates, missing data and source identity; post-computation checks examine equation-of-motion residuals, energy or equilibrium relations, result file read-back and target response reconstruction. Pre-QA prevents erroneous input from entering the solver; post-QA checks whether the solve and export maintain definitions. Both passing still only establishes evidence at the program and data packet level.

Independent solver cross-verification requires common mass, stiffness, damping, loads and integration settings. If two implementations use different load mapping or modal truncation, result differences cannot be attributed to the solver. Cross-verification should first select a linear model and one response target, making differences clearly locatable, before progressively adding complexity. No such complete external cross-verification exists yet, so v0.15 program gates cannot be described as independent structural validation.

After real upstream force data packets enter, benchmark and real-input pathways should be maintained in parallel. The benchmark continues to check program regression; the real-input pathway evaluates source-constrained response. If the real-input run changes the program version, the benchmark must be re-run to confirm the modification has not broken existing internal consistency. Conversely, benchmark passing also cannot substitute for the quality audit of real input. The division of labor between the two pathways avoids conflating software regression testing with scientific application.

The Researcher's key decisions for Structural include whether the structural model represents the target problem, which upstream forces may enter, which response quantities need to be computed, which method comparisons have interpretive value, and which results may be handed off to DesignLoad. Decision records must point to specific objects. Permitting a given benchmark to serve externally as method material is not equivalent to approving its numerical values as engineering responses; permitting real input for exploration is also not equivalent to approving its entry into candidate design expressions.

This chapter achieving figure, table and equation gates only shows that the presentation objects for methods, parameters and results are sufficiently complete. The number of figures does not change input evidence identity; the numerical values in tables do not change benchmark scope; equation closure does not change physical validation gaps. All current output remains model-internal benchmark, not design response; any subsequent level elevation requires real upstream

data, independent solver cross-verification and the Researcher's per-response judgment.

The purpose of model-internal benchmarks is determined by their reproducible condition sets. As long as structural and load parameters, random seed strategy and statistical definitions remain frozen, the benchmark can continuously serve software regression and method comparison; once any of these conditions change, it should be recorded as a new benchmark branch. Maintaining old branches in parallel rather than overwriting helps distinguish changes from program evolution versus changes from scientific input, and avoids automatically treating the latest version as a higher evidence level.

21.3.3 Current Scientific Limitations and Future Research

The conclusions remain constrained to the scope of linear elastic, quasi-steady force, constant C_d, uniform modulation, CAARC beam-18, and one set of Chen-Letchford parameters. PETWLoadPacket write-out verification is not closed; the data packet primarily supports the TDA pathway. There is no end-to-end closed loop with a real upstream PETW load packet, nor external independent solver cross-verification. Nonlinear effects, crosswind response, torsional response, local member pressure response and joint sensitivity to storm parameters are all unclosed.

Future research requires completing the following specific items: closing PETWLoadPacket write-out verification and extending EPSD and extremes pathway data packet support, driving an end-to-end run with a real upstream force data packet, introducing external independent solver cross-verification for the same structural model, and item-by-item extension of nonlinear, crosswind, torsional, local member pressure response and joint sensitivity to storm parameters. Until these conditions are completed, Structural Agent output retains its candidate research interface status, and ESWL candidate quantities and G_{GFF} output continue to be labeled as method verification quantities that do not enter design adoption.

Evidence and Formula Boundaries of Structural Agent Chapter 21 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "Structural Agent." Content without physical equations belongs to the research methods, knowledge or coordination layer, and formalized formulas are not used to substitute for missing physical evidence. Chapter 21 remains a source-constrained synthesis and cannot support validated design wind, design loads, code recommendations, or scientific conclusions approved by PI

22 DesignLoad Agent

DesignLoad Agent is the Step 5 final engineering translation layer in the PETW pipeline, responsible for connecting mechanism-aware wind hazard, aerodynamic loads and structural response to interpretable candidate design load expressions. Since 2026-06-16, the original Step 5 design load and original code interface steps have been merged into a single scope of responsibility, with code-provision variable ownership, applicability boundaries and code interface wording all incorporated into this stage. This chapter reports the state of the agent as of 2026-08-12 in seven sections covering scientific purpose, knowledge base, Researcher involvement and decisions, workflow and interactions, completed work and results, key formulas, and current scientific limitations and future research. All output retains candidate interpretation, not_approved and not_for_design status; no final η values, C-GFF design coefficients, design wind speeds, design loads, code provision recommendations or design guidance exist.

22.1 Scientific Questions, Knowledge, and Researcher Decisions

22.1.1 Scientific Purpose

The DesignLoad Agent does not generate original meteorological, CFD, wind tunnel or structural experimental results, but to receive source-bound upstream evidence, organize it into mechanism-specific, structure-specific and response-specific candidate load expressions, and manage source, units, applicability domain and claim ceiling. Translation products include candidate chains from mechanism-conditional hazard to wind field, pressure, aerodynamic force and structural response, GFF (gust-front factor) and G-GFF (generalized gust-front factor) factor decomposition and response ratios, constrained forms of response-specific ESWL, and parallel interfaces for the ASCE 7-22 baseline branch and the PETW mechanism branch.

The core theoretical reference is Kareem et al. [2019] and the generalized wind load chain (DOI 10.1061/(ASCE)ST.1943-541X.0002376, Zotero key 3R5RN3GT). This framework extends the Davenport stationary wind load chain to non-stationary wind, expressing relationships among wind, force and response using time-frequency and evolutionary power spectral density representations. This literature provides a theoretical anchor for the organizational approach of this stage and does not itself support any numerical coefficient or code recommendation of this study. Factor decomposition methodology references include the Kwon and Kareem [2009] gust-front factor framework and the Kwon and Kareem [2013] generalized gust-front factor computational framework.

22.1.2 Knowledge Base

The knowledge assets of DesignLoad comprise Wiki, publication-type reviews and program archives. The designload_step5 namespace currently contains 97 files, covering 37 canonical cards, 35 detailed readings, 412 registry records and 94 SQLite/FTS documents. All 159 governed synchronization tests passed; these results only demonstrate knowledge layer consistency and do not establish scientific approval. Table 32 presents the knowledge asset classification overview.

Table 32: DesignLoad Agent knowledge asset overview. Test passes demonstrate knowledge layer consistency and do not establish scientific approval.

Asset category	Count
Wiki files	97
Canonical cards	37
Detailed readings	35
Registry records	412
SQLite/FTS documents	94
Governance synchronization tests	159/159 passed

The latest publication-type review is v1.2 from 2026-07-24, with 21 pages of Chinese main text and 22 pages in English, containing 35 cited unique keys, 118 citation commands, 7 governed equations and 5 figures; scientific status is not_scientifically_approved. This version predates the program implementation in 2026-08 and is treated as a review of GFF/G-GFF theory, PETW framework and code interfaces, not a carrier for the August (month 8) program results.

22.1.3 Researcher Involvement and Decisions

The Researcher's (Liang Hu / PI involvement in this stage is concentrated on scope boundaries, benchmark selection and evidence gating. PI decided on 2026-06-16 to merge the original Step 5 design load and original code interface steps, so that translation from wind, force, response to design expression and code variable ownership are governed by the same entity. For benchmark selection,

PI decided to use ASCE 7-22 as the primary code benchmark, with ASCE 7-16 serving only as a historical version audit object.

When common-event complete pressure evidence is not yet available, PI authorized an independent assumption-driven dimensional method demonstration branch, constrained to program chain checking and assumption sensitivity exposure, not to be interpreted as observed event results. After PI reviewed the Wiki decision package on 2026-07-23, the knowledge layer entered governed synchronization; calibration screening stopping before optimization likewise follows the established contract, because parameter fitting has no basis without a governed common-event target. These decisions maintain the same evidence level: software and mathematical closure does not equal physical validation, and any candidate expression entering the next stage requires new common-event evidence and PI scientific decision.

22.2 Methods, Results, and Formulas

22.2.1 Workflow and Interactions

DesignLoad uses packetized and source-bound evidence as its sole entry point. Upstream EVA provides mechanism-conditional $\hat{V}_{3s}$ (unit: m/s), return-level evidence and occurrence rate λ_k (unit: events/year); Structural Response provides response metrics, load effects and candidate response factors or ESWL definitions; literature-side supports code provision and literature interpretation. Upstream packets must simultaneously carry field quantities, spectral quantities, spatial coherence, phase, duration, source identity and applicability domain, with Windfield candidate point EPSD and spatial coherence from cross-source candidates, and zero phase as an explicit assumption.

Each piece of evidence must be explicitly bound to mechanism k (dimensionless index), structural archetype a (dimensionless index) and response quantity r (dimensionless index). Response factors for displacement, base shear, overturning moment and member internal force cannot substitute for each other. The ASCE baseline branch and PETW mechanism branch are calculated separately, then compared under common hazard, structural and response definitions, without mechanical cross-multiplication. The velocity pressure height variation coefficient K_z (dimensionless), directionality factor K_d (dimensionless), gust effect factor G/G_f (dimensionless) and pressure terms of ASCE 7-16 and 7-22 must be edition-bound and not mixed across editions.

Candidate GFF and G-GFF response ratios are organized by factor decomposition as

$$G_{G\text{-}F} = I_1\, I_2\, I_3\, I_4 \tag{37}$$

Each factor is dimensionless, but their denominators, physical meanings and evidence owners differ. I_1 represents the kinematic difference between the gust-front profile and the boundary layer profile; I_2 represents the dynamic effect of a finite-duration pulse; I_3 aggregates three candidate pathways: long pulse, point EPSD and full time-frequency; I_4 is currently implemented as $I_4 = C_{D,GF}/C_D$, where $C_{D,GF}$ (dimensionless) and C_D (dimensionless) are the drag coefficients under gust-front conditions and boundary layer conditions respectively.

Governance adopts the one-effect-one-owner principle to prevent double counting: the direct pressure pathway already includes aerodynamic transfer, so subsequent generic pathways must not additionally multiply by I_4, a second C_d, G_f or other non-stationary amplification terms. C-GFF is an internal classification concept; η is a governed diagnostic schema; neither is an already-calibrated design coefficient. The output facing the Researcher is evidence completeness, candidate expressions, applicability domain, uncertainty and not-for-use labels; no quantity is automatically

elevated to design status at this stage.

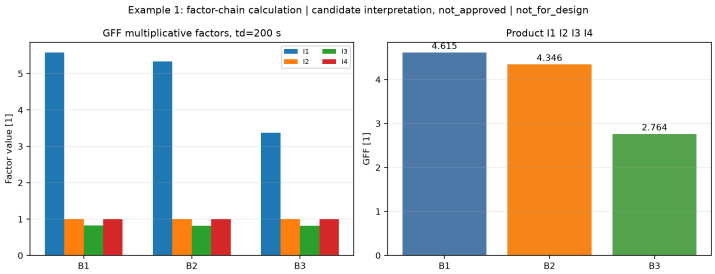


Figure 95: GFF/G-GFF factor chain and denominator basis for each factor (I_1 through I_4 candidate response ratio decomposition). Source: PETW DesignLoad full inventory 2026-08-12 (PETW original figure); candidate method demonstration, not a design value.

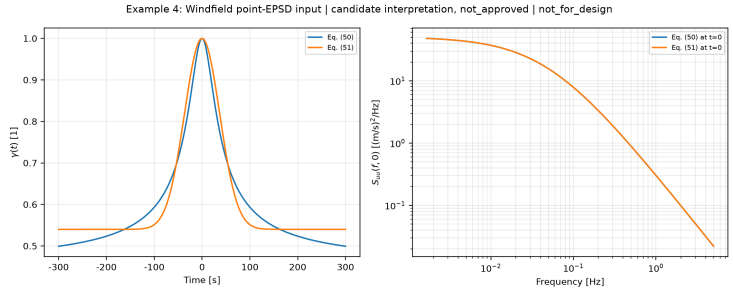


Figure 96: Windfield point EPSD candidate temporal modulation and spectral shape (12-case point EPSD benchmark). Source: PETW DesignLoad full inventory 2026-08-12 (PETW original figure); candidate method demonstration, not a design value.

22.2.2 Completed Work and Results

Completed executable assets include the I_1 through I_4 GFF/G-GFF research calculator, ASCE 7-22 research-type baseline, Windfield point EPSD integration, B2 bounded full time-frequency response, B2 assumption-driven dimensional method demonstration, C0 pre-calibration screening, and P1 synthetic direct pressure example. In the 2026-08-12 fresh execution, all 29/29 unit tests, B2 benchmark 9/9 checks, calibration screening 10/10 checks and P1 example 9/9 checks passed; calibration screening stopped before optimization per contract. The same-date 36-page inventory covers 9 case groups, 29 equation/align environments, 29 figure references and 16 newly added calculation figures; 14 key numerical read-backs, 12 content checks and 13 independent-audit correction checks all passed.

The B2 dimensional demonstration uses a separable evolutionary PSD as the load model:

$$S_{Q_1Q_1}(f,t) = g^4(t)\, S_{Q_1Q_1,0}(f) \tag{38}$$

where $S_{Q_1Q_1}(f,t)$ (unit: N^2/Hz) is the time-varying generalized force spectral density, $g(t)$ (dimensionless) is the temporal modulation function, $S_{Q_1Q_1,0}(f)$ (unit: N^2/Hz) is the base stationary generalized force spectral density, and f (unit: Hz) is frequency. This model declares assumptions of $C_d = 1$, zero phase and unity horizontal joint acceptance. Table 33 presents the main outputs of the B2 dimensional demonstration.

The P1 example uses a synthetically prescribed $p(z,t)$ (unit: Pa) to walk through the minimal chain from surface pressure to line load, aerodynamic force, generalized force $Q_1(t)$ (unit: N) and first-mode SDOF response; input status is synthetic_prescribed_not_observed. Table 34 presents the main outputs of the P1 synthetic example.

Conditioned verification shows that the I_2 numerical integration under the $\zeta_1 = 0.01$ (damping

Table 33: Main outputs of the B2 assumption-driven dimensional demonstration. Conditions: Kwon and Kareem hypothetical B2 building, $C_d = 1$, zero phase, unity horizontal joint acceptance; all numerical values are assumption-driven code output, not observed event results or design values.

Output quantity	Value	Unit
Peak mean generalized force	4.40902	MN
Generalized force standard deviation	1.03537	MN
Peak mean roof displacement	0.290713	m
Roof displacement standard deviation	0.0811783	m
First-mode inertial base shear	2.88619	MN
First-mode inertial overturning moment	192.413	MN·m
Source-style roof displacement envelope	0.520020	m

Table 34: Main outputs of the P1 synthetic direct pressure example. Input is synthetically prescribed pressure $p(z, t)$, not observed data; direct aerodynamic force and inertial recovery quantities have different physical definitions and are not interchangeable.

Output quantity	Value	Unit
Peak direct force	0.857185	MN
Direct overturning moment	49.9998	MN·m
Generalized force	0.499998	MN
Roof displacement	0.0329414	m

ratio, dimensionless) literature table basis differs from the analytical form by at most 0.0378%; the auxiliary $\zeta_1 = 0.001$ pathway has a maximum difference of 1.389%; accuracy claims must therefore be bound to damping conditions. In the full time-frequency pathway, the maximum response ratio of Roncallo-Solari Eq. 51 relative to Eq. 50 is 1.01417 to 1.01694, with peak time shifting 2.5 to 2.625 s; these differences show only model sensitivity, not that one method is superior to another.

Under fixed Exposure C 7-22 and default Davenport I_3, the GFF diagnostic values for three building benchmarks appear in Table 35.

Table 35: GFF diagnostic values for three building benchmarks under Exposure C conditions. The I_3 and GFF of B3 are approximately 3.6% and 3.0% lower than the published comparison targets respectively; this deviation may be non-conservative in direction and cannot be called conservative. All values are candidate method demonstrations, not design values.

Benchmark	I_3	GFF	Evidence status
B1	–	4.615130	candidate
B2	–	4.346416	candidate
B3	lower by approx. 3.6%	2.764084	candidate

The Exposure C internal consistency check yields $K_z(10\,\mathrm{m}) = 0.999$, $K_z(60.4\,\mathrm{m}) = 1.441$, $K_z(100\,\mathrm{m}) = 1.598$ and $K_z(200\,\mathrm{m}) = 1.840$. For benchmark maintenance, 12-case point EPSD, 8-case full time-frequency and independent RK4 verification all remain passing, with spatial, temporal and frequency grid variations all below 1%.

The C0 pre-calibration screening yields a rank of 4 for the local elasticity matrix of six outputs against four parameters (C_d, c_1 multiplier, coherence decay, t_d (unit: s)), with a condition number of 74.9542; the closest parameter pair is coherence decay and t_d. These are model structure diagnostics, not observation-driven identifiability or confidence interval evidence. Calibration screening stopped before optimization per contract, because parameter fitting has no basis without a governed common-event target.

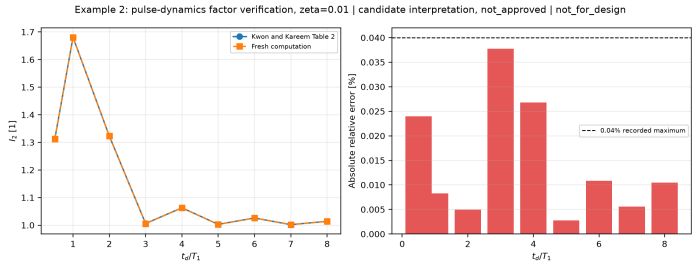


Figure 97: I_2 conditioned numerical verification: RK4 numerical integration compared with literature table basis ($\zeta_1 = 0.01$). Source: PETW DesignLoad full inventory 2026-08-12 (PETW original figure); candidate method demonstration, not a design value.

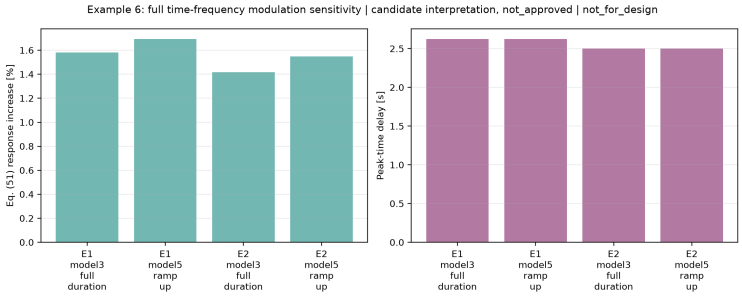


Figure 98: B2 full time-frequency 8-branch comparison: normalized response variation and peak time shift ($f \leq 0.5\,\text{Hz}$, $\Delta z \leq 100\,\text{m}$), not dimensional structural response. Source: PETW DesignLoad full inventory 2026-08-12 (PETW original figure); candidate method demonstration, not a design value.

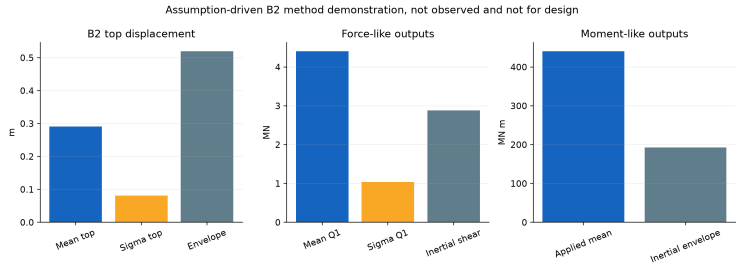


Figure 99: B2 assumption-driven dimensional demonstration faceted output (displacement, force, moment, Kwon and Kareem hypothetical B2 building). Source: PETW DesignLoad full inventory 2026-08-12 (PETW original figure); candidate method demonstration, not a design value.

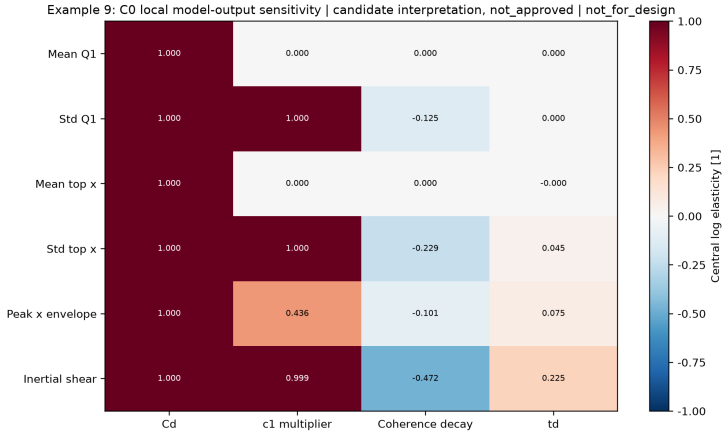


Figure 100: C0 pre-calibration screening local elasticity heatmap (C_d, c_1 multiplier, coherence decay, t_d). Source: PETW DesignLoad full inventory 2026-08-12 (PETW original figure); supports parameter order screening only, candidate method demonstration, not a design value.

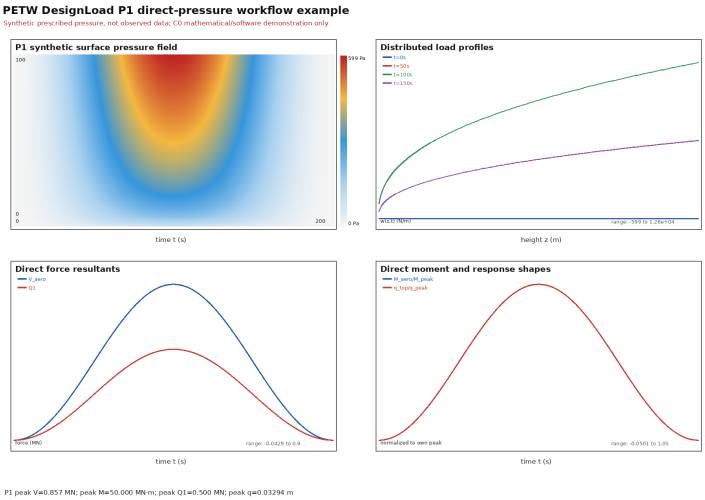


Figure 101: P1 synthetic direct pressure workflow: chain from synthetically prescribed $p(z,t)$ to line load and aerodynamic force; input is not observed. Source: PETW DesignLoad full inventory 2026-08-12 (PETW original figure); candidate method demonstration, not a design value.

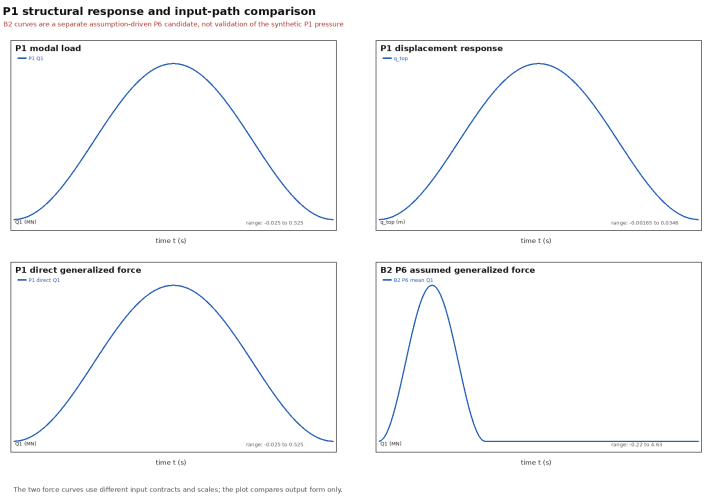


Figure 102: P1 first-mode response recovery: direct aerodynamic force and inertial recovery quantities have different physical definitions and are not interchangeable. Source: PETW DesignLoad full inventory 2026-08-12 (PETW original figure); candidate method demonstration, not a design value.

22.2.3 Key Formulas

When organizing the above assets into candidate design expressions, the general form adopted by this stage is

$$V_{\text{design}} = F[V_k(T),\ \lambda_k,\ \eta_k,\ \theta] \tag{39}$$

where $V_k(T)$ (unit: m/s) is the wind speed corresponding to return period T (unit: year) under mechanism k, λ_k (unit: events/year) is the occurrence rate of mechanism k, η_k (dimensionless) is the structural response modification, and θ is the parameter and applicability condition set. This form is a general framework for organizing upstream evidence, not an already-validated design formula.

In the P1 direct pressure pathway, surface pressure $p(z,s,t)$ (unit: Pa) is integrated along the building perimeter s to form line load $f(z,t)$ (unit: N/m):

$$f(z,t) = \oint_C p(z,s,t)\,\hat{n}_x(s)\,ds \tag{40}$$

where C is the cross-section perimeter path and $\hat{n}_x(s)$ (dimensionless) is the x-component of the cross-section outward normal. The line load is projected through modal decomposition to obtain the first-mode generalized force $Q_1(t)$ (unit: N):

$$Q_1(t) = \sum_{j=1}^{N_{\text{node}}} \phi_{1,j}\, f(z_j,t)\,\Delta z_j \tag{41}$$

where $\phi_{1,j}$ (dimensionless) is the first-mode shape value at node j, z_j (unit: m) is the node height, and Δz_j (unit: m) is the height segment corresponding to that node. The first-mode SDOF response equation is

$$m_1\,\ddot{q}_1(t) + c_1\,\dot{q}_1(t) + k_1\,q_1(t) = Q_1(t) \tag{42}$$

where m_1 (unit: kg) is the first modal mass, c_1 (unit: N·s/m) is the first modal damping, k_1 (unit: N/m) is the first modal stiffness, and $q_1(t)$ is the first generalized coordinate.

The ASCE 7-22 research-type simplified baseline is assembled as follows. The velocity pressure is

$$q_z = 0.613\,K_z\,V_{3\text{s}}^2 \tag{43}$$

where q_z (unit: Pa) is the velocity pressure at height z, K_z (dimensionless) is the velocity pressure exposure coefficient, $V_{3\text{s}}$ (unit: m/s) is the 3 s gust wind speed, and the constant 0.613 has units of kg/m^3, corresponding to $\frac{1}{2}\rho$ at standard atmospheric density $\rho = 1.225$ kg/m^3. The base design pressure is

$$p_{\text{base}} = q_z\,K_d\,G_{\text{ASCE}}\,C_p \tag{44}$$

where $K_d = 0.85$ (dimensionless) is the directionality factor, G_{ASCE} (dimensionless) is the ASCE gust effect factor, $C_p = 1.3$ (dimensionless) is the external pressure coefficient (diagnostic input), and $K_{zt} = 1$ (dimensionless, topographic factor) and $K_e = 1$ (dimensionless, ground elevation factor) are fixed simplified values. The GFF-modified force is

$$F_{\text{GFF}} = F_{\text{base}}\,K_{z,k}\,G_{G\text{-}F} \tag{45}$$

where F_{base} (unit: N) is the ASCE baseline force, $K_{z,k}$ (dimensionless) is the height modification under mechanism k, and $G_{G\text{-}F}$ (dimensionless) is the generalized gust-front factor. This baseline is maintained in parallel with the PETW mechanism branch and is not a complete code-compliant procedure.

22.3 Evidence Audit, Interfaces, and Limitations

22.3.1 Evidence Ownership, Closure Levels, and Audit of Candidate Expressions

The entry point to DesignLoad is not a single unified result file but a set of evidence objects owned by different stages. Events and EVA own event identity, sample composition, mechanism occurrence rate and candidate extreme value statistical source descriptions; Windfield owns wind speed

quantities, profiles, time-frequency characteristics and scale conversion conditions; WindTunnel and CFD own pressures or aerodynamic forces and their comparison ranges; Structural owns the structural model, response targets and response evidence. DesignLoad is limited to checking whether these objects can be combined under common mechanism, common event, common structure and common response definitions; it does not take over scientific ownership of upstream variables. Ownership separation prevents a downstream expression from self-supplying missing variables when sources are incomplete.

Event identity is the first gate for combination. Variables from different events, different simulations or different assumption branches may be used for method demonstration, but cannot be juxtaposed and written as a complete chain for the same event. B2, P1, EPSD, Structural benchmark and CFD constrained cases each have independent input identities. They collectively demonstrate that different computational modules can run, but cannot be merged into a set of event-level results. The DesignLoad audit must retain case identity alongside each output to avoid producing a visual impression of end-to-end closure from the consolidated display of figures and tables.

Wind speed identity is the second gate. Observed gusts, simulated grid peak values, Windfield converted quantities, EVA candidate return levels and code-benchmark wind speeds have different time-averaging, spatial support, height, mechanism conditions and statistical semantics. Symbol similarity does not imply substitutability. Candidate $V_k(T)$ can enter Eq. 39 only when sample, wind speed definition and return period semantics are complete; EVA has not yet provided such final input. Consequently, this equation expresses only variable relationships and audit positions, and does not produce usable design wind values.

Aerodynamic force identity is the third gate. The direct pressure pathway already includes building surface aerodynamic effects in the pressure integration; the generic pathway using force coefficients represents aerodynamic conversion through coefficients. Intermediate quantities from the two pathways cannot be arbitrarily superimposed. If the direct pressure pathway has already produced line loads, further multiplying by a drag coefficient would double-count aerodynamic conversion; if the Structural input already contains random fluctuations, superimposing a second fluctuating amplification term would likewise cause double counting. The one-effect-one-owner principle requires each physical effect to appear only once in one pathway, marked with its owner in the closure matrix.

Structural response identity is the fourth gate. Displacement, acceleration, base shear, overturning moment and member internal force have different response functionals. Response modification, G-GFF and ESWL must be bound to the target response. The Structural roof displacement benchmark can support checks of the roof displacement method chain, but does not support a same-value coefficient for base bending moment or member internal force. The DesignLoad output schema therefore must simultaneously store structural archetype, response quantity, load direction, statistical definition and baseline; storing only an identity-free factor is not permitted.

Closure levels serve to indicate how far a given chain has been checked, not to show that results are adoptable for design. C0 corresponds to mathematical and software-level executability and internal verification; C1 requires upstream physical quantities under a common event to be closed in identity and units; C2 requires comparison with independent physical evidence; C3 involves design governance and applicability scope review. The current B2 assumption-driven branch has reached C0; it cannot be elevated to C1, C2 or C3 simply because its units are complete, its

equations solvable and its tests passing. Closure level is an audit coordinate, not a quality score.

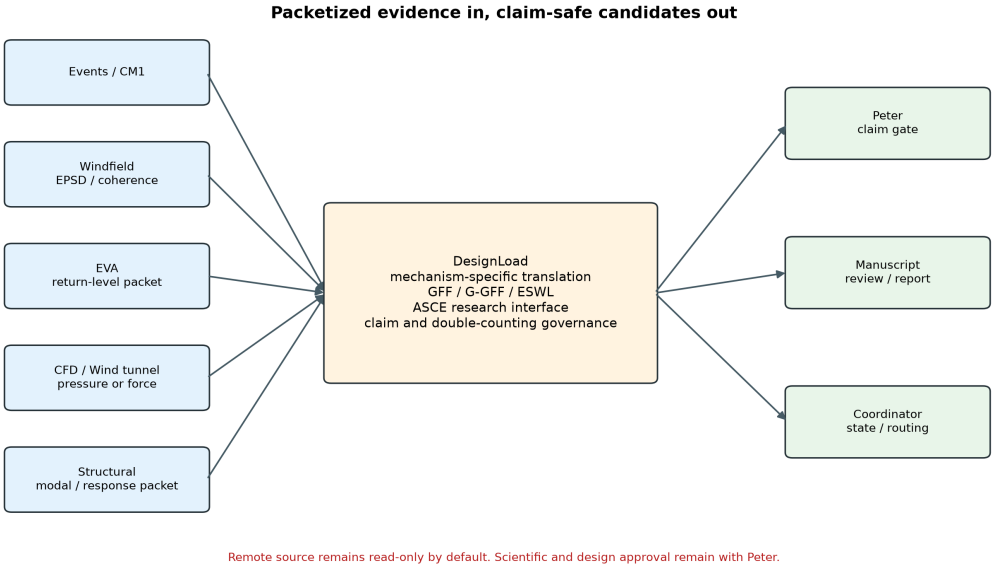


Figure 103: DesignLoad upstream interface architecture. PETW original project figure; interface connectivity represents variable relationships and does not show that common-event evidence has been closed.

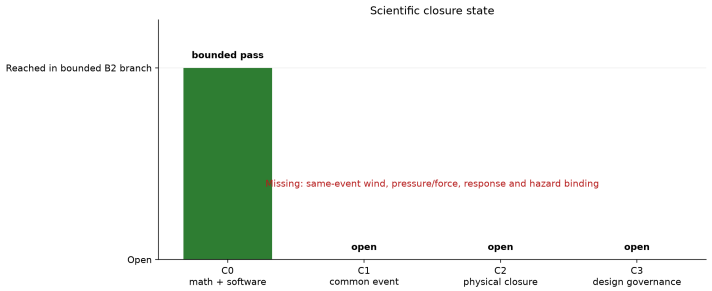


Figure 104: C0 through C3 closure levels and their evidential meaning. PETW original project figure; mathematical and software closure of the current method demonstration does not establish physical or design closure.

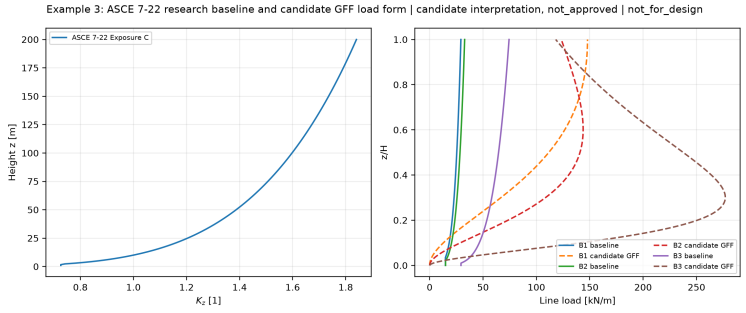


Figure 105: Height coefficients and line load diagnostics in the ASCE 7-22 research-type baseline. PETW original project figure; this branch is used for method verification and is not a complete code-compliant procedure.

Figures 103 through 106 serve, respectively, evidential roles for interfaces, closure levels, code baseline diagnostics and time-frequency pathway sensitivity. The interface figure answers where variables come from; the closure ladder answers what level of checking has been reached; the code baseline figure answers how the research-type implementation is organized along height; the twelve-case figure answers how current assumption variations affect candidate results. The four cannot be merged into design adoption evidence, nor can any single figure independently support final coefficients.

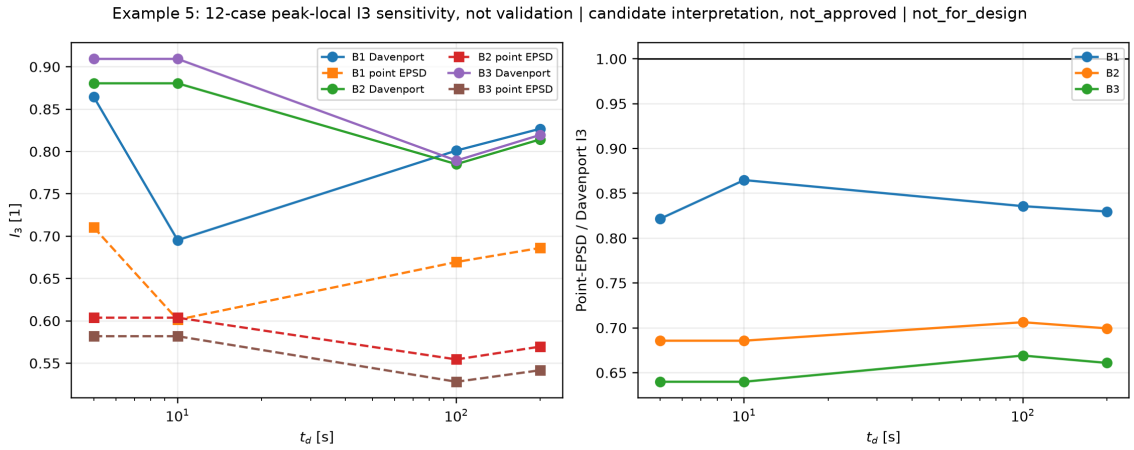


Figure 106: Point EPSD twelve-case I_3 candidate results. PETW original project figure; case comparison represents model-internal sensitivity, not observational calibration.

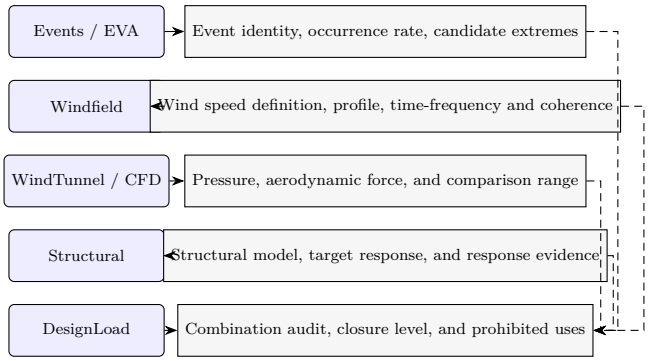


Figure 107: DesignLoad evidence ownership and combination audit relationships. Left column: owners of scientific objects; right column: evidence content each owner is responsible for; dashed lines indicate source-bound objects entering combination audit, not ownership transfer. PETW original project TikZ figure; status is candidate interpretation, not_approved, not_for_design.

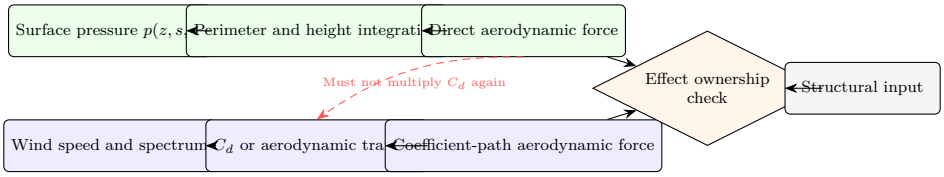


Figure 108: One-effect-one-owner check for the direct pressure pathway and coefficient pathway. The two pathways can each form candidate structural inputs, but when direct pressure integration already includes aerodynamic transfer, the drag coefficient must not be applied again. PETW original project TikZ method figure; status is candidate interpretation, not_approved, not_for_design.

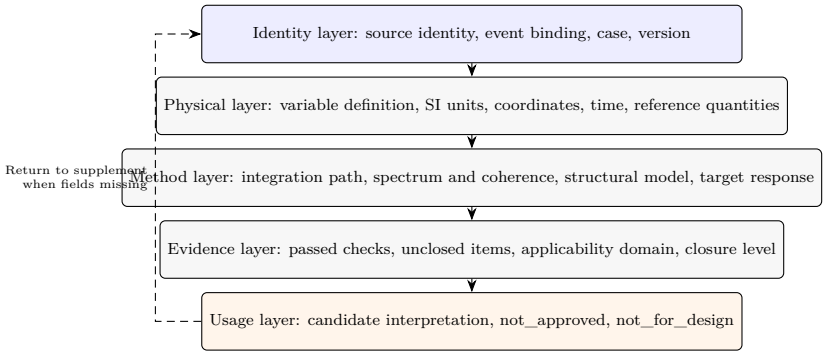


Figure 109: Five-layer interface structure of the candidate DesignLoad packet. Numerical arrays constitute auditable input only when stored jointly with identity, physical definitions, methods, evidence status and machine-readable prohibited uses. PETW original project TikZ interface figure; this does not show that the common-event chain has been closed.

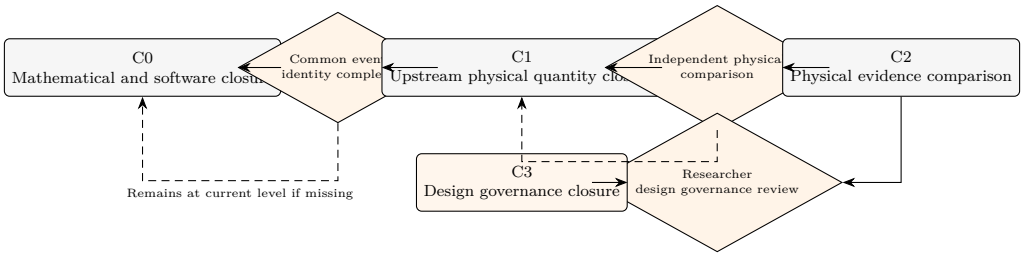


Figure 110: Evidence elevation gates from C0 to C3 for candidate expressions. Each elevation respectively requires common event identity, independent physical comparison and Researcher design governance review; the current B2 method demonstration is at C0 only. PETW original project TikZ evidence figure; status is candidate interpretation, not_approved, not_for_design.

The four original project figures respectively constrain evidence ownership, aerodynamic effect deduplication, packet completeness and evidence elevation. Their scientific content is not mutually substitutable: the ownership figure cannot prove that variables are combinable; the deduplication figure cannot prove that inputs represent a common event; the packet figure cannot prove that field contents are correct; and the elevation gate figure cannot rewrite internal checks as physical validation. The figures only make existing methods and governance interfaces in the text explicit, without adding any numerical results, scientific conclusions or design approvals.

The ASCE baseline branch provides provide edition-bound research comparison targets. Implementation requires binding the provisions used, coefficients, exposure categories, directions, pressure coefficients and load combinations to the same edition. The current simplified branch has fixed several inputs and omitted other checks in a complete compliance procedure; therefore it can only be used for verifying unit, along-height variation and candidate response ratio calculation interfaces. Presenting the simplified branch alongside the PETW mechanism branch is designed to expose variable differences between the two, not to mechanically append mechanism modifications to code results.

The GFF factor chain requires auditing numerators, denominators and owners. The profile ratio of I_1 must state comparison height and reference profile; the dynamic ratio of I_2 must state pulse and structural time scales; the candidate pathways of I_3 must state long pulse, point EPSD or full time-frequency representation; the drag coefficient ratio of I_4 must state the aerodynamic source. If any factor lacks a common baseline, the product, even if computable, does not carry unified physical semantics. Current implementation can support checks of the calculations and condition sensitivity of each branch, but no common-event observation yet provides consistent input for all four factors.

The value of the B2 dimensional demonstration lies in carrying the unit chain through to displacement, force and moment, and exposing the positions where assumptions control results. It uses a hypothetical building, prescribed drag coefficient, zero phase and horizontal joint acceptance assumptions. Its results can support checks of the mathematical relationships among generalized force, spectral integration, structural response and recovery quantities, but cannot show that a real thunderstorm event, real building or code case would have the same output. Assumption-driven does not mean useless; it requires that the use be constrained to method audit and input requirement identification.

The P1 direct pressure example provides an alternative minimal chain. The prescribed $p(z,t)$ undergoes perimeter integration, along-height discretization and modal projection to form $Q_1(t)$, which then enters the first-mode response equation. The audit focus of this pathway is the physical distinction among pressure, line load, generalized force, direct aerodynamic force and inertial

recovery quantity. Direct aerodynamic force comes from external load integration; inertial recovery quantity comes from structural motion; the two can be related through the equation of motion but cannot be interchanged simply because they share the same units. The P1 input is explicitly synthetic prescribed rather than observed; this identity must be retained with all derived figures and tables.

The full time-frequency branch is used to check the coupling between non-stationary spectra and structural frequency response. Retained variance ratios, frequency upper limits, spatial discretization and phase assumptions jointly constrain results. Analytical ratios with and without zero frequency use different integration bases and should not be mixed within the same comparison. The current independent RK4 and grid variation checks support internal consistency of the numerical implementation, but cannot substitute for physical validation of the upstream spectrum, coherence and phase.

Pre-calibration screening serves parameter identifiability diagnostics, not parameter estimation. The rank and condition number of the local elasticity matrix describe the local response structure of current outputs to candidate parameters and cannot provide observation-driven confidence intervals. Continuing optimization without a common-event target would only find parameter combinations relative to the assumed target, and the results cannot be interpreted as physical calibration. Stopping before optimization per contract thus preserves the method boundary and avoids mistaking algorithmic convergence for observational support.

Numerical verification also needs to be separated from scientific verification. Unit tests check whether functions satisfy programmatic assertions under prescribed input; benchmarks check whether the current implementation reproduces frozen output; independent calculations check whether different numerical approaches agree under common equations. All three can increase software evidence strength but do not answer whether the input model represents real wind fields, aerodynamic forces or structures. The DesignLoad report presents both these checks and scientific gaps to prevent “reproducible” from being read as “validated.”

Candidate expression output should include values, units, conditions, sources, closure levels and prohibited uses. For dimensionless factors, the numerator and denominator definitions should also be stored; for ESWL, the target response; for hazard or return levels, the sample and mechanism conditions; for code branches, the edition and omitted items. Isolated numerical values lacking these fields should not enter downstream files or external figures.

The Researcher’s judgment is distributed at entry, combination and exit. Entry judgment determines whether upstream evidence has a usable identity; combination judgment determines whether different objects are permitted to be connected in a common expression; exit judgment determines whether candidate results are usable for method comparison, subsequent research, or higher-level review. The three judgments cannot be merged into a single blanket approval. A specific branch passing mathematical checks can only support the corresponding C0 use; a given upstream data packet passing identity checks does not automatically approve the final candidate expression.

Near-term closure work should start from common-event evidence rather than continuing to expand isolated method branches. The minimal object requires the wind speed identity, surface pressure or governed generalized aerodynamic force, structural model, target response and comparison baseline from the same event. Subsequently, checking units, coordinates, time, spectrum and coherence, completing Structural response and returning to DesignLoad. This object still requires

physical comparison and Researcher review before it can potentially move from C0 to a higher level. Current sources are sufficient to define this minimal pathway but have not yet provided the common-event input it requires.

Discussion of candidate design implications must remain conditioned. Different methods under common assumptions can be compared regarding how they organize variables; missing evidence that would block certain inferences can be identified; but final η, C-GFF, design wind, design loads or code adoption conclusions cannot be given. The scientific contribution of DesignLoad at this stage is to make missing items and double-counting risks explicit, and to bind each candidate quantity to an auditable evidence chain.

22.3.2 Conditioned Interpretation of Branch-Level Results

B2, P1, point EPSD, full time-frequency and pre-calibration screening should be viewed as five interrelated but evidence-identity-independent branches. B2 checks the dimensional generalized load-to-response chain; P1 checks the direct pressure-to-response chain; point EPSD checks simplified non-stationary spectral representation; full time-frequency checks richer time-frequency coupling; pre-calibration screening checks the local relationship between candidate parameters and outputs. The five branches share some code and theoretical background but have no common-event input; the most complete variables from each branch cannot be spliced into a single closed chain.

B2 output displacement, force and moment come from the same assumption-driven model; unit consistency permits their mutual verification within the equation of motion and recovery relationships. This consistency cannot prove that hypothetical building parameters, aerodynamic force model or non-stationary input represent actual objects. The report retains specific numerical values so that program reconstruction and independent verification are executable; the assumptions, units and candidate status alongside each value are equally part of the result, and citing without these conditions changes the evidence identity.

The P1 direct pressure pathway has an interface form closer to upstream physical quantities, but the current pressure remains prescribed input. Pressure integration along the perimeter requires consistency between outward normals and path direction; along-height discretization requires complete nodal segment coverage; modal projection requires explicit mode shape normalization. First-mode response recovery further requires that modal mass, damping and stiffness use the same model as the projection. Although any of these can be supplied by code default values, they cannot be omitted in scientific interpretation. P1 serves to expose the fields that a real pressure packet will be required to carry in the future.

The point EPSD branch separates temporal modulation and base spectrum, offering higher computational efficiency while depending more heavily on separability assumptions. The twelve cases are used to examine the effect of modulation, spectral or structural parameter variation on I_3; the current comparison describes only the model-internal response surface. If the frequency shape of a real wind field varies with time, the separable form may be insufficient; in that case, full time-frequency representation or redefined modulation would be needed. Before selecting a more complex branch, it remains necessary to check whether upstream evidence supports its additional degrees of freedom.

The full time-frequency branch incorporates time, frequency and spatial correlation more directly into the integration and is therefore more sensitive to grid, truncation and phase. The frequency upper limit and height interval constrain the verified domain; results outside this domain

cannot be extrapolated by the current convergence checks. The independent RK4 pathway provides cross-verification of equation implementation; response ratios and peak time differences provide branch sensitivity; but no observed response is available to determine which branch is closer to physical processes.

Retaining the two variance ratio bases illustrates that the integration domain must be fixed in audits. When zero frequency is included, deterministic or near-zero frequency components enter the total; when excluded, only selected frequency bands are counted. Both ratios may be correct under their respective definitions; mixing them creates a spurious inconsistency. DesignLoad should store the frequency domain, discretization and normalization denominator in the result schema to avoid storing only percentages.

Full rank in the calibration screening only shows that four candidate parameters have distinguishable directions against six outputs under the current local linearization; it does not indicate global uniqueness, nor that the parameters can be stably estimated from real data. The condition number reflects the degree of local ill-conditioning; the closest parameter pair indicates potential correlation. Without an observation target, error model and prior ranges, continuing optimization cannot produce auditable physical parameters. The stopping point itself is a governance outcome, not a deficiency of incomplete computation.

The ASCE research-type branch needs to maintain its benchmark identity. Along-height variation of height coefficients can check whether the implementation conforms to frozen formulas and parameters; line loads can check units and integration; response ratios can check the interface with the PETW branch. However, a complete code application also includes provisions not yet implemented or awaiting authoritative original text verification. Benchmark figures and tables must not use "compliant," "code-passing" or synonymous expressions, to avoid writing a research-type implementation as a formal design procedure.

Comparison between the direct pressure pathway and the coefficient pathway is meaningful only under common geometry, common wind field, common response and common reference. The two pathways currently use different inputs; output differences cannot be used to judge method superiority. After a future common-event object is established, the two pathways can first be compared for consistency in mean effects, then extended to fluctuating and time-frequency responses; if differences exist, they should be localized level by level through pressure integration, coefficient definition, spatial coherence and structural projection.

The closure matrix should simultaneously record "existing objects" and "missing objects." Existing objects include programs, formulas, assumed inputs and internal checks; missing objects include common-event pressures, governed aerodynamic forces, EVA candidate statistics, physical comparison and design governance. Displaying only existing objects tends to produce over-interpretation of progress; displaying only missing objects fails to convey the established method foundation. The purpose of the matrix is to show what evidence each candidate claim still requires, not to rank lanes.

Management of η must distinguish variable schema from numerical assignment. The schema can specify how this quantity is bound to mechanism, structure and response, and can specify numerator and denominator; numerical assignment requires common-event data and Researcher review. Only the schema and candidate computational pathways can be discussed; no approvable η value exists. C-GFF similarly can serve as an internal classification concept organizing pathways but does not carry a calibrated design coefficient identity.

Uncertainty propagation cannot be absorbed by a single safety factor. Uncertainties in event classification, extreme value sample, wind field conversion, aerodynamic force, structural model and response method have different sources and correlations. Compressing them into a single sourceless factor loses auditability. DesignLoad should first maintain itemized entries and conditions; only after evidence and risk frameworks are clear can the question of how to combine them be studied. This combination is not executed at this stage.

The captions of result figures serve as local prompts for claim ceilings. B2 figures must state assumption-driven; P1 figures must state synthetic pressure; EPSD figures must state model-internal cases; ASCE figures must state research-type simplified branch; calibration figures must state screening rather than fitting. Figures separated from the main text should still retain these prompts to prevent misinterpretation in presentation slides or downstream reports.

Data in tables also require the same discipline. Value columns must be juxtaposed with unit columns; conditions are written into captions or adjacent text; evidence status indicates candidate or missing. Horizontal rules and decimal precision in tables do not elevate evidence levels. Reprinting the same output does not add independent evidence; only comparisons from different definitions, different implementations or independent physical sources can potentially add verification strength.

The future DesignLoad interface should support machine-readable prohibited uses. Beyond source identity and units, each variable should also record not_for_design, not_approved, response binding, event binding and closure level. Program checks of these fields before combination can prevent clearly incompatible data from being automatically used, but machine checks cannot substitute for the Researcher's judgment on scientific applicability.

The PI claim gate is positioned before any candidate expression elevation. Review materials should include upstream sources, common-event identity, formulas and implementation, physical comparison, sensitivity, uncertainty, applicability scope and prohibited uses. Unit tests, figure counts, successful compilation or internal numerical consistency alone are all insufficient for requesting elevation. If materials still have source conflicts, the conflicts should be retained as review objects rather than eliminated in the summary.

The most substantive completed state currently is that multiple method branches have reached executable and auditable states, rather than that final design quantities have been formed. Writing this status accurately allows subsequent work to start from known interfaces and gaps, and prevents model output from being adopted prematurely. The entire chapter continues to maintain candidate interpretation, not_approved, not_for_design; any higher-level statement exceeds the frozen evidence.

Maintaining the simultaneous presence of the three status terms has substantive significance. Candidate interpretation shows that the current interpretation may still change with new evidence; not_approved shows that the Researcher has not yet given scientific approval; not_for_design shows that even if internally executable, entry into engineering adoption is not permitted. The three constrain different stages and cannot be replaced by a blanket "preliminary results," nor can any of the three be removed due to program updates or additional figures.

22.3.3 Current Scientific Limitations and Future Research

The independent audit identifies four scientific limitations that must be retained. The first term of g_R and simplified R_L for the ASCE flexible path require authoritative original text verification before the correct form can be confirmed. The G_{GLF} metadata returned by G-GFF is inconsistent

with the internal values actually used in the load calculation; this field cannot serve as machine-readable evidence for the calculation denominator. The retained variance ratio of the full-time-frequency branch is 77.9381% and cannot be mixed with the 80.1561% analytic ratio that includes zero-frequency. The 599.400 Pa in P1 is the highest midpoint-grid pressure, not the analytical roof pressure. Additionally, four groups of source numerical conflicts remain unresolved: 47.0 versus 49.0 m/s, 1988–2017 versus 1998–2017, approximately 1.37 versus 1.99, and 2160 versus 2220 s.

Table 36 presents the major evidence gaps and closure status.

Table 36: DesignLoad Agent major evidence gaps and closure status. In the C0 through C3 research structure, only the B2 assumption-driven branch has reached C0 mathematical and software closure.

Gap ID	Content	Current status
C1	Common-event closure: surface pressure, aerodynamic generalized force, line load, force cross-EPSD, and generalized-force EPSD from the same event	Missing
C2	Physical model validation	Not completed
C3	Design governance closure	Not completed
B2 domain limit	Full time-frequency applicable only for $f \leq 0.5$ Hz and $\Delta z \leq 100$ m	Domain-limited
Adapters	Generic direct-pressure, Q_1, force cross-EPSD, and response-specific ESWL adapters	Incomplete

Only the B2 assumption-driven branch has reached C0 mathematical and software closure. The ASCE 7-22 pathway still requires authoritative original text verification of g_R, resonant terms, K_{zt}, K_e, C_p, internal pressure, directional cases, C&C, load combinations, minimum loads and reliability procedure. Subsequent work is prioritized by evidence: first obtain same-event surface pressure or governed generalized aerodynamic forces, obtain mechanism-conditional hazard input, complete ASCE original text verification, then extend adapters and governed calibration.

Evidence and Formula Boundaries of DesignLoad Agent Chapter 22 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "DesignLoad Agent." Content without physical equations belongs to the research methods, knowledge or coordination layer, and formalized formulas are not used to substitute for missing physical evidence. Chapter 22 remains a source-constrained synthesis and cannot support validated design wind, design loads, code recommendations, or scientific conclusions approved by PI

23 Knowledge Agents

Knowledge Agents form PETW's literature-identity and external-knowledge layer. They comprise three agents: Zotero Literature, PETW LitWiki, and Web and WeChat Knowledge. They provide source identity verification, claim-level evidence extraction, thematic relationship organization and external knowledge supplementation to all scientific agents. The three agents operate independently; identity records, reading extraction and external information belong to different agents, preventing a single agent from certifying the quality of its own processing. All output of the knowledge layer constitutes research input and evidence organization records and does not produce scientific conclusions; all content in this chapter remains `not_scientifically_approved`.

23.1 Evidence, Results, and Interpretation

23.1.1 Common Purpose and Inter-Layer Division of Labor

The shared purpose of the three knowledge agents is to provide four categories of knowledge supply to all scientific agents of PETW. The first category covers formal literature identity: every reading record can be traced back to a definite publication, version and page number, preventing unverified sources from entering scientific work. The significance of formal literature identity lies in eliminating source ambiguity and making citation behavior auditable. The second category covers claim-level evidence: specific statements in the literature are bound to their provenance, which allows subsequent citations to be grounded in specific paragraphs and data rather than remaining at the level of generalized citation of an entire publication. The granularity of claim-level evidence ensures citation precision; the provenance of a single claim can be pinpointed to a specific section, table row or equation number of the publication. The third category covers thematic relationships: knowledge organized by topic, method, variable and source supports cross-lane reuse and method comparison, enabling the treatment of the same research question across different sources to be examined in parallel. Thematic relationships are the core feature distinguishing the knowledge layer from a simple literature list; they organize discrete sources into an evidence network searchable by research question. The fourth category covers external knowledge: covering information sources such as web pages and WeChat that have not entered the formal literature system, supplementing background clues and progress information beyond formal literature.

The division of labor among the three maintains deliberate separation. Zotero Literature serves as the sole authoritative record layer for source identity, preserving formal literature identity, PDFs and bibliographic information; its input is literature discovery results and PDF files, and its output is verified literature records and unique identifiers. PETW LitWiki undertakes full-text reading and claim extraction; its input is literature identities verified by Zotero, and its output is claim records organized by topic and relationship mappings, with each claim attached to a verified literature identity and annotated with source paragraph. Web and WeChat Knowledge preserves web pages, WeChat messages and other external materials; its input is web search results and message sources, and its output is structured external knowledge records that need to interface with Zotero identity verification before entering formal citation. A direct consequence of separating the three is that each agent is limited to the processing quality within its scope of responsibility: Zotero is responsible for the pass rate of identity verification, LitWiki is responsible for the coverage and claim granularity of reading extraction, and Web and WeChat Knowledge is responsible for the structured recording of external sources.

Identifying a publication and understanding a publication are two different cognitive tasks; identity records and reading records therefore belong to different agents. Project state and result reports produced by PETW's own work are likewise stored in the knowledge layer, making the research process itself a traceable asset. For thunderstorm wind research, literature sources span multiple disciplinary domains including event classification, ground observation, mesoscale simulation, downburst modeling, wind field characterization, extreme value analysis, wind tunnel test design and structural loads; no single search approach can cover all source types. The knowledge layer uses the division of labor among the three agents to place the above sources into a unified governance framework: each source undergoes identity verification, reading extraction and thematic classification through three stages, with processing status preserved alongside the source. Downstream citations can trace from the claim back to the reading record, and then from the

reading record back to the literature identity, without a missing link in between. The core value of this inter-layer division of labor lies not in the extraction efficiency of individual claims, but in the completeness and traceability of the evidence chain: any cited scientific statement can be traced back to its original source along a defined pathway.

23.1.2 The Cycle of Knowledge Entering Research

The sequence for knowledge entering research is a fixed six-stage cycle: source discovery, identity verification, full-text reading and claim extraction, review synthesis, namespace mounting, and Researcher judgment. The sequential relationship of the six stages is a structural constraint that cannot be skipped or shortened.

Source discovery is performed by Gemini and Web scout, outputting a preliminary list of candidate sources and access paths. The scope of source discovery covers academic databases, publisher platforms and open-access channels; discovery results include basic bibliographic information and access feasibility assessments of candidate publications. Identity verification is performed by Zotero Literature, verifying the literature identity, version, publication information and PDF availability of each candidate source. Sources that pass verification receive a unique identifier and are placed in the Zotero library; the identifier serves as the sole citation key for that source in all subsequent stages. Sources that do not pass are annotated as pending supplementation or unavailable, with annotation reasons preserved alongside the record. The pass rate of identity verification directly reflects the quality of the source discovery stage; an excessively low pass rate indicates insufficient candidate precision in the discovery stage.

Full-text reading and claim extraction are performed by PETW LitWiki. LitWiki conducts paragraph-by-paragraph reading of publications with verified identities, extracting the specific statement, source paragraph, method elements and applicable conditions of each claim. Extracted claim records are attached to the verified literature identity; the relationship between claim and source is fixed at the time of extraction and cannot be subsequently modified in source reference. Claim granularity is controlled at the level of a single scientific statement: one claim corresponds to one independently citable method, parameter, conclusion or boundary statement in the literature. Claim extraction simultaneously records applicable conditions and known limitations, which allows subsequent review synthesis to filter by conditions rather than aggregating solely by topic.

Review synthesis is conducted after reading records are aggregated by topic. Each lane compares the methods, variables, and conclusions from multiple sources under the same topic during the review stage, forming method cards and boundary statements. The value of synthesis depends not on numerical coverage but on the comparative relationships among sources: the parameter choices, applicable ranges and known limitations of the same method across different sources are recorded in parallel for subsequent research design reference. The output of review synthesis is structured method comparisons and boundary statements, not summaries of individual sources. The final form of knowledge is method cards and applicability boundary tables following review synthesis; they are anchored by literature identity, and any synthetic conclusion can be decomposed back to the sources that constitute it.

Namespace mounting is the channel through which knowledge assets enter a lane. Each namespace represents a complete processing chain from identity verification to claim organization for a source collection, carrying its own source scope, processing status and evidence categories. Mounting is delivered at the namespace level; when a lane makes citations during the review stage, the citation object is a specific record within the namespace rather than the namespace as a

whole. New reading records first enter the namespace before becoming visible to the corresponding lane, with processing status maintained together with the namespace. Researcher judgment is the final stage of the cycle and the defined gate between knowledge assets and scientific conclusions. Whether a claim supports a particular scientific conclusion, and whether a source is applicable to a specific lane's research design, are always decisions for the Researcher. All output of the knowledge layer retains the `requires_peter_review` attribute.

Knowledge quality assurance is distributed across every stage of the cycle. The quality of the source discovery stage is reflected by the pass rate of identity verification; the quality of identity verification is quantified by retrieval verification and source-semantic verification, where retrieval verification confirms that a publication can be located and accessed, and source-semantic verification confirms that the publication content is consistent with its bibliographic information; the quality of claim extraction is ensured by the direct correspondence between claims and source paragraphs; the quality of review synthesis is demonstrated by the completeness of multi-source comparison. These quality mechanisms operate independently at each stage and are not lowered at upstream stages due to downstream stage requirements.

23.1.3 Relationship with the Researcher

The relationship between Knowledge Agents and the Researcher is defined by evidence boundaries. The three agents are limited to source discovery, identity verification, full-text reading, claim extraction and thematic relationship organization. Whether a claim supports a particular scientific conclusion, and whether a source is applicable to a specific lane's research design, are always judgments for the Researcher. The `requires_peter_review` attribute is the defined gate between knowledge layer output and scientific conclusions: this attribute shows that output has been completed only to the evidence organization level, and whether it may enter a specific lane's scientific judgment requires item-by-item confirmation by the Researcher. Knowledge assets mounted to a lane therefore have only citation value, not endorsement value.

The relationship between the knowledge layer and the Researcher is summarized as supply and adjudication. Agents are responsible for organizing evidence to be traceable, locatable, and citable; the Researcher is responsible for deciding how evidence is used, the scope of adoption and the mode of interpretation; there is no automatic transition between the two. The Researcher's involvement is placed at the end of the cycle; when knowledge assets enter a specific lane's scientific work, the scope of adoption and mode of interpretation are decided by the Researcher, and the knowledge layer only ensures that assets are traceable, locatable, and citable. This boundary is maintained consistently in governed namespaces: the `scientifically_admissible` count in WindTunnel's 55-source evidence governance remains 0/55, and the `scientific_approval_status` of `wrf_mesoscale_modeling` remains `not_approved`; both numbers reflect the independence between knowledge organization completeness and scientific admission. The knowledge layer continues to accumulate and organize, but does not make adoption decisions for any lane, nor does the completeness of its own organization change the scientific status of any claim. Knowledge organization completeness and scientific admission are two independent judgments: the former describes the state of evidence organization work, the latter rests with the Researcher's adjudication on the scientific applicability of evidence, and the distance between the two judgments is filled by the Researcher's independent review, which the knowledge layer neither participates in nor shortens.

The version discipline of knowledge records further maintains this distinction. When literature identity updates, PDF version changes or reading extraction revisions occur, old records should

not be overwritten without trace; new records need to state whether the change occurred in bibliographic information, full text content, page location or claim interpretation. Claims already used by downstream lanes still point to their frozen identity at the time of use; subsequent updates trigger re-verification rather than automatic replacement. This makes it possible to determine whether a change in the text originates from a new source, a new version, or a reinterpretation of the same source. Version traceability can only support auditing; it cannot automatically prove that a new record is superior to an old one.

External web pages and group messages retain their clue identity when entering the knowledge layer. They can be used for discovering formal sources, indicating research progress or locating project files, but do not support formal scientific citation before completing literature identity verification and full-text reading. Web content is mutable and message context may also be incomplete; therefore, the structured records of Web and WeChat Knowledge must preserve access location and applicable scope, and hand over content that needs to enter the formal evidence chain to Zotero Literature and LitWiki. The serial connection of the three is not redundant labor, but separates discovery, identity and semantics for independent verification.

Namespace scale likewise cannot be used as a source quality indicator. File counts, claim counts and relationship counts reflect organization scope and granularity, and cannot substitute for source authority, full-text availability, method applicability and claim-level local support. A given namespace being larger in scale may result from the research question having more dimensions, or may contain records pending exclusion or serving only as background. When the Researcher assesses a specific claim, they need to return to that claim's source, location and conditions, rather than making judgments based on namespace totals.

The principal deliverable of the knowledge layer to downstream is checkable limitations. Each method card, in addition to recording source support, should also record objects not covered by the source, conclusions that cannot be drawn, and conflicts with other sources. If sources disagree on variable definitions, parameter ranges or method applicability, LitWiki retains side-by-side records, review synthesis explains the conditions of difference, and the Researcher decides whether to return to the original text or add sources. Knowledge agents do not substitute physical and method judgment with surface agreement among majority sources.

Accordingly, what Knowledge Agents have currently completed is evidence infrastructure and governed knowledge supply. They have made the source identities, reading records and thematic relationships of multiple lanes traceable, but have not provided automatic admission for any scientific claim. The honest output when sources are insufficient is to annotate partial, metadata-only or not available and limit textual detail, rather than filling gaps with adjacent sources or repeated summaries. This discipline makes shortfalls in the knowledge layer visible and gives subsequent supplementation work clear targets.

Evidence and Formula Boundaries of Knowledge Agents Chapter 23 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "Knowledge Agents." Content without physical equations belongs to the research methods, knowledge or coordination layer, and formalized formulas are not used to substitute for missing physical evidence. Chapter 23 remains a source-constrained synthesis and cannot support validated design wind, design loads, code recommendations, or scientific conclusions approved by PI

24 Writing and Communication Agents

Writing and Communication Agents include Manuscript, Figures, Daily Briefing, and Collaboration Test Bot and Claude Analysis Bot located in the Lark Buildings group. They are responsible for organizing source-identified scientific content into Chinese and English reports, producing charts and figures, providing progress information to the Researcher, and serving as the opinion channel for external reviewers. All writing and communication output constitutes presentation derivatives; scientific status remains `not_scientifically_approved`.

24.1 Limitations, Interfaces, and Researcher Judgment

24.1.1 Manuscript and Figures

Manuscript organizes source-identified scientific content into Chinese and English reports, maintaining the completeness of citations, equations, figures, tables and limitation statements throughout the organization process, without diluting or upgrading evidence levels due to language conversion. Chinese and English versions maintain the same sources and evidence boundaries; page count differences arise from language typesetting. All Chinese reviews in PETW's review family are produced by Manuscript: CM1 two families, CFD AutoCFD v9, Windfield REV5 (53 pages in Chinese, 76 pages in English), EVA (21 pages in Chinese, 25 pages in English), WindTunnel R07 (92 pages in Chinese, 109 pages in English), Structural (45 pages each in Chinese and English), DesignLoad v1.2 (21 pages in Chinese, 22 pages in English); evidence status for all is engineering preparation. These reviews are candidate research documents whose role in the pipeline is organization and presentation, arranging the source-identified scientific content already produced by each lane in a unified structure for the Researcher's judgment. Manuscript does not generate new scientific content; its work boundary is to present existing content while maintaining its evidence level unchanged, and content with missing sources or unclear boundaries does not enter the text. The scale of each family varies with the evidence requirements of each lane; scale differences arise from differences in the organization scope of each lane and do not establish quality or conclusion comparisons.

Figures produces architecture diagrams, schematic diagrams, and data-driven result figures, with fully English internal labels for the English version. Architecture diagrams express the organizational relationships of PETW, schematic diagrams express the qualitative structure of methods or mechanisms, and data-driven result figures directly present the output of each lane, with in-figure numerical values and coordinate conditions annotated according to source evidence level. The Chinese-English dual versions of the main architecture diagram have been completed, covering SVG, PDF and PNG formats, with PNG width no less than 4000 px. Completeness of English internal labels is part of the bilingual publication discipline; label replacement does not change the numerical values and conditions of figures. Data-driven result figures require confirmation that their data source report already exists before production; in-figure numerical values are taken from the source report, and Figures does not itself supply numbers for figures. Figures constitute presentation derivatives; the production process follows the same source and evidence boundaries as the main text.

24.1.2 Daily Briefing and External Collaboration Channels

Daily Briefing provides compact progress information to the Researcher, covering the day's progress and pending items for each lane. Numbers and conclusions in the briefing are based on the corresponding result reports; the briefing itself does not establish a scientific record, nor does it

enter the knowledge layer's evidence supply. The coverage of the briefing includes items completed, in progress and pending for the day, with each information item corresponding directly to a source report; the briefing only indicates status and location, and the Researcher returns to the report itself when details are needed. This division of labor places progress information and scientific results in different channels, avoiding the version inconsistency that arises from maintaining the same conclusion in both a summary and the main text.

Collaboration Test Bot and Claude Analysis Bot are located in the Lark Buildings group, constituting the external opinion entry point of PETW. External Reviewers submit opinions through this group. Collaboration Test Bot receives requests explicitly directed at PETW and routes them through a defined path to the PETW Coordinator. Claude Analysis Bot reads files accessible within the group and completes in-depth comparative analysis and reviewer-style comments. There is an explicit human judgment step between external opinions and scientific work: reviewer opinions are transformed into specific scientific work by the Researcher and Coordinator before entering the corresponding lane for execution. Reviewer opinions enter through the group and do not gain access to any profile. Collaboration Test Bot only receives requests that explicitly target PETW; other group content does not enter the PETW pipeline. Claude Analysis Bot can only read files accessible within the group; its output exists in the form of reviewer-style comments. Transforming comments into scientific work requires matching the opinion to a specific lane's current design; this matching is performed by the Researcher and Coordinator, and the two bots are limited to the channel and comparative analysis.

24.1.3 Researcher Involvement

The Researcher decides the writing scope, figure selection, external sharing, external review responses and final text. Writing scope determines which scientific content enters the report, figure selection determines which figures are presented alongside the text, external sharing determines which reports may leave the project boundary, external review response determines which opinions are transformed into scientific work, and final text determines the version for release. These five categories of decisions together constitute the exit gate of the writing layer: writing scope and figure selection occur before writing begins, external sharing and review response occur after the text is formed, and the final text decision covers all output. Output from Manuscript and Figures is presented externally only after the Researcher's selection; whether external opinions received by Collaboration Test Bot are transformed into scientific work is jointly decided by the Researcher and Coordinator. No writing agent may independently publish or externally share content, nor change existing evidence levels in the text.

The writing process must maintain consistency among sources, main text, figures, and tables, and status terms. If numerical values from source reports are compressed into summaries, conditions, units and case identity must still be preserved; when figures are moved from original reports into integrated reports, original data paths, generation methods and claim limits are transferred together; language polishing may only adjust syntax and terminology, and cannot change "planned," "candidate" or "diagnostic" to completed or validated. If scope differences arise between the Chinese and English versions, the differences must be explained by explicit writing decisions and cannot be independently produced by the translation process.

The work boundary of Figures also includes preventing figure visual structure from causing evidence escalation. Flow arrows indicate information or plan transmission, not that physical causality has been closed; side-by-side panels indicate comparison on a common page, not that

data come from the same event; color hierarchy indicates classification or status, not statistical significance. Coordinate axes, units, time windows and sources of result figures must be readable; method figures should state in the caption that they are original project syntheses. Decorative figures without data sources do not enter the scientific figure count.

Daily Briefing provides only pointers and pending items for the same result, without maintaining a second set of scientific narrative. If the briefing is inconsistent with the result report, the source report and subsequent Researcher decisions take precedence, and the pointer is corrected in the next briefing. This avoids progress text becoming unconditional conclusions after multiple forwards. The briefing serves to reduce location costs, not to substitute for reading source materials.

External reviewer opinions retain their opinion identity upon entry. Factual corrections need to return to the source for verification, method suggestions need to be passed to the relevant lane for evaluation, claim challenges need to check the evidence ceiling, and supplementary sources need to enter the knowledge workflow. An opinion coming from outside does not automatically elevate its authority level; internal personnel likewise cannot skip verification because they disagree. The Researcher decides the response approach; Manuscript only implements accepted modifications and preserves the records on which the modifications are based.

Pre-final-text checks include citation key existence, figure paths and sources, equation variables and units, prohibited expressions, claim boundaries and chapter structure. Passing the checks only shows that the text is deliverable at the contract level. Whether scientific content is accepted, which quantitative results are permitted to be retained, and which figures may be sent externally are still decided by the Researcher after independent review. Writing and Communication Agents are therefore responsible for expressing evidence accurately, concisely and auditably, not for conferring final authority on evidence.

Evidence and Formula Boundaries of Writing and Communication Agents Chapter 24 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "Writing and Communication Agents." Content without physical equations belongs to the research methods, knowledge or coordination layer, and formalized formulas are not used to substitute for missing physical evidence. Chapter 24 remains a source-constrained synthesis and cannot support validated design wind, design loads, code recommendations, or scientific conclusions approved by PI

25 Computation, Remote and Project Support Agents

The Computation, Remote and Project Support Agents layer includes seven agents: MATLAB, ComputeOps, RemoteOps, CAD, ModelOps, IncidentOps and Vault Manager. This layer does not produce scientific content; it provides numerical computation execution, remote reading and operational support, geometry preparation, model availability maintenance, service recovery and project file management, enabling the work of scientific agents to be executed and to remain traceable. All output of the support layer remains at the engineering preparation and record level; all content in this chapter remains `not_scientifically_approved`.

25.1 Evidence, Results, and Interpretation

25.1.1 Common Role and Functional Classification

The seven agents are classified into three functional categories. The first category covers execution and conversion, comprising MATLAB, ComputeOps, RemoteOps and CAD, which convert

evidence into executable inputs and convert jobs into readable receipts. The second category covers system availability maintenance, comprising ModelOps and IncidentOps, which address model invocation and service recovery for the entire system. The third category covers project asset organization, undertaken by Vault Manager, which addresses persistent management of project files and records. The three categories correspond to three evidence types: execution and conversion produces engineering preparation and generated inputs (scripts, time series, geometric models, data files); system availability maintenance produces operational records (model status, recovery operation logs); project asset organization produces project management records (file indices, directory structures). None of these evidence types leads to scientific results: job status proves the existence of the execution chain, file integrity proves the existence of artifacts, and neither proves the establishment of any scientific conclusion. The support layer therefore holds no entry point for scientific judgment; there is a category boundary between its work records and scientific conclusions that cannot automatically be crossed.

25.1.2 Support Relationships with Scientific Agents

The support relationships of the seven agents with scientific agents appear in Table 37. From the distribution of support relationships, computationally intensive lanes (CM1, WRF, CFD) predominantly receive support from ComputeOps and RemoteOps, because numerical simulation requires computation coordination and remote data synchronization. Data organization and analysis lanes (Windfield, Structural, DesignLoad, WindTunnel) receive scripting, post-processing and plotting support through MATLAB, because the work of these lanes involves time series processing, statistical analysis and result visualization. Geometry-related lanes (CFD, WindTunnel) receive geometry preparation support from CAD, because mesh generation and model fabrication require geometric input. ModelOps, IncidentOps and Vault Manager serve all lanes, because model availability, service recovery and project records are project-level common prerequisites.

Support relationships simultaneously carry authorization conditions: MATLAB executes only within the given script and data scope and does not itself modify scientific assumptions; ComputeOps and RemoteOps coordinate and read only within the Researcher's authorized scope, with HPC submission and remote write-access requiring explicit authorization; CAD's geometric output is premised on geometric key assumptions confirmed by the Researcher, and changes to geometric assumptions require the Researcher's re-confirmation. This support matrix describes the division of responsibilities and authorization conditions, and does not establish endorsement of any lane's scientific results.

25.1.3 Researcher Authorization and Decisions

All actions of the support layer are executed and reported within the Researcher's authorized scope. Six categories of actions retain human decision points, with no automatic pass-through.

Remote write-access determines which content may enter the remote workspace, involving data writes to external systems. The remote workspace is the location of the PETW source of truth; any write changes the authoritative record status of the project. Accordingly, RemoteOps requires the Researcher's explicit authorization before executing remote writes; the authorization scope includes the write target path, write content and expected effect. HPC submission determines whether computation jobs enter the cluster, involving the allocation and use of computational resources. Once an HPC job is submitted it consumes computational resources, and job output may become input for downstream lanes; ComputeOps requires the Researcher to confirm job configuration and resource requirements before submission. Facility action determines the use

Table 37: Support relationships of computation, remote and project support agents with scientific agents. Support relationships describe the division of responsibilities and authorization conditions and do not establish endorsement of any lane's scientific results. Compiled from the 2026-08-12 support architecture audit record.

Support agent	Served scientific agents	Support content	Authorization conditions
MATLAB	Windfield, Structural, DesignLoad, WindTunnel	Scripting, post-processing, plotting, target history generation	Given script and data scope
ComputeOps	CM1, WRF, CFD	Local and remote computation coordination, scheduling, job status monitoring	Researcher authorized scope; HPC requires explicit authorization
RemoteOps	CM1, WRF, CFD	Remote reading and operational support, data synchronization	Researcher authorized scope; remote write-access requires explicit authorization
CAD	CFD, WindTunnel	Geometry preparation, model construction, measurement point layout drawings	Researcher confirms geometric key assumptions
ModelOps	All lanes	Model availability maintenance, version management	Model version changes require Researcher decision
IncidentOps	All lanes	Service recovery, incident recording	Recovery operations recorded on file
Vault Manager	All lanes	Project file organization, asset indexing, record management	Does not modify file content

authorization for wind tunnels and other facilities, involving the operation of physical assets; facility use requires joint authorization from the Researcher and the facility provider, and the support layer does not possess independent authority for facility use. Geometric key assumptions determine the preconditions for CAD output, involving basic parameters such as cross-section proportions, scale ratios, measurement point density and locations. Changes to geometric assumptions alter the input boundaries of CFD meshes and WindTunnel models, which in turn affect the comparability of downstream computations and experiments; CAD requires the Researcher's re-confirmation when geometric assumptions change. Model changes determine the model versions adopted by each lane, involving the configuration basis of scientific lanes; version changes may alter the premises of a lane's scientific output, and ModelOps records the change content during version changes with the Researcher deciding whether to adopt. Formal project status changes determine project-level status updates, involving the authoritative expression of project records; Vault Manager does not itself modify project status records.

These six categories of actions involve, respectively, external systems, computational resources, physical assets, model premises, scientific configuration and project records; the impact scope of each category extends beyond the responsibility boundary of a single agent, and therefore all retain human decision points. Authorization and reporting constitute the two disciplines of the support layer: authorization determines whether an action occurs, reporting determines how the result of the action is recorded, and both are constrained by the Researcher's established scope. Agents may not itself expand their permissions, may not itself change project status, and may only execute and report results within the authorized scope. The definition and adjustment of authorization scope are led by the Researcher; the action space of the support layer remains synchronized with the Researcher's authorization scope.

The support layer therefore does not produce any content that can automatically be elevated to scientific conclusions. Its output remains at the engineering preparation or record level; scientific judgment for entering the next stage is made solely by the Researcher. This arrangement makes all actions of the support layer traceable within the authorized scope, while ensuring that actions beyond the authorized scope do not automatically occur. The evidence categories of the support layer permanently remain at three levels: engineering preparation, execution records and project management; the category boundary with scientific conclusions is maintained by the six categories of human decision points and does not automatically dissolve due to the degree of completion of support work.

Execution records also need to distinguish among planned, submitted, running, completed, and scientifically received. Planned shows that resources and inputs are prepared; submitted shows that the job has entered the scheduling system; running shows that the computation process has started; completed shows that the process has returned a status; scientific reception requires the corresponding lane to check the output and evidence boundaries. The first four are recorded by support agents; the last belongs to the scientific agent and the Researcher. A normal job completion status does not mean that the output satisfies physical diagnostics; queue or resource status also does not enter scientific conclusions.

Generated artifacts from MATLAB and CAD constitute input conversion products. Script read-back can check whether file formats, array lengths, unit fields and geometric dimensions are preserved, but cannot judge whether the target time history represents a real wind field, nor whether the geometric choice is most suitable for the experiment or CFD. Hashes, versions and parameters before and after conversion should be saved so that lanes can locate input changes; however, these operational metadata are retained in the ledger and do not need to occupy scientific body text with internal configuration details.

The authorization minimization of ComputeOps and RemoteOps simultaneously protects resources and evidence identity. The authorization scope of a job should include the target, configuration, resources and output location; the authorization scope of remote transfer should include source, destination and read-write direction. Retries, parameter changes or writes beyond the scope require new decisions. This prevents operational convenience from altering scientific configuration and enables determination of whether unrecorded input changes exist when results show discrepancies.

The version records of ModelOps serve reproducibility, but model updates should not automatically propagate to ongoing scientific chains. If the language model, analysis program or dependency versions change, relevant lanes need to judge whether old output is still comparable, whether regression checks are needed, and whether the new version introduces different premises. When IncidentOps restores services, it prioritizes restoring the established configuration and does not change scientific tasks through temporary substitution; if this cannot be maintained, differences should be explicitly reported and a decision awaited.

The indexing of Vault Manager needs to maintain separation between file content and project status. File existence, path locatability and hash stability can prove artifact identity but cannot prove content correctness. Formal project status can only be updated through designated governance pathways and cannot automatically be changed based on newly appearing result files in a directory. Indexing should likewise leave traces for deletions, moves or replacements, making citation path changes traceable without breaking historical sources.

The maturity of support relationships should be measured by whether interfaces are clear, authorization is clear and receipts are traceable, rather than by execution volume. More submissions, more synchronizations or more generated files may reflect workload, but may also amplify unresolved input problems. The support layer should prioritize ensuring that each action has a clear source, boundary and recipient, avoiding batch execution with no downstream use.

Handoff with scientific agents adopts a minimally sufficient data packet. Input preparation delivers parameters, units, format and sources; computation coordination delivers configuration, environment, exit status, and result paths; remote synchronization delivers source and target identity; CAD delivers geometry version and key assumptions. The downstream confirms object completeness upon receipt and then independently performs scientific diagnostics. Support agents do not fill in scientific pass status on behalf of the recipient.

When a support action fails, the record should state the failure location and artifact impact, not write a service failure as a scientific failure. Input files not generated, jobs not submitted or synchronization incomplete will block corresponding scientific steps; a job completed but failing physical diagnostics belongs to the scientific lane. Separating the two types of failure allows repair actions to target infrastructure or the method itself, rather than masking scientific problems through repeated submission.

Completed deliveries cited in this chapter are all limited to frozen records. Target time history read-back, computation coordination, geometric models and asset indexing prove that the corresponding support objects exist and are locatable, and do not show that facilities, HPC or remote write-access have received authorization beyond the recorded scope. Future actions continue to request Researcher decisions item by item, particularly for physical facility use, long-duration computation and remote writes.

The final quality criterion of the support layer is to enable scientific lanes to answer three factual questions: where the input came from, under what authorization and configuration the action occurred, and where the output is and how it was received. If all three facts are complete, scientific personnel can reproduce or audit the process; if any one is missing, even if result files exist, they should not enter downstream use. This standard maintains the division of labor between engineering preparation and scientific interpretation.

Accordingly, what Computation, Remote and Project Support Agents provide is controlled execution capability, not autonomous action authority. MATLAB, ComputeOps, RemoteOps, CAD, ModelOps, IncidentOps and Vault Manager each operate only on their designated objects; all changes affecting external systems, resources, facilities, key assumptions, models or formal status are returned to the Researcher. This boundary allows automation to improve continuity while preventing the automation level from being mistaken for scientific maturity.

Evidence and Formula Boundaries of Computation, Remote and Project Support Agents Chapter 25 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "Computation, Remote and Project Support Agents." Content without physical equations belongs to the research methods, knowledge or coordination layer, and formalized formulas are not used to substitute for missing physical evidence. Chapter 25 remains a source-constrained synthesis and cannot support validated design wind, design loads, code recommendations, or scientific conclusions approved by PI

Part V
Integrated PETW Workflows and Evaluation

26 Complete workflow from events to design loads

PETW's scientific pipeline begins from event classification, proceeds through meteorological simulation, wind field characterization, statistical modeling, engineering conversion and structural response, and ultimately arrives at candidate design load expressions. Each step of the pipeline is carried out by one or more scientific agents, and adjacent steps are connected through clearly defined interface variables. Interface variables serve both as the scientific content of upstream outputs and as constraint conditions for downstream inputs. This chapter describes, in the order of the scientific chain, the pipeline's interface structure, evidence ceilings, failure and return conditions, uncertainty propagation mechanisms, and the positions of Researcher intervention. The pipeline is the scientific framework of PETW, and agents are the organizational means for implementing the scientific framework; the framework determines what scientific questions each step needs to answer, and agents determine who organizes evidence and executes computations. All content in this chapter remains `not_scientifically_approved`.

26.1 Evidence, results and interpretation

26.1.1 Pipeline interface structure and transfer conditions

The pipeline contains six steps, and a set of interface variables exists between each pair of adjacent steps. The definitions of interface variables precede any computations or observational results, because they constrain what the upstream must provide and what the downstream may expect.

The output interface for Step 0 (Events) contains event identity, mechanism label $k \in \{\text{Path C}, \text{Path M}, \text{Path S}, \dots\}$, observational evidence and mechanism hypotheses. The mechanism label is the classification index for all subsequent steps: different values of k correspond to different simulation strategies, different wind field feature extraction schemes, and different extreme value sample definitions. Event identity must be accompanied by station identification, observation height, reporting time interval and observation quantity definition. The BESSIE event is the current observational reference case, and the Level 2 reflectivity and radial velocity data from three NEXRAD stations KFDR, KTLX and KVNX, together with the Oklahoma Mesonet surface station 5-minute observation records, provide the multi-source fusion input required for event identification. The evidence ceiling for Step 0 is: observational records support event identification and mechanism hypotheses but do not establish mechanism confirmation or completion of the event catalog. Mechanism labels at this stage are candidate labels, and their confirmation depends on evidence from subsequent steps and Researcher judgment. Failure conditions for Step 0 include untraceable event identity, missing observation sources, or contradiction between mechanism hypotheses and observational records. When any failure condition is triggered, the event is not transferred to Step 1 but instead returns to the event identification stage for re-evaluation.

The input interface for Step 1 (CM1 and WRF) comes from the event identity and mechanism labels of Step 0. For mechanism-oriented simulation problems, CM1 generates idealized downbursts using a cooling-source parameterization scheme, with interface inputs including the pre-convective environmental profile and cooling-source geometric parameters. For real event environment reconstruction, WRF reconstructs the convective environment at the time of event occurrence using a mesoscale numerical model, with interface inputs including reanalysis initial fields and boundary

conditions. The output interface for Step 1 is a time series of high-resolution three-dimensional wind fields, containing horizontal and vertical velocity components, temperature and humidity perturbations. The transfer condition for this output is: the simulation has been completed, the physical units and coordinates of output quantities have been specified, and the simulation scheme parameters and boundary conditions have been recorded. The 24/9/3/2/2 distribution of CM1 is a registry-derived model-internal Level-1 screening result, not a validated downburst simulation. The Schumacher 2023 reconstruction of WRF maintains PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION status. Failure conditions for Step 1 include simulation non-convergence, output physical quantities exceeding reasonable ranges, or untraceable simulation schemes. Upon failure, the process returns to the simulation configuration stage, and undiagnosed simulation outputs are not transferred to Step 2.

Step 2 (Windfield) extracts engineering-usable wind field features from the simulated wind fields of Step 1: vertical profiles (10 m to 500 m wind shear and low-level jet position), horizontal footprints (outflow radius and spatial decay), near-surface turbulence characteristics (turbulence intensity, gust factor, integral scale) and temporal evolution (gust duration, rising and falling edges). The extraction results constitute a feature vector, and each storm type has an independent feature distribution. The output interface for Step 2 is directed toward three different downstream recipients: the output toward EVA consists of wind speed quantity definitions and mechanism labels compatible with extreme value analysis; the output toward CFD consists of profile parameters and turbulence statistics required for inlet boundary conditions; the output toward WindTunnel consists of target velocity time histories and scaling conditions. The evidence ceiling for Step 2 depends on whether the upstream simulation resolution is sufficient to support the extracted features. When the resolution is insufficient to support a particular feature, that feature is annotated as unavailable and is not transferred to the downstream. Source conflicts (differences in definitions or numerical values for the same feature across different literature sources) are preserved as unresolved in the Windfield stage, and scale conversion has not yet been closed.

Step 3 (EVA) determines candidate extreme value analysis routes after wind speed quantity definitions, mechanism classifications and event independence are established. The input interface consists of the feature vector provided by Step 2 and the event independence information provided by Step 0. The output interface is the candidate return-period wind speed $V_k(T)$ and its uncertainty range. The transfer condition for this output is: the composition process and detection conditions of the extreme value sample have been recorded, the fitting method and parameter estimation approach have been described, and confidence intervals or uncertainty ranges have been quantified. The evidence ceiling for EVA is: there is currently no final event sample, no final return-period level, no final $\hat{V}_{3s}$, and no design wind speed. Candidate analysis results do not enter downstream design expressions before Researcher review.

Step 4 is jointly constituted by three agents forming the engineering conversion and structural response evidence layer. WindTunnel (Step 4a) provides validation data and calibration references from controlled experiments, with its current status being DESIGN_ONLY_NOT_ACTIVE, and no physical experimental results are available. CFD (Step 4b) receives the inlet conditions from Step 2 and computes the flow field around buildings and surface pressure time histories, with transfer conditions requiring that case identity, grid, turbulence closure model, computational domain and time range are all recorded. CFD values are valid within the scope of case, grid, closure, domain and time range, and do not establish independent numerical method verification. Structural (Step 4c) receives floor force time histories provided by CFD or WindTunnel and computes

displacements, accelerations and overall response. The output of Structural depends on input load quality, structural model, damping and nonlinearity assumptions, and does not establish design loads.

Step 5 (DesignLoad) receives outputs from all upstream steps and organizes mechanism-conditional statistics, event rates and response evidence into candidate design expressions. Its general form can be expressed as

$$V_{\text{design}} = f(V_k(T),\ \eta_k,\ \lambda_k) \tag{46}$$

where $V_k(T)$ is the candidate return-period wind speed corresponding to mechanism k (Step 3 output), η_k is the response correction coefficient corresponding to mechanism k (Step 4 output), and λ_k is the event occurrence rate of mechanism k (joint output of Step 0 and Step 3). The specific form of $f(\cdot)$ depends on the choice of design framework (gust response factor method, equivalent static wind load method, or time-domain direct method) and is determined by the Researcher. Equation (46) is the general interface of candidate design expressions, not an approved design value or code recommendation. All outputs of DesignLoad maintain `candidate interpretation, not_approved` and `not_for_design` status.

The corresponding methodological relationship summarizes the pipeline's interface structure and transfer conditions.

26.1.2 Current case combinations and end-to-end coverage

The evidence at each step currently comes from combinations of different cases, and no single event traverses the entire chain. Table 38 lists the coverage and evidence status of major cases across steps.

Table 38: Coverage and evidence status of current major cases across PETW pipeline steps. "✓" shows that the case has produced evidence materials at that step, "—" shows that no evidence from that case exists at that step. The evidence status column annotates the scientific limitations of each case at its covered steps. The combination of cases illustrates the current evidence status and connection pattern at each step, and does not establish a single verified end-to-end chain.

Case	S0	S1	S2	S3	S4	S5	Evidence status
BESSIE	✓	—	—	—	—	—	Observational reference, not mechanism confirmation
Moore 40-case	—	✓	—	—	—	—	Level-1 screening, not validation
Schumacher 2023	—	✓	—	—	—	—	PARTIAL_ALIGNMENT
D420 CM1-to-CFD	—	✓	—	—	✓	—	NOT_CONSISTENT comparison
Case011Up	—	—	—	—	✓	—	ABL maintenance diagnostics
exact146	—	—	—	—	—	✓	Model-internal benchmark
B2/P1/EPSD	—	—	—	—	—	✓	Method demonstration, not design values

BESSIE provides the observational reference for Step 0, including multi-station radar data, surface station network observations and ground lightning location fusion information. Moore 40-case provides the CM1 model-internal Level-1 screening for Step 1: the 24/9/3/2/2 distribution is a registry-derived model-internal screening result, where 24 represents the number of members classified into a certain primary category, and the remainder are members of other categories. Schumacher 2023 provides the WRF mesoscale reconstruction for Step 1, with status PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION, where the reconstructed convective structure and reflectivity field partially align with observations in spatial position but exhibit differences from the reference data in temporal evolution and the tail phase. D420 provides the CM1-to-CFD interface demonstration from Step 1 to Step 4, but the comparison between M0/M1 configurations is in NOT_CONSISTENT_CONTROLLED_COMPARISON status. Case011Up

provides the ABL maintenance analysis for Step 4, including inlet profile versus target comparisons, streamwise drift diagnostics and iteration evolution, but ABL maintenance diagnostics do not establish solver verification. exact146 provides the structural response benchmark from Step 4c to Step 5, including the knowledge base of 192 Zotero identities and 146 detailed readings, but the output depends on input load quality and structural model assumptions. B2/P1/EPSD provide method demonstrations for Step 5: B2 provides a dimensioned benchmark demonstrating the gust response factor computation path, P1 demonstrates the direct pressure method route, and EPSD demonstrates the non-stationary time-frequency response path; all three are method demonstrations rather than design values.

The cases cannot be written as a single verified end-to-end chain. The observational reference of BESSIE cannot directly be connected to the model screening results of Moore 40-case, because the initial fields of Moore 40-case come from different environmental profiles. The CM1-to-CFD interface demonstration of D420 lacks controlled comparison conditions between the M0/M1 configurations. The input loads used by the structural response benchmark of exact146 do not come from actual outputs of upstream steps in the PETW pipeline. The method demonstrations of B2/P1/EPSD use independent parameter inputs without direct linkage to upstream steps. End-to-end traversal requires the same event to begin from Step 0 classification and observation, proceed through meteorological simulation for that event, wind field feature extraction based on that simulation, statistical analysis based on those features, engineering conversion and structural response based on those statistics, and finally arrive at candidate design expressions based on that response. This condition is currently not satisfied.

The corresponding methodological relationship shows the coverage distribution of the current case combinations in the pipeline.

26.2 Limitations, interfaces and Researcher judgment

26.2.1 Distinction between reproducibility and the scientific framework

The reproducibility of the pipeline is defined at two levels. At the computational level, reproducibility requires that the input parameters, software versions, configuration files and random seeds of each step are recorded, so that the same inputs in the same environment produce the same outputs. Computational reproducibility is the execution-level responsibility of agents, recorded by each agent upon task completion. At the scientific level, reproducibility requires that the scientific framework of the pipeline (the scientific questions, interface definitions, evidence ceilings and transfer conditions of each step) be independent of the specific agent implementation. In other words, if all agents in PETW were replaced by manual execution, the scientific framework and its interface variable definitions, evidence ceiling constraints and transfer condition judgments should remain unchanged. The practical implication of this distinction is: the scientific validity of the pipeline depends on the completeness of framework definitions and the accuracy of Researcher judgment, not on the execution efficiency or degree of automation of agents. The framework is PETW's structured answer to thunderstorm wind engineering scientific questions, and agents are the organizational means for decomposing that answer into executable work.

End-to-end reproducibility also requires that each interface can be independently read back. The upstream should be able to reconstruct the source, units, coordinates, time and applicability conditions of variables from the handoff package, and the downstream should be able to state which fields were used, how they were converted, and which fields were rejected. If only the final arrays are saved while interface decisions are lost, even if the same program can be run again,

it is impossible to determine whether the two runs used the same scientific definitions. Interface read-back is therefore equally necessary as code replay.

The sparsity of the current case combinations dictates that near-term work should prioritize closing one minimal evidence chain rather than continuing to add unconnected examples. The minimal chain begins from one event identity, selects a simulation or observation route compatible with that event, preserves Windfield transfer conditions, and then propagates the same input to engineering conversion and a single structural response target. Each step can halt when a higher evidence level has not been reached, and return the gap to the upstream. Only by maintaining the same event and the same variable identity will additional computations increase end-to-end information.

Interface return is not an ancillary operation after failure, but a routine mechanism of the pipeline. If upstream data lacks units, time averaging or spatial support, the downstream should not guess default values; if the downstream target requires coherence or phase information not provided by the upstream, an explicit data request should be issued. The return record specifies the missing fields, scientific consequences and the minimum permissible repair, enabling the upstream to make localized supplements rather than re-executing the entire chain.

Researcher intervention also needs to be bound to interface objects. Event selection, simulation configuration, scale conversion, sample admission, experimental execution, CFD transfer-out, structural target and candidate expressions each produce different decisions. A specific decision is valid only for the specified object and does not propagate to other steps as a general approval. This granularity allows human judgment to be auditable and prevents a single authorization from being interpreted by the downstream as an evidence level elevation.

Uncertainty at interfaces transfers both numerical values and identities. Numerical ranges can describe parametric or statistical errors, while identity fields specify whether the uncertainty originates from classification, modeling, conversion or methodology. The downstream can propagate existing ranges within its computations but cannot unconditionally merge uncertainties from different sources. Current data are insufficient to establish a full-chain joint distribution; therefore this study adopts itemized preservation and conditioned interpretation.

The three-segment engineering chain of WindTunnel, CFD and Structural particularly requires a common target quantity. Wind tunnel tests may provide point pressures, CFD may provide field pressures or floor forces, and Structural requires nodal forces or generalized forces. Comparison and transfer must specify the integration or mapping from measurement points to pressure bands, from field pressures to floors, and from floors to nodes. If the common target is only the top displacement, intermediate objects must still be preserved so that the physical or numerical source can be located when discrepancies arise.

The candidate expression of DesignLoad sits at the end of the chain but does not possess authority to repair upstream gaps. When EVA has no candidate sample, DesignLoad cannot substitute from simulation peaks; when WindTunnel has no physical measurements, a design scheme cannot substitute for validation; when Structural has only benchmarks, its response ratios cannot substitute for results from real inputs. Maintaining vacancies at the chain end protects subsequent research better than filling in incompatible values.

Accordingly, the complete workflow described in this chapter characterizes scientific dependencies and evidence discipline, not a project pipeline that has already been run to completion. Existing cases demonstrate that certain interfaces, diagnostics and method branches can be orga-

nized, but no single line of cases covers Step 0 through Step 5. This genuine shortfall should be preserved in the contract gate records and cannot be offset by the number of charts or richness of results from different cases.

The acceptance object for the end-to-end chain also needs to be fixed to a single response target. Although event identity, meteorological fields, Windfield packet, pressures or floor forces, and structural model cover different physical layers, only under the same target response can one determine whether the information provided by each interface is sufficient. If the target is top displacement, load-to-height mapping, structural mode shapes and time synchronization constitute the minimum check set; if the target changes to base bending moment or torsion, the required spatial resolution, moment arms and facet information change accordingly. The minimal chain is therefore not the file collection with the least content, but rather the evidence object that enables a source-constrained target response to be completely read back from input to output.

Chain verification can simultaneously employ forward propagation and backward tracing. Forward propagation checks whether upstream permitted uses are inherited after each handoff, and backward tracing proceeds from the candidate response back to structural parameters, aerodynamic forces, wind fields and event identity, checking whether default values or identity substitutions have occurred in between. If the two directions yield different variable definitions at a given interface, the discrepancy should be recorded as a return condition. This approach does not elevate the evidence level of existing cases, but it can locate the reasons for incomplete connectivity to specific fields, specific conversions and specific decisions.

After the minimal chain is completed, it should still be preserved in parallel with the existing benchmarks of each lane. Benchmarks are used to check whether program evolution has broken internal consistency, while the common-event chain is used to check source-constrained scientific propagation; the two have different input identities and acceptance objects. When common-event results show anomalies, benchmarks should first be used to rule out program regression, and then the investigation should return to interfaces and physical inputs for localization; benchmark passage can only narrow the scope of investigation and cannot prove that common-event results are correct.

Evidence and formula boundaries of the complete workflow from events to design loads
Chapter 26 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of the “complete workflow from events to design loads.” Content without physical equations belongs to the research methodology, knowledge or coordination layer, and formalized formulas do not substitute for missing physical evidence. Chapter 26 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations or scientific conclusions approved by PI

27 Complete workflow of knowledge, computation, writing and external review

The operation of the scientific pipeline depends on four supporting workflows: the knowledge workflow provides literature sources and evidence organization, the computation workflow executes numerical tasks and result extraction, the writing workflow transforms scientific content into reviewable technical text, and the external review workflow introduces independent checks and feedback. The relationship between the four workflows and the scientific pipeline is: the pipeline defines the structure and transfer conditions of scientific questions, and the four workflows provide

operational support for the pipeline within their respective scopes of responsibility. Completeness of the supporting workflows does not equal evidence completeness of the scientific pipeline. All content in this chapter remains `not_scientifically_approved`.

27.1 Evidence, results and interpretation

27.1.1 Knowledge workflow

The knowledge workflow spans six stages from source discovery to Researcher judgment, with each stage having specified inputs, outputs, and quality assurance mechanisms.

Source discovery is performed by Gemini or Web scout, producing a preliminary list of candidate sources and search paths. The scope of source discovery covers academic databases, publisher platforms, open access channels and patent databases. Discovery results contain basic bibliographic information and access feasibility assessments for candidate literature, but do not include content reading. The quality of the source discovery stage is indirectly reflected by the pass rate of subsequent identity verification.

Identity verification is performed by Zotero Literature, confirming the literature identity, version, publication information, DOI and PDF availability of each candidate source. Sources that pass verification receive a unique identifier (Zotero key), which serves as the sole reference key for that source in all subsequent stages, preventing ambiguity in source identity during reading, extraction and citation. Sources that fail verification are annotated with reasons (duplicate records, unavailable PDF, unconfirmable identity, etc.) and do not proceed to the next stage. Identity verification and full-text reading are performed by different agents, because identifying a piece of literature and understanding a piece of literature are two different cognitive tasks. The result of identity verification is that the source exists and can be located, which is not equivalent to the source having been understood or its content having been extracted.

Full-text reading and claim extraction are performed by PETW LitWiki. LitWiki conducts paragraph-by-paragraph reading of identity-verified literature, extracting the specific statement, source paragraph, methodological elements and applicability conditions of each claim. Claim granularity is controlled at the level of a single scientific statement: one claim corresponds to one independently citable method, parameter, conclusion or boundary specification in the literature. Extracted claim records are attached to verified literature identities, and the mapping relationship between claims and sources is fixed at the time of extraction. Extraction simultaneously records applicability conditions and known limitations, which allows subsequent citations to judge applicability ranges by condition. Review synthesis is conducted after reading records are aggregated by topic; each lane compares the methods, parameter choices and applicability boundaries of multiple sources on the same topic against each other, forming method cards and boundary specifications. The value of synthesis lies in the comparative relationships among sources: parameter choices, applicability ranges and known limitations of the same method across different sources are recorded in parallel for research design reference.

Namespace mounting is the channel through which knowledge assets enter a lane. Each namespace represents a complete processing chain from identity verification to claim organization for a set of sources, carrying its own source scope, processing status and evidence category. Each scientific agent obtains evidence supply commensurate with its lane through its own namespace. The namespace `step1_cm1` of CM1 contains 45 source records, with 45/45 passing source-semantic verification. The namespace `eva_vhat3s` of EVA contains 154 files, supporting the organization of the extreme value analysis method family. The namespace `windfield_step2` of Windfield

contains 138 files, with 120 paper pages of reading coverage. The namespace `step4c_struct` of Structural contains 413 files, covering 192 Zotero identities and 146 detailed readings. Delivery is at the namespace level, and when a lane makes references, the referenced objects are specific records within the namespace rather than the namespace as a whole. Researcher judgment is the final stage of the cycle: whether a claim supports a particular scientific conclusion and whether a source is applicable to a particular lane's research design are always decisions that rest with the Researcher.

Failure and return conditions of the knowledge workflow are distributed across the stages. Source discovery failure (searches not covering the target domain) triggers search strategy adjustment. Identity verification failure (literature does not exist, is duplicated, or PDF is unavailable) causes the source to remain in candidate status. Full-text reading failure (literature content is irrelevant to the research question) records the reading result as inapplicable. Claim extraction failure (no independently citable claims in the literature) records the extraction result as an empty set. Failure at any stage does not propagate to the next stage: sources that fail verification do not enter reading, and sources whose reading does not produce claims do not enter review synthesis. Failure records are preserved in the knowledge system as references for subsequent work, avoiding redundant retrieval of the same source.

27.1.2 Computation workflow

The computation workflow advances in five steps from scientific question to result interpretation. The Researcher determines the scientific question and scope of the computation, including the target event, configuration, output quantities and resource boundaries. The scientific agent prepares methods and inputs, including configuration file writing, input data conversion, diagnostic scheme design and pre-run verification. ComputeOps or RemoteOps executes the computation within the authorized scope, with execution records containing environment version, timestamps, resource usage and exit status. The agent analyzes outputs, including post-processing script execution, diagnostic quantity extraction and result comparison. The Researcher interprets results, distinguishing among model outputs, diagnostic evidence and scientific inferences.

The authorization boundary for computation is determined before task initiation. When remote writes, HPC submissions and facility actions are involved, the Researcher's explicit decision precedes execution. The setting of authorization boundaries distinguishes the computation workflow from an unconstrained automated pipeline: agents do not possess the authority to bypass authorization boundaries and initiate computations on their own. After computation is completed, the output enters the corresponding step of the scientific pipeline accompanied by a complete execution record, so that result reproducibility does not depend on verbal agreements. Failure conditions for the computation workflow include environment unavailability (remote resources unauthorized or unreachable), computation non-convergence (abnormal solver exit status or iterations not reaching criteria) and incomplete output (partial output files missing or corrupted). Each type of failure returns to the agent's diagnostic stage, where the agent provides failure reasons and remediation plans, and the Researcher decides whether to retry, adjust parameters or terminate the computation task.

27.1.3 Closed-loop handling of failures, conflicts and review comments

Failure records are a shared scientific asset across the four workflows. Knowledge retrieval failure delineates the boundary of current source coverage, computation failure delineates limitations of configuration, environment or numerical methods, writing failure indicates insufficient source

density or claim ceiling, and unresolved review comments show that external questions still lack evidence. Failure objects should preserve trigger conditions, content already checked, current impact and the preconditions for the next permitted retry. Preserving only successful products causes subsequent personnel to repeat infeasible routes and also creates the impression that project progress exceeds actual evidence progress.

Source conflicts and computation conflicts cannot automatically be eliminated by majority vote. When two pieces of literature use different definitions for the same quantity, the knowledge workflow should first distinguish definitions and domains of applicability; when two implementations give different outputs for the same equation, the computation workflow should fix inputs, discretization and comparison quantities; the main text reports only discrepancies that have been located and their impacts. Agreement among language models can increase check coverage but cannot substitute for original sources or independent computations. Researcher decisions should be based on materials where the conflict structure has been made explicit.

Closing the loop on external review requires preserving the original comment text, target object, verification path, response decision and version result. Factual corrections can return to the knowledge or computation workflow, methodological challenges can return to the corresponding scientific agent, claim challenges return to writing and source verification, and new sources first undergo identity verification before entering the main text. After modifications are completed, they should point to specific derived versions, enabling the reviewer to confirm where the comments took effect. If a comment is not adopted for the time being, the evidentiary reasoning and unresolved conditions should also be recorded, rather than treating silence as closure of the issue.

The common acceptance criterion for the four supporting workflows is therefore readability, not file completeness. Any conclusion in the main text should be traceable back to a source or computational object, any figure or table should be traceable back to data and scripts, any Researcher decision should be traceable back to the evidence visible at the time, and any external comment should be traceable back to the processing result. Once the read-back structure is established, the workflows can proceed in parallel without confusing identities; the current structure still serves candidate materials and does not confer scientific approval on any result.

Read-back also requires that relationships be preserved when objects are deleted or replaced. When a source is judged inapplicable, a computation is superseded by a new version, a figure is withdrawn due to scope error, or a main text claim is downgraded, the old object should no longer serve as an active basis, but its status change and replacement object need to be auditable. This allows distinguishing among currently used evidence, evidence retained solely for historical audit, and evidence that has been withdrawn, preventing the downstream from reintroducing invalidated materials from old catalogs.

Archival checks should also confirm that derived objects have not become detached from their upstream versions. If main text, figures or review responses still point to old computations, they should be updated or the retention reason explicitly stated before the release candidate. This check belongs to version consistency control and does not change the scientific evidence level of the object.

Evidence and formula boundaries of the complete workflow of knowledge, computation, writing and external review Chapter 27 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of the “complete workflow of knowledge, computation, writing and external review.” Content without physical equations be-

longs to the research methodology, knowledge or coordination layer, and formalized formulas do not substitute for missing physical evidence. Chapter 27 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations or scientific conclusions approved by PI

28 Practical value of the agentic approach for PETW

The evaluation of the value of the agentic approach is based on the actual work of PETW, not on tool counts, agent counts or the automation level itself as evaluation criteria. PETW is an engineering research project spanning seven scientific domains: event classification, meteorological simulation, wind field characterization, statistical modeling, engineering conversion, structural response and design load expression. The complexity of this research lies not in the depth of any single domain, but in the interface structure between domains and the continuity of evidence transfer. The agentic approach derives value from its organizational capability for this complexity, not in the elimination of complexity. The distinction between the scientific framework (the pipeline structure, interface variables, evidence ceilings and transfer conditions defined by PETW) and the implementation method (multi-agent human-in-the-loop architecture) is maintained throughout the discussion in this chapter. All content in this chapter remains `not_scientifically_approved`.

28.1 Problems, concepts and methods

28.1.1 Practical role in the continuity of scientific steps

PETW's scientific pipeline contains six steps, each carried out by one or more scientific agents. The fixed assignment of agents means that the knowledge state, configuration history and result records of each lane are continuously maintained across tasks: the CM1 agent preserves records of parameter selections and result comparisons from simulation tasks at different times, and the Windfield agent preserves the unresolved status of source conflicts and version changes of feature extraction methods. A practical consequence of continuity is the reduction of information loss between tasks, which allows subsequent work to continue from where the previous work stopped rather than re-establishing context.

In cross-lane handoffs, the handoff mechanism between agents provides clearly defined interface transfers: the output of the upstream agent is transferred to the downstream agent together with its evidence level, applicability conditions and known limitations. Handoff records make the continuity of the evidence chain auditable. In traditional manual execution modes, information transfer across steps often relies on personal memory or unstructured communication, and the completeness and accuracy of information degrade with time and changes in participants. Structured handoffs between agents migrate transfer content from personal memory to persistable records. The handoff mechanism guarantees the completeness of information transfer, not the scientific correctness of the information content. Erroneous upstream output remains erroneous output after complete transfer; the value of the handoff lies in making errors traceable rather than nonexistent.

28.1.2 Researcher focus

The agentic approach allows the Researcher to focus on scientific judgment rather than process management. Programmatic verification tasks (format checking, citation completeness verification, limitation statement compliance checking, unit consistency verification) are handled by agents, and issues requiring scientific judgment are explicitly returned to the Researcher in the form of decision requests. The content of decision requests includes all background information needed for judgment: relevant evidence materials, applicability conditions, known limitations and candidate

options. After receiving a decision request, the Researcher makes a scientific judgment, and the judgment result is recorded and executed.

A practical consequence of focused attention is that the frequency of Researcher judgment is decoupled from the frequency of pipeline operation. In traditional modes, the Researcher needs to personally participate in every stage of every task, including substantial routine work that does not require scientific judgment. In the agentic mode, routine work is continuously executed by agents, and the Researcher intervenes at judgment points. This mode enables the Researcher to address more issues requiring scientific judgment within the same period, but does not change the scientific rigor requirements of each judgment itself. The Researcher does not lower judgment standards due to efficiency gains, and agents do not acquire scientific judgment authority due to automation capabilities.

28.1.3 Currently supportable value conclusions

The conclusions supportable by existing project materials are limited to the organizational level. Fixed lanes, source ledgers, interface records and layered status help maintain continuity in a multi-year project; programmatic checks help identify inconsistencies among citations, units, figures and text; explicit decision gates help prevent agents from self-elevating claims. Existing materials do not support comparison of the general superiority or inferiority of the agentic approach versus traditional teams in terms of scientific discovery quality, verification speed or design reliability, because the project has not established a controlled organizational method comparison, nor has it completed a full-chain scientific cycle.

Subsequent evaluation of method value can rely on the existing work records, but evaluation metrics must be aligned with scientific objectives. Metrics that can be examined include recovery time for the same object, interface field missing rate, conflict localization time, proportion of redundant computations, types of claim modifications after review, and rework count for decision packages. These quantities can only describe organizational performance and still cannot substitute for physical result quality. If the complete event chain is closed in the future, one can further observe whether uncertainties maintain source distinction after multiple handoffs and whether the Researcher can trace backward from the terminal expression to each upstream object.

Accordingly, the stage-specific evidence from PETW regarding the agentic approach is a bounded case record. It shows how a complex scientific engineering project can organize responsibilities, interfaces, sources and human decisions into a read-back structure, and it also exposes risks such as recording costs, summary loss, metric bias and review burden. This case has not proven autonomous science, nor has it proven that this organizational approach is applicable to all research questions. All value judgments continue to be constrained by the current project stage, infrastructure and unclosed scientific chain, and remain `not_scientifically_approved`.

This evaluation also does not set manual execution as a default disadvantage. Human teams can equally achieve continuity and traceability through rigorous ledgers, version control and review; the agentic structure merely allocates part of the maintenance work to sustainably executing roles. Whether it holds practical value depends on whether recording quality, interface discipline, error detection and Researcher judgment are improved, not on role names or automated execution itself.

This boundary also serves as the baseline for subsequent evaluation.

Evidence and formula boundaries of the practical value of the agentic approach for PETW Chapter 28 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of the “practical value of the agentic approach for PETW.”

Content without physical equations belongs to the research methodology, knowledge or coordination layer, and formalized formulas do not substitute for missing physical evidence. Chapter 28 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations or scientific conclusions approved by PI

Part VI
Concluding Remarks and Outlook

29 Current understanding

This study has formed stage-specific understanding at three levels: the scientific framework, the evidence accumulation at each work stage, and the research methodology. All understanding remains in candidate status and has not yet passed independent scientific review by the Researcher. The discussion below describes, level by level, the content that has been established, the boundaries that have been identified, and the questions that remain unresolved.

29.1 Limitations, interfaces and Researcher judgment

29.1.1 Scientific framework level

PETW's scientific framework takes the classification-first principle as its starting point, organizing thunderstorm wind engineering research into a six-step evidence chain from event classification to candidate design load expressions. Classification-first means that the mechanism identity k (such as Path C cooling-source downburst, Path M mesoscale convective system outflow, Path S supercell-type outflow, etc.) is determined at Step 0 and then carried through all subsequent steps: different values of k correspond to different simulation strategies, different wind field feature extraction schemes, different extreme value sample definitions and different response corrections. This principle elevates classification from post-processing to the pipeline entrance, making the method choices, parameter spaces and evidence ceilings of each step indexed by mechanism. The division into Step 0 through Step 5 creates a set of interface variables between each pair of adjacent steps, and the definitions of interface variables precede computations or observational results, constraining what the upstream must provide and what the downstream may expect. The propagation discipline of nine categories of uncertainty (classification uncertainty, observational uncertainty, simulation uncertainty, wind field feature extraction uncertainty, statistical uncertainty, engineering conversion uncertainty, structural model uncertainty, design expression uncertainty and transfer-accumulated uncertainty) requires that uncertainty be transferred downstream along the chain without being truncated or hidden, so that the evidence ceiling of downstream conclusions is determined by upstream evidence rather than by tool capabilities.

The framework describes the classification and evidence status of each event using a six-layer state vector:

$$\mathbf{s}_e = (F_s,\ C_e,\ M_w,\ \phi_w,\ R_e,\ D_u) \tag{47}$$

where F_s is the parent storm family, C_e is the surface event class, M_w is the local wind mechanism, ϕ_w is the engineering wind-field feature, R_e is the evidence reliability, and D_u is the downstream use status. $\mathbf{s}_e$ is an information organization scheme rather than a probabilistic model; its function is to make event identity, mechanism hypotheses and evidence levels traceable throughout the entire pipeline. The key constraint of the current understanding is: the values of R_e for most events

remain at candidate or partially confirmed stages, and D_u is in unapproved status for all events. The structure of the framework itself has become relatively stable, but structural completeness of the framework does not equal completeness of the evidence filled into the framework.

29.1.2 Evidence status of each work lane

The evidence accumulation at each work lane is at different levels of maturity. The discussion below describes, in the order of the scientific chain, the completed work, established evidence ceilings and unresolved questions for each lane.

Events has established multi-source event data and observational references. The observational basis for the BESSIE event includes Level 2 reflectivity and radial velocity data from three NEXRAD stations KFDR, KTLX and KVNX, as well as Oklahoma Mesonet surface station 5-minute observation records. 879 records were used for data recovery and event information structure description, but 879 records do not represent 879 independent events. The Events synthesis program report contains ten data tables, covering the BESSIE surface observation sequence, storm-scale evidence and display indicators. The observed gust value of 34.9 m/s retains the complete semantics of station identification, 10 m observation height, 5-minute reporting interval and 3-second observed maximum. The evidence ceiling remains as follows: observational records support event identification and mechanism hypotheses but do not establish mechanism confirmation or completion of the event catalog. The event catalog has not yet been established, and the Researcher needs to decide which events can enter the scientific input of subsequent steps.

CM1 has established the 40-case mechanism screening and D420 result presentation. The 24/9/3/2/2 distribution in the 40-case registry is a registry-derived model-internal Level-1 screening result, with systematic sensitivity organization based on the variation range of the cooling-source parameterization scheme. The D420 CM1-to-CFD result presentation report covers M0 and M1 states and comparison conditions, but the comparison status is NOT_CONSISTENT_CONTROLLED_CO meaning that the comparison conditions across different cases have not yet been uniformly controlled. The evidence ceiling remains as follows: CM1 output is model-internal screening, not a validated downburst simulation or engineering inflow. CM1 peak values are not $\hat{V}_{3\mathrm{s}}$, nor are they validated 10 m 3 s gusts. Transfer from CM1 to engineering application requires Windfield's resolution judgment and the Researcher's method decision.

WRF has established the real-case reconstruction for the Schumacher 2023 event. The reconstruction work includes domain and time configuration, convective structure and reflectivity reconstruction, WSPD10MAX extraction, vertical cross-section analysis, and numerical comparison of four-point locations with results from the published study. The reconstruction results exhibit partial consistency with the structures reported in the published study, but quantitative differences exist. The formal status is PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION. The evidence ceiling remains as follows: the WRF reconstruction is not a complete reproduction of the results from the published study, nor is it independent verification. The physics parameterization schemes, initial field sources and boundary condition processing used in the reconstruction have all been recorded, but whether they are fully consistent with the exact configuration of the published study has not been completely confirmed.

Windfield has established the multi-scale wind field feature model and packet specification. 15 TikZ vector figures cover multi-scale components, method families, forcing routes, footprint and temporal semantics, profile families, stationarity levels, time-frequency information preservation, spatial coherence, evidence route ceilings, wind-speed-to-load interfaces, conditional factor hierar-

chies and observational coverage lower bounds. 13 comparison tables cover method family inputs, outputs and transfer ceilings. The Coordinator summary records the work scope of Windfield, completed content and source conflicts. The evidence ceiling remains as follows: source conflicts (differences in definitions or numerical values for the same wind field feature across different literature sources) are preserved as unresolved, and the conversion procedure from meteorological grid to engineering scale has not yet been given a closed definition. The Researcher needs to decide the conflict resolution approach and the conditioned scheme for scale conversion.

EVA has established the method framework but maintains zero-input status. The Step 3 framework summary identifies six core distinctions: observed gusts are not true event maxima, classification labels are not ground truth, confirmed-only samples are not the true population, CM1 peak values are not $\hat{V}_{3s}$, baseline return levels are not design winds, and sensitivity analyses are not code recommendations. The literature synthesis covers Gumbel, GEV, POT/GPD, XIMIS, Method of Independent Storms, regional frequency, directionality and mixed-climate methods. The $\hat{V}_{3s}$ Wiki entry list and coverage status have been established, but the existence of the entry list does not establish completion of the $\hat{V}_{3s}$ definition or numerical availability. The evidence ceiling remains as follows: there is currently no final event sample, no final return-period level, no final $\hat{V}_{3s}$ and no design wind speed. Candidate analysis results of EVA do not enter downstream design expressions before Researcher review.

WindTunnel has completed the full design of accelerating-flow wind tunnel experiments but has no physical experimental results. The experimental design covers 5:1 scale models (M11, M31, M51, with cross-section widths of 60, 180 and 300 mm respectively, all with a depth of 60 mm), pressure tap layouts (draft 26/60/62 taps), PIV measurement windows, six groups of test matrices YM01 through YM06 (each group defining steady-state wind speed U_s, transient wind speed U_t, acceleration parameter a_p and rise time T_{rise} as well as nominal Reynolds number), ten stage gates G0 through G9 (with completion and stop conditions defined for each gate), synchronization schemes and verification paths. The design review bilingual documentation covers apparatus configuration, scaling and similarity criteria, measurement systems, test matrices, sampling and data processing schemes, and quality control frameworks. The WT-M1 review's 14 figures cover inflow families, target velocity time histories $U(t)$, evidence architecture, comparison ceilings, mechanism schematics, temporal cells and acceleration-deceleration directions. The evidence ceiling remains as follows: the DOWN branch maintains DESIGN_ONLY_NOT_ACTIVE status, and no completed physical experiments, facility actions, hardware automation, measured pressures or measured PIV results exist. The transition from design to experiment requires joint authorization from the Researcher and the facility.

CFD has formed model outputs for limited cases. The ABL maintenance analysis for Case011Up (job 3400033) includes five diagnostics: final profile versus target comparison, streamwise drift diagnostics, iteration evolution, resolvable height and building-upstream profile. The floor force result analysis provides force heat maps, key profiles and derived indicators. The nine data tables for D420 cover source traceability, execution QA, variable and unit semantics, boundary surface and flux contracts, case identity and comparison axes, current result status, evidence categories and claim boundaries, and the figure-to-source matrix. The Coordinator summary records the work scope of CFD and the current status of AutoCFD. The evidence ceiling remains as follows: CFD values are valid within the scope of case, grid, closure, domain and time range but do not establish independent numerical method verification; ABL maintenance diagnostics do not establish solver verification, nor do they constitute a completed comparison with experiments; long-duration HPC

computations have not yet been completed; direct conflicts render single-factor attribution untenable. Floor forces require Researcher judgment on input load quality and applicability boundaries before entering Structural.

Structural has established benchmark-level response paths. The exact146 detailed case study provides 13 result figures covering structure and modes, mean wind and zero-plane displacement d_0, spectra and coherence, representative time histories, force and equivalent static wind load (ESWL) profiles, ensemble statistics, cross-metric ratios, evolutionary power spectral density (EPSD) and background-resonant decomposition, generalized gust effect factor (GGFF) decomposition, verification errors, extreme value analysis and multi-modal multi-degree-of-freedom (MDMF) analysis. The Step 4c structural response review covers load chain division of labor, input object hierarchy, wind-speed-to-pressure route classification, response computation method classification and applicability boundaries. The evidence ceiling remains as follows: results depend on input load quality, structural model selection, damping assumptions and nonlinearity treatment, and belong to model-internal benchmarks rather than design responses.

DesignLoad has formed method demonstrations and candidate design expressions. The audit report covers the B2 dimensioned benchmark, P1 direct pressure case study, EPSD integration benchmark, full time-frequency I3 independent verification, GFF factor chain, load workflow closure matrix, and the C0 through C3 four-level research structure. 10 JSON data sources provide the raw data for all computations, and 33 figures cover results and processes. The evidence ceiling remains as follows: all outputs maintain candidate interpretation, not_approved and not_for_design status. The closure levels in the closure matrix are not design conclusions, eta values and C-GFF values have not been determined, and the formation of any design wind speed, design load or code recommendation requires Researcher approval after the upstream evidence chain is complete.

29.1.3 Research methodology level

The role of the Multi-Agent Human-in-the-loop method in this study can be understood along three dimensions. Execution continuity dimension: the division of labor among nine scientific agents and fifteen support/collaboration agents enables the knowledge bases, configuration files and computational results of each lane to accumulate continuously, without depending on memory retention within a single conversation. Check traceability dimension: source registers, claim matrices, handoff records, reading ledgers, figure ledgers and citation ledgers enable each significant statement in the main text to be traced back to specific data files, computational outputs or literature entries. Judgment explicit-return dimension: matters requiring scientific judgment (mechanism selection, method decisions, claim elevation, design expression approval) are explicitly returned to the Researcher through decision points and are not decided by agents on their own. The method's value arises not from tool counts or the automation level, but in making the organization and transfer of evidence checkable. The method does not produce new physical evidence, the Researcher's scientific responsibility has not been transferred, and the Coordinator does not possess scientific judgment authority. All products currently maintain `not_scientifically_approved` status.

The corresponding methodological relationship summarizes evidence maturity and ceilings for each work lane.

The maturity diagram needs to be read along vertical evidence chains rather than horizontal product counts. The observation that a particular lane possesses more literature, figures or programs only shows that the method preparation and recording for that lane are more complete, and does not show that its outputs possess higher physical credibility. The zero-input status of EVA,

the design-only status of WindTunnel, the case-limited status of CFD and the candidate status of DesignLoad collectively determine that the current chain terminus cannot form design quantities. The understanding at this stage therefore comprises two parts: the established framework, interfaces and diagnostics can continue to be reused, while unclosed event samples, scale conversions, measurement comparisons and per-response evidence must be maintained as vacancies. Any subsequent status elevation should point to specific objects and new evidence, and cannot automatically be triggered by version updates, product counts or program pass rates.

Evidence and formula boundaries of the current understanding Chapter 29 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "current understanding." Content without physical equations belongs to the research methodology, knowledge or coordination layer, and formalized formulas do not substitute for missing physical evidence. Chapter 29 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations or scientific conclusions approved by PI

30 Near-term research directions

The near-term research directions of PETW are determined by the scientific dependency order, meaning that the initiation or deepening of downstream steps must await sufficient evidence input from upstream steps. The discussion below describes, in dependency order from upstream to downstream, the specific content, required evidence, preconditions, Researcher decision items and the changes that advancing each direction may produce in subsequent claims.

30.1 Evidence, results and interpretation

30.1.1 Expansion and verification of event types

Current observational references are concentrated on BESSIE and a small number of events, and the coverage of event types is limited. The near-term direction is to expand coverage to more types of convective storm events (including different parent storm families F_s and surface event classes C_e) and to establish multi-source observational evidence for each event type. The expansion work includes: identifying meteorological radar, satellite and surface observation records for more candidate events; establishing standardized descriptions of event start and end times, spatial extent and observation sources; and verifying the consistency of mechanism hypotheses and classifications for existing events. The precondition is that existing multi-source data access paths continue to be available and that the data recovery process of the Events agent can handle new events. The Researcher needs to decide: which event types belong to the near-term priority scope; by what criteria mechanism hypotheses for candidate events enter subsequent steps; and whether the rules for determining event independence need to be systematized. If event type coverage is expanded, the output of Step 0 will expand from observational references for a small number of events to a structured information collection for multiple event types, which will directly affect the range of simulation object selection for Step 1 and the basis for extreme value sample composition for Step 3.

30.1.2 Conditional definition of Windfield scale conversion

A scale gap exists between the output grid resolution of meteorological simulations (typically on the order of hundreds of meters) and the spatial resolution required for engineering applications (typically at the building scale, on the order of meters to tens of meters). The near-term direction for Windfield is to provide a conditional definition of this scale conversion process, specifically to clarify: the assumptions on which the conversion depends (local homogeneity, terrain correction,

roughness treatment, etc.), the range of conditions under which the conversion is applicable (which wind field features retain physical meaning after conversion and which are not extractable at the current resolution), and how the uncertainty introduced by the conversion is quantified and transferred to the downstream. The precondition is that Step 2 has identified the key wind field features of each storm type and that the Researcher has decided how to handle source conflicts present in the literature definitions of these features. The Researcher needs to decide: the method family selection for scale conversion (statistical downscaling, physical downscaling or hybrid schemes); the identity definition of profile model parameters (from observational fitting, simulation extraction or literature recommended values); and the applicability boundary of stationarity assumptions in non-stationary thunderstorm wind scenarios. Once the conditional definition of scale conversion is completed, the interface from Step 2 to Step 3 and Step 4 will expand from the current feature vector to a complete packet with attached conversion conditions and uncertainty records, which will change the composition method of EVA samples and the generation method of CFD inlet conditions.

30.2 Limitations, interfaces and Researcher judgment

30.2.1 Near-term handling of uncertainties and conflicts

Near-term work should prioritize separating uncertainties by source rather than rushing to form a full-chain single range. Classification uncertainty is held by Events, simulation and scale conversion uncertainties are held by CM1, WRF and Windfield respectively, sample and model selection uncertainties are held by EVA, and experimental measurement, CFD numerical and structural model uncertainties are held by the corresponding lanes. DesignLoad can check whether these itemized components are complete but cannot compress them into a single correction coefficient when the relevant structure is absent.

Existing source conflicts should also be converted into controlled research questions. For definition conflicts, the comparison object, averaging time, height and units should first be fixed; for numerical conflicts, it should be checked whether they arise from different events, different versions or different processing; for methodological conflicts, parallel branches should be preserved under common input and common output metrics. The Researcher's decision may select one branch as the near-term baseline, but the branch set aside temporarily should still retain its identity and applicability conditions, so that baseline selection is not mistakenly written as resolution of the source conflict.

Status updates after near-term work is completed need to be executed on a per-claim basis. Obtaining more events does not automatically establish an EVA sample, completing a pilot does not automatically approve the full experimental matrix, CFD approaching agreement with a single condition does not automatically validate all cases, and real loads entering Structural do not automatically produce design responses. Status changes should specify which interpretation the new evidence supports, which alternative interpretations remain, which variables can be propagated downstream and which uses continue to be prohibited. This granularity enables the advancement of research directions to genuinely increase usable evidence rather than merely increasing products.

Resource allocation should follow this evidence increment logic. Work that can eliminate interface identity conflicts, obtain measured references, or establish a minimal common-event object is typically closer to chain closure than adding more outputs on isolated branches. Specific priorities are still determined by the Researcher based on data availability, facility conditions and scientific risk; computational resource availability does not automatically constitute justification

for initiation.

At the end of each round of near-term work, the dependency graph and blocking records should also be updated to confirm whether new evidence has changed the entry conditions for subsequent tasks. If it has not, the results should continue to be labeled as method preparation, diagnostic supplement or source refinement, avoiding the misreading of scheduling status as scientific status. This update keeps resource decisions in correspondence with evidence changes and also provides a read-back basis for the next round of selection.

Evidence and formula boundaries of near-term research directions Chapter 30 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "near-term research directions." Content without physical equations belongs to the research methodology, knowledge or coordination layer, and formalized formulas do not substitute for missing physical evidence. Chapter 30 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations or scientific conclusions approved by PI

31 Improvements to the Human-in-the-loop approach

During the course of using the Multi-Agent Human-in-the-loop method, this study has identified several improvement directions. These improvements target the implementation level of the method and do not change the fundamental principle that the Researcher decides scientific statements. The discussion below describes, across six areas of Researcher interface, cross-agent variable consistency, knowledge base and source management, figure reuse, multi-model comparison and external collaboration, the identified limitations, improvement ideas and expected effects. All improvement ideas are directions rather than already realized capabilities.

In the area of Researcher interface, the decision points currently returned by agents include scientific question descriptions, candidate options, existing evidence summaries and suggested actions. The improvement direction is to make the presentation of decision points more compact and structured, which allows the Researcher to locate key information more quickly during review. Proposed measures include: organizing the content of decision points into standard fields (problem statement, evidence basis, option list, impact scope and blocking conditions) so that decision points returned by different lanes are consistent in format; arranging multiple related decision points by dependency relationship within the same interface so that the Researcher can see the upstream-downstream connections among decisions; and embedding references to the reading ledger and claim matrix within decision points so that the Researcher can quickly trace back to supporting evidence. The expected outcome of improvement is to reduce the time overhead for the Researcher to understand decision point content, but not to reduce the number of judgments the Researcher needs to make.

In the area of cross-agent variable consistency, different lanes may currently use different symbols, unit conventions or semantic scopes when defining the same physical quantity. The improvement direction is to establish a cross-agent variable definition table so that the semantics of the same scientific variable are fully aligned across Events, CM1, Windfield, EVA, CFD, Structural and DesignLoad. Proposed measures include: each variable's definition entry contains symbol, name, physical quantity type, SI units, upstream source agent, downstream using agent, applicability range and the mapping path from variable to data file; the variable definition table is maintained at the Coordinator level, and any lane adding or modifying a variable definition

must pass a consistency check; interface handoffs explicitly reference variable definition entries so that the receiving party can verify the variable's meaning and units. The expected outcome of improvement is to reduce misreadings at interfaces caused by variable semantic inconsistencies, making cross-lane result comparison and propagation more reliable.

In the area of knowledge base and source management, current knowledge base updates depend on each agent actively consulting and recording during task execution. The improvement direction is to make the entry of new sources and new results more timely and systematic. Proposed measures include: establishing trigger mechanisms for source registration so that newly discovered literature, newly generated data files or newly completed computational results enter the reading ledger as pending verification entries at the time of production; automatically triggering source re-verification before important decisions, comparing whether the currently cited source version is consistent with the latest available version; and making the knowledge base update records themselves traceable so that the Researcher can audit which sources entered the knowledge base at what time and on what basis. The expected outcome of improvement is to reduce claim risk caused by outdated or omitted sources, but not to change the Researcher's final authority to approve claims.

In the area of figure reuse, result figures produced by upstream lanes currently require manual copying or regeneration when entering downstream reports. The improvement direction is to enable upstream result figures to enter downstream reports directly while preserving complete source information. Proposed measures include: each figure's entry in the figure ledger contains original data path, generation script path, original figure number, evidence class and claim limit, and these attributes are automatically inherited rather than re-annotated when the downstream makes references; the conversion process from English figure labels to Chinese annotations is standardized using the project's unified terminology correspondence table; and figures referenced across lanes retain the original figure number prefix and source lane identifier in downstream reports. The expected outcome of improvement is to reduce information loss and redundant work when figures are transferred across lanes.

In the area of multi-model comparison, current multi-model usage is primarily reflected in different agents selecting different language models for different tasks. The improvement direction is to make independent checking of the same scientific question by different models routine and to ensure that check results and discrepancy locations enter the records. Proposed measures include: for key claims or key computations, review is independently completed by at least two different models, with consistent portions and differing portions of results recorded separately; discrepancy locations are annotated with the nature of the discrepancy (wording difference, numerical difference, method selection difference or claim boundary judgment difference), and the Researcher needs to review only the differing portions rather than the full text; model selection basis and task matching records enter the model performance tracker so that subsequent tasks can reference historical performance rather than evaluating from scratch. The expected outcome of improvement is to increase the check coverage of key claims and reduce the risk of single-model bias.

In the area of external collaboration, External Reviewer comments currently enter the project workflow through the Coordinator. The improvement direction is to make the conversion path from external comments to scientific work more direct and more structured. Proposed measures include: classifying external review comments into four categories of factual corrections, method suggestions, claim challenges and supplementary literature, with each category having a corresponding

processing workflow and response timeline; processing results of external comments entering the review record so that the Researcher can track the status of each comment; and the communication path for external collaboration entering the Coordinator's fixed request path through the Collaboration Test Bot already configured in the Lark group, reducing fragmentation of communication channels. The expected outcome of improvement is to allow the scientific value of external collaboration to enter the project more fully, but adoption of external comments is still determined by the Researcher.

The improvements across these six areas share a common objective: reducing discontinuities and misreadings in the information transfer process, making evidence organization and checking more efficient. These improvements do not change the authoritative position of the Researcher in scientific judgment, do not change the fundamental principle that agent products remain in candidate status, and do not change the propagation discipline that evidence chain completeness determines claim ceilings.

31.1 Limitations, interfaces and Researcher judgment

31.1.1 Decision gate grading and consolidation

Decision gates need to be graded by scientific impact. Decisions affecting event admission, model form, experimental execution, result transfer-out and design implications have higher impact and should receive direct Researcher review of the complete evidence package; decisions affecting only file naming, plotting style or presentation-layer choices that do not change numerical values can be handled according to frozen rules. Grading does not diminish human authority but rather reduces the interference of low-risk routine matters with scientific review. Any automated handling rule must state its scope of applicability, and when source conflicts, variable semantic changes or claim ceiling changes are encountered, it must return to human judgment.

Related decisions can also be consolidated for presentation according to common objects. The mechanism identity, simulation route and Windfield conversion for one event are interrelated; if they appear separately as context-free requests, the Researcher needs to repeatedly reconstruct the background. A consolidated decision package should preserve the independent result of each decision while displaying dependency relationships, reversibility and impacts on the downstream. Consolidation should not compress multiple approvals into one blanket approval, and in particular should not allow resource execution authorization to substitute for scientific transfer-out judgment.

Decision records need to include the evidence version visible at the time. Subsequent source or computation updates may change the basis for judgment, and old decisions should still be preserved and annotated as to whether re-review is needed. If only the current conclusion is preserved, the project cannot distinguish whether a judgment change came from new evidence or from wording adjustments. Traceable decision versions also enable external reviewers to understand why a particular candidate branch was preserved or stopped at the time.

31.1.2 Boundary between automated checks and scientific review

Automated checks are suitable for processing formalizable conditions, such as citation key existence, unit field completeness, time axis monotonicity, array shape matching, figure source locatability and prohibited status propagation with packets. These checks can prevent obvious errors before objects enter the downstream and summarize exceptions for the Researcher. They cannot judge whether a particular parameter represents the target event, whether a particular model is suitable for the target mechanism, whether a particular experimental deviation is acceptable, or whether a candidate conclusion has design implications.

An improved system should classify check results as pass, fail and undeterminable. Undeterminable is used for conditions lacking scientific context or requiring original evidence interpretation, preventing the program from giving a pass with default values when information is insufficient. For failed items, the system reports the specific field, affected object and permissible repair; for undeterminable items, the system generates a bounded decision request. This distinction can reduce the risk of interpreting "no error detected" as "scientifically verified."

Cross-agent consistency checks should also allow intentional differences. Different lanes may use different time averaging, spatial support or normalization schemes because of different physical objects, and the objective is not to force all variables to adopt the same representation but rather to make differences explicit and convertible. The variable table should record equivalence conditions and prohibited conversions, and interface checks should execute conversions when conditions are met and otherwise preserve multiple objects. This can reduce situations where surface consistency masks erasure of physical semantics.

31.1.3 Verification of improvement effectiveness

Methodological improvements need to be evaluated through actual work records rather than solely by interface completion or program deployment. One can check whether the rework reasons for decision points have decreased, whether missing interface fields are detected earlier, whether source information is complete when figures are reused across lanes, whether multi-model discrepancies are accurately routed, and whether external comments can be traced back to specific versions. The evaluation object is the read-back capability and error localization ability of the organizational process, not the correctness rate of scientific conclusions.

Any improvement should first be piloted on a limited set of objects. One can select an existing benchmark or a candidate packet that does not involve facility actions and compare recording completeness, manual review burden and anomaly localization paths before and after the improvement. Pilot passage only supports expanding the scope of method usage and does not support changing scientific claims. If new tools introduce additional summary layers or automated conversions, a dedicated check should also verify whether conditions and prohibited uses are lost during compression and conversion.

The improvement objective of Human-in-the-loop is therefore not to reduce human judgment itself, but to ensure that each judgment faces materials with clear boundaries, complete sources and concentrated discrepancies. Agents handle preparation, checking, routing and recording, while the Researcher is responsible for scientific applicability and responsibility judgments that cannot be decided by formal rules. Both types of responsibilities are grounded in objects and versions, which enables maintaining the candidate status of all products while improving work continuity.

Evidence and formula boundaries of improvements to the Human-in-the-loop approach Chapter 31 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "improvements to the Human-in-the-loop approach." Content without physical equations belongs to the research methodology, knowledge or coordination layer, and formalized formulas do not substitute for missing physical evidence. Chapter 31 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations or scientific conclusions approved by PI

32 Extensions beyond PETW

The Multi-Agent Human-in-the-loop method employed in this study is not limited to the field of thunderstorm wind engineering. The core structure of the method is: the research problem is decomposed into multiple ordered steps, each step is executed by an agent with clearly defined responsibility boundaries, evidence and uncertainty are transferred between steps through clearly defined interface variables, and the Researcher maintains judgment authority at critical nodes. The applicability of this structure depends on whether the research problem itself possesses a set of structural preconditions. The discussion below examines these preconditions, potentially applicable fields, adaptation work required for migration, and the limitations of PETW as a case study.

32.1 Evidence, results and interpretation

32.1.1 Structural conditions for method applicability

The method is applicable to scientific engineering research possessing the following structural conditions. First, the research problem involves evidence articulation across multiple specialized stages, and conclusions from a single stage are insufficient to support the final judgment. The six-step pipeline of PETW is an instance of this condition: event classification, meteorological simulation, wind field characterization, statistical modeling, engineering conversion and design load expression respectively require knowledge from meteorology, computational fluid dynamics, structural engineering and statistics, and conclusions from any single step are insufficient to form design-level judgments. Second, each stage has clearly definable inputs, outputs and verification conditions, making the definition and checking of interface variables possible. The interface variables of PETW (event identity, simulated wind fields, wind field feature vectors, extreme value samples, inlet conditions, floor force time histories, candidate design expressions) are instances of this condition. Third, uncertainty needs to be transferred along the chain rather than truncated or hidden at a particular stage, so that the evidence ceiling of downstream conclusions is determined by upstream evidence. The nine-category uncertainty propagation of PETW is an instance of this condition. Fourth, the evidence ceiling of the final conclusion is determined by the weakest link and cannot be compensated by strengthening any single stage. Current status of PETW clearly embodies this condition: even though the method demonstrations of DesignLoad are quite complete, because the EVA sample has not yet been established and WindTunnel experiments have not yet been executed, candidate design expressions still cannot become design recommendations. Fifth, human judgment cannot be omitted, because cross-stage evidence integration involves scientific interpretation and responsibility attribution.

These five conditions are interrelated: multi-stage articulation generates the need for interface definitions, interface definitions generate the need for uncertainty transfer, uncertainty transfer generates the need for weakest-link constraints, and weakest-link constraints generate the need for human judgment at critical nodes. Research problems satisfying these five conditions are structurally suited to the organizational approach of agent division of labor, interface transfer and Researcher decision gates. Research problems that do not satisfy some of these conditions may not require such a complex organizational structure, or may be amenable to simpler automation schemes.

32.1.2 Potentially applicable research fields

Based on the structural conditions above, the following types of research fields structurally possess the possibility of applying the method, but applicability needs to be verified by actual evidence from each field, and cannot be inferred solely from structural similarity.

Cross-scale natural hazard engineering research is the category most structurally similar to PETW. In earthquake engineering, the chain from source mechanism, wave propagation, site effects, structural response to design spectra is structurally similar to the scientific chain of PETW, similarly involving multi-disciplinary stages, interface variables, uncertainty transfer and design-level judgments. In flood engineering, the chain from rainfall fields, hydrological models, flood routing, levee response to flood protection design also possesses similar structure. The common characteristic of these fields is: observational data and numerical simulations belong to different specialized domains, the conversion from physical phenomena to engineering design involves multiple scale crossings, and final conclusions affect safety design and therefore cannot be fully automated.

Complex materials and manufacturing process research may also possess applicable conditions. The chain from microstructure characterization, constitutive model calibration, manufacturing process simulation, quality inspection to performance prediction involves cross-disciplinary articulation of materials science, mechanics, manufacturing engineering and statistics; interface variables (microstructural parameters, constitutive parameters, process parameters, inspection indicators) can be clearly defined; and uncertainty progressively accumulates in the transfer from microscale to macroscale.

In environmental and ecosystem assessment, the chain from emission sources, transport models, exposure assessment, dose-response to risk characterization similarly involves multi-disciplinary evidence articulation, interface definition, uncertainty propagation and human judgment at the decision level. In such research, data sources for each stage are dispersed, methodological choices significantly affect conclusions, and final conclusions involve public health or policy decisions, thus requiring explicit evidence traceability and judgment nodes.

In multi-physics coupled engineering problems (such as thermal-structural-fluid coupling, electromagnetic-thermal-mechanical coupling), different physical fields are handled by different specialized tools and models, and the definition of coupling interfaces, data transfer and convergence checking constitute natural interface variables and verification conditions, while uncertainty propagation during coupling iterations requires explicit management.

The commonality between the fields listed above and PETW lies in the similarity of structural conditions, not in the similarity of scientific content. Method migration is not about replacing the agent names of PETW with terminology from other fields, but about identifying the pipeline structure, defining interface variables, establishing uncertainty propagation rules and determining the positions of Researcher judgment in the new field.

32.1.3 Limitations of PETW as a case study

PETW is one case study for the method, but PETW cannot on this basis be written as having proven the method to be a universally applicable autonomous science system. The case-study nature of PETW implies the following limitations.

The pipeline structure of PETW is designed for the thunderstorm wind engineering problem. Although the organizational principles of the pipeline (classification-first, interface transfer, uncertainty propagation, Researcher decision gates) possess generality, the specific step divisions,

interface variable definitions and evidence status vocabulary of the pipeline reflect the specific needs of the wind engineering field. These specific implementations cannot directly be transplanted to other fields.

The agent division of labor of PETW is implemented on a specific technical infrastructure (specific language models, specific project management tools, specific computing platforms). The effectiveness of the method may be influenced by infrastructure capabilities. Under different technical conditions, the capability boundaries, reliability and efficiency of agents may differ, and the applicability of the method may consequently change.

PETW is currently in the construction phase of the scientific chain and has not yet completed complete evidence transfer from events to design loads. The performance of the method on a complete evidence chain (including the accumulated magnitude of uncertainty after full-chain propagation, the convergence of cross-lane consistency checks after multiple iterations, and the judgment burden on the Researcher during full-chain review) has not yet been tested in practice. Accordingly, the validation of the method by PETW is limited to the organizational and recording capabilities during the construction phase, and does not include method performance over a complete scientific cycle.

Whether other fields are suited to adopting the method requires each field to independently assess the degree to which the structural conditions are satisfied and to accumulate evidence of method applicability through actual research in each field. PETW can provide references for organizational principles, interface design patterns and Researcher intervention strategies, but cannot substitute for each field's own method validation.

Evidence and formula boundaries of extensions beyond PETW Chapter 32 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "extensions beyond PETW." Content without physical equations belongs to the research methodology, knowledge or coordination layer, and formalized formulas do not substitute for missing physical evidence. Chapter 32 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations or scientific conclusions approved by PI

33 Concluding remarks

PETW organizes thunderstorm wind engineering research from single-step tool usage into a multi-step, multi-stage, mutually constrained evidence network through the classification-first principle and the scientific chain structure of Step 0 through Step 5. In this framework, event classification determines mechanism identity, mechanism identity is carried through all stages of meteorological simulation, wind field characterization, statistical modeling, engineering conversion and design load expression, and the propagation discipline of nine categories of uncertainty ensures that the evidence ceiling of each stage is determined by upstream inputs rather than by tool capabilities. Each stage is currently at a different level of maturity: Events has established observational references but the event catalog has not yet been formed, CM1 has completed mechanism screening but its outputs are model-internal, WRF has achieved a partially consistent reconstruction but it is not a complete reproduction, Windfield has established feature models but source conflicts are unresolved and scale conversion is not closed, EVA has established the method framework but maintains zero-input status, WindTunnel has completed the experimental design but has no physical experimental results, CFD has formed limited-case outputs but lacks controlled experimental verification, Structural has established benchmark-level response paths but depends on input load

quality, and DesignLoad has formed method demonstrations but all outputs maintain candidate status. Existing cases are distributed across different steps and cannot be assembled into end-to-end results for a common event; the near-term scientific priority is to close one minimal object with clear sources, continuous variable identity and a binding to a single response target, and to allow failure conditions, interface returns and unresolved conflicts to be preserved with the object. All products maintain `not_scientifically_approved` status, and the formation of any design wind speed, design load or code recommendation requires independent Researcher approval after the evidence chain is complete.

The Multi-Agent Human-in-the-loop method has provided three aspects of support in the organization and implementation of the above scientific framework: execution continuity (nine scientific agents and fifteen support/collaboration agents enable the knowledge, configurations and results of each stage to accumulate continuously), check traceability (source registers, claim matrices, handoff records and the ledger system enable main text statements to be traced back to data, computations and literature), and explicit return of judgment (matters requiring scientific judgment are returned to the Researcher through decision points rather than decided by agents on their own). The value of the method lies in making the organization and transfer of evidence checkable, not in tool counts or the automation level. Its organizational benefits are accompanied by risks including version maintenance, interface synchronization, summary condition loss and review burden, and therefore require joint control through object identity, failure records, prohibited uses and decision versions. Near-term application should also prioritize preserving common-event identity and per-response boundaries, avoiding the formation of a false chain impression from the complete outputs of different cases, and ensuring that conditions are traceable. The method does not produce new physical evidence, does not change the attribution of the Researcher's scientific responsibility, does not establish proof of an autonomous science system, and does not allow model agreement, program passage or product completeness to automatically become scientific approval. The scientific progress of PETW is ultimately determined by the physical evidence, observational data, experimental results and independent judgment of the Researcher at each stage, and the role of the multi-agent structure is to make the process of evidence organization, transfer and review more reliable and traceable.

Evidence and formula boundaries of Concluding remarks Chapter 33 retains only content directly related to the scientific questions, methods, evidence, limitations and interfaces of "Concluding remarks." Content without physical equations belongs to the research methodology, knowledge or coordination layer, and formalized formulas do not substitute for missing physical evidence. Chapter 33 remains a source-limited synthesis and cannot support verified design winds, design loads, code recommendations or scientific conclusions approved by PI

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